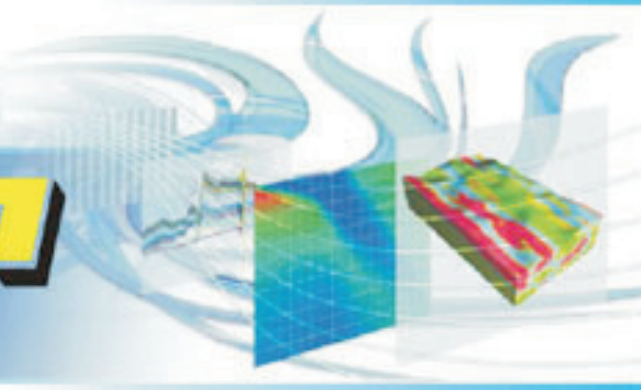
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DATA INTERPRETATION & SUFFICIENCY





Haripal Rawat

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DATA INTERPRETATION & SUFFICIENCY

By
Haripal Rawat

UPKAR PRAKASHAN, AGRA-2

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Data Interpretation & Data Sufficiency

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1 Introduction

Now a days, Data interpretation is an important aspect of every competitive examination. Usually, a table or a graph or a diagram is given with some facts or the required information and candidates are required to answer the questions that follow for the test of their ability of analysing the given information in the form of facts and figures.

Data—Data are the assemblage of facts at any one centered place. Generally, the facts are given in the form of a diagram whether it may be a figure of rows and columns or a form of a graph or a circular form or diagram.

For examples, the facts or the required information may be given in any form as follows—

(A) Study the following information which is a form of a data.

"In an organization consisting of 750 employees, the ratio of males to females is 8 : 7 respectively. All the employees work in five different departments *viz*. HR, Management, PR, IT and Recruitment, 16% of the females work in Management department, 32% of males are in HR department. One fifth of the females are in the department of recruitment. The ratio of males to females in the management department is 3 : 2 respectively, 20% of the total numbers of employees are in PR department; Females working in recruitment are 50% of the males working in the same department 8% of the males are in IT department. The remaining males are in PR department, 22% of the females work in HR department and the remaining females are working in IT department."

On the above information, any question or questions may be asked, *e.g.*—

What is the total number of females working in the IT and recruitment department together ?

(A) 147 (B) 83
(C) 126 (D) 45
(E) None of these

Data based on the facts or the information as above, will be discussed in detail in the chapter 6 : caselet.

(B) In the form of 'rows and columns' which is a tubular form of a data, *e.g.*—

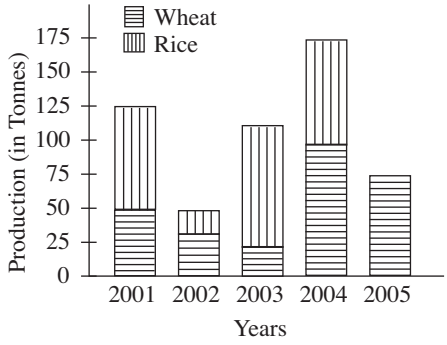
Number of Girls in Four Streams of a College Over the Years

Years	Streams			
	Arts	Science	IT	Commerce
2005	250	150	50	60
2006	300	125	55	70
2007	280	170	40	55
2008	350	120	35	50
2009	300	180	60	70

Questions based on the tabular form of data will be discussed in detail in the chapter 2 : Table.

(C) Any other form of a graphical or non graphical diagram, *e.g.*—

(1) **A graphical diagram of a data**—



On the above information, questions may be followed as—

(a) In which year, the production of rice is low ?

(A) 2002 (B) 2001
(C) 2005 (D) 2003
(E) 2004

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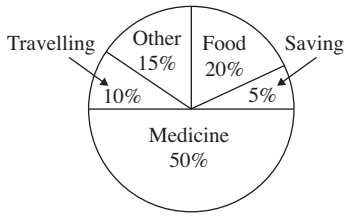
(b) What is the average production of wheat all over the years ?

(A) 25 tonnes (B) 50 tonnes

(C) 40 tonnes (D) 62 tonnes

(E) None of these

(2) **Pie diagram of a data—**



Monthly income = Rs. 20,000

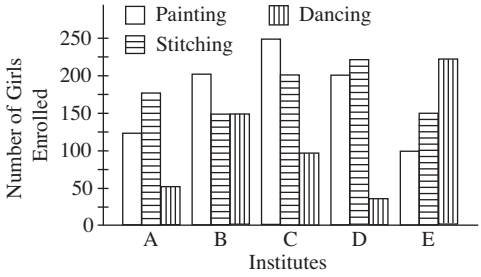
The above diagram shows the expenditure of the monthly income of a man—Different kinds of data and their relevant questions will be discussed in detail in their corresponding chapters. Now, we are discussing what *Data Interpretation is* ?

Data Interpretation—By the word 'Data-Interpretation' we mean understanding, organising and drawing appropriate conclusions from the given Data.

Actually, Data Interpretation is an act of extracting useful information and conclusions from the given data.

For example, Here we have a data in the form of following diagram.

Number of Girls Enrolled in Different Hobby Classes in Various Institutes in a Year—



By this diagram, we can find the important information or the conclusions easily, such as—

(i) The total number of girls in all the institutes.

(ii) The number of girls in the painting or the stitching or the dancing in all the institutes.

(iii) The respective ratio of total number of girls enrolled in painting, stitching and dancing from all the institutes together.

(iv) Number of girls enrolled in stitching in institute B forms what per cent of the total number of girls enrolled in stitching in all the institutes together.

(v) The other relevant conclusions that can be found from the diagram.

The act of finding important conclusions or the information from the above diagram is An Example of Data-interpretation.

Classification of Data—Generally, Data can be classified as—

(i) Tables
(ii) Graphs
(iii) Pie charts
(iv) Combination of diagrams
(v) Venn Diagram
(vi) Number Diagram
(vii) Caselets
(viii) Network Diagram
(ix) Scatter Diagram

Points to Remember

- For finding appropriate information or the conclusions from the given data, first of all we must have a cursory glance over the given data or the information figure and digest quickly what the diagram or the data represents.
- Take special care of units and points indicated in the graphical diagram.
- Read the questions that follow the data or the diagram carefully and answer accordingly.
- Many questions will be there which can be solved just by looking at the diagram or the data.
- Use mathematical means or the formulas, if necessary to collect the appropriate conclusions.

●●

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2 Table

Table—A table is the easier form used to summarise data in a meaningful way, it presents the data systematically in the form of rows and columns.

In the tabular form of the data, information or the facts are arranged in alphabetical or the chronological order.

Points to Remember

- Study the title of the table carefully that gives you a description of the contents of the table, kinds of data and the period for which it occurred.
- A dash or the blank indicates that corresponding data is not available.
- If you are arranging data in the form of a table, remember that the zero is always indicated by 0. A dash or the blank should never be indicated as zero.

Exercise on the Tabular Form of the Data

Exercise 1

Directions—Study the following table carefully and answer the questions given below it—

Crimes Registered in 2009 in the Various States

(Incidence and Rate per 100000 Population)

Crimes/States		UP	MP	Delhi	Bihar
Dacoity	Incidence	8800	2650	500	7800
	Rate	6·2	4·0	4	5·6
Murder	Incidence	9200	892	480	8200
	Rate	7·0	2·0	4·5	6·2
Rape	Incidence	7800	582	138	2850
	Rate	6·2	3·2	0·4	2·8

1. What is the average rate per hundred population of murder for all the given states ?
   (A) 0·00492 (B) 4·92
   (C) 0·492 (D) 49·2
   (E) None of these
2. What is the difference between the number of murder for UP and the murder of rape for Delhi ?
   (A) 1562 (B) 9262
   (C) 9062 (D) 962
   (E) None of these
3. What is the maximum number of the incidence of crimes per lac population for a which state ?
   (A) 24700 (B) 25800
   (C) 27500 (D) 26800
   (E) None of these
4. What is the percentage difference of incidence of dacoity in UP as compared with Bihar ?
   (A) 13% (B) 11%
   (C) 14% (D) 15%
   (E) None of these
5. Which state has the minimum rate of incidence for the crime of rape ?
   (A) MP (B) UP
   (C) Bihar (D) Delhi
   (E) None of these

Answers with Explanation

1. (A) Required average
$= \frac{7{\cdot}0 + 2{\cdot}0 + 4{\cdot}5 + 6{\cdot}2}{4}$
$= \frac{19{\cdot}7}{4}$
$= 4{\cdot}92$ per lac population
$\therefore$ Per hundred population
$= \frac{4{\cdot}92}{100000} \times 100$
$= 0{\cdot}00492$

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2. (C) The required difference
   = 9200 – 138 = 9062

3. (B) Number of the incidence of crimes in UP
   = 8800 + 9200 + 7800
   = 25800
   Number of the incidence of crimes in MP
   = 2650 + 892 + 582
   = 4124
   Number of the incidence of crimes in Delhi
   = 500 + 480 + 138
   = 1118
   Number of the incidence of crimes in Bihar
   = 7800 + 8200 + 2850
   = 18850
   ∴ Clearly the maximum number of incidence of the crimes has occurred in UP, *i.e.*, 25800.

4. (A) The required % difference
   $= \left(\frac{8800 - 7800}{7800}\right) \times 100$
   = 13% Approx.

5. (D) Dehli, *i.e.*, 0·4

Exercise 2

Directions—Study the following table carefully and answer the questions that follow—

The Aggregate 1003 Runs in the Tests Made by Sachin Tendulkar in the Year 2001

Opposition	Tests	Inning	Runs	Highest Score	Average	100s	50s
Australia	3	6	304	126	50·67	1	2
Zimbabwe	2	4	199	74	66·33	0	2
South Africa	2	4	193	155	64·33	1	0
England	3	4	307	103	76·75	1	2
Total	10	18	1003	155	62·60	3	6

Note—The average is calculated on as many innings in which the batsman loses his wicket.

1. What is the approximate ratio of the average runs of Australia to the average runs of Zimbabwe made by Sachin Tendulkar ?
   (A) 15 : 22 (B) 12 : 15
   (C) 17 : 22 (D) 22 : 17
   (E) None of these

2. How many percentage are the runs of England with the comparison to the total aggregate runs ?
   (A) 30% (B) 35%
   (C) 40% (D) 25%
   (E) None of these

3. For which apposition did Sachin Tendulkar had the minimum average of runs ?
   (A) Australia
   (B) Zimbabwe
   (C) South Africa
   (D) England
   (E) None of these

4. The approximate ratio of runs made by Sachin Tendulkar between England and South Africa is—
   (A) 15 : 7 (B) 11 : 7
   (C) 7 : 11 (D) 7 : 15
   (E) None of these

Answers with Explanation

1. (C) $\frac{A}{Z} = \frac{50·67}{66·33} = \frac{17}{22}$
   ⇒ A : Z = 17 : 22 (Approx.)

2. (A) The required percentage
   $= \frac{307 \times 100}{1003}$
   = 30% (Approx.)

3. (A) 30% Australia

4. (B) The required ratio
   $= \frac{\text{England}}{\text{S. Africa}}$
   $= \frac{307}{193} \Rightarrow \frac{4}{7}$
   ⇒ 11 : 7 (Approx.)

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Exercise 3

Directions—Study the following table carefully and answer the questions given below—

Number of Bales of Wool Processed by 5 Woolen Mills

Month	Name of the Mill				
	Polar	Shephered	Kiwi	Warmwear	Comfy
January	900	850	350	1000	850
Feburary	800	700	1050	1100	850
March	1050	800	1000	1100	950
April	800	850	850	1100	850
May	950	900	1050	1150	850
Total	4500	4100	4900	5450	4350

1. Which mill has the processing of wool in March the highest percentage of the total processing by that mill during the five months period ?
   (A) Polar (B) Shephered
   (C) Kiwi (D) Warmwear
   (E) Comfy
2. The wool processing by Warmwear in April is what per cent of its wool processing in the month of January ?
   (A) 91 (B) 110
   (C) 115 (D) 10
   (E) 11
3. Which of the five mills has the highest ratio of wool processing done in April to that done in February ?
   (A) Polar (B) Shephered
   (C) Kiwi (D) Warmwear
   (E) Comfy
4. In the case of which mill is the wool processing in February and March together the lowest among the five mills processing during the same period ?
   (A) Comfy (B) Warmwear
   (C) Kiwi (D) Shephered
   (E) Polar
5. The total of wool processing done by Kiwi during the given period is approximately what per cent of that done by Shephered ?
   (A) 80 (B) 87
   (C) 8 (D) 108
   (E) 120

Answers with Explanation

1. (A) Percentage processing of wool in the month of March by different mills—

$$\text{Polar} = \frac{1050 \times 100}{4500} = 23{\cdot}33\%$$

$$\text{Shephered} = \frac{800 \times 100}{4100} = 19{\cdot}51\%$$

$$\text{Kiwi} = \frac{1000 \times 100}{4900} = 20{\cdot}40\%$$

$$\text{Warmwear} = \frac{1100 \times 100}{5450} = 20{\cdot}18\%$$

$$\text{Comfy} = \frac{950 \times 100}{4350} = 21{\cdot}83\%$$

∴ The highest percentage is of the mill Polar.

2. (B) The required %

$$= \frac{1100 \times 100}{1000} = 110\%$$

3. (B) Seeing the table, we find that only Shephered shows less processing in February in comparison to the month of April. So, it gives the maximum ratio.
4. (D) Shephered shows the lowest processing in the month of February and March.
5. (E) The required%

$$= \frac{4900 \times 100}{4100} = 120\% \text{ (Approx.)}$$

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Exercise 4

Directions—The table given below shows a survey carried out at a railway station for the arrivals and departures of trains for the month of January 2000. Study the table and answer the following question—

Deley (in Min.)	Number of Arrivals	Number of Departures
0	1250	1400
0—30	114	82
30—60	31	5
Over 60	5	3
Total	1400	1490

1. The total number of late arrivals of trains is—
   (A) 90 (B) 95
   (C) 145 (D) 150
   (E) None of these
2. The total number of late departures of trains is—
   (A) 85 (B) 87
   (C) 90 (D) 150
   (E) None of these
3. The percentage of number of trains arriving late at the station is—
   (A) 6% (B) 10·4%
   (C) 10·7% (D) 10·9%
   (E) None of these
4. If the punctuality of railways is defined as the number of occasions on which trains arrived or departed in time as a percentage of total number of arrivals and departures from the station, then the punctuality for the month under observation is—
   (A) 94·3% (B) 91·7%
   (C) 89·2% (D) 75·0%
   (E) None of these

Answers with Explanation

1. (D) Total number of late arrivals
   = 1400 – 1250
   = 150
2. (C) Total number of late departures
   = 1490 – 1400
   = 90
3. (C) The required %
   $= \frac{150 \times 100}{1400}$
   = 10·7%
4. (B) The required %
   $= \left(\frac{1250 + 1400}{1400 + 1490}\right) \times 100$
   $= \frac{2650}{2890} \times 100$
   = 91·7%

Exercise 5

Directions—Study the following table and answer the questions that follow—

Yearly Production (in thousand) of Scooters in Different Factories

Factory	1985	1986	1987	1988	1989
P	20	15	24	13	17
Q	16	23	41	20	15
R	14	21	30	16	12
S	25	17	15	12	22
T	40	32	39	41	35
Total	115	108	149	102	101

1. In which year, the production of scooters of all factories was equal to the yearly average number of scooters produced during 1985-1989 ?
   (A) 1985 (B) 1986
   (C) 1987 (D) 1988
   (E) None of these
2. Which factory/factories showed a decreases of 25% in the—
   (A) P (B) S
   (C) Q and R (D) P and T
   (E) None of these
3. The ratio of the production of scooters by factory P to that by factory T in 1985 is—
   (A) 2 : 3 (B) 1 : 2
   (C) 3 : 2 (D) 2 : 1
   (E) None of these
4. In which year was the total production of scooters the maximum ?
   (A) 1989 (B) 1986
   (C) 1987 (D) 1985
   (E) None of these

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5. In which year was the total production of scooters of all factories 20% of the total production of scooters during 1985-1989 ?
   (A) 1988 (B) 1985
   (C) 1986 (D) 1989
   (E) None of these

Answers with Explanation

1. (A) The required average
$$= \frac{115 + 108 + 149 + 102 + 101}{5}$$
$$= \frac{575}{5} = 115$$
Hence, it was the year of 1985, when the production of scooter of all factories was equal to the above average.

2. (C) There are only three factories Q, R and T which showed decrease in the production in 1989 as compared to 1988
Percentage decrease in Q
$$= \frac{20 - 15}{20} \times 100$$
$$= 25\%$$
Percentage decrease in R
$$= \frac{16 - 12}{16} \times 100 = 25\%$$
Percentage decrease in T
$$= \frac{41 - 35}{41} \times 100$$
$$= 14{\cdot}63\%$$
$\therefore$ The factories showing a decrease of 25% in 1989 are Q and R only.

3. (B) The required ratio
$$= \frac{20}{40} = \frac{1}{2}$$
$\Rightarrow$ 1 : 2

4. (C) 1987

5. (B) The total production of scooters during 1985 – 1989
$= 115 + 108 + 149 + 102 + 101$
$= 575$
$\therefore$ 20% of 575
$$= \frac{20 \times 575}{100}$$
$= 115$
Hence, it was the year of 1985.

Exercise 6

Directions—Study the following table and answer the questions that follow—

Age Group (in years)	Magazines Read						Total Sample Surveyed (Including non-readers)	
	Sports		Film		Both			
	M	F	M	F	M	F	M	F
10—15	40	30	30	20	10	15	100	120
16—35	160	120	180	100	80	65	240	150
36—60	50	40	40	50	30	20	200	430

Note—M $\Rightarrow$ Male, F $\Rightarrow$ Female.

1. The number of people who read atleast one type of magazine and are over 35 years in age, is—
   (A) 36 (B) 130
   (C) 230 (D) 180
   (E) None of these

2. The number of people in the age group 10-15, who read only one type of Magazine, is—
   (A) 25 (B) 70
   (C) 95 (D) 120
   (E) None of these

3. The number of females in the age group 16-35 who do not read 'sports' Magazine is—
   (A) 120 (B) 90
   (C) 60 (D) 30
   (E) None of these

4. The number of males in the age group 16-35 who do not read 'Film' Magazine is—
   (A) 60 (B) 80
   (C) 140 (D) 190
   (E) None of these

5. What per cent of people over 35 years do not read either type of Magazine ?
   (A) 14% (B) 50·27%
   (C) 54% (D) 63·49%
   (E) None of these

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Answers with Explanation

1. (C) The required number of people
   = 50 + 40 + 40 + 50 + 30 + 20
   = 230
2. (D) The required number
   = 40 + 30 + 30 + 20
   = 120
3. (D) The required number
   = 150 – 120 = 30
4. (A) The required number
   = 240 – 180
   = 60
5. (D) Total people including non-readers over 35 years
   = 200 + 430
   = 630
   Total readers over 35
   = 50 + 40 + 40 + 50 + 30 + 20
   = 230
   ∴ Total readers over 35 years do not read either type of Magazine
   = 630 – 230
   = 400
   ∴ 400 out of 630 ⇒ 63·49%

Exercise 7

Directions—The following table showing expenditure details of a family during the years 1991 to 1995. Study the table carefully and answer the questions that follow—

S. No.	Item of Expenditure	Expenditure (in Rs. '000)					
		1991	1992	1993	1994	1995	Total
1.	Food	800	900	1050	1200	1400	5350
2.	House Rent	150	150	210	240	300	1050
3.	Clothing	75	100	130	170	250	725
4.	Fuel & Electricity	30	40	50	60	70	250
5.	Education	150	170	200	260	300	1080
6.	Medical Services	75	90	100	110	150	525
7.	Miscellaneous	220	250	260	360	430	1520
	Total	1500	1700	2000	2400	2900	10500

1. What is the per cent increase in expenditure on education from 1991 to 1995 ?
   (A) 50 (B) 75
   (C) 100 (D) 150
   (E) None of these
2. Considering the total expenditure for all the five years together, what is the per cent expenditure on House rent ?
   (A) 15 (B) 12
   (C) 10 (D) 8
   (E) None of these
3. There is no increase in expenditure in 1992 as compared to 1991 on item—
   (A) Food (B) House rent
   (C) Clothing (D) Medical services
   (E) None of these
4. In the light of the total expenditure for 1991, 1992, 1993, 1994 and 1995, what will be the likely expenditure in 1996 ?
   (A) Rs. 3000000 (B) Rs. 3200000
   (C) Rs. 3500000 (D) Rs. 3700000
   (E) None of these
5. Which item of expenditure accounted for the maximum part of total expenditure in all the five years ?
   (A) Clothing (B) Education
   (C) House rent (D) Food
   (E) None of these

Answers with Explanation

1. (C) The required % $= \frac{300 - 150}{150} \times 100$
   $= 100\%$

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2. (C) The required % $= \frac{1050 \times 100}{10500}$
$= 10\%$

3. (B) House rent

4. (C) Total expenditure follows the pattern—
$+200, +300, +400, +500$
$\therefore$ For the year of 1996, It follows + 600
The likely expenditure
$= 2900 + 600$
$= 3500$
$\Rightarrow$ Rs. 35,00,000

5. (D) Food

Exercise 8

Directions—Study the following table carefully to answer the questions that follow—

Populations (in thousands) of Six States Over the Years

Years	State					
	A	B	C	D	E	F
1992	125	210	85	150	98	138
1995	135	225	89	170	110	152
1998	142	240	93	180	130	160
2001	148	250	99	215	140	175
2004	155	270	105	230	145	190
2007	160	290	110	240	160	198

1. What was the average population of all the states together in 1998 ?
(A) 157500 (B) 175000
(C) 157200 (D) 172500
(E) None of these

2. Population of the state C in 2001 is approximately what per cent of the total population of all states together in the year ?
(A) 12 (B) 11
(C) 10 (D) 8
(E) 13

3. Approximately what is the per cent rise in population of state C in 2007 from 1995 ?
(A) 29 (B) 30
(C) 28 (D) 20
(E) 24

4. Which state had the highest per cent rise in population from 2001 to 2004 ?
(A) C (B) B
(C) D (D) F
(E) None of these

5. What is the average population of state D for all the years together ?
(A) 195700 (B) 197500
(C) 175900 (D) 179500
(E) None of these

Answers with Explanation

1. (A) Average population
$= \frac{142 + 240 + 93 + 180 + 130 + 160}{6}$ ths.
$= \frac{945}{6}$ ths.
$= 157500$

2. (C) Required % $= \frac{99 \times 100}{1027}\%$
$= 9{\cdot}64\%$
$\Rightarrow$ 10% (App.)

3. (E) Required rise % $= \frac{110 - 89}{89} \times 100$
$= 23{\cdot}59\%$
$\Rightarrow$ 24% (App.)

4. (D) For A% rise $= \frac{155 - 148}{148} \times 100$
$= 4{\cdot}73\%$
For B% rise $= \frac{270 - 250}{250} \times 100\%$
$= 8\%$
For C% rise $= \frac{105 - 99}{99} \times 100\%$
$= 6{\cdot}06\%$
For D% rise $= \frac{230 - 215}{215} \times 100\%$
$= 6{\cdot}98\%$
For E% rise $= \frac{145 - 140}{140} \times 100\%$
$= 3{\cdot}57\%$
For F% rise $= \frac{190 - 175}{175} \times 100\%$
$= 8{\cdot}57\%$
$\therefore$ State for highest % rise = F.

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5. (B) Average population

$$= \frac{150 + 170 + 180 + 215 + 230 + 240}{6} \text{ ths.}$$

$$= \frac{1185000}{6}$$

$$= 197500$$

Exercise 9

Directions—Study the table carefully to answer the questions that below—

Number of Workers Working During Six Months in Various Factories

(Number in Hundreds)

Months	Factories				
	A	B	C	D	E
January	65	41·2	72·8	63·5	83
February	78	30	61	60	74
March	42	65	71·6	76	70·3
April	51	72·8	83·5	21·8	66
May	60	68·2	61·6	80·2	56·9
June	63·5	52·5	73·2	57	44·7

1. What is the difference in the total number of workers working in various months from Factory A and the total number of workers working in various months from Factory E ?
   (A) 3540 (B) 3940
   (C) 3290 (D) 4230
   (E) None of these

2. What is the respective ratio of the total number of workers from Factories B and C working in the month of March and the total number of various working in the same month from Factories A and D ?
   (A) 5 : 6 (B) 238 : 345
   (C) 59 : 69 (D) 683 : 590
   (E) None of these

3. What is the total of the average of number of workers working in the month of January from all the Factories and the average of number of workers working in the month of April from all the Factories ?
   (A) 10098 (B) 11290
   (C) 12404 (D) 13516
   (E) None of these

4. What is the average number of workers working in various months from factory C ?
   (A) 70·55 (B) 7055
   (C) 6780 (D) 67·80
   (E) None of these

5. The total number of workers from Factory B is approximately what per cent of the total number of workers working from Factory D ?
   (A) 56 (B) 65
   (C) 76 (D) 84
   (E) 92

Answers with Explanation

1. (A) Total number of workers working in various months from Factory A
   = 359·5 (in hundred)
   Total number of workers working in various months from Factory E
   = 394·9 (in hundreds)
   Required difference = 394·9 – 359·5
   = 35·4 hundred
   = 3540

2. (D) Total number of workers from Factories B and C in March = 65 + 71·6
   = 136·6 (in hundreds)
   = 13660
   Total number of workers from Factories A and D in March
   42 + 76 = 118 × 100
   = 11800
   $\therefore$ Required ratio $= \frac{13660}{11800}$
   $= \frac{683}{590}$
   $\Rightarrow$ 683 : 590

3. (C) Average of number of workers working in January in all Factories

$$= \frac{65 + 41·2 + 72·4 + 63·5 + 83}{5}$$

$$= \frac{325·1}{5} = 65·02 \text{ hundreds}$$

   Average of number of workers working in April in all Factories

$$= \frac{51 + 72·8 + 83·5 + 21·8 + 66}{5}$$

$$= \frac{295·1}{5} = 59·02 \text{ (in hundreds)}$$

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Total of average of number of workers
= 65·02 + 59·02
= 124·04 hundreds
= 12404

4. (B) Average number of workers working in various months in Factory C

$$= \frac{72{\cdot}4 + 61 + 71{\cdot}6 + 83{\cdot}5 + 61{\cdot}6 + 73{\cdot}2}{6}$$

$$= \frac{423{\cdot}3}{6} = 70{\cdot}55 \text{ hundreds}$$

$\Rightarrow$ 7055

5. (E) Total number of workers from Factory B
= 329·7 hundreds
$\Rightarrow$ 32970
Total number of workers from Factory D
= 358·5 hundreds
$\Rightarrow$ 35850

$$\therefore \quad \text{Required } \% = \frac{32970 \times 100}{35850}\%$$

$$= \frac{65940}{717}\% = 91{\cdot}96\%$$

$\Rightarrow$ 92% (App.)

Exercise 10

Directions—Study the table carefully to answer the questions that follow—

Number of Students Appeared (A) and Qualified (Q) in an Examination from Various Institutes Over the Years

Institute	Years									
	2003		2004		2005		2006		2007	
	A	Q	A	Q	A	Q	A	Q	A	Q
B	1545	1240	1654	1566	1684	1500	1440	1165	1564	1462
C	1647	1106	1897	1689	1550	1278	1390	1072	1575	1388
D	1765	1567	1574	1024	1754	1210	1364	1145	1510	1214
E	1530	1234	1886	1542	1806	1586	1478	1388	1654	1296
F	1605	1356	2004	1930	1666	1498	1560	1389	1690	1480

1. Percentage of candidates qualified over appeared from Institute D is the lowest during which of the following years ?
(A) 2003 (B) 2004
(C) 2005 (D) 2007
(E) None of these

2. Approximately what is the percentage of candidates qualified over appeared from all the institutes together in 2007 ?
(A) 68 (B) 55
(C) 74 (D) 92
(E) 86

3. What is the difference between the number of students appeared but not qualified in the exam. from institute B in the year 2004 and the number of students appeared but not qualified in the exam. from the same institute in the year 2006 ?
(A) 187 (B) 88
(C) 275 (D) 373
(E) None of these

4. What is the approximate average number of candidates appeared for the exam. from institute E over the years ?
(A) 1759 (B) 1586
(C) 1671 (D) 1924
(E) 1837

5. What is the percentage of the candidates qualified over the number of candidates appeared for the exam in the year 2005 from all institutes together ?
(A) 92·34 (B) 73·47
(C) 66·94 (D) 83·59
(E) None of these

Answers with Explanation

1. (B) Year wise percentage of candidates qualified over appeared from institute D

$$2003 \Rightarrow \frac{1567 \times 100}{1765}$$

$$= 88{\cdot}78\%$$

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$$2004 \Rightarrow \frac{1024 \times 100}{1574}$$
$$= 65{\cdot}06\%$$
$$2005 \Rightarrow \frac{1210 \times 100}{1754}$$
$$= 68{\cdot}98\%$$
$$2006 \Rightarrow \frac{1145 \times 100}{1364}$$
$$= 83{\cdot}94\%$$
$$2007 \Rightarrow \frac{1214 \times 100}{1510}$$
$$= 80{\cdot}39\%$$

$\therefore$ Lowest percentage is in the year 2004.

2. (E) Number of students appeared in examination from all institutes in 2007
$= 1564 + 1575 + 1510 + 1654 + 1690$
$= 7993$
Number of students qualified from all institutes in 2007
$= 1462 + 1388 + 1214 + 1296 + 1480$
$= 6840$
$\therefore$ Required % of candidates
$$= \frac{6840 \times 100}{7993}$$
$= 85{\cdot}57\%$
$\Rightarrow$ 86% (App.)

3. (A) Number of students of institute B appeared but not qualified in 2004
$= 1654 - 1566 = 88$
Number of students of institute B appeared but not qualified in 2006
$= 1440 - 1165 = 275$
$\therefore$ Required difference
$= 275 - 88 = 187$

4. (C) Number of candidates appeared for exam from institute E over the years
$= 1530 + 1886 + 1806 + 1478 + 1654$
$= 8354$
$\therefore$ Required average
$$= \frac{8354}{5} = 1670{\cdot}8$$
$\Rightarrow$ 1671 (APP.)

5. (D) Number of candidates from all institutes appeared for exam in the year 2005
$= 1684 + 1550 + 1754 + 1806 + 1666$
$= 8460$
Number of candidates from all institutes qualified for exam in the years 2005
$= 1500 + 1278 + 1210 + 1586 + 1498$
$= 7072$
$\therefore$ Required % of the candidates
$$= \frac{7072 \times 100}{8460}$$
$= 83{\cdot}59\%$

Exercise 11

Directions—Study the table given below to answer the questions that follow—

Income (Rs.)	Tax (Rs.)
0—4000	1% of income
4000—6000	40 + 2% of income over 4,000
6000—8000	80 + 3% of income over 6000
8000—10,000	140 + 4% of income over 8000
10,000—15,000	220 + 5% of income over 10,000
15,000—25,000	470 + 6% of income over 15,000
25,000—50,000	1070 + 7% of income over 25,000

1. How much tax is due on an income of Rs. 7500 ?
(A) Rs. 80 (B)Rs. 125
(C) Rs. 150 (D)Rs. 225
(E) None of these

2. If your income for a year is Rs. 26,000. You receive a raise so that next year your income will be Rs. 29,000. How much more will you pay in taxes next year if the tax remains the same ?
(A) Rs. 70 (B) Rs. 180
(C) Rs. 200 (D) Rs. 210
(E) Rs. 250

3. Vibhav paid Rs. 100 as tax. If X is his income, then which of the following statements in true ?
(A) $0 < X < 4000$
(B) $4000 < X < 6000$
(C) $6000 < X < 8000$
(D) $8000 < X < 10,000$
(E) None of these

4. Town X has a population of 50,000. The average income of a person who lives in the town X is Rs. 3,700 per year. What is the

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total amount paid in taxes by the people of town X ?

(Assume that each person pays tax on Rs. 3,700)

(A) Rs. 37
(B) Rs. 3,700
(C) Rs. 1,85,000
(D) Rs. 18,50,000
(E) None of these

5. A person, whose income is Rs. 10,000, pays what per cent of his or her income on taxes approximately ?

(A) 1
(B) 2
(C) 3
(D) 4
(E) None of these

Answers with Explanation

1. (B) $80 + 3\%$ of $1500 = 80 + \frac{3 + 1500}{100}$
$= 80 + 45$
$=$ Rs. 125

2. (D) 7% of $3000 = \frac{7 \times 3000}{100}$
$=$ Rs. 210

3. (C) $6000 < X < 8000$

4. (D) $50,000 \times (1\%$ of $3700)$
$= 50,000 \times 37$
$=$ Rs. 18,50,000

5. (B) Income tax paid on Rs. 10,000
$=$ Rs. 220, which is
$\frac{220}{10,000} \times 100 =$ 2·2% of the income
$\Rightarrow$ $=$ 2% (App.)

Exercise 12

Directions—Study the following table care-fully to answer the questions that follow—

Distribution of Marks Obtained by 100 Students in Papers I, II and III Out of 50

Number of Students and Obtained Marks					
Paper	**40 and above**	**30 and above but less than 40**	**20 and above but less than 30**	**10 and above but less than 20**	**Less than 10**
I	12	18	42	20	8
II	16	19	38	17	10
III	11	24	44	15	6
Avg. of I, II and III	14	20	43	16	7

1. How many students have secured less than 30 marks in paper II ?

(A) 65 (B) 27
(C) 38 (D) 48
(E) None of these

2. How many students will pass if they one required to obtain minimum 60% only as average marks of three papers ?

(A) 14
(B) 20
(C) 21
(D) Cannot be determined
(E) None of these

3. How many students will definitely pass if it is compulsory to obtain minimum 20% marks in each paper ?

(A) 92
(B) 94
(C) 90
(D) Cannot be determined
(E) None of these

4. Minimum how many students will pass if they are required to obtain minimum 40% marks either in paper-I or in paper-III ?

(A) 72 (B) 73
(C) 77 (D) 79
(E) None of these

5. How many students will pass if it is compulsory to pass only in paper II with minimum 40% marks ?

(A) 38 (B) 73
(C) 35 (D) 16
(E) None of these

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Answers with Explanation

1. (A) $38 + 17 + 10 = 65$

2. (E) Passing marks $= 60\%$ of 50
$= \frac{60 \times 50}{100}$
$= 30$
$\therefore$ Number of students who got more than 30 average marks of three papers
$= 20 + 14$
$= 34$

3. (C) 90 because 10 students failed in paper-II.

4. (A) Minimum passing marks
$= \frac{50 \times 40}{100}$
$= 20$
$\therefore$ For the paper I, Number of students
$= 12 + 18 + 42$
$= 72$
For the paper-III, Number of students
$= 11 + 24 + 44$
$= 79$
$\therefore$ Minimum number $= 72$

5. (B)

Exercise 13

Directions—The following table shows the percentage population of six states below poverty line and the proportion of male and female. Study the table carefully and answer the questions that follow—

State	Percentage Population Below Poverty Line	Proportion of Male and Female	
		M : F Below Poverty Line	M : F Above Poverty Line
A	12	3 : 2	4 : 3
B	15	5 : 7	3 : 4
C	25	4 : 5	2 : 3
D	26	1 : 2	5 : 6
E	10	6 : 5	3 : 2
F	32	2 : 3	4 : 5

1. The total population of state A is 3000, then what is the approximate number of females above poverty line in the state ?
(A) 1200 (B) 2112
(C) 1800 (D) 1950
(E) None of these

2. The total population of the state C and the State D together is 18000, what is the total number of females below poverty line in the above states ?
(A) 5000 (B) 5500
(C) 4800 (D) Data inadequate
(E) None of these

3. The population of males below poverty line in state A is 3000 and that in state E is 6000, then what is the ratio of the total population of state A and E ?
(A) 3 : 4 (B) 4 : 5
(C) 1 : 2 (D) 2 : 3
(E) None of these

4. If the population of males below poverty line in state B is 500, what is the total population of that state ?
(A) 14,400 (B) 6000
(C) 8000 (D) 7600
(E) None of these

5. If in state E population of females above poverty line is 19,800, what is the population of males below poverty line in that states ?
(A) 5500 (B) 3000
(C) 2970 (D) Data inadequate
(E) None of these

Answers with Explanation

1. (E) Number of females above poverty line
$= 3000 \times (100 - 12)\% \times \frac{3}{7}$
$= \frac{3000 \times 88 \times 3}{100 \times 7}$
$= 1131$ (App.)

2. (D) Since we cannot find the population of states C and D separately, therefore we cannot find the required value.

3. (E) Population of the state A below poverty line
$= 3000 \times \frac{5}{3}$
$= 5000$

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$\therefore$ Total population of the state A

$$= \frac{5000 \times 100}{12}$$

The population of the state E below poverty line

$$= 6000 \times \frac{11}{6}$$

$$= 11{,}000$$

$\therefore$ Total population of state E

$$= \frac{11000 \times 100}{10}$$

$\therefore$ Required Ratio $= \frac{5}{12} \times \frac{10}{11} = \frac{25}{66}$

$\Rightarrow$ 25 : 66

4. (C) Total population of the state B

$$= 500 \times \frac{12}{5} \times \frac{100}{15}$$

$$= 8000$$

5. (B) Population of state E

$$= 19800 \times \frac{5}{2} \times \left(\frac{100}{100 - 10}\right)$$

$$= 55{,}000$$

$\therefore$ Population of males below poverty line

$$= 55000 \times \frac{10}{100} \times \frac{6}{11}$$

$$= 3000$$

Exercise 14

Directions—Study the table carefully to answer the questions that follow—

Number of Items Manufactured (M) and Sold (S) (in millions) by Six Different Companies Over the Years

Year	Company											
	A		B		C		D		E		F	
	M	S	M	S	M	S	M	S	M	S	M	S
2003	8·5	5·3	7·3	6·6	8·0	6·0	7·6	5·2	7·5	6·1	7·8	4·5
2004	8·3	6·2	7·9	6·2	8·1	5·8	8·3	5·7	8·0	6·6	7·8	5·0
2005	6·5	3·1	6·9	4·8	7·8	4·3	7·8	4·5	8·5	6·8	8·4	5·4
2006	7·2	5·2	8·3	5·3	7·9	4·6	7·9	4·8	6·7	5·4	8·2	6·2
2007	7·1	5·8	8·0	5·9	7·9	4·9	6·8	5·0	7·7	4·9	8·7	6·0
2008	8·0	6·2	8·2	6·1	7·6	6·0	7·5	6·1	7·9	4·9	6·5	4·2

1. What is the respective ratio of total number of items sold by Company A over all the years together to those sold by Company D over all the years together ?
   (A) 351 : 323 (B) 313 : 318
   (C) 289 : 296 (D) 291 : 263
   (E) None of these

2. Total number of items **not sold** by Company B over all the years together is **approximately** what per cent of total number of items manufactured by it over all the years together ?
   (A) 25 (B) 38
   (C) 12 (D) 42
   (E) 6

3. Number of items sold by Company E in the years 2006 and 2007 together is what per cent of the number of items manufactured by it in these years ? (rounded off to the nearest integer)
   (A) 61 (B) 35
   (C) 56 (D) 72
   (E) None of these

4. Which Company manufactured the highest number of items over all the years together ?
   (A) C (B) E
   (C) F (D) B
   (E) None of these

5. What is the number of items **not** sold by Company C in the year 2003 ?
   (A) 2000 (B) 20,00,000
   (C) 2,00,000 (D) 20,000
   (E) None of these

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Answers with Explanation

1. (E) Required ratio

$$= \frac{\text{Total number of items sold by company A over all the years}}{\text{Total number of items sold by company D over all the years}}$$

$$= \frac{(5{\cdot}3 + 6{\cdot}2 + 3{\cdot}1 + 5{\cdot}2 + 5{\cdot}8 + 6{\cdot}2) \text{ in million}}{(5{\cdot}2 + 5{\cdot}7 + 4{\cdot}5 + 4{\cdot}8 + 5{\cdot}0 + 6{\cdot}1) \text{ in million}}$$

$$= \frac{31{\cdot}8 \text{ million}}{31{\cdot}3 \text{ million}} = 318 : 313$$

2. (A) Required percentage

$$= \frac{\text{Total number of items \textbf{not} sold by company B over all the years}}{\text{Total number of items manufactured by company B over all the years}}$$

$$= \frac{(46{\cdot}6 - 34{\cdot}9)}{46{\cdot}6} \times 100\%$$

$$= \frac{11{\cdot}7 \times 100}{46{\cdot}6}\%$$

$$= 25{\cdot}107\% \simeq 25\%$$

3. (D) Required percentage

$$= \frac{\text{Total number of items sold by company E in the years 2006 and 2007}}{\text{Total number of items manufactured by it in these years}}$$

$$= \frac{(5{\cdot}4 + 4{\cdot}9) \text{ million}}{(6{\cdot}7 + 7{\cdot}7) \text{ million}} \times 100\%$$

$$= \frac{10{\cdot}3 \times 100}{14{\cdot}4} \times 100\%$$

$$= 71{\cdot}527\%$$

$$\simeq 72\% \quad \text{(Rounded to nearest integer)}$$

4. (C) Total number of items manufactured over all the years, by—

Company A
= 8·5 + 8·3 + 6·5 + 7·2 + 7·1 + 8·0
= 45·6 million

Company B
= 7·3 + 7·9 + 6·9 + 8·3 + 8·0 + 8·2
= 46·6 million

Company C
= 8·0 + 8·1 + 7·8 + 7·9 + 7·9 + 7·6
= 47·3 million

Company D
= 7·6 + 8·3 + 7·8 + 7·9 + 6·8 + 7·5
= 45·9 million

Company E
= 7·5 + 8·0 + 8·5 + 6·7 + 7·7 + 7·9
= 46·3 million

Company F
= 7·8 + 7·8 + 8·4 + 8·2 + 8·7 + 6·5
= 47·4 million

Hence, the highest number of items over all the years together, is manufactured by Company F.

5. (B) Total number of items **not** sold by company C in the year 2003.
= (8·0 – 6·0) million = 2 million
= 20,00,000

Exercise 15

Directions—Study the following table carefully to answer the questions that follow—

Table Giving Number of Candidates Appeared in the Examination and Percentage of Students Passed from Various Institutes Over the Years

Year	Institute											
	A		B		C		D		E		F	
	App.	% Pass	App.	% Pass	App.	% Pass	App.	% Pass	App.	% Pass	App.	% Pass
2001	450	60	540	40	300	65	640	50	600	45	680	60
2002	520	50	430	70	350	60	620	40	580	70	560	70
2003	430	60	490	70	380	50	580	50	680	70	700	66
2004	400	65	600	75	450	70	600	75	720	60	780	70
2005	480	50	570	50	400	75	700	65	700	48	560	50
2006	550	40	450	60	500	68	750	60	450	50	650	60
2007	500	58	470	60	470	60	720	70	560	60	720	50

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1. What is the ratio between the number of students passed from institute F in 2003 and the number of students passed from institute B in 2005 respectively ?
   (A) 95 : 154 (B) 154 : 95
   (C) 94 : 155 (D) 155 : 94
   (E) None of these

2. What is the ratio between the average number of students appeared from institute A for all the years and that from institute D respectively ?
   (A) 463 : 353 (B) 353 : 463
   (C) 461 : 333 (D) 333 : 461
   (E) None of these

3. What is the total number of students passed from all institutes together in year 2006 ?
   (A) 1895 (B) 1985
   (C) 1295 (D) 1465
   (E) None of these

4. What is the overall percentage of students passed from all institutes together in 2004 ? (rounded off to nearest integer)
   (A) 68 (B) 70
   (C) 69 (D) 71
   (E) None of these

5. **Approximately,** what is the overall percentage of students passed from institute C for all the years ?
   (A) 60 (B) 70
   (C) 75 (D) 55
   (E) 65

Answers with Explanation

1. (B) Reqd. ratio
$$= \frac{66 \square 700}{100} : \frac{50 \square 570}{100}$$
$$= 154 : 95$$

2. (D) Reqd. ratio
$$= \frac{3330}{7} : \frac{4610}{7}$$
$$= 333 : 461$$

3. (A) Reqd. number
$= 220+270+340+450+225+ 390$
$= 1895$

4. (C) Reqd. $\% = \frac{2453}{3550} \square 100\%$
$= 69{\cdot}09\%$
$\simeq 69\%$

5. (E) Reqd. $\% = \frac{1832 \square 100}{2850}\%$
$= 64{\cdot}28\%$
$\simeq 65\%$

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3 Bar Graph

The Dictionary defines the 'BAR' as a long piece of a thick wood or a metal. For our purpose, A 'bar' is, actually a thick line whose width is shown only for the attention by which we can observe the given figure easily.

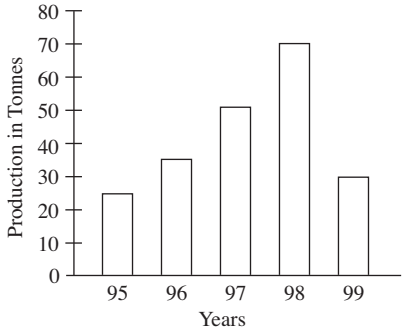
Bars are really just one dimensional as only the length of the bar matters important, not the width and may be horizontal or the vertical.

A bar graph is a well defined diagram of various bars depended on the given data. Generally, the respective figures are written at the end of each bar to facilitate the interpretation-easily, otherwise the figures are written only on the parallel axis. Mainly the bar graphs are of three types. These are—

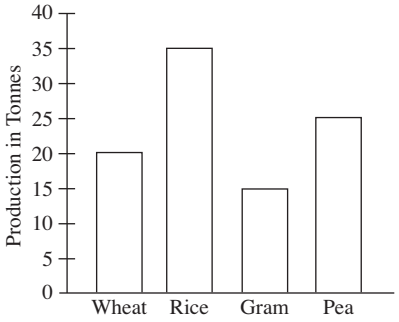
1. Simple Bar Graph
2. Component Bar Graph
3. Multiple Bar Graph

(1) **Simple Bar Graph**—In simple bar graph, one bar represents only one variable or one component, *viz*., one bar for only one item or matter or the number. Each and every bar remains separate to the other one.

For example 1.



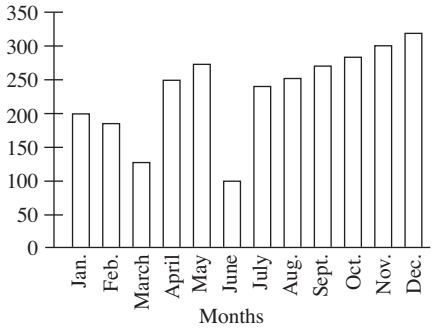
For example 2. The following simple Bar graph shows the production of wheat, rice, gram and pea in tonnes in the year of 2007.



For example 3. Production of scooters by a company in various months of a year is shown by this simple Bar graph.

Production of Scooters

(In thousands)

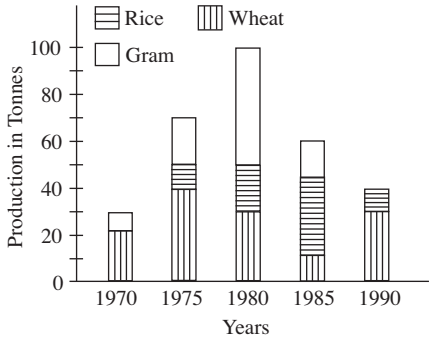


(2) **Component Bar Graph**—In component Bar Graph, the total magnitude of a bar is to be divided into two or more than two parts of sub classes. The bars are drawn proportional in length to the total and divided in the ratios of their components, *viz*., one bar for two or more than two items, or the matters, but each and every bar remains separate to the other one.

Component Bar Graph is also called sub divided Bar Graph.

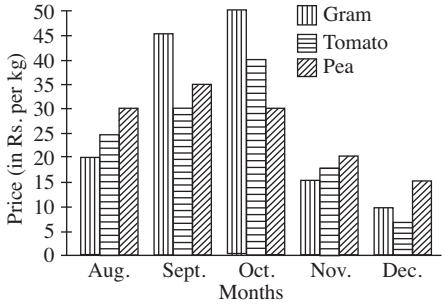
For example—The following diagram is a example of component Bar graph or the sub-divided Bar Graph of a town.

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(3) **Multiple Bar Graph**—In multiple Bar Graph, two or more than two bars make a unit compound of bars of the different items or the components by meeting each other with their respective magnitudes. A unit compound of bars remains a definite separation to the another unit of compound.

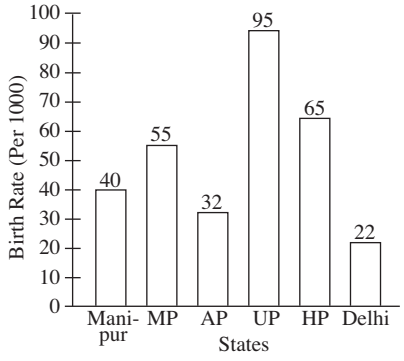
For example—The following multiple Bar graph shows the condition of different commodities during the last 5 months of the years 2008.



Exercise 1

Directions—Study the following graph carefully and answer the questions that follow—

Birth Rates of Different States



1. Which of the following state has 20% less the rate of birth than that of HP ?
   (A) AP (B) Manipur
   (C) MP (D) UP
   (E) None of these
2. The ratio of the state having highest birth rate to the state having lowest birth rate is—
   (A) 95 : 22 (B) 22 : 95
   (C) 5 : 7 (D) 7 : 5
   (E) None of these
3. What is the average Birth rates of all the states excepting UP ?
   (A) 40 (B) 43 (App.)
   (C) 45 (D) 44
   (E) None of these
4. The average birth rate is by what per cent greater or lower than the birth rate of UP ?
   (A) 43 (B) 50
   (C) 46 (D) 48
   (E) None of these
5. The pair of the birth rates of which of the following states is equal ?
   (A) Manipur and MP; UP
   (B) MP and AP; HP
   (C) HP and Delhi; MP
   (D) MP and Delhi; UP
   (E) None of these

Answers with Explanation

1. (E) HP $\Rightarrow$ 65

   $\therefore$ 20% of HP $= \frac{65 \times 20}{100}$

   $= 13$

   20% less than that of HP

   $65 - 13 = 52$

2. (A) The required Ratio $\Rightarrow$ UP/Delhi

   $= \frac{95}{22}$

   $\Rightarrow$ 95 : 22

3. (B) The average Birth rate excepting UP

   $= \frac{40 + 55 + 32 + 65 + 22}{5}$

   $= \frac{214}{5}$

   $= 43$ (App.)

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4. (C) Birth rate of UP = 95
The average birth rate = 51·50
$\Rightarrow$ The average birth rate < UP
$\therefore$ The lower = 95 – 51·50
= 43·50
$\therefore$ The lower % = $\frac{43{\cdot}50 \times 100}{95}$
= 46% (App.)

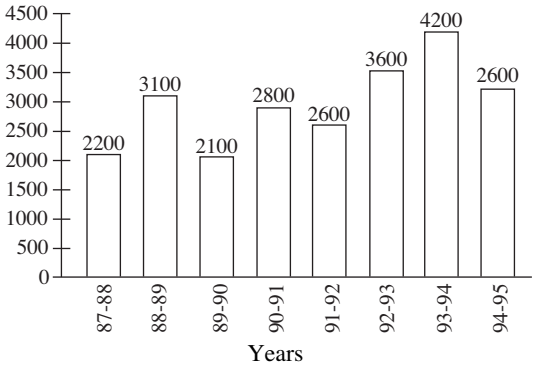
5. (A) Manipur and MP; UP.

Exercise 2

Directions—Study the following graph carefully and answer the questions that follow—

Trade Deficit of a Country

(In Rs. crores)



1. The deficit in 93-94 was roughly how many times the deficit in 90-91 ?
(A) 1·4 (B) 1·5
(C) 2·5 (D) 0·4
(E) None of these

2. The increase in deficit in 93-94 was how much per cent of the deficit in 89-90 ?
(A) 200 (B) 150
(C) 100 (D) 210
(E) None of these

3. In which of the following years, the per cent increase of deficit was highest over its preceding year ?
(A) 92-93 (B) 90-91
(C) 93-94 (D) 88-89
(E) None of these

4. The ratio of the number of years, in which the trade deficit is above the average deficit, to those in which the trade deficit is below the average deficit is—
(A) 3 : 5 (B) 5 : 3
(C) 4 : 4 (D) 3 : 4
(E) None of these

5. The deficit in 92-93 was approximately how much per cent of the average deficit ?
(A) 150 (B) 140
(C) 125 (D) 90
(E) None of these

Answers with Explanation

1. (B) If it is x times,
$4200 = x \times 2800$
$\Rightarrow x = \frac{4200}{2800} = \frac{3}{2}$
$= 1{\cdot}5$

2. (A) Let it is P% of deficit in 89-90
$\Rightarrow 4200 = \frac{P \times 2100}{100}$
$\Rightarrow P = 200$

3. (D) Per cent increase in deficit 92-93
$= \frac{1000}{2600} \times 100 = \frac{500}{13}$
$= 38\frac{6}{13}\%$
Per cent increase in 90-91
$= \frac{700}{2100} \times 100$
$= 33\frac{1}{3}\%$
In 93-94 $\frac{600}{3600} \times 100 = 16\frac{2}{3}\%$
In 88-89, $\frac{900}{200} \times 100 = 40\frac{10}{11}\%$

4. (A) Average deficit
$= \frac{2200 + 3100 + 2100 + 2800 + 2600 + 3600 + 4200 + 2600}{8}$
$= \frac{23200}{8}$
$= 2900$
In three years, the trade deficit is above 2900, and in the five years, it is below 2900.
$\therefore$ Required ratio = 3 : 5

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5. (C) If this $x\%$, then

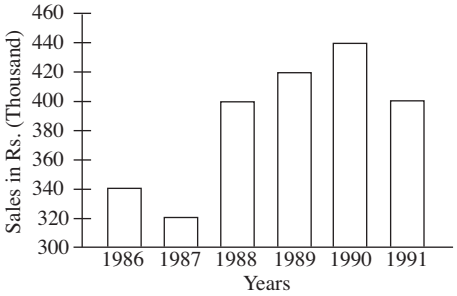
$$3600 = \frac{x \times 2900}{100}$$

$$\therefore \quad x = \frac{3600}{29}$$

$$= 125 \text{ (App.)}$$

Exercise 3

Directions—Study the following graph carefully and answer the questions that follow—



1. By how much amount are the sales in 1989 more than that in 1987 ?
   (A) Rs. 100 (B) 10,000
   (C) Rs. 1,00,000 (D) Rs. 10,00,000

2. The sales in 1987 are how many times to that in 1988 ?
   (A) 0·8 (B) 1·25
   (C) 8 (D) 0·25

3. In which year do the sales show the least per cent increase over those in the previous year ?
   (A) 1986 (B) 1988
   (C) 1989 (D) 1990

4. The ratio of the number of years for which the sales were above average to the number of years for which the sales were below average is—
   (A) 2 : 1 (B) 3 : 2
   (C) 4 : 3 (D) 1 : 2

5. What are the approximate average sales (in thousands) for the years 1988 to 1991 ?
   (A) 420 (B) 425
   (C) 430 (D) None of these

Answers with Explanation

1. (C) Sales in 1989 = Rs. 420 ths.
   Sales in 1987 = Rs. 320 ths.

   $\therefore$ Required amount
   = Rs. (420 – 320) × 1000
   = Rs. 1,00,000

2. (A) Let the required value is x,
   then $320 = x \times 400$

   $$\Rightarrow \quad x = \frac{320}{400}$$

   $$= 0{\cdot}8$$

3. (D) Increase from
   (i) 1987 to 1988 = 25%
   (ii) 1988 to 1989 = 5%

   (iii) 1989 to 1990 $= \frac{20 \times 100}{420}$

   $= 4{\cdot}76\%$

4. (A) The average sales

   $$= \frac{340 + 320 + 400 + 420 + 440 + 400}{6}$$

   $$= \frac{2320}{6}$$

   $$= 386{\cdot}66$$

   Sales are above average in 1988, 1989, 1990, 1991 and are below 1986, 1987

   $\therefore$ Required ratio = 4 : 2
   = 2 : 1

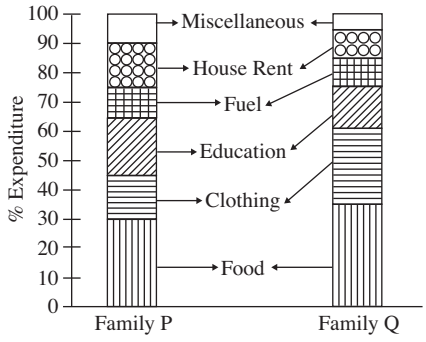
5. (D) Average $= \frac{400 + 420 + 440 + 400}{4}$

   $$= \frac{1660}{4}$$

   $$= 415$$

Exercise 4

Directions—Study the following graph carefully and answer the questions that follow—



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1. What fraction of the total expenditure is spent on education in family P ?

   (A) $\frac{13}{21}$ (B) $\frac{2}{3}$

   (C) $\frac{9}{13}$ (D) $\frac{1}{5}$

   (E) None of these

2. If the total expenditure on family Q is Rs. 1,000, then money spent on clothes by this family during the year is—

   (A) Rs. 200 (B) Rs. 600

   (C) Rs. 2000 (D) Rs. 6000

   (E) None of these

3. If the total annual expenditure of family P is Rs. 30,000, the money spent on food, clothes and house rent is—

   (A) Rs. 18,500 (B) Rs. 18,000

   (C) Rs. 21,000 (D) Rs. 15,000

   (E) None of these

4. If both the families have the same expenditure, which one spends more on education and miscellaneous together ?

   (A) P (B) Q

   (C) Both spends equal amount

   (D) Data inadequate

   (E) None of these

5. What percentage is Q's expenditure on food over P's expenditure on food, taking equal total of expenditure ?

   (A) 10% (B) 70%

   (C) 133·33% (D) 75%

   (E) 80%

Answers with Explanation

1. (D) Money spent on education in family P
   = 65 – 45
   = 20% of total expenditure
   = $\frac{1}{5}$ of the total expenditure

2. (C) Money spent on clothes by family Q
   = (60 – 40)% of total expenditure
   = Rs. $\frac{20}{100} \times 10{,}000$
   = Rs. 2000

3. (B) Money spent by P on food, clothes and House rent
   = [30 + (45 – 30) + (90 – 75)]% of total expenditure
   = 60% of Rs. 30,000
   = Rs. $\left(\frac{60}{100} \times 30{,}000\right)$
   = Rs. 18,000

4. (A) Money spent by P on education and miscellaneous
   = [(65 – 45) + (100 – 90)]%
   = 30%
   Money spent by Q on education and miscellaneous
   = [(75 – 60) + (100 – 95)]%
   = 20%
   ∴ Family P spends more on these heads.

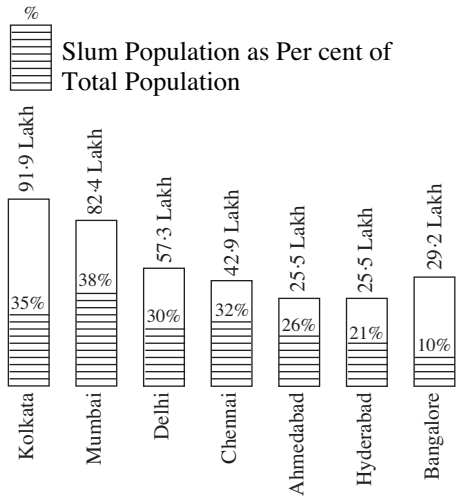
5. (C) Q's expenditure on food
   = 40%
   P's expenditure food
   = 30%
   Q's percentage over P's
   = $\left(\frac{40}{30} \times 100\right)$%
   = 133·33%

Exercise 5

Directions—Study the following graph carefully and answer the questions that follow—

Slum Population in Metropolis 1991

(in Lakh)



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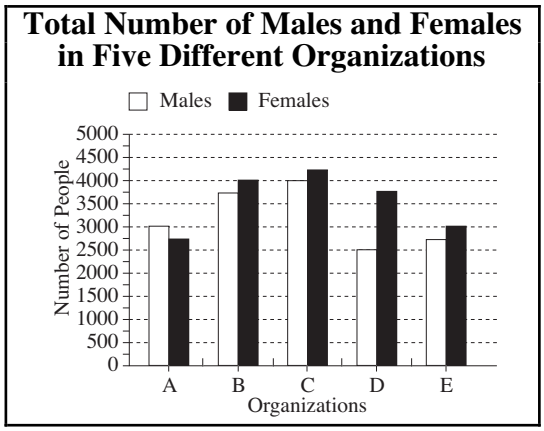
1. The total slum population of Kolkata in 1991 was approximately—
   (A) 30 lakh (B) 31 lakh
   (C) 32 lakh (D) 33 lakh
   (E) None of these
2. The difference in the slum population of Bangalore and Hyderabad was—
   (A) 4·1 lakh (B) 3·71 lakh
   (C) 2·43 lakh (D) 2 lakh
   (E) None of these
3. The city with the highest slum population was—
   (A) Mumbai (B) Kolkata
   (C) Delhi (D) Chennai
   (E) None of these
4. Two cities with nearly equal slum population were—
   (A) Ahmedabad and Hyderabad
   (B) Delhi and Chennai
   (C) Hyderabad and Bangalore
   (D) Mumbai and Kolkata
   (E) None of these
5. The slum population of Delhi was more than 3 times the slum population of—
   (A) Hyderabad (B) Ahmedabad
   (C) Bangalore (D) Chennai
   (E) None of these
6. The slum population of all the seven cities nearly equalled the total population of—
   (A) Kolkata and Bangalore
   (B) Delhi and Chennai
   (C) Delhi and Hyderabad
   (D) Mumbai and Ahmedabad
   (E) None of these
7. The ratio of slum population to total population in Kolkata was what times the same ratio in Bangalore ?
   (A) 3 (B) 3·5
   (C) 4 (D) 5
   (E) None of these
8. In terms of slum population, the second city with the least population was—
   (A) Delhi (B) Bangalore
   (C) Ahmedabad (D) Hyderabad
   (E) None of these

Answers with Explanation

1. (C) 35% of $91{\cdot}9 = \frac{35}{100} \times 91{\cdot}9$
   $= 32$ lakh (App.)
2. (C) 21% of $25{\cdot}5 - 10\%$ of $29{\cdot}2$
   $= 5{\cdot}355 - 2{\cdot}920$
   $= 2{\cdot}435$ lakh
3. (B) Slum population
   In Kolkata = 32·165 lakh
   In Mumbai = 31·312 lakh
   In Delhi = 17·190 lakh
   In Chennai = 13·728 lakh
   In Ahmedabad = 6·630 lakh
   In Hyderabad = 5·355 lakh
   In Bangalore = 2·920 lakh
4. (D) 5. (A)
6. (D) Total slum population
   = 109·3 lakh
   Mumbai + Ahmedabad
   = 107·9 lakh
7. (B) Let it is x times, then
   $\frac{32{\cdot}165}{91{\cdot}9} = x \times \frac{2{\cdot}92}{29{\cdot}2}$
   $\therefore \quad x = \frac{32{\cdot}165 \times 29{\cdot}2}{91{\cdot}9 \times 2{\cdot}92}$
   $= 3{\cdot}5$
8. (D)

Exercise 6

Directions—Study the following graph carefully to answer the questions that follow—



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1 What is the average number of females from all the Organizations together ?
(A) 3800 (B) 3550
(C) 3300 (D) 3150
(E) None of these

2. The number of males from Organization A is **approximately** what per cent of the total number of males from all the Organizations together ?
(A) 18 (B) 28
(C) 11 (D) 31
(E) 36

3. What is the difference between the total number of females and the total number of males from all the Organizations together ?
(A) 1500 (B) 1750
(C) 1800 (D) 2050
(E) None of these

4. What is the respective ratio of number of females from Organizations C to the number of females from Organization E ?
(A) 14 : 17 (B) 17 : 14
(C) 14 : 15 (D) 15 : 14
(E) None of these

5. The total numbers of males from Organizations A & B together are **approximately** what per cent of the total number of males from Organizations C, D and E together ?
(A) 58 (B) 75
(C) 69 (D) 83
(E) 52

Answers with Explanation

1. (E) Reqd. average
$= \frac{(2750 + 4000 + 4250 + 3750 + 3500)}{5}$
$= \frac{18250}{5}$
$= 3650$

2. (A) Reqd. %
$= \frac{3000 \times 100}{(3000 + 3750 + 4000 + 2500 + 3250)}$
$= \frac{3000 \times 100}{16500}\%$
$= 18{\cdot}18\% \simeq 18\%$

3. (B) Required difference
$= 18250 - 16500$
$= 1750$

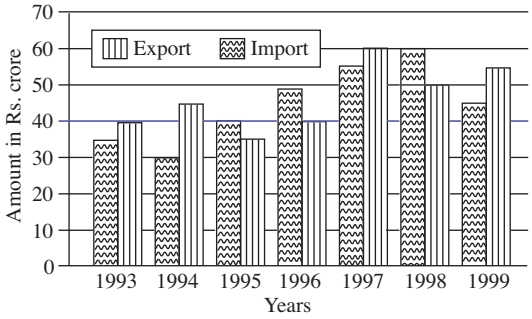
4. (B) Reqd. ratio $= \frac{4250}{3500}$
$= 17 : 14$

5. (C) Reqd. % $= \frac{6750 \times 100}{9750}\%$
$= 69{\cdot}23\%$
$\simeq 69\%$ (App.)

Exercise 7

Directions—Study the following graph carefully to answer the questions that follow—

Import and Export of Spare Parts by an Automobile Company Over the Given Years



1 During which year the percentage rise/fall in imports from the previous year is the lowest ?
(A) 1994 (B) 1998
(C) 1997 (D) 1995
(E) None of these

2. What is the ratio of total imports to total exports for all the given years together ?
(A) 31 : 35 (B) 35 : 31
(C) 65 : 63 (D) 63 : 65
(E) None of these

3. In which of the following pairs of years the total import is equal to total export in the same pair of years ?
(A) 1996-1997 (B) 1993-1998
(C) 1998-1999 (D) 1995-1996
(E) None of these

4. The total exports in the years 1995, 1996 and 1999 together are what per cent of the total

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import during the same period ? (up to two decimal places).

(A) 107·41 (B) 107·14
(C) 93·33 (D) 93·67
(E) None of these

5. Which of the following pairs of years and the per cent increase in the export over the previous year is correctly matched ?

(A) 1996-14·29 (B) 1997-10
(C) 1995-33·33 (D) 1994-11·11
(E) None of these

Answers with Explanation

1. (B) According to the graph.

2. (D) Total imports in the given years
= 35 + 30 + 40 + 50 + 55 + 60 + 45
= 315 crores
Total exports in the given years
= 40 + 45 + 35 + 40 + 60 + 50 + 55
= 325 crores
Hence, required ratio
$= \frac{315}{325} = \frac{63}{65} = 63 : 65$

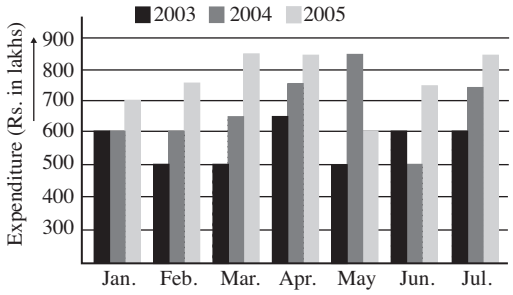
3. (C) Obvious from the graph.

4. (E) Total exports in the years 1995, 1996 and 1999
= 35 + 40 + 55
= 130 crores
Total imports in the years 1995, 1996 and 1999
= 40 + 50 + 45
= 135 crores
Now required % $= \frac{130 \times 100}{135}$
= 96·29%

5. (A) In 1996, % increase in export
$= \frac{5}{35} \times 100$
$= \frac{100}{7}$
= 14·29%

Exercise 8

Directions—The following graph gives expenditure of a company in the years 2003, 2004 and 2005 for the months January to July. Read the graph and answer the questions—



1. What is the total expenditure (Rs. in lakhs) of the company during the period January to July in the year 2003 ?

(A) 3,800 (B) 3,950
(C) 4,600 (D) 5,350

2. What is the average monthly expenditure (Rs. in lakhs) from January to July during the year 2005 ?

(A) 658·3 (B) 766·7
(C) 764·3 (D) 657·1

3. By what per cent is the expenditure in April, 2005 higher than that in the same month in 2004 ?

(A) $15\frac{5}{13}$ (B) $30\frac{10}{13}$
(C) $26\frac{2}{3}$ (D) $13\frac{1}{3}$

4. By what per cent is the expenditure in February, 2004 ?

(A) 20 (B) 25
(C) 13·3 (D) 23

Answers with Explanation

1. (B) Total expenditure from Jan. to July in 2003
= Rs. (600 + 500 + 500 + 650 + 500 + 600 + 600) lakh
= Rs. 3950 lakh

2. (C) Reqd. Average expenditure
$= \text{Rs.} \frac{700 + 750 + 850 + 850 + 600 + 750 + 850}{7}$ lakh

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$= \text{Rs. } \frac{5350}{7} \text{ lakh}$

$= \text{Rs. } 764{\cdot}3 \text{ lakh (Approx.)}$

3. (D) Reqd. higher %

$$= \frac{850 - 750}{750} \times 100\%$$

$$= 13\frac{1}{3}\%$$

4. (A) Reqd. lower %

$$= \frac{750 - 600}{750} \times 100\%$$

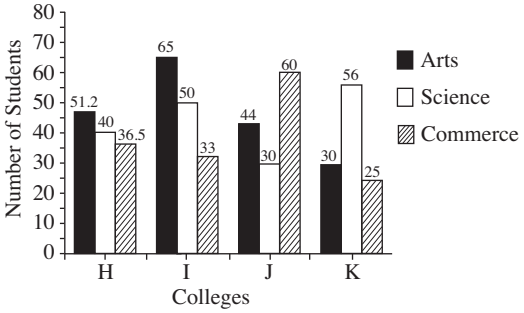
$$= 20\%$$

Exercise 9

Directions—Study the following graph carefully and answer the questions given below it.

Number of Students Studying in Various Colleges from Various Faculties

(Number in thousands)



1. What is the difference between the total number of students studying in college H and those studying in college K ?
   (A) 16100 (B) 15800
   (C) 16300 (D) 16700
   (E) None of these

2. What is the total number of students studying in all the colleges together ?
   (A) 520900 (B) 520700
   (C) 610200 (D) 510800
   (E) None of these

3. What is the respective ratio of the students from the faculty of Science from colleges H and I together to the students from the same faculty from colleges J and K together ?
   (A) 43 : 45 (B) 41 : 43
   (C) 45 : 43 (D) 43 : 41
   (E) None of these

4. The number of students from the faculty of Science from college I are **approximately** what per cent of the total number of students studying in that college ?
   (A) 34% (B) 36%
   (C) 80% (D) 40%
   (E) 42%

5. What is the average number of students from the faculty of Commerce from all the colleges together ?
   (A) 36825 (B) 38655
   (C) 35625 (D) 36585
   (E) None of these

Answers with Explanation

1. (D) Reqd. difference
   $= [(51{\cdot}2 + 40 + 36{\cdot}5) \sim (30 + 56 + 25)]$ thousand
   $= (127{\cdot}7 \sim 111)$ thousand
   $= 16{\cdot}7$ thousand
   $= 16700$

2. (B) Total number of students
   $= [51{\cdot}2 + 40 + 36{\cdot}5 + 65 + 50 + 33 + 44 + 30 + 60 + 30 + 56 + 25]$ thousand
   $= (127{\cdot}7 + 148 + 134 + 111)$ thousands
   $= 520{\cdot}7$ thousands
   $= 520700$

3. (C) Reqd. ratio $= \frac{(40 + 50)}{(30 + 56)}$

   $= \frac{90}{86} = 45 : 43$

4. (A) Reqd. % $= \frac{50 \times 100}{148}\%$

   $= 33{\cdot}78\%$

   $\simeq 34\%$ (App.)

5. (E) Reqd. average number

   $= \frac{36{\cdot}5 + 33 + 60 + 25}{4}$ thousand

   $= \frac{154{\cdot}5}{4}$ thousand

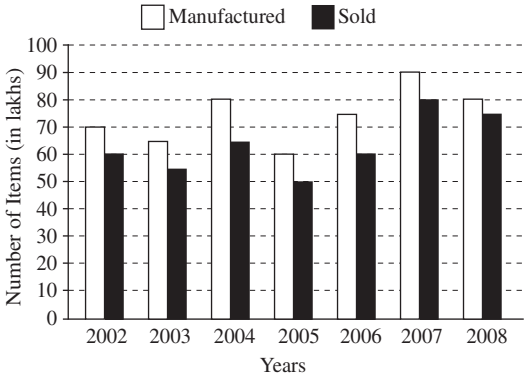
   $= 38625$

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Exercise 10

Directions—Study the following graph carefully to answer these questions—

Number of Items (in lakhs) Manufactured and Sold by a Company Over the Years



1. **Approximately** what is the average number of items unsold for all the years together ?
   (A) 10,50,000 (B) 10,55,000
   (C) 10,43,000 (D) 10,40,000
   (E) 10,70,000

2. **Approximately** what is the average number of items sold for all the years together ?
   (A) 60 lakhs (B) 61 lakhs
   (C) 63 lakhs (D) 67 lakhs
   (E) 69 lakhs

3. Number of items manufactured in 2007 is what per cent of the total number of items manufactured in all the years together ? (Rounded off to two digits after decimal)
   (A) 17·31 (B) 13·71
   (C) 17·03 (D) 13·97
   (E) None of these

4. What is the ratio between total number of items sold and the total number of items manufactured respectively in all the years together ?
   (A) 87 : 104
   (B) 89 : 102
   (C) 87 : 102
   (D) 89 : 104
   (E) None of these

5. During which year the percentage of items unsold was the highest ?
   (A) 2004 (B) 2006
   (C) 2008 (D) 2002
   (E) None of these

Answers with Explanation

1. (E) Average
$= \frac{(10 + 10 + 15 + 10 + 15 + 10 + 5)}{7}$ lakh
$= \frac{75}{7}$ lakh = 10·714 × 100000
$\simeq$ 1070000 (App.)

2. (C) Average
$= \frac{(60 + 55 + 65 + 50 + 60 + 80 + 75)}{7}$
$= \frac{445}{7}$ lakhs = 63·57 lakhs
$\simeq$ 63 lakhs (App.)

3 (A) Reqd. % $= \frac{90 \times 100}{520}\%$
$= 17{\cdot}307\%$
$= 17{\cdot}31\%$ (App.)

4. (D) Reqd. ratio = 445 : 520
= 89 : 104

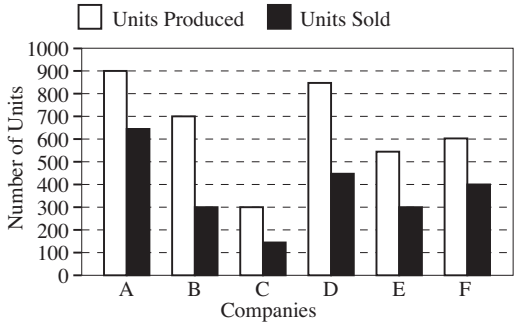
5. (B) % in 2002 $= \frac{10 \times 100}{70}\%$
$= 14{\cdot}3\%$
% in 2003 $= \frac{10 \times 100}{65}\%$
$= 15{\cdot}38\%$
% in 2004 $= \frac{15 \times 100}{80}\%$
$= 18{\cdot}75\%$
% in 2005 $= \frac{10 \times 100}{60}\%$
$= 16{\cdot}67\%$
% in 2006 $= \frac{15 \times 100}{75}\%$
$= 20\%$

Exercise 11

Directions—Study the following graph carefully to answer the questions that follow—

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Production and Sale of Printers of Various Companies in a Month



1 What is the average number of Units sold by all the Companies together ?
(A) 360 (B) 390
(C) 375 (D) 410
(E) None of these

2. Which Company had the highest percentage of sale with respect to its production ?
(A) D (B) B
(C) E (D) A
(E) None of these

3. What is the average number of Units produced by all the Companies together ?
(A) 675 (B) 650
(C) 625 (D) 600
(E) None of these

4. The total units sold by the Companies A, B and C together is **approximately** what per cent of the total units produced by these Companies ?
(A) 62 (B) 50
(C) 76 (D) 84
(E) 58

5. What is the respective ratio of the total production of companies D and E to the total sale of the same Companies ?
(A) 28 : 15 (B) 9 : 5
(C) 15 : 11 (D) 2 : 3
(E) None of these

Answers with Explanation

1. (C) Total units sold by all six companies
= (650 + 300 + 150 + 450 + 300 + 400)
= 2250

∴ Average number of units sold by all six companies
$= \frac{2250}{6} = 375$

2. (D) Percentage of sale with respect to its production

$A \rightarrow \frac{650 \times 100}{900}\% = 72{\cdot}2\%$

$B \rightarrow \frac{300 \times 100}{700}\% = 42{\cdot}8\%$

$C \rightarrow \frac{150 \times 100}{300}\% = 50\%$

$D \rightarrow \frac{450 \times 100}{850}\% = 52{\cdot}9\%$

$E \rightarrow \frac{300 \times 100}{550}\% = 54{\cdot}5\%$

$F \rightarrow \frac{400 \times 100}{600}\% = 66{\cdot}6\%$

∴ Company A had the highest percentage.

3. (B) Total units produced by all six companies
= (900 + 700 + 300 + 850 + 550 + 600)
= 3900

∴ Average number of units produced by all companies
$= \frac{3900}{6} = 650$

4. (E) Total units sold by A, B, C
= (650 + 300 + 150)
= 1100
Total units produced by A, B, C
= (900 + 700 + 300)
= 1900
∴ Required percentage
$= \frac{1100 \times 100}{1900}\%$
= 57·89%
≃ 58% (App.)

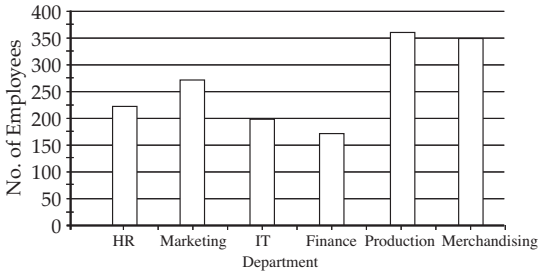
5. (A) Total production of companies D and E
= 850 + 550 = 1400
Total sale of the companies D and E
= 450 + 300 = 750
∴ Required ratio $= \frac{1400}{750} = \frac{28}{15}$
= 28 : 15

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Exercise 12

Directions—Study the graph carefully to answer the questions that follow—

Number of Employees Working in Different Departments of an Organization and the Ratio of Males to Females



Department	Males	Females
HR	9	16
Marketing	3	2
IT	9	31
Finance	2	3
Production	11	4
Merchandising	4	3

1. What is the total number of Males working in all Departments together ?
   (A) 755 (B) 925
   (C) 836 (D) 784
   (E) None of these

2. What is the number of Females working in the HR department ?
   (A) 158 (B) 128
   (C) 136 (D) 144
   (E) None of these

3. What is the respective ratio of total number of employees working in the production department to those working in the Merchan dising department ?
   (A) 15 : 14 (B) 8 : 7
   (C) 14 : 15 (D) 7 : 8
   (E) None of these

4. In which Department are the lowest number of Females working ?
   (A) Marketing (B) Production
   (C) HR (D) Finance
   (E) None of these

5. What is the total number of employees from all Departments together in the Organization ?
   (A) 1500 (B) 1575
   (C) 1525 (D) 1625
   (E) None of these

Answers with Explanation

1. (C) Reqd. number
   = 81 + 165 + 45 + 70 + 275 + 200
   = 836

2. (D) No. of Females in HR deptt.
   $= \frac{225 \times 16}{(9 + 16)}$
   = 144

3. (A) Reqd. ratio $= \frac{375}{350}$
   = 15 : 14

4. (B) No. of females in HR
   $= \frac{16}{25} \times 225 = 144$
   No. of females in Marketing
   $= \frac{2}{5} \times 275 = 110$
   No. of females in IT
   $= \frac{31}{40} \times 200 = 155$
   No. of females in Finance
   $= \frac{3}{5} \times 175 = 105$
   No. of females in Production
   $= \frac{4}{15} \times 375$
   = **100** (Lowest)
   and No. of females in Merchandising
   $= \frac{3}{7} \times 350$
   = 150

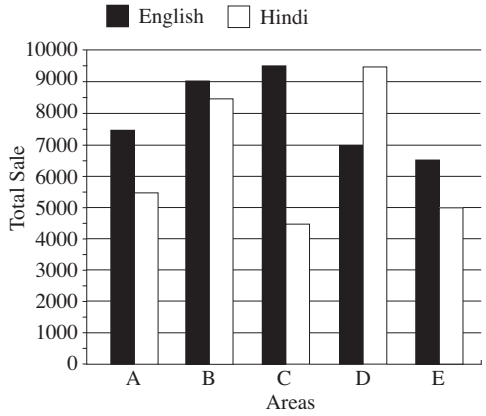
5. (E) Reqd. number
   = 225 + 275 + 200 + 175 + 375 + 300
   = 1550

Exercise 13

Directions—Study the following graph carefully to answer the questions that follow—

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Total Sale of English and Hindi Newspaper in Five Different Localities of a City



1. The sale of English Newspaper in Localities B & D together is **approximately** what per cent of the sale of English Newspaper in Localities A, C and E together ?
   (A) 162 (B) 84
   (C) 68 (D) 121
   (E) 147

2. What is the difference between the total sale of English Newspapers and the total sale of Hindi Newspapers in all the Localities together ?
   (A) 6000 (B) 6500
   (C) 7000 (D) 7500
   (E) None of these

3. The sale of English Newspaper in Locality A is **approximately** what per cent of the total sale of English Newspapers in all the Localities together ?
   (A) 527 (B) 25
   (C) 111 (D) 236
   (E) 19

4. What is the average sale of Hindi Newspaper in all the Localities together ?
   (A) 6600 (B) 8250
   (C) 5500 (D) 4715
   (E) None of these

5. What is the respective ratio of the sale of Hindi Newspaper in Locality A to the sale of Hindi Newspaper in Locality D ?
   (A) 11 : 19 (B) 6 : 5
   (C) 5 : 6 (D) 19 : 11
   (E) None of these

Answers with Explanation

1. (C) Reqd. % $= \frac{(9000 + 7000) \times 100}{(7500 + 9500 + 6500)}\%$

$= \frac{1600000}{23500}\% = 68{\cdot}08\%$

$\simeq$ 68% (App.)

2. (B) Reqd. difference = 39500 – 33000

= 6500

3. (E) Reqd. % $= \frac{7500 \times 100}{39500}\%$

$= 18{\cdot}987\%$

$\simeq$ 19% (App.)

4. (A) Reqd. average

$= \frac{(5500 + 8500 + 4500 + 9500 + 5000)}{5}$

$= \frac{33000}{5} = 6600$

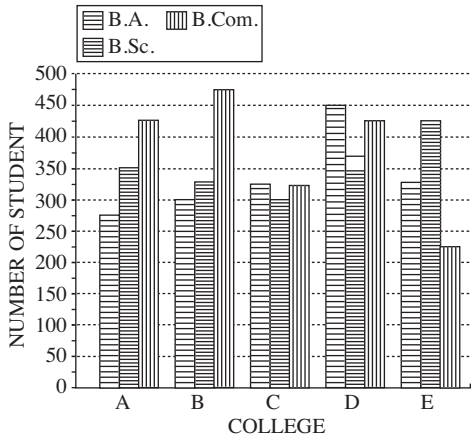
5. (A) Reqd. ratio = 5500 : 9500

= 11 : 19

Exercise 14

Directions—Study the following graph carefully to answer the questions that follow—

Number of Students Enrolled in Three Different Disciplines in Five Different Colleges



1. What is the total number of students studying B.Sc. in all the Colleges together ?
   (A) 1825 (B) 1975
   (C) 1650 (D) 1775
   (E) None of these

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2. What is the respective ratio of total number of students studying B.Sc. in the Colleges C and E together to those studying B.A. in the Colleges A and B together ?
   (A) 24 : 23 (B) 25 : 27
   (C) 29 : 23 (D) 29 : 27
   (E) None of these

3. What is the respective ratio of total number of students studying B.Sc. and B.Com. in all the Colleges together ?
   (A) 71 : 67 : 75 (B) 67 : 71 : 75
   (C) 71 : 68 : 75 (D) 75 : 71 : 68
   (E) None of these

4. Number of students studying B.Com. in College C forms **approximately** what per cent of the total number of students studying B.Com. in all the Colleges together ?
   (A) 39 (B) 21
   (C) 44 (D) 33
   (E) 17

5. Number of students studying B.A. in College B forms what per cent of total number of students studying all the disciplines together in that College ? (rounded off of two digits after decimal)
   (A) 26·86 (B) 27·27
   (C) 29·84 (D) 32·51
   (E) None of these

Answers with Explanation

1. (D) Required number
   = 350 + 325 + 300 + 375 + 425
   = 1775

2. (C) Required ratio $= \frac{300 + 425}{275 + 300} = \frac{725}{525}$
   = 29 : 23

3. (A) Required ratio = 1775 : 1675 : 1875
   = 71 : 67 : 75

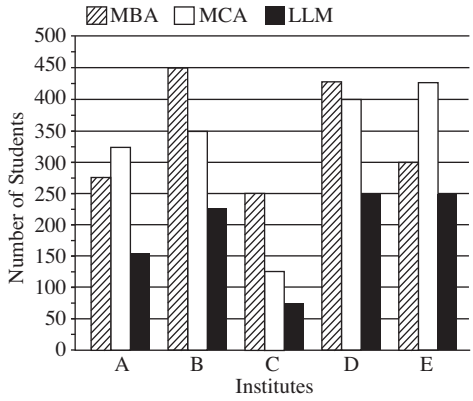
4. (E) Required % $= \frac{325 \times 100}{1875}$
   = 17

5. (B) Required % $= \frac{300 \times 100}{300 + 325 + 475}$
   $= \frac{30000}{1100}$
   = 27·27

Exercise 15

Directions—Study the following graph carefully to answer the questions that follow—

Number of Students Enrolled in Three Different Disciplines in Five Different Institutes



1. Number of students studying MCA in Institute D forms what per cent of total number of students studying all the disciplines together in that Institute ?
   (Rounded off to two digits after decimal)
   (A) 38·85 (B) 40·48
   (C) 37·21 (D) 36·36
   (E) None of these

2. Number of students studying MCA in Institute E forms **approximately** what per cent of the total number of students studying MCA in all the Institutes together ?
   (A) 42 (B) 26
   (C) 38 (D) 12
   (E) 20

3. What is the respective ratio of total number of students studying LLM in the Institutes C and E together to those studying MBA in the Institutes A and B together ?
   (A) 2 : 5 (B) 7 : 6
   (C) 2 : 1 (D) 13 : 29
   (E) None of these

4. What is the total number of students studying MBA in all the Institutes together ?
   (A) 1800 (B) 1725
   (C) 1875 (D) 1650
   (E) None of these

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5. What is the respective ratio of total number of students studying MBA, MCA and LLM in all the Institutes together ?
   (A) 68 : 65 : 38 (B) 68 : 38 : 65
   (C) 68 : 61 : 38 (D) 68 : 38 : 61
   (E) None of these

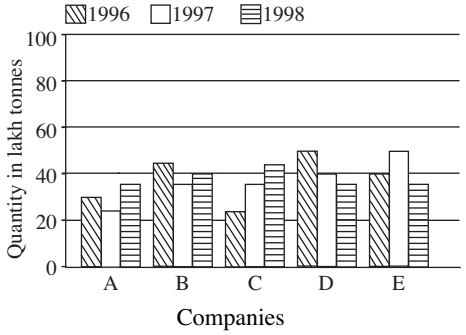
Answers with Explanation

1. (C) 2. (B)
3. (D) Required ratio $= \frac{75 + 250}{275 + 450} = \frac{325}{725}$
   $= 13 : 29$
4. (E) Required number
   $= 275 + 450 + 250 + 425 + 300$
   $= 1700$
5. (A) Required ratio $= 1700 : 1625 : 950$
   $= 68 : 65 : 38$

Exercise 16

Directions—Study the following graph carefully to answer the following questions—

The Production of Fertilizer in Lakh Tonnes by Different Companies for Three Years 1996, 1997 and 1998



1. The total production by five companies in 1998 is what per cent of the total production by companies B and D in 1996 ?
   (A) 100% (B) 150%
   (C) 95% (D) 200%
   (E) None of these
2. What is the ratio between average production by Company B in three years to the average production by company C in three years ?
   (A) 6 : 7 (B) 8 : 7
   (C) 7 : 8 (D) 7 : 6
   (E) None of these
3. For which of the following companies the rise or fall in production of fertilizer from 1996 to 1997 was the maximum ?
   (A) A (B) B
   (C) C (D) D
   (E) E
4. What is the per cent drop in production by Company D from 1996 to 1998 ?
   (A) 30 (B) 43
   (C) 50 (D) 35
   (E) None of these
5. The average production for three years was maximum for which of the following companies ?
   (A) B only (B) D only
   (C) E only (D) B and D both
   (E) D and E both

Answers with Explanation

1. (D) Required percentage
   $= \frac{35 + 40 + 45 + 35 + 35}{45 + 50} \times 100$
   $= \frac{190}{95} \times 100$
   $= 200\%$
2. (B) Average production by B
   $= \frac{45 + 35 + 40}{3}$
   $= 40$
   Average production by C
   $= \frac{25 + 35 + 45}{3} = 35$
   Ratio $= (40 : 35) = 8 : 7$
3. (C) **Quicker Approach**—Maximum difference 10 lakh tonnes for the three companies C, D and E. So, our answer should be the company for which the production is least in 1996. Because to calculate the % increase or decrease our denominator is the production in 1996.
4. (A) Percentage drop $= \frac{50 - 35}{50} \times 100$
   $= 30\%$
5. (E)

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4 Line Graph

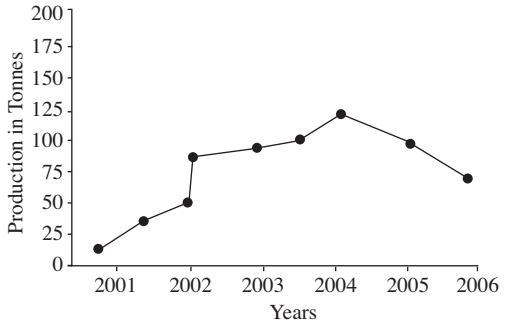
Line Graph—Line Graph represents a pictorial presentation of the given data. It is also called a cartesian graph of pictorial representations.

Generally, a line graph indicates the variation of a quantity or a magnitude with respect to two parameters caliberated on the axes X and Y respectively.

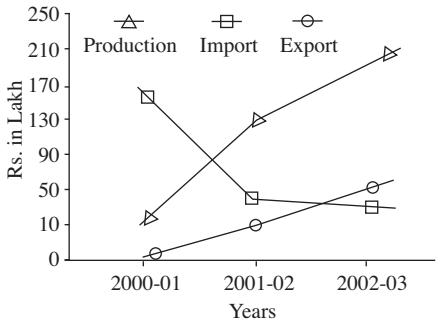
If it is drawn with the help of only a single line, It is called a **Single Line Graph** or a **Simple Line Graph**.

If the graph has at least two or more than two drawee lines, it is called a **Multiple Line Graph**.

Example 1. The following graph is an example of a single line graph.



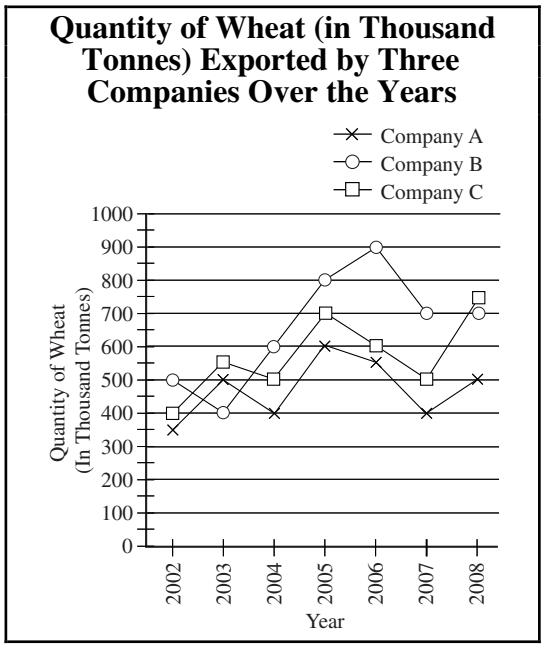
Example 2. The following graph is an example of a multiple line graph.



The Pictorial Lines Show the Trends in Production, Import and Export

Exercise 1

Directions—Study the following graph carefully and answer the questions that follow—



1. What is the per cent increase in exports of company C from 2004 to 2008 ?
   (A) 50 (B) 33·33
   (C) 150 (D) 133·33
   (E) None of these

2. Total exports of company A for all the years are **approximately** what per cent of the total exports of company B for all the years ?
   (A) 75 (B) 128
   (C) 139 (D) 68
   (E) 72

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3. Per cent rise in exports from the previous year was the maximum during which year for company 'B' ?
   (A) 2005 (B) 2004
   (C) 2006 (D) 2008
   (E) None of these

4. What are the average exports of company B for all the years ? (in thousand tonnes rounded off to two digits after decimal)
   (A) 766·67 (B) 667·14
   (C) 657·14 (D) 756·57
   (E) None of these

5. What is the ratio between total exports of the three companies in 2003 and 2006 respectively ?
   (A) 41 : 29 (B) 51 : 29
   (C) 29 : 51 (D) 29 : 41
   (E) None of these

Answers with Explanation

1. (A) % Increase $= \frac{750 - 500}{500} \times 100\%$
   $= 50\%$

2. (E) Reqd. % $= \frac{3300 \times 100}{4600}\% = 71{\cdot}74\%$
   $\simeq$ 72 (App.)

3. (B) % Increase in 2005 from the previous year
   $= \frac{800 - 600}{600} \times 100\%$
   $= 33\frac{1}{3}\%$
   % increase in 2004 from the previous year
   $= \frac{600 - 400}{400} \times 100\%$
   $= 50\%$
   % increase in 2006 from the previous year
   $= \frac{900 - 800}{800} \times 100\%$
   $= 12\frac{1}{2}\%$
   % increase in 2008 from the previous year = 0.
   Hence, maximum % rise in export was during 2004.

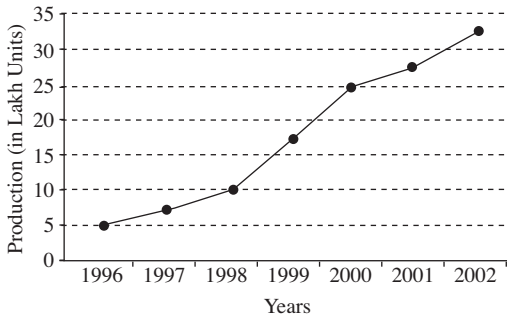
4. (C) Reqd. average $= \frac{4600}{7}$
   $= 657{\cdot}14$ thousand tonnes

5. (D) Reqd. ratio = 1450 : 2050
   = 29 : 41

Exercise 2

Directions—Study the following graph carefully and answer the questions that follow—

Production of a Company (in Lakh Units) Over the Years



1. The production in 2002 is what per cent of production in 1996 ?
   (A) 650% (B) 550%
   (C) 329% (D) 320%
   (E) None of these

2. What is the approximate average production (in lakhs) for the given years ?
   (A) 18 (B) 19
   (C) 20 (D) 18·5
   (E) 17

3. Which of the following is the highest difference in production between two adjacent years ?
   (A) 5 lakhs (B) 10 lakhs
   (C) 9 lakhs (D) 7·5 lakhs
   (E) None of these

4. Which year had the highest per cent increase in production over the previous year ?
   (A) 2000
   (B) 1999
   (C) 2002
   (D) 1997
   (E) None of these

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Answers with Explanation

1. (A) Production in 1996 = 5 lakh units
Production in 2002 = 32·5 lakh units
$\therefore$ The required percentage = $\frac{32{\cdot}5}{5} \times 100$
= 650%

2. (A) Average production
$= \frac{5 + 7{\cdot}5 + 10 + 17{\cdot}5 + 25 + 27{\cdot}5 + 32{\cdot}5}{7}$
$= \frac{125}{7} = 17{\cdot}8$
$\Rightarrow$ 18 lakh units

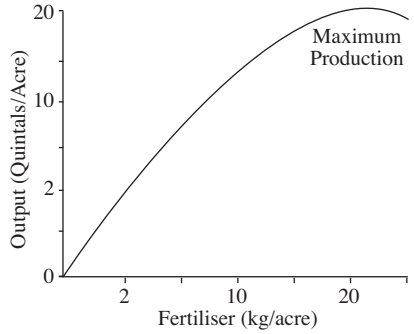
3. (D) This is obvious by the graph.

4. (B) Per cent increase in 1999
$= \frac{17{\cdot}5 - 10}{10} \times 100 = 75$
Per cent increase in 2000
$= \frac{25 - 17{\cdot}5}{17{\cdot}5} \times 100 = 42{\cdot}86$
$\Rightarrow$ In 1999, It is the highest.

Exercise 3

Directions—Study the following graph carefully and answer the questions that follow—

Relationship between Fertilizer Consumed in kg per Acre to Output in Quintals Per Acre



1. If a farmer is having 5 acres of land and only 50 kg of fertilizer, which of the following will give the best yield ?
(A) 10 kg per acre
(B) 20 kg in one acre and the remaining 30 kg over 4 acres
(C) 20 kg each in two acres and remaining in three acres
(D) All of the above will give the same yield

2. What is the angle that the limited portion of the graph is making with the X–axis ?
(A) 30° (B) 45°
(C) 60° (D) 80°

3. What is the angle that the later part of the graph is making with the Y–axis ?
(A) 45° (B) 30°
(C) 60° (D) 90°

4. Increasing the fertilizer use, stops showing an improvement in productivity after—
(A) 10 kg per acre
(B) 20 kg per acre
(C) Above 20 kg per acre
(D) 2 kg per acre

5. If a farmer has only 10 acres of from land and only 100 kg of fertilizer, what should be his maximum yield in quintals ?
(A) 50 (B) 100
(C) 150 (D) 200

6. The correlation between the output (production) and the fertilizer usage (till at least upto 20 kg per acre) can be said to be—
(A) Positive and close to 1
(B) Positive and small
(C) Negative and small
(D) Negative and close to 1

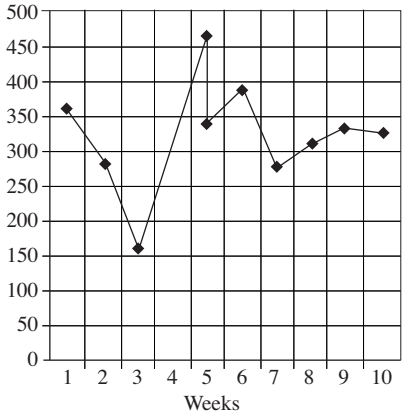
Answers

1. (D) 2. (B) 3. (D) 4. (B) 5. (B) 6. (A)

Exercise 4

Directions—Study the following graph carefully and answer the questions that follow—

Sales Forecast for the Next Ten Weeks



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1. If the forecasted demand is met by having uniform production during the weeks at an average level, the number of weeks during which demand will not be met is—
   (A) 2
   (B) 3
   (C) 4
   (D) None of these
2. If the production is uniform what should be the minimum capacity of the storage space to store the units in excess of demand ?
   (A) 25
   (B) 50
   (C) 100
   (D) 200
3. If the maximum production capacity is 300 units, the unmet demand will be—
   (A) 225
   (B) 275
   (C) 175
   (D) All the demand will be met

Answers with Explanation

1. (C) The average forecast sales

$$= \frac{362{\cdot}5 + 275 + 162{\cdot}5 + 462{\cdot}5 + 337{\cdot}5 + 387{\cdot}5 + 275 + 312{\cdot}5 + 330 + 325}{10}$$

$$= \frac{3225}{10}$$

$$= 322{\cdot}5$$

∴ The number of week is 4.

2. (D)
3. (A) The maximum production

$$= 362{\cdot}5 + 275 + 162{\cdot}5 + 462{\cdot}5 + 337{\cdot}5 + 387{\cdot}5 + 275 + 312{\cdot}5 + 330 + 325$$

$$= 3225$$

∴ The unmet demand

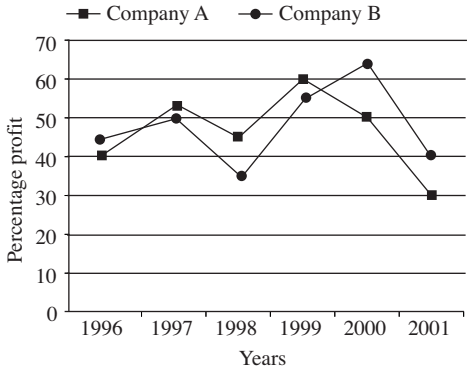
$$= 3225 - 3000$$

$$= 225$$

Exercise 5

Directions—Study the following Graph carefully and answer the questions that follow—

Percentage Growth in Population of Six States from 1998 to 1999 and 1999 to 2000



1. The population of the state 'Q' in the year of 1999 was what per cent of its population in the year of 2000 ?
   (A) $66\frac{2}{3}\%$ (B) $47\frac{1}{3}\%$
   (C) 130% (D) 37%
   (E) $62\frac{2}{3}\%$
2. The population of the state 'O' in the year of 1998 was 8 lakh, then what was its approximate population in the year of 2000 ?
   (A) 24 lakh (B) 26 lakh
   (C) 14 lakh (D) 23 lakh
   (E) None of these
3. If the population of the states 'M' and 'R' in 1998 are in the ratio 3 : 2 and the population of the state 'M' in 1999 was 126 lakh, then what was the population of the state 'R' in 2000—
   (A) 70 lakh (B) 93·60 lakh
   (C) 152 lakh (D) 65 lakh
   (E) None of these
4. In 1998 the population of the states 'N' and 'P' were equal and the population of the state 'P' in 2000 was 62 lakh, then what was the population of the state 'N' in the year of 2000 ?
   (A) 50 lakh (B) 70 lakh
   (C) 58·20 lakh (D) 67·20 lakh
   (E) 68·20 lakh

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5. The population of the state 'M' in 2000 was what fraction of its population in 1998 ?

(A) $\frac{20}{49}$ (B) $\frac{10}{19}$

(C) $\frac{49}{20}$ (D) $\frac{19}{10}$

(E) None of these

Answers with Explanation

1. (A) Let the population of the state 'Q' in 1999 $= 100$

$\therefore$ Population in 2000 $= 150$

$\therefore$ The required % $= \frac{100}{150} \times 100$

$= 66\frac{2}{3}\%$

2. (E) The population of the state 'O' in the year of 2000

$= 8 \times \frac{180}{100} \times \frac{160}{100}$

$= 23$ lakh

3. (B) Let the population of the states 'M' and 'R' in 1998 is

$= 3x$ and $2x$ respectively

$\therefore \; 3x \times \frac{140}{100} \Rightarrow x = 30$

$\therefore$ Population of the state 'R' in 1998

$= 30 \times 2$

$= 60$ lakh

and in 2000 $= 60 \times 1{\cdot}3 \times 1{\cdot}2$

$= 93{\cdot}60$ lakh

4. (D) The population of the state 'P' in 1998

$= 62 \times \frac{100}{125} \times \frac{100}{124}$

$= 40$ lakh

$\therefore$ Population of state 'N' in 1998

$= 40$ lakh

and the population in 2000

$= 40 \times 1{\cdot}2 \times 1{\cdot}4$

$= 67{\cdot}20$ lakh

5. (C) The required fraction

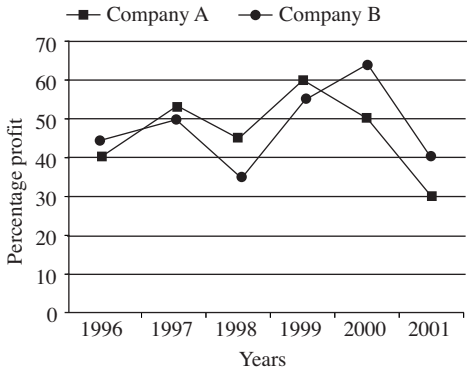
$= \frac{245}{100}$

$= \frac{49}{20}$

Exercise 6

Directions—Study the following graph carefully and answer the questions that follow—

Percentage Profit Earned by Two Companies Over the Given Years



1. If the income of Company A in 1998 was equal to its expenditure in 2000, what was the ratio between Company's expenditure in the years 1998 and 2000 respectively ?

(A) 29 : 20

(B) 20 : 29

(C) 19 : 20

(D) Cannot be determined

(E) None of these

2. If the income of Company B in 1999 was Rs. 18·6 lakhs and ratio of incomes of Companies A and B in 1999 was 2 : 3, what was the expenditure of Company A in 1999 (in Rs. lakhs) ?

(A) 12 (B) 12·4

(C) 7·75 (D) 9·75

(E) None of these

3. If the total expenditure of the two Companies in 2001 was Rs. 18 lakhs and expenditures of Companies A and B in that year were in the ratio of 4 : 5 respectively, then what was the income of Company B in that year (in Rs. lakh) ?

(A) 8

(B) 10

(C) 10·4

(D) Cannot be determined

(E) None of these

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4. If the income of Company A in 1999 was equal to the expenditure of Company B in 2000, then what was the ratio of expenditure of Company A in 1999 to the income of Company B in 2000 ?
   (A) 25 : 66 (B) 66 : 25
   (C) 10 : 13 (D) 13 : 10
   (E) None of these

5. If the total income of Company A in all the years together was equal to the total expenditure of Company B in all the years together, which was Rs. 265 lakhs, what was the total percentage profit earned by Company A for all the years together ?
   (A) 45
   (B) 137
   (C) 52
   (D) Cannot be determined
   (E) None of these

Answers with Explanation

1. (B) $E_{98} : E_{2000} = I_{98}\left(\frac{100}{145}\right) : E_{2000}$
$= 100 : 145 \ (\because I_{98} = E_{2000})$
$= 20 : 29$

2. (C) According to the given information,
$\frac{\text{Income of company A in 1999}}{\text{Income of company B in 1999}} = \frac{2}{3}$
$\Rightarrow$ Income of Company A in 1999 $= \frac{2}{3} \times 18{\cdot}6$
$I_{A99} = 12{\cdot}4$ lakhs
$\Rightarrow E_{A99} = 12{\cdot}4\left(\frac{100}{160}\right)$
$= 7{\cdot}75$ lakhs

3. (E) Suppose expenditures of A and B in the year 2001 are $4x$ and $5x$ respectively. Then
$4x + 5x = 18$ lakhs
$\therefore \quad x = 2$ lakhs
$4x = 8$ lakhs
$5x = 10$ lakhs
$I_{nB} = 10\left(\frac{140}{100}\right)$
$= 14$ lakhs

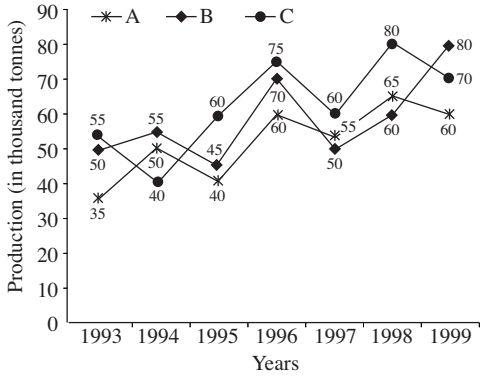
4. (A) $I_{nA99} = E_{B2000}$ (given)
Now, $E_{A99} : I_{nB2000} = I_{nA99}\left(\frac{100}{160}\right) : E_{B2000}\left(\frac{165}{100}\right)$
$= 100 \times 100 : 160 \times 165$
$= 25 : 66$

5. (D) We cannot find the expenditure of company A in the given years separately. So, we cannat find the profit of the company.

Exercise 7

Directions—Study the following Graph carefully and answer the questions that follow—

Production of Sugar (in thousand tonnes) by Three Sugar Factories Over the Given Years



1. In which of the following years for company 'A' the per cent rise/fall from the previous year is the maximum ?
   (A) 1996 (B) 1993
   (C) 1995 (D) 1998
   (E) None of these

2. Average production per year for company 'B' is approximately what per cent of the average production per year of company C ?
   (A) 105% (B) 85%
   (C) 107% (D) 93%
   (E) 97%

3. What is the per cent rise in production of company 'C' in 1996 from 1995 ?
   (A) 20% (B) 22%
   (C) 18% (D) 15%
   (E) None of these

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4. What is the difference between the average production of the three companies together in 1999 (in thousand tonnes) ?
   (A) 20 (B) 15
   (C) 17 (D) 22
   (E) None of these

5. For which of the following period of years the total production of the three companies together is equal ?
   1. 1993-94 2. 1995-97
   3. 1996-98 4. 1994-95
   (A) 2 only (B) only 3
   (C) 4 only (D) Both 3 and 4
   (E) Both 2 and 3

Answers with Explanation

1. (A) Per cent rise or fall from the previous year of the company A as—

1994	1995	1996	1997	1998	1999
+42·85	–20	+50	–8·33	+18·18	–7·69

2. (D)

3. (E) Per cent rise for the company C from 1995 to 1996

$$= \frac{75 - 60}{60} \times 100$$

$$= 25\%$$

4. (B) Average production in 1997

$$= \frac{50 + 55 + 60}{3}$$

$$= 55 \text{ thousands tonnes}$$

Average production in 1999

$$= \frac{80 + 70 + 60}{3}$$

$$= 70 \text{ thousand tonnes}$$

$\therefore$ Required difference 70 – 55

$$= 15 \text{ thousand tonnes}$$

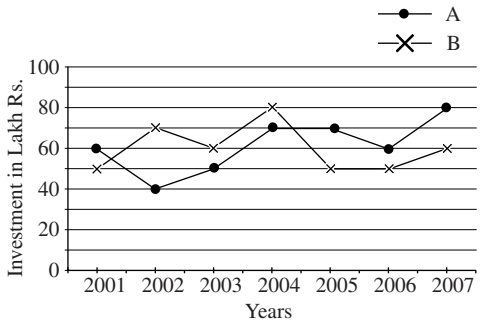
5. (D) The total production of the three companies

1993	1994	1995	1996	1997	1998
140	145	145	205	165	205

Exercise 8

Directions—Study the following graph carefully to answer the questions that follow—

Investments (in lakh Rs.) of Two Business Partners A and B Over the Year



1. What was the per cent rise in A's investment in 2004 from the previous year ?
   (A) 25% (B) 20%
   (C) $33\frac{1}{3}\%$ (D) $33\frac{2}{3}\%$
   (E) None of these

2. What was the per cent rise in investment of B in 2004 from 2001 ?
   (A) 45·6 (B) 37·5
   (C) 30 (D) 60
   (E) None of these

3. What was the per cent rise/fall in the total investment of A and B together from 2002 to 2005 ? (Rounded off to two digits after decimal)
   (A) 8·33% fall (B) 9·09% rise
   (C) 8·33% rise (D) 9·09% fall
   (E) None of these

4. What is the ratio between total investment of A in 2001, 2002 and 2003 together and the total investment of B in these three years together respectively ?
   (A) 5 : 6 (B) 6 : 5
   (C) 15 : 17 (D) 17 : 15
   (E) None of these

5. Investment of B in 2003 is **approximately** what per cent of his total investment for all the years together ?
   (A) 12 (B) 18
   (C) 20 (D) 17
   (E) 14

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Answers with Explanation

1. (E) Reqd. % rise $= \frac{70 - 50}{50} \times 100\%$
$= 40\%$

2. (D) Reqd. % rise $= \frac{80 - 50}{50} \times 100\%$
$= 60\%$

3. (B) Reqd. % rise
$= \frac{(50 + 70) - (40 + 70)}{(40 + 70)} \times 100\%$
$= \frac{10 \times 100}{110}\% = 9{\cdot}09\%$

4. (A) Reqd. ratio $= \frac{(60 + 40 + 50)}{(50 + 70 + 60)}$
$= \frac{150}{180} = 5 : 6$

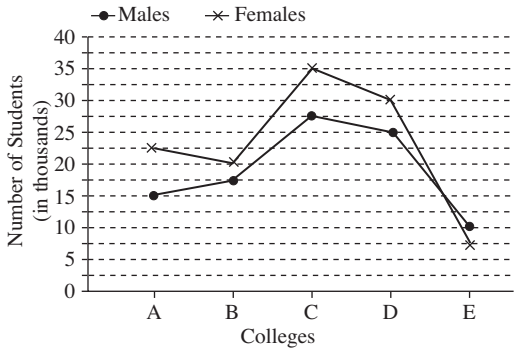
5. (E) Reqd % $= \frac{60}{(50 + 70 + 60 + 80 + 50 + 50 + 60)} \times 100\%$
$= \frac{60}{420} \times 100\%$
$= 14{\cdot}28\% \simeq 14\%$ (App.)

Exercise 9

Directions—Study the following graph carefully and answer the questions below it.

Number of Students (Males and Females) Passed Out from Various Colleges in a Year

(Number in thousands)



1. What is the average number of students (Males and Females) passed out from all the colleges together ?
(A) 38000 (B) 48000
(C) 42000 (D) 51000
(E) None of these

2. The number of females passed out from college C is **approximately** what per cent of the total number of females passed out from all the colleges together ?
(A) 28 (B) 30
(C) 36 (D) 25
(E) 40

3. What is the difference between the total number of students passing out from college A and the total number of students passing out from college E ?
(A) 20,500 (B) 21,000
(C) 10,500 (D) 10,000
(E) None of these

4. What is the respective ratio of the total number of males to the total number of females passed out from all the colleges together ?
(A) 19 : 23 (B) 18 : 25
(C) 23 : 19 (D) 25 : 18
(E) None of these

5. The number of males passing out from colleges A and B together is what per cent of the number of females passing out from colleges C and D together ?
(A) 45 (B) 40
(C) 35 (D) 50
(E) None of these

Answers with Explanation

1. (C) Reqd. average
$= \frac{(15 + 22{\cdot}5 + 17{\cdot}5 + 20 + 27{\cdot}5 + 35 + 25 + 30 + 7{\cdot}5 + 10)}{5}$
$= 42000$

2. (B) Reqd % $= \frac{35 \times 100}{115}$
$= 30{\cdot}43\%$
$\simeq 30\%$ (App.)

3. (E) Reqd. difference
$= (15 + 22{\cdot}5) - (7{\cdot}5 + 10)$ thousand
$= (37{\cdot}5 - 17{\cdot}5)$ thousands
$= 20000$

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4. (A) Reqd. ratio

$$= \frac{(15 + 17{\cdot}5 + 27{\cdot}5 + 25 + 10{\cdot}0)}{(22{\cdot}5 + 20 + 35 + 30 + 7{\cdot}5)}$$

$$= \frac{95}{115} = 19 : 23$$

5. (D) Reqd. % $= \frac{(15 + 17{\cdot}5)}{(35 + 30)} \times 100\%$

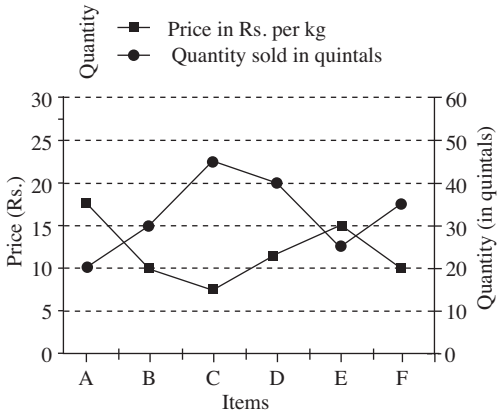
$$= \frac{32{\cdot}5}{65} \times 100\%$$

$$= 50\%$$

Exercise 10

Directions—Study the following graph carefully to answer the questions that follow—

Quantity of Various Items Sold and Price Per kg



1. If the quantity sold of item D increased by 50% and the price reduced by 10%. What was the total value of the quantity sold for item D ?
   (A) Rs. 675 (B) Rs. 6,750
   (C) Rs. 67,550 (D) Rs. 67,500
   (E) None of these

2. **Approximately**, what is the average price per kg of items A, B and C ?
   (A) Rs. 9·50 (B) Rs. 8
   (C) Rs. 7·50 (D) Rs. 9
   (E) Rs. 10·50

3. What is the ratio between the total values of quantity sold for items E and F respectively ?
   (A) 15 : 14 (B) 3 : 2
   (C) 5 : 7 (D) 7 : 5
   (E) None of these

4. Total value of the quantity sold for item C is what per cent of the total value of the quantity sold for item E ?
   (A) 111 (B) 85
   (C) 90 (D) 87·5
   (E) None of these

5. If the price as well as the quantity sold is increased by 20% for item A, what is the total value of quantity sold for item A ?
   (A) Rs. 48,500 (B) Rs. 49,000
   (C) Rs. 42,000 (D) Rs. 50,400
   (E) None of these

Answers

1. (D) 2. (E) 3. (A) 4. (C) 5. (D)

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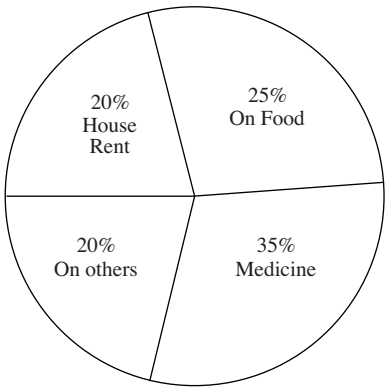
5 Pie Chart

Pie chart—Pie chart or the Pie graph is a complete circle or a Pie in which the total quantity or the magnitude of the given question is distributed over the various parts of an angle of 360°.

In the Pie chart or the pie graph, the data can be plotted with respect to any one parameter, therefore its usage is restricted. It is the best use to show the shares of various parties having a particular quantity for the distribution among themselves in various parts or the percentage.

Data interpretation by the Pie chart or the Pie graph is very useful for representing the shares or proportions or the percentage of various components or the elements with respect to the total quantity or the magnitude. Questions in the examinations are formally asked either in the form of a **simple pie chart** that is a form of a single Pie graph or in the form of the **Multiple Pie chart** that is a form of two or more than two Pie charts together. Generally, two diagrams of Pie chart are given to refer the conditions of the questions.

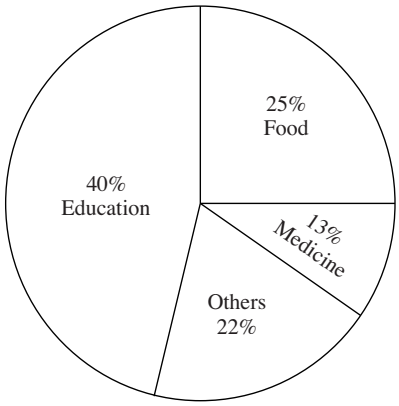
Example 1. The following example refers to the **Simple pie chart** or the **Single pie chart** showing the expenditure pattern of a person out of his total income.



Total Income Rs. 15,000 per month.

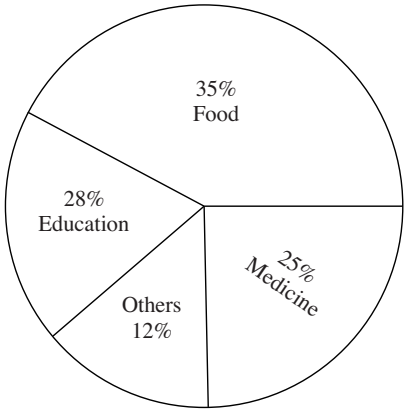
Example 2. The following example refers a question of **Multiple pie chart** showing the expenditure on various items by two families.

Family A



Total Expenditure Rs. 12,000 per month.

Family B



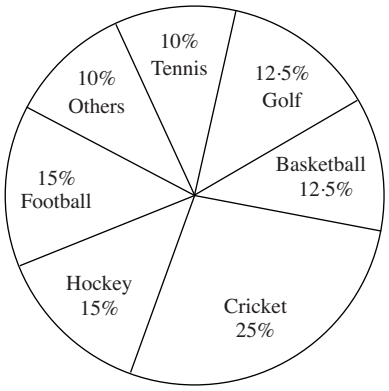
Total Expenditure Rs. 15,000 per month.

Exercise 1

Directions—Study the following pie graph carefully and answer the questions that follow—

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Per cent of Amount Spent by a Country on Various Sports for One Year



1. If the total amount spent on sports during the year was Rs. 1,50,00,000, then the amount spent on Cricket and Hockey together was—
   (A) Rs. 60,00,000 (B) Rs. 50,00,000
   (C) Rs. 37,50,000 (D) Rs. 75,00,000
2. If the total amount spent during the year was Rs. 1,20,00,000, how much was spent on Basketball ?
   (A) Rs. 12,50,000 (B) Rs. 10,00,000
   (C) Rs. 12,00,000 (D) Rs. 15,00,000
3. The ratio of the total amount spent on Football to that spent on Hockey was—
   (A) 1 : 15 (B) 1 : 1
   (C) 15 : 1 (D) 3 : 2
4. The graph shows that the most popular game is—
   (A) Hockey (B) Football
   (C) Cricket (D) Basketball
5. The country spent the same amount of money on—
   (A) Hockey and Tennis
   (B) Golf and Tennis
   (C) Golf and Basketball
   (D) Cricket and Football
   (E) Hockey and Golf

Answers with Explanation

1. (A) Money spent on Cricket = 25%
   Money spent on Hockey = 15%
   Cricket and Hockey together = 25 + 15
   = 40%

$$\therefore \ 40\% \text{ of } 1,50,00,000 = \frac{40 \times 1,50,00,000}{100}$$

$$= \text{Rs. } 60,00,000$$

2. (D) Spent on Basketball = 12·5%

$$\therefore \quad \frac{12{\cdot}5 \times 1,20,00,000}{100} = \text{Rs. } 15,00,000$$

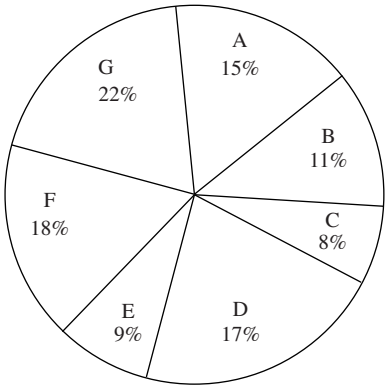
3. (B) The required ratio $= \frac{15}{15} = 1 : 1$
4. (C) Cricket
5. (B) Golf and Basketball (12·5% for each).

Exercise 2

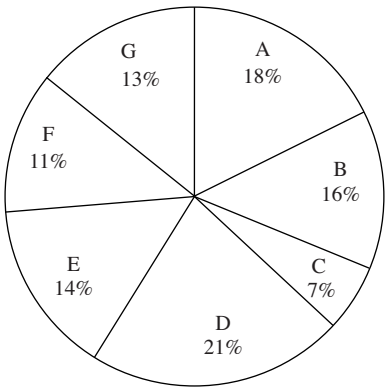
Directions—Study the following pie graph carefully and answer the questions that follow—

Classification of Appeared Candidates in a Competitive Test from Different States and Qualified Candidates from These States

Appeared Candidates 45,000



Qualified Candidates 9000



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1. What is the ratio of the number of appeared candidates from states C and E together to that of the appeared candidates from states A and F together ?
   (A) 17 : 33 (B) 11 : 13
   (C) 13 : 27 (D) 17 : 27
   (E) None of these
2. In which state, the percentage of qualified candidates with respect to that of appeared candidates is minimum ?
   (A) C (B) F
   (C) D (D) E
   (E) G
3. What is the difference between the number of qualified candidates of states D and those of G ?
   (A) 690 (B) 670
   (C) 780 (D) 720
   (E) None of these
4. What is the percentage of qualified candidates with respect to appeared candidates from state B and C taken together ? (rounded to two decimal places)
   (A) 23·11 (B) 24·21
   (C) 21·24 (D) 23
   (E) None of these
5. What is the ratio between the number of candidates qualified from states B and D together to the number of candidates appeared from state 'C' respectively ?
   (A) 8 : 37 (B) 11 : 12
   (C) 37 : 40 (D) 7 : 37
   (E) None of these

Answers with Explanation

1. (A) Required ratio $= \frac{8+9}{15+18}$

$= \frac{17}{33}$

$\Rightarrow$ 17 : 33

2. (E) The graphs show the ratio of % qualified candidates with respect to the appeared is the least for the state G.
3. (D) The required difference

$= (21 - 13)\%$ of 9000

$= 720$

4. (B) Required % $= \frac{(16+7)\% \text{ of } 9000}{(11+8)\% \text{ of } 45000} \times 100$

$= \left(\frac{\frac{23 \times 9000}{100}}{\frac{19 \times 45000}{100}}\right) \times 100$

$= \frac{23 \times 100}{19 \times 5}$

$= 24{\cdot}21\%$

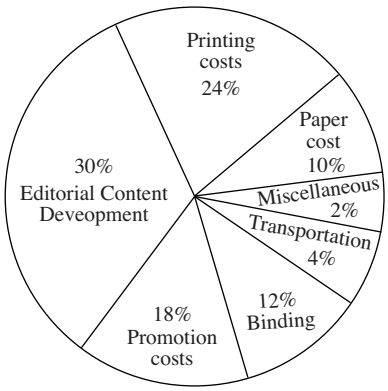
5. (C) Required ratio $= \frac{(16+21)\% \text{ of } 9000}{8\% \text{ of } 45000}$

$= \frac{37}{40} \Rightarrow 37 : 40$

Exercise 3

Directions—Study the following diagram of Pie chart carefully and answer the questions that follow—

Expenditure Increase in Printing a Magazine



1. What is the angles for the sector representing paper cost ?
   (A) 10° (B) 36°
   (C) $23\frac{1}{2}°$ (D) 45°
2. What should be the centre angle of the sector representing transportation charges ?
   (A) 4° (B) 8·4°
   (C) 12·4° (D) 14·4°
3. If the editorial content development cost is Rs. 30,000 then the cost of transportation can be expected to be—
   (A) Rs. 4000 (B) Rs. 400
   (C) Rs. 12,000 (D) Rs. 2000

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4. For a given issue of the magazine, the miscellaneous cost is Rs. 2000 and the print run is 12,500 copies. What should be the sale price if the publisher desires a profit of 5% ?
   (A) Rs. 5 (B) Rs. 7·50
   (C) Rs. 8 (D) Rs. 8·40

5. If for the same data as given in the previous question, the print-run were to be 50,000 copies, the sale price per copy would have been—
   (A) Rs. 5 (B) Rs. 2
   (C) Rs. 2·10 (D) 2·20

6. If the promotional costs for a given issue of the magazine is Rs. 9000, then the total expenditure in bringing out that issue of the magazine is—
   (A) Rs. 50,000 (B) Rs. 1,00,000
   (C) Rs. 45,000 (D) Rs. 60,000

7. For the same data as given in the previous question, what is the cost of editorial content development ?
   (A) Rs. 45,000 (B) Rs. 30,000
   (C) Rs. 15,000 (D) Rs. 20,000

Answers with Explanation

1. (B)

2. (D) If $100\% \rightarrow 360°$

$$\therefore \quad 4\% \rightarrow \frac{360°}{100} \times 4 = 14{\cdot}4°$$

3. (A) On $30\% \rightarrow$ Rs. 30,000

$$\therefore \quad \text{On } 4\% = \text{Rs. } \frac{30{,}000}{30} \times 4$$
$$= \text{Rs. } 4000$$

4. (D) $2\% \rightarrow$ Rs. 2000

$\therefore$ Total cost = Rs. 1,00,000

$\therefore$ Cost price per copy

$$= \frac{1{,}00{,}000}{12{,}500}$$
$$= \text{Rs. } 8$$

$\therefore$ Selling price per copy

$$= \frac{\text{C.P.}(100 + 5)}{100}$$
$$= \frac{8(100 + 50)}{100} = \frac{8 \times 105}{100}$$
$$= \text{Rs. } 8{\cdot}40$$

5. (C) $$\text{C.P. per copy} = \frac{1{,}00{,}000}{50{,}000}$$
$$= \text{Rs. } 2$$
$$\therefore \quad \text{S.P. per copy} = \frac{2(100 + 3)}{100}$$
$$= \frac{2 \times 105}{100}$$
$$= \text{Rs. } 2{\cdot}10$$

6. (A) $18\% \rightarrow 9000$

$$\therefore \quad \text{Total cost} \rightarrow \frac{9000 \times 100}{18}$$
$$= \text{Rs. } 50{,}000$$

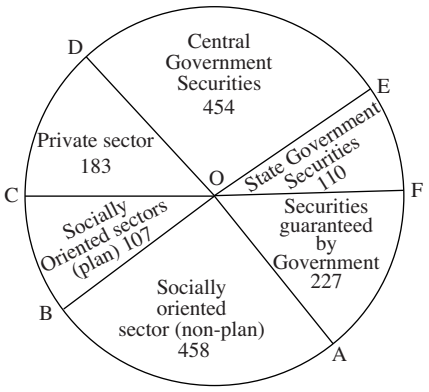
7. (C) $18\% \rightarrow 9000$

$$\therefore \quad 30\% \rightarrow \frac{9000 \times 30}{18}$$
$$= \text{Rs. } 15{,}000$$

Exercise 4

Directions—Study the following diagram of Pie chart carefully and answer the questions that follow—

The Gross Investments of Life Insurance Corporation of India (In Crores of Rupees) in Different Sectors are Shown



1. The percentage of gross investments in State Government securities is nearly—
   (A) 7·1% (B) 7·8%
   (C) 8·6% (D) 9·2%

2. The magnitude of $\angle$ AOC is nearly—
   (A) 123° (B) 132°
   (C) 126° (D) 115°

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3. The investment in socially oriented sector (Plan and Non-plan) is … than the investment in Government securities (central and state) by……
   (A) More, 4 crore (B) More, 1 crore
   (C) More, 111 crore (D) Less, 106 crore
4. The investment in private sectors is nearly …… per cent higher than the investments in State Government securities.
   (A) 66 (B) 54
   (C) 46 (D) 40
5. The ratio of the area of the circle above COF to the area of the circle below it is nearly—
   (A) 1 : 2 (B) 35 : 37
   (C) 83 : 88 (D) 88 : 83

Answers with Explanation

1. (A) The required %
$$= \frac{110 \times 100}{458 + 107 + 183 + 454 + 110 + 227}$$
$$= \frac{11,000}{1539}$$
$$= 7{\cdot}1\%$$

2. (B) The magnitude of
$$\angle AOC = \frac{458 + 107}{1539} \times 360°$$
$$= \frac{565 \times 360°}{1539}$$
$$= 132°$$

3. (B) More, and
$$(458 + 107) - (454 + 110) = 1 \text{ crore}$$

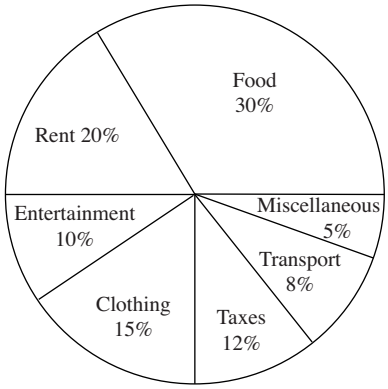
4. (A) Investment in private sectors
$$\frac{183 - 110}{110} \times 100 = \frac{73 \times 100}{110}$$
$$= 66\%$$

5. (C) The required ratio
$$= \frac{183 + 454 + 110}{107 + 458 + 227}$$
$$= \frac{747}{792} = \frac{83}{88}$$
$$\Rightarrow \quad 83 : 88$$

Exercise 5

Directions—Study the following diagram of Pie chart carefully and answer the questions below it.

Expenditure Distribution of a Family



1. If the family spends Rs. 6,500 per month, how much are its taxes ?
   (A) Rs. 7,800 (B) Rs. 9,360
   (C) Rs. 9,800 (D) Rs. 10,080
2. How many degrees should there be in the central angle showing clothing, taxes and transportation combined ?
   (A) 100 (B) 110
   (C) 120 (D) 126
3. How much more money per month is spent by the family on food as compared to the rent, if the family spends Rs. 6,500 per month ?
   (A) Rs. 650 (B) Rs. 700
   (C) Rs. 750 (D) Rs. 800
4. If the expenditure budget of the family is raised to Rs. 8,000 per month and distribution on various item remain the same, then the monthly expenses on both, the entertainment and the transport, will be—
   (A) Rs. 1,800 (B) Rs. 1,600
   (C) Rs. 1,440 (D) Rs. 1,220

Answers with Explanation

1. (B) $\text{Taxes} = \frac{12 \times 650}{100}$
$$= \text{Rs. } 780 \text{ per months}$$
$$\therefore \quad \text{Re. } 780 \times 12 = \text{Rs. } 9360 \text{ per year}$$

2. (D) Clothing, taxes and transportation consumed 35%
$$\therefore \quad 100\% \rightarrow 360°$$
$$\therefore \quad 35\% \rightarrow \frac{360° \times 35}{100}$$
$$= 126$$

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3. (A) 10% of Rs. 6500

$$= \frac{10 \times 6500}{100}$$

= Rs. 650 per month

4. (C) 18% of Rs. 8000

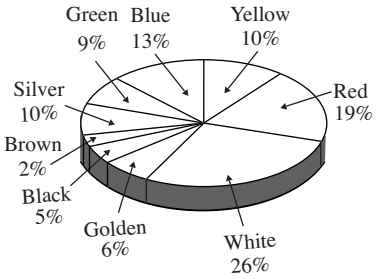
$$= \frac{18 \times 8000}{100}$$

= Rs. 1440

Exercise 6

Directions—Study the following diagram of Pie chart carefully and answer the questions below—

Selling of the Car in UK According to the Colours



1. 50% of all the cars consisted of which colours of car ?
   (A) Black, Golden, Blue, Red
   (B) Blue, Black, Red, Silver
   (C) White, Golden, Blue, Black
   (D) None of these

2. Cars of which colour are 20% less popular than white coloured cars ?
   (A) Black (B) Golden
   (C) Red (D) Blue
   (E) None of these

3. Cars of which colour are 13% less popular than white cars ?
   (A) Blue (B) Green
   (C) Silver (D) Yellow
   (E) None of these

4. Cars of which colour when increased by two per cent and then combined with that of red cars will make 30% of the total ?
   (A) Golden (B) Blue
   (C) Black (D) Yellow
   (E) None of these

5. If in a certain period the total production of all cars was 95400 than how many more blue cars were sold than green ?
   (A) 2580 (B) 3618
   (C) 2850 (D) 3816
   (E) None of these

Answers

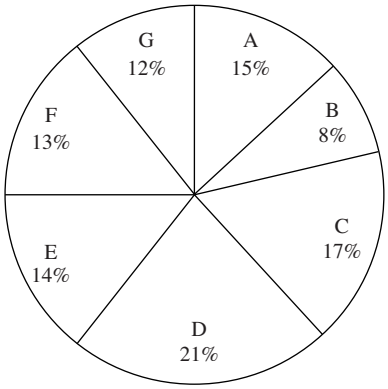
1. (A) 2. (B) 3. (A) 4. (E) 5. (D)

Exercise 7

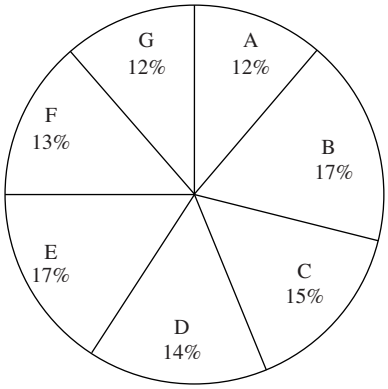
Directions—Study the following diagram carefully and answer the questions that follow—

Distribution of Candidates Studying Arts and Commerce from Seven Different Institutes A, B, C, D, E, F and G

Total Number of Students Studying Arts = 3800



Total Number of Students Studying Commerce = 4200



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1. What is the ratio between the number of students studying Arts from Institute E and the number of students studying Commerce from institute B respectively ?
   (A) 17 : 19 (B) 19 : 27
   (C) 14 : 19 (D) 19 : 21
   (D) None of these
2. What is the total number of students studying Arts from institutes A and G together ?
   (A) 1102 (B) 918
   (C) 966 (D) 1130
   (E) None of these
3. How many students are studying Commerce from institutes B and D together ?
   (A) 1158 (B) 1302
   (C) 1232 (D) 1272
   (E) None of these
4. How many students are studying Arts and Commerce from Institute B ?
   (A) 1418 (B) 2000
   (C) 1018 (D) 1208
   (E) None of these
5. What is the ratio between the numbers of students studying Arts and Commerce respectively from Institute E ?
   (A) 19 : 27 (B) 17 : 29
   (C) 19 : 29 (D) 17 : 27
   (E) None of these

Answers with Explanation

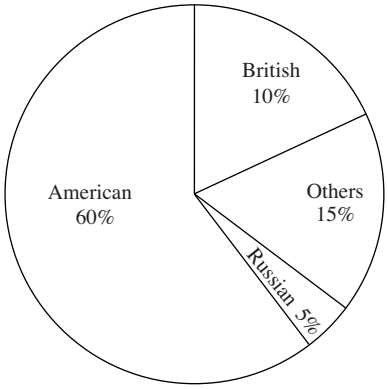
1. (D) The required ratio $= \frac{14\% \text{ of } 3800}{14\% \text{ of } 4200}$
   $= \frac{19}{21}$
   $\Rightarrow$ 19 : 21
2. (E) The required number
   = 27% of 3800 = 1026
3. (B) The required number
   = 31% of 4200 = 1302
4. (C) The required number
   = 8% of 3800 + 17% of 4200
   = 304 + 714
   = 1018
5. (E) The required ratio
   $= \frac{14\% \text{ of } 3800}{17\% \text{ of } 4200} = \frac{19}{17}$
   $\Rightarrow$ 19 : 17

Exercise 8

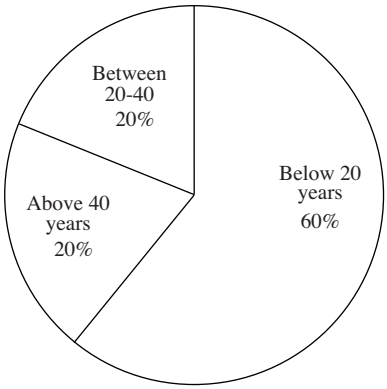
Directions—Study the following diagram carefully and answer the questions that follow—

Characteristics of Foreign Tourists Visiting India During a Year

Countrywise Distribution



Age-wise Distribution



1. If in a given year, 1,00,000 tourists visited India and the age-wise distribution data applies to all countrie the number of American tourists who visited India during the year and are in the age group of 20-40 years is—
   (A) 12,000 (B) 20,000
   (C) 40,000 (D) 60,000
2. With the same data given in the previous question, what would be the number of

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Russian tourists who are below 20 years of age ?
(A) 3000 (B) 300
(C) 330 (D) 3500

3. With the same data give above, the number of British tourists between 20 and 40 years of age would be—
(A) 400 (B) 4000
(C) 4400 (D) 440

4. With the same data, how many tourists were below 20 years, but neither American, nor Russian nor British ?
(A) 900 (B) 1900
(C) 9000 (D) 60,000

5. What is the ratio of British tourists below 20 years to the Russian tourists above 40 years ?
(A) 1 : 2 (B) 12 : 1
(C) 3 : 4 (D) 4 : 3

Answers with Explanation

1. (A) Number of Americans who visited India
= 60% of 1,00,000
= 60,000
Number of Americans in the age group of 20 – 40 years who visited India
= 20% of 60,000
= 12,000

2. (A) No. of Russian Tourists
= 5000
No. of Russian Tourists below 20 years of age
= 60% of 5000
= 3000

3. (B) No. of British Tourists
= 20,000
No. of British Tourists between 20 to 40 years of age
= 20% of 20,000
= 4000

4. (C) No. of other tourists
= 15,000
No. of other tourists below 20 years of age
= 60% of 15,000
= 9000

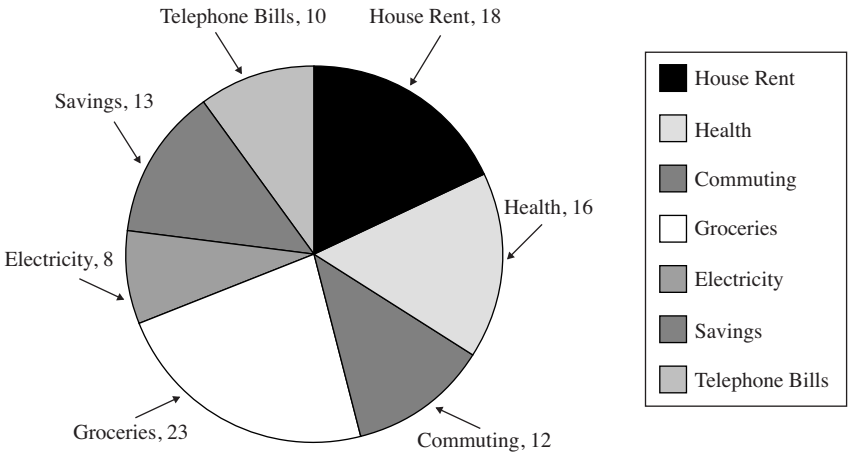
5. (B) $\frac{\text{British tourists below 20 years}}{\text{Russian tourists above 40 years}}$
$= \frac{60\% \text{ of } 20{,}000}{20\% \text{ of } 5000}$
$= \frac{12{,}000}{1000}$
$= \frac{12}{1}$
$\Rightarrow$ 12 : 1

Exercise 9

Directions—Study the following diagram carefully and answer the questions that follow—

Percentage wise Break up of Spending Pattern of a Family in a Month

Total Amount Spent in a Month = Rs. 60,000



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1. What is the amount spent by the family on Commuting ?
   (A) Rs. 9600 (B) Rs. 8400
   (C) Rs. 7200 (D) Rs. 6000
   (E) None of these
2. What is the total amount spent by the family on Telephone Bills, Health and Electricity together ?
   (A) Rs. 13,800 (B) Rs. 18,600
   (C) Rs. 17,400 (D) Rs. 20,400
   (E) None of these
3. What is the respective ratio of amount spent by family on Groceries to the amount spent on House rent ?
   (A) 23 : 18 (B) 13 : 28
   (C) 18 : 23 (D) 28 : 13
   (E) None of these
4. Amount invested by the family on Savings forms what per cent of amount spent on Health ?
   (A) 123 (B) 81·25
   (C) 120·50 (D) 85·75
   (E) None of these
5. Total amount spent by the family on Commuting and Telephone Bills together forms **approximately** what per cent of the amount spent on Groceries ?
   (A) 153 (B) 148
   (C) 135 (D) 112
   (E) 96

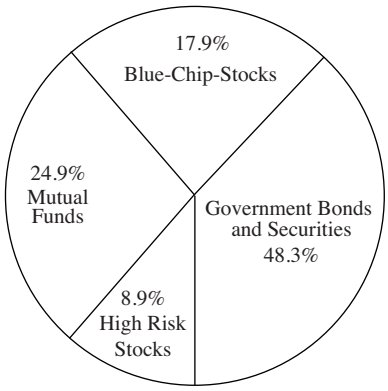
Answers with Explanation

1. (C) Expenditure on commuting
$$= \frac{60,000 \square 12}{100}$$
$$= \text{Rs. } 7200$$
2. (D) Required exp.
$$= \frac{(10 + 16 + 8) \square 60,000}{100}$$
$$= \text{Rs. } 20,400$$
3. (A) 4. (B) 5. (E)

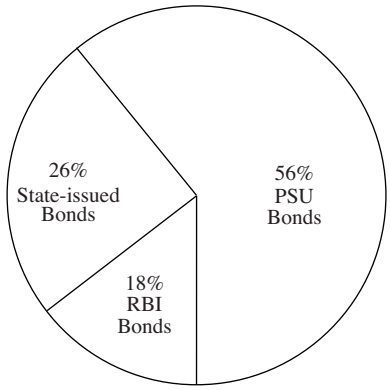
Exercise 10

Directions—Study the following diagrams carefully and answer the questions that follow—

Investment Portfolio
Total Investment Profile
Rs. 5·4 crore



Government Bonds and Securities



1. Approximately, how much money of the investment portfolio has been invested in high-risk stocks ?
   (A) Rs. 48,06,000
   (B) Rs. 51,30,000
   (C) Rs. 54,00,000
   (D) Rs. 36,00,000
2. Approximately, how much money has been invested in state-issued bonds ?
   (A) Rs. 65,20,500
   (B) Rs. 67,81,320
   (C) Rs. 62,59,680
   (D) Rs. 52,16,400
3. The ratio of money invested in Mutual Funds and State-issued Bonds is approximately—
   (A) 1 : 1 (B) 2 : 1
   (C) 1 : 3 (D) 3 : 1

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4. Which of the following earned the least amount of money for the investment portfolio ?
   (A) PSU Bonds
   (B) Mutual Funds
   (C) Blu-chip Stocks
   (D) Cannot be determined

Answers with Explanation

1. (A) 8·9% of 5·4 core = Rs. 48,06,000

2. (B) Investment
   In Government Bonds and Securities
   = 48·3% of 5·4 crore
   = $\frac{48·3}{100} \times 5·4$ crore
   = Rs. 483 × 54000
   In state-issued Bonds
   = 26% of (48·3% of 5·4 crore)
   = $\frac{26}{100} \times 483 \times 54000$
   = 26 × 483 × 540
   = Rs. 67,81,320

3. (B) Money invested in Mutual Funds
   = 24·9% of 5·4 crore
   = 1·3446 crore
   = Rs. 1,34,46,000
   Money invested in state issued Bonds
   = Rs. 67,81,320 (by Q. 2)
   ∴ The required ratio
   = $\frac{1,34,46,000}{67,81,320}$
   = 2 : 1 (App.)

4. (C) Mutual Funds : 1,34,46,000
   PSU Bonds : 56% of (48·3% of 5·4 crore)
   = 1·460592 crore
   = 1,46,05,920
   Blue-chip Stocks : 17·9% of 5·4 crore
   = 96,66,000

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6 Caselet

Caselet—A caselet is a complete paragraph full of numerical information that provides the required Data in order to answer the questions that follow the given information. Generally, in this type of data, A paragraph that contains some facts or the numerical information is given to us and we are required to answer the questions that follow the numerical information.

To calculate the required answers easily, first of all, we must transfer the given information into a tabular form of data.

It would be a wrong strategy for the given caselet, to proceed without forming a table. It may seem a bit tedious to prepare the table but one it is made, all the answers will be self-evident. Therefore, study, first of all, the given paragraph of information carefully and prepare the required table to answer the questions.

Exercise 1

Directions—Study the following caselet carefully and answer the questions that follow—

Mr. Ramchandran has recently acquired four companies—A, B, C and D. He noticed that the sales of the company D are half that of company A, whereas the profits of the company A are double that of company D. The expenses of company C are Rs. 2 crores less than that of company D. Whereas the profit of the company B is Rs. 1 crore less than that of company C. The expenses of company A are two times that of company II. It is also known that the sales of the company C are Rs. 12 crores or one-fourth that of company B. An insider further informs Mr. Ramchandran that the sales of the company D are Rs. 10 crores more than that of company C and the expenses of company A are 80% of its own sales.

Note—1. All figures are for the years 2005-2006.

2. Profit = Sales – Expenses.

1. What is the total sale of all the four companies ?
   (A) Rs. 126 crores (B) Rs. 150 crores
   (C) Rs. 117 crores (D) Rs. 125 crores
   (E) None of these
2. The expenses of the company A exceed that of the company C by—
   (A) Rs. 17·6 crores (B) Rs. 19·6 crores
   (C) Rs. 18·6 crores (D) Rs. 50·8 crores
   (E) None of these
3. Which company had the maximum profit ?
   (A) B (B) C
   (C) D (D) A
   (E) None of these
4. The expenses of the company B exceed the profit of the company A by—
   (A) Rs. 44·8 crores (B) Rs. 56·2 crores
   (C) Rs. 43·8 crores (D) Rs. 62·2 crores
   (E) None of these
5. Which company was running in the maximum loss ?
   (A) C (B) B
   (C) A (D) D
   (E) None of these

Answers with Explanation

Based on the above information, the facts may be simplified as—

(i) Sales of the company D

$$= \frac{1}{2} \times \text{sales of company A}$$

$\Rightarrow$ Sales of company A

$$= 2 \times \text{sales of company D}$$

(ii) Profit of the company D

$$= \frac{1}{2} \times \text{profit of company A}$$

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⇒ Profit of the company A
= 2 × profit of the company D

(iii) Expenses of the company C
= Expenses of the company D – 2 crores

(iv) Profit of the company B
= Profit of the company C – 1 crore

(v) Expenses of company A
= 2 × expenses of the company D

(vi) Sales of the company C
= Rs. 12 crores or $\frac{1}{4}$ × sales of the company B

⇒ Sales of the company B
= 4 × 12 crores
= Rs. 48 crores

(vii) Sales of the company D
= Rs. 10 crores
+ Sale of the company C

(viii) Expenses of the company A
= 80% of the own sales

(ix) Profit
= Sales – expenses

⇒ Expenses
= Sales – profit

Now, we can calculate the answers of the questions or we may also make the required table as—

Company	Sales (in crore)	Expenses (in crore)	Profit (in crore)
A	44	35·2	+ 8·8
B	48	52·6	–4·6
C	12	15·6	–3·6
D	22	17·6	+ 4·4
Total	126	121·0	5·0

Now, By seeing the table, the answers of the questions are—

1. (A) Rs. 126 crores
2. (B) Rs. 35·2 crores – Rs. 15·6 crores
= Rs. 19·6 crores
3. (D) Company A
4. (C) Rs. 52·6 crore – Rs. 8·8 crores
= Rs. 43·8 crores
5. (B) Company B

Exercise 2

Directions—Study the following caselet carefully and answer the questions that follow—

Four students—A, B, C and D appeared in a law examination which had six semesters—s_1, s_2, s_3, s_4, s_5 and s_6.

In each semester, there were 5 papers—Paper I, Paper II, Paper III, Paper IV, Paper V and Full marks for each Paper is 100.

Students A obtained the marks in the Ist semester in all the five Papers—45, 62, 48, 56 and 55 respectively, whereas the student B obtained the marks in the same semester and papers—48, 47, 58, 57 and 49. Student C obtained the marks in the Ist semester in all the five Papers—62, 48, 49, 50 ad 60, whereas student D obtained the marks in the same semester and Papers—45, 58, 46, 49 and 65. Further, students A obtained the marks in the 2nd semester and in all Papers—48, 64, 56, 58 and 52, whereas the students B, C and D obtained the marks in the same semester and Papers respectively,

B : 50, 55, 59, 56 and 51

C : 60, 50, 50, 55 and 70

D : 47, 60, 47, 53 and 65.

Marks obtained by the four students in the 3rd semester and all the five papers respectively,

A : 49, 60, 60, 60, 55

B : 52, 52, 63, 58 and 52

C : 55, 52, 51, 60 and 67

D : 50, 62, 49, 55 and 62.

Again, students A, in the 4th semester and in all papers, obtained the following marks—47, 65, 64, 61 and 55 respectively, whereas in the same semester and papers, the remaining students had their performance as—50, 48, 52, 60 and 55, 58, 55, 52, 65 and 70, 52, 63, 51, 50 and 63 respectively. For the 5th semester, the students had their performance in all the papers as—

A : 48, 70, 62, 63 and 54

B : 54, 50, 61, 62 and 56

C : 60, 55, 55, 62 and 63

D : 52, 65, 53, 60 and 70 respectively.

For the last semester, All the students, in all the papers, had their performance as—

A : 50, 72, 65, 65 and 57

B : 55, 55, 60, 60 and 60

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C : 62, 58, 57, 63 and 68

D : 55, 67, 55, 65 and 55

1. In which semester had the student A the best performance in the 5th paper ?
   (A) s_6 (B) s_3
   (C) s_1 (D) s_2
   (E) None of these
2. Which student had the best performance in the 3rd paper of the 4th semester ?
   (A) B (B) A
   (C) D (D) C
   (E) None of these
3. In how many papers have the students shown a regular better performance ?
   (A) 6 (B) 4
   (C) 5 (D) 7
   (E) None of these
4. Which student had shown the best performance in the sixth semester exams ?
   (A) D (B) Either A or C
   (C) A (D) C
   (E) None of these
5. What is the percentage difference between B and D in the third semester exams ?
   (A) 1 (B) 2
   (C) 0·1 (D) 0·2
   (E) None of these
6. In which paper had B obtained the maximum marks ?
   (A) III (B) V
   (C) IV (D) Either III or IV
   (E) None of these
7. How many semesters of the students A for the paper III and student B for the same paper show below average performance ?
   (A) 6 (B) 2
   (C) 3 (D) 5
   (E) None of these

Answers with Explanation

On the above information given in the caselet, we can simplify the given facts in the information as—

	Student A					Student B				
Papers	**I**	**II**	**III**	**IV**	**V**	**I**	**II**	**III**	**IV**	**V**
S1	45	62	48	56	55	48	47	58	57	49
S2	48	64	56	58	52	50	55	59	56	51
S3	49	60	60	60	55	52	52	63	58	52
S4	47	65	64	61	55	50	48	52	60	55
S5	48	70	62	63	54	54	50	61	62	56
S6	50	72	65	65	57	55	55	60	60	60
	Student C					**Student D**				
Papers	**I**	**II**	**III**	**IV**	**V**	**I**	**II**	**III**	**IV**	**V**
S1	62	48	49	50	60	45	58	46	49	65
S2	60	50	50	55	70	47	60	47	53	65
S3	55	52	51	60	67	50	62	49	55	62
S4	58	55	52	65	70	52	63	51	50	63
S5	60	55	55	62	63	52	65	53	60	70
S6	62	58	57	63	68	55	67	55	65	55

According to the table, the answers are as—

1. (A) Semester 6.
2. (B) Student A.
3. (C) Student A → IV, student B → V, student C → III and student D → II and III are the desirable five papers.
4. (C) Student A.
5. (D) 0·2%
6. (D) Either III or IV.
7. (E) None of these

   Average of A in III

$$= \frac{48 + 56 + 60 + 64 + 62 + 65}{6}$$

$$= \frac{353}{6} = 58{\cdot}8$$

   Average of B in III

$$= \frac{58 + 59 + 63 + 52 + 61 + 60}{6}$$

$$= \frac{353}{6}$$

$$= 58{\cdot}8$$

   $\therefore$ For A → S_1 and S_2 and for B → S_1 and S_4 are the four desirable semesters.

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Exercise 3

Directions—Study the following information given in the caselet carefully and answer the questions that follow—

Ever since the decontrol of phosphatic and potassic fertilisers came into force in 1992—while retaining urea under price control regime with a heavy subsidy component, there has been a steep increase in the farm gate prices of these complex plant nutrients resulting in a slowdown in their consumption.

Following decontrol, consumption of phosphates declined from 3·32 million tonnes in 1991-92 to 2·87 million tonnes the next year and further to 2·67 million tonnes in 1993-94. It recovered by nine per cent to 2·93 million tonnes in 1995-96 and fall agian to 2·89 million tonnes in 1996-97.

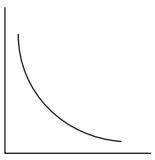
As for potassic fertiliser, the consumption slumped from 1·36 million tonnes in 1991-92 to 9·4 takh tonnes in 1992-93—31 per cent drop. The next year it dipped further to 8·9 lakh tonnes. In 1994-95, consumption was 1·12 million tonnes and since then, it inched forward to 1·15 million tonnes in 1995-96 and 1·18 million tonnes next year. In contrast, the consumption of urea steadily rose 8·05 million tonnes in 1991-92 to 10·1 million tonnes in 1996-97.

1. The paragraph *inter alia* implies—
   I. Not much change in price of urea.
   II. Continuous increase in consumption of urea.
   (A) Only I is true
   (B) Only II is true
   (C) Both I and II are true
   (D) Both I and II are false

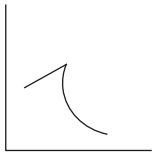
2. Decline in the consumption of potassic and phosphatic fertilisers is primarily due to—
   (A) Increase in price
   (B) Decontrol
   (C) Subsidy given to urea
   (D) Changed requirement

3. Which of the following graphs best describes the consumption of phosphates during 1991-92 to 1996-97 ?

(A) 

(B) 

(C) 

(D) 

4. During the period, the consumption of potassic fertilizer was minimum in—
   (A) 1992-93 (B) 1993-94
   (C) 1994-95 (D) 1995-96

5. Suppose a cultivator uses fertilizers in the following ratio—
   Urea (N) : Phosphates (P) : Potash (K)
   = 4 : 2 : 1
   Prices per tonne in 1996-97 were Rs. 300 for urea, Rs. 900 for phosphates and Rs. 600 for potash. How much money he had to spend for 1400 tonnes of fertilizer ?
   (A) Rs. 8·4 lakh
   (B) Rs. 7·2 lakh
   (C) Rs. 6·6 lakh
   (D) None of these

Answers with Explanation

1. (C) According to the first para of the given caselet.

2. (A) According to the first para of the given caselet.

3. (A) According to the figures given about the consumption of phosphates in the second para of the given caselet.

4. (B) According to the third para of the given caselet. It is 8·9 lakh tonnes.

5. (B) Suppose quantities of Urea, Phosphates and Potash used are 4K, 2K and K tonnes respectively.

$$\therefore \quad 4K + 2K + K = 1400$$
$$7K = 1400$$
$$\Rightarrow \quad K = 200$$

$\therefore$ Quantity of Urea used = 800 tonnes

Quantity of Phosphates used = 400 tonnes

Quantity of Potash used = 200 tonnes

$\therefore$ Expenditure for 1400 tonnes of fertilizer

$= 800 \times 300 + 400 \times 900 + 200 \times 600$

$= 720000 \Rightarrow 7{\cdot}2$ lakh

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Exercise 4

Directions—Study the following information given in the caselet carefully and answer the questions that follow—

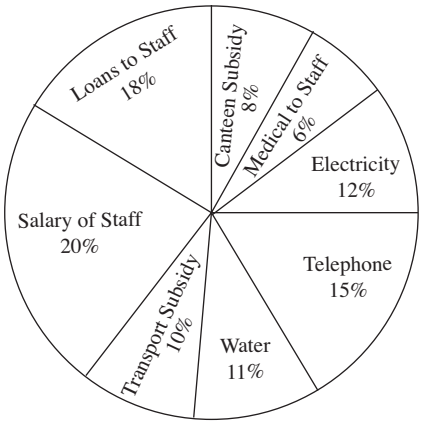
Mr. Dev established an organisation in the year of 1995. He observed that in the years of 1995, he had to spend on the various heads of the organisation as Rs. 18,50,000. He spent the amount of Rs. 18,50,000 on the various heads as 12% on electricity, 15% on telephone, 11% on water, 10% on transport, 20% on the salary to staff, 18% loans to staff, 8% on canteen subsidy and 6% on the medical to the staff.

1. What is the difference between the expenditure on salary to staff and loans to staff ?
   (A) Rs. 37,200 (B) Rs. 35,700
   (C) Rs. 37,500 (D) Rs. 35,000
   (E) None of these
2. What was the total expenditure on Electricity and Water together ?
   (A) Rs. 4,25,000 (B) Rs. 4,25,500
   (C) Rs. 4,22,500 (D) Rs. 4,25,800
   (E) None of these
3. What is the amount spent on Transport subsidy and Canteen subsidy together ?
   (A) Rs. 3,34,000 (B) Rs. 3,43,000
   (C) Rs. 3,30,000 (D) Rs. 3,33,000
   (E) None of these
4. Amount spent of medical to staff is what per cent of the amount spent on Salary ?
   (A) 30% (B) 33%
   (C) 25% (D) 22%
   (E) None of these
5. What is the amount spent on Telephone ?
   (A) Rs. 2,75,500
   (B) Rs. 2,70,500
   (C) Rs. 2,77,500
   (D) Rs. 2,77,000
   (E) None of these

Answers with Explanation

The information that has been given in the above caselet may be simplified either in the form of a Pie chart or in the tabular form of the data, as—

Expenditure on Various Heads
Total Expenditure Rs. 18,50,000



Or as—

Various Heads	Expenditure %	Expenditure (Rs.)
Electricity	12	$\frac{1850000 \times 12}{100} = 2,22,000$
Telephone	15	$\frac{1850000 \times 15}{100} = 2,77,500$
Water	11	$\frac{1850000 \times 11}{100} = 2,03,500$
Transport	10	$\frac{1850000 \times 10}{100} = 1,85,000$
Salary (Staff)	20	$\frac{1850000 \times 20}{100} = 3,70,000$
Loans (Staff)	18	$\frac{1850000 \times 18}{100} = 3,33,000$
Canteen Subsidy	8	$\frac{1850000 \times 8}{100} = 1,48,000$
Medical	6	$\frac{1850000 \times 6}{100} = 1,11,000$
Total expenditure = Rs. 18,50,000		

Now, the answers of the questions can be got easily, as—

1. (E) Required difference
$$= 18,50,000 \times \left(\frac{20}{100} - \frac{18}{100}\right)$$
$$= 18,500 \times 2$$
$$= \text{Rs. } 37,000$$
2. (B) Required expenditure
$$= 18,50,000 \times \left(\frac{12}{100} + \frac{11}{100}\right)$$

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$= 18,500 \times 23$

$= \text{Rs. } 4,25,500$

3. (D) Total expenditure

$= 18,50,000 \times \left(\frac{10}{100} + \frac{8}{100}\right)$

$= 18,500 \times 18$

$= \text{Rs. } 3,33,000$

4. (A) Required $\% = \frac{6}{20} \times 100\%$

$= 30\%$

5. (C) Required amount $= \frac{15}{100} \times 1850000$

$= \text{Rs. } 2,77,500$

Exercise 5

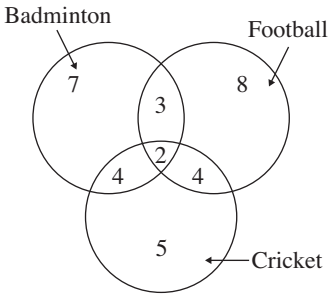
Directions—Study the following information carefully and answer the questions that follow—

Students of a class play only one or two or three games out of the three games—Badminton, Football and Cricket. 5 students play only Cricket, 8 students play only Football and 7 students play only Badminton. 4 students play only two games—Cricket and Football, 3 students play only two games—Badminton and Football and other 4 students play only two games Badminton and Cricket. 2 students play all the three games.

1. How many students play Badminton ?
   (A) 14 (B) 17
   (C) 12 (D) 13
   (E) None of these
2. How many students play Football ?
   (A) 8 (B) 17
   (C) 15 (D) 14
   (E) None of these
3. How many students play Cricket with Badminton ?
   (A) 9 (B) 10
   (C) 4 (D) 6
   (E) None of these
4. How many students play Cricket with Football ?
   (A) 7 (B) 4
   (C) 6 (D) 15
   (E) None of these
5. How many students are there in the class ?
   (A) 33 (B) 31
   (C) 36 (D) 35
   (E) None of these

Answers with Explanation

The information can be simplified as—



1. (E) The number of the students who play Badminton

$= 7 + 3 + 2 + 4$

$= 16$

2. (B) The number of the students who play Football

$= 3 + 8 + 4 + 2 = 17$

3. (D) The number of students who play Cricket with Badminton $= 4 + 2$

$= 6$

4. (C) The number of the students who play Cricket with Football

$= 2 + 4$

$= 6$

5. (A) The total number of the students

$= 7 + 3 + 8 + 4 + 2 + 4 + 5$

$= 33$

Exercise 6

Directions—Study the following caselet carefully and answer the questions that follow—

A survey was conducted among 770 people who speak one or more languages from among Hindi, English and Urdu. It was also found that 500 people speak Hindi, 400 English and 300 Urdu.

(i) 30% of the Urdu-speaking people speak all three languages, which is 10% less than those who speak Hindi and English both but not Urdu.

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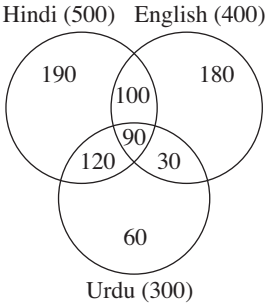
(ii) Number of people who speak Hindi and Urdu both but not English is $33\frac{1}{3}\%$ less than the no. of people who speak only English.

(iii) Number of people who speak English and Urdu both but not Hindi is 30.

1. How many people speak only Hindi ?
   (A) 190
   (B) 170
   (C) 120
   (D) Cannot be determined
   (E) None of these
2. How many people speak only English ?
   (A) 190
   (B) 100
   (C) 90
   (D) Cannot be determined
   (E) None of these
3. How many people speak Hindi and Urdu both but not English ?
   (A) 180
   (B) 120
   (C) 90
   (D) 150
   (E) None of these
4. By what per cent the no. of people who speak only Urdu is less than those who speak Hindi and English both but not Urdu
   (A) $66\frac{2}{3}\%$
   (B) $33\frac{1}{3}\%$
   (C) 40%
   (D) Cannot be determined
   (E) None of these
5. By what per cent the no. of people who speak only English is more than those who speak Hindi and Urdu but not English ?
   (A) 40%
   (B) $66\frac{2}{3}\%$
   (C) 50%
   (D) Cannot be determined
   (E) None of these

Answers with Explanation

The information in the given caselet can be transferred as—



(i) 30% of Urdu = 30% of 300
= 90

Number of people who speak Hindi and English both not Urdu = 100.

(ii) Number of people who speak English and Urdu but not Hindi = 30

Therefore, no. of people who speak only

English = 400 – (100 + 90 + 30)
= 180 …(A)

(iii) Now, with the help of (A),

Number of people who speak Hindi and Urdu both but not English = 120 …(B)

Therefore, no. of people who speak only Urdu

= 300 – (120 + 90 + 30)
= 60 …(C)

Similarly, no. of people who speak only Hindi

500 – (100 + 90 + 120) = 190 …(D)

1. (A) From the question II.
2. (E) From the equation A.
3. (B) From the equation B.
4. (C) Number of people who speak only Urdu
   = 300 – (120 + 90 + 30)
   = 60
   $\therefore$ Required less % = $\frac{100 - 60}{100} \times 100$
   = 40%
5. (C) Required more % = $\frac{180 - 120}{120} \times 100$
   = 50%

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Exercise 7

Directions—Study the following information carefully and answer the questions that follow—

Five horses, Red, White, Grey, Black and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets Rs. 3000, Rs. 2000 and Rs. 1000 on Red, White and Black horses respectively and ends up with no profit and no loss.

1. Which of the following cannot be true ?
   (A) At least two horses finished before Spotted
   (B) Red finished last
   (C) There were three horses between Black and Spotted
   (D) There were three horses between White and Red
   (E) Grey came in second
2. Suppose, in addition, it is known that Grey came in fourth. Then which of the following cannot be true ?
   (A) Spotted came in first
   (B) Red finished last
   (C) White came in second
   (D) Black came in second
   (E) There was one horse between Black and White

Answers with Explanation

1. (D) There were three horses between White and Red.
2. (C) White came second.

Exercise 8

Directions—Study the following information carefully and answer the questions that follow—

Mr. David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is Rs. 30. On the other hand, the cost, in rupees, of producing x unit is $240 + bx + cx^2$, where b and c are some constants. Mr. David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by $66\frac{2}{3}\%$. However, an increase in daily production from 40 to 60 units result in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr. David can sell as much as he can produce. His objective is to maximize the profit.

1. How many units should Mr. David produce daily ?
   (A) 130 (B) 100
   (C) 70 (D) 150
   (E) None of these
2. What is the maximum daily profit, in rupees, that Mr. David can realize from his business ?
   (A) 620 (B) 920
   (C) 840 (D) 760
   (E) Cannot be determined

Answers with Explanation

1. (B) Cost function $c(f) = 240 + bx + cx^2$

When production changes from 20 to 40

$$[c(40)^2 + b(40) + 240] - [c(20)^2 + b(20) + 240] = \frac{2}{3}[c(20)^2 + b(20) + 240]$$

$$\Rightarrow (1600c + 40b + 240) - (400c + 20b + 240) = \frac{2}{3}(400c + 20b + 240)$$

$$\Rightarrow 1200c + 20b = \frac{2}{3}(400c + 20b + 240)$$

$$\Rightarrow 3600c + 60b = 800c + 40b + 480$$

$$\Rightarrow 2800 + 20b = 480$$

$$\Rightarrow 140c + b = 24 \quad \ldots(1)$$

When production changes from 40 to 60

$$[c(60)^2 + b(60) + 240] - [c(40)^2 + b(40) + 240] = \frac{1}{2}[c(40)^2 + b(40) + 240]$$

$$\Rightarrow 2400c = 240$$

$$\Rightarrow c = \frac{1}{10}$$

On substituting in equation (1)

$$140 \times \frac{1}{10} + b = 24$$

$$14 + b = 24$$

$$b = 10$$

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Profit $p(x)$ = Sales – Cost

$$p(x) = 30x - \frac{x^2}{10} + 10x + 240$$

$$p(x) = -\frac{x^2}{10} + 20x - 240$$

On differentiating and putting equal to zero

$$-\frac{2x}{10} + 20 = 0$$

$$\Rightarrow \quad x = 100$$

Profit $p(x)$ at 100 $= -1000 + 2000 - 240$

$= 760$

Ans. 100

2. (D) Maximum daily profit

As we have solved in previous question that if he produces 100 units daily then he can gain maximum profit.

The maximum daily profit

$= f(100)$

$= -1000 + 2000 - 240$

$= 760$

Exercise 9

Directions—Answers the questions on the basis of the information given below—

Ram and Shyam run a race between points A and B, 5 km apart. Ram starts at 9 a.m. from A at a speed of 5 km/hr, reaches B, and returns to A at the same speed. Shyam starts at 9 : 45 a.m. from A at a speed of 10 km/hr, reaches B and comes back to A at the same speed.

1. At what time do Ram and Shyam first meet each other ?

(A) 10 a.m. (B) 10 : 10 a.m.

(C) 10 : 20 a.m. (D) 10 : 30 a.m.

(E) None of these

2. At what times does Shyam overtake Ram ?

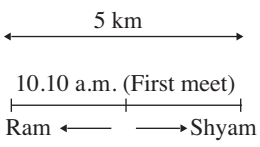
(A) 10 : 20 a.m. (B) 10 : 30 a.m.

(C) 10 : 40 a.m. (D) 10 : 50 a.m.

(E) None of these

Answers with Explanation

1. (B) Distance covered by Ram in 45 minute



$$= \frac{3}{4} \times 5$$

$= 3{\cdot}75$ km at 9·45 a.m.

Distance covered by Shyam when Ram reached P.

$$B = 10 \times \frac{1}{4} = 2{\cdot}5$$

Distance between Ram and Shyam at 10 a.m.

$5 - 2{\cdot}5 = 2{\cdot}5$ km

Now, time taken by Ram and Shyam to meet each other

$$= \frac{2{\cdot}5}{\text{Relative speed}}$$

$$= \frac{2{\cdot}5}{5 + 10}$$

= 10 minute

$\therefore$ Ram and Shyam will first meet

10 hr + 10 m = 10 : 10 a.m.

2. (B) They first meet each other at 10 : 10 a.m.

Time taken by Shyam to reach point B

$$= \frac{\frac{5}{6}}{10}$$

$$= \frac{1}{12}$$

= 5 minute

Now, distance between Ram and Shyam when Shyam reached point (B)

$$\frac{5}{6} + \frac{1}{12} \times 5 = \frac{15}{12} \text{ km}$$

Time taken by Shyam to overtake Ram

$$\frac{15/12}{\text{Relative speed}} = \frac{15/12}{10 - 5} = \frac{3}{12} \text{ hr}$$

That is 15 minute

Time 10 : 10 + 5 minute + 15 minute

= 10 : 30 a.m.

Exercise 10

Directions—Study the following caselet carefully and answer the questions that follow—

A survey was conducted involving 300 organisations regarding website and management of E-Commerce in their organisations. The question of management of E-Commerce was relevant to those organisations who already had their websites. The results of the survey are shown ahead—

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Question asked : Does your Organisation have Website ?

Percentages of Different Responses

Yes	61%
No. Planning with 3 years	4%
No. Planning with 2-3 years	15%
No. Planning next year	10%
No. Planning this years	10%

Question asked : Who manages Electronic commerce in your organisation ?

Percentage of Different Responses

IT/MIS Dept.	65%
Special Task force	10%
Senior Management	7%
Sales/Marketing	15%
Customer Services	3%

1. How many organisations already have their websites ?
   (A) 61 (B) 300
   (C) 183 (D) 200
2. How many organisations plan to introduce website this year ?
   (A) 10 (B) 30
   (C) 60 (D) 12
3. Amongst the organisations, having their websites already, there are a few where management of E-Commerce is looked after by IT/MIS department. Number of such organisations is (approx.)—
   (A) 65 (B) 183
   (C) 151 (D) 119
4. Share or organisations where E-Commerce is looked after by special task force to the total sample surveyed is about—
   (A) 10% (B) 6%
   (C) 8% (D) 5%
5. Which of the following definitely emerges from the study ?
   (A) Website is becoming popular among various organisations
   (B) Website is managed primarily by IT/MIS departments
   (C) Within 3 years, the website will be introduced by all the organisations covered by the survey
   (D) It is better to ask Sales/Marketing Division to manage E-Commerce activities

Answers with Explanation

1. (C) 61% of 300 = 183
2. (B) 10% of 300 = 30
3. (B) 65% of 183 = 119
4. (B) 61% of 300 = 183
   ⇒ 183 organisations have websites, 10% of the websites are being looked by special Task Force, *i.e.*, 18·3 ⇒ 18% (App.) which is 6% of 300.
5. (A)

Exercise 11

Directions—The following caselet shows some data about the cricket matches played between India and New Zealand. Study the information given in the caselet carefully and answer the questions that follow—

India Vs. New Zealand
Matches Played : 42

Won by India : 24 Won by NZ : 18

Highest Innings Totals :

India	289-3(50)	Delhi	1994-95
NZ	348-(50)	Nagpur	1995-96

Lowest Innings Totals :

India	113(44·2)	Perth	1984-86
NZ	126(35)	Bombay	1995-96

Highest Match Aggregates : 597(89·3)

NZ	348-8(50)	
India	249(39·3) at Nagpur	1995-96

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Lowest Match Aggreagates : 228(84·3)

NZ	115-7(40.1)	
India	113(44.2)at Perth	1985-86

Centuries :

For India

117	S. R. Tendulkar	(Bangalore)	14·05·97
115	S. R. Tendulkar	(Baroda)	28.10.94
108*	M. Azharuddin	(Baroda)	17.12.88
103*	S. M. Gavaskar	(Nagpur)	31..10.87
102*	M. Amarnath	(Sharjah)	27.03.88

For New Zealand :

114*	G. M. Turner	(Manchester)	14.06.75
114	N. J. Astle	(Nagpur)	26.11.95
108	K. R. Rutherford	(Baroda)	28.10.94
107*	M. D. Crowe	(Jamshedpur)	15.11.95
104	M. D. Crowe	(Dunedin)	01.03.90
103	C. L. Cairns	(Pune)	24.11.95

5 wickets in an innings for India :

5-26 J. Srinath	(Visakhapatnam)	10.12.88
5-32 J. Srinath	(Indore)	15.12.88
5-33 A. Kumble	(Wellington)	30.03.94
5-33 M. Prabhakar	(Amritsar)	18.11.95

For New Zealand :

5-23 R. O. Collinge	(Christchurch)	21.02.76
5-32 R. J. Hadlee	(Perth)	09.12.80

Most Economical Bowling :

For India 10-2-17-0 R. J. Shastri	(Perth)	85-86
For NZ 10-5-13-0 E. J. Chatfield	(Adeliade)	80-81

Most Expensive Bowling :

For India 10-0-74-0 S. K. Sharma	(Baroda)	88-89
For NZ 10-0-70-0 J. V. Coney	(Brisbane)	80-81

1. If the number of matches won by either side was to be shown on the pie-chart, what would be the angle subtended at the number of matches won of New Zealand ?
   (A) 120° (B) 180°
   (C) 154° (D) 130°
2. In the same diagram, the angle for India would be—
   (A) 180° (B) 240°
   (C) 230° (D) 206°
3. The data given here is based on the matches played between India and New Zealand over a period of approximately—
   (A) 5 years (B) 50 years
   (C) 21 years (D) 10 years
4. The ratio of the number of matches in which centuries were made to the number of matches won by New Zealand is—
   (A) 5/18 (B) 6/18
   (C) 11/42 (D) 3/18
5. The ratio of number of matches in which centuries were made to the number of matches won by India is—
   (A) 2/7 (B) 3/8
   (C) 1/5 (D) 5/24
6. The ratio of the number of matches played to the number in which centuries were made by either side is—
   (A) 42/11 (B) 42/20
   (C) 4/1 (D) 3/5

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7. Approximately, how many years differences is there between the year in which India scored its lowest innings total land the year in which New Zealand did the same ?
   (A) 1 year (B) 3 years
   (C) 10 years (D) 20 years

8. Which of the following is out of place in the group ?
   (A) S. R. Tendulkar (B) M. Amarnath
   (C) M. Azharuddin (D) Saurav Ganguly

9. Which of the following is out of place ?
   (A) G. M. Turner (B) M. D. Crowe
   (C) R. J. Hadlee (D) N. J. Astle

10. Which of the following is out of place ?
    (A) R. J. Shastri (B) J. Srinath
    (C) A. Kumble (D) M. Prabhakar

11. Which of the following would belong to the same category as S. R. Tendulkar ?
    (A) G. M. Turner (B) M. D. Crowe
    (C) N. J. Astle (D) C. L. Cairns

12. Which of the following would belong to the same category as E. J. Chatfield ?
    (A) R. J. Shastri (B) A. Kumble
    (C) J. Srinath (D) M. Prabhakar

13. What is the ratio of the most economical to the most expensive bowling for New Zealand ?
    (A) 1 : 3 (B) 13 : 70
    (C) 10 : 10 (D) 1 : 2

14. What is the ratio of the most expensive to the most economical bowling for India ?
    (A) 74 : 17 (B) 17 : 74
    (C) 1 : 3 (D) 2 : 1

15. India's lowest innings score was in a match played at—
    (A) Christchurch (B) Wellington
    (C) Perth (D) Sharjah

Answers with Explanation

1. (C) $42 \rightarrow 360° \Rightarrow 18$
$= \frac{360°}{42} \times 18$
$= 154{\cdot}28°$
$\Rightarrow 154°$

2. (D) $360° - 154° = 206°$

3. (C) A period of 1975 to 1996.

4. (B) Centuries were made in 6 matches by New Zealand out of total Matches played 18.
$\therefore$ Required ratio $= \frac{6}{18}$

5. (D) The required ratio $= \frac{5}{24}$.

6. (A) 7. (C)

8. (D) Saurav Ganguly was not a member of the team.

9. (C) R. J. Hadlee is a bowler.

10. (B)

11. (B) Crowe and Tendulkar scored two centuries each.

12. (A) Most economical bowlers from either side.

13. (B) 14. (A) 15. (C)

Exercise 12

Directions—Answer the questions on the basis of the information given below—

In an examination, there are 100 questions divided into three groups A, B and C such that each group contains atleast one question. Each question in group A carries 1 mark, each question in group B carries 2 marks and each question in group C carries 3 marks. It is known that the questions in group A together carry atleast 60% of the total marks.

1. If group B contains 23 questions, then how many questions are there in group C ?
   (A) 1
   (B) 2
   (C) 3
   (D) Cannot be determined

2. If group C contains 8 questions and group B carries atleast 20% of the total marks, which of the following best describes the number of questions in group B ?
   (A) 11 or 12
   (B) 12 or 13
   (C) 13 or 14
   (D) 14 or 15

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Answers with Explanation

1. (A) Group B contains 23 questions which carry 46 marks

   If group C contains 1 question which will carry 3 marks

   ∴ Group A will contains 76 questions which will cary 76 marks

   ∴ Total marks = 125

   Now 76 marks of 125 marks are = 60·8%

   Hence, group C will contain only 1 question.

2. (C) In group C there are 8 questions
   → 24 marks

   If in group B there are 14 questions
   → 28 marks

   ∴ In group A there are 78 questions
   → 78 marks

   Total mark = 130

   ∴ % marks in group B $= \frac{28 \times 100}{130}$

   $= 21{\cdot}54$

   If in group B there are 13 questions → 26 marks

   ∴ mark of group C = 24

   and marks of group A = 79

   ∴ % marks in group B $= \frac{26 \times 100}{129}$

   $= 20{\cdot}15\%$

   Hence, group B contains either 13 or 14 questions.

●●

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7 Combination of Diagrams

'Combination of Diagrams' means a combination of two or more than two diagrams or the graphs at one place. In a combination of diagrams, a question has at least two diagrams or the graphs of different kinds-showing the various conditions of the question.

Generally, the questions have the following combination of diagrams—

(1) Pie Chart and Table.

(2) Pie Chart and Bar Graph.

(3) Pie Chart and Line Graph.

(4) Bar Graph and Table.

(5) Line Graph and Table

(6) A pair of pie charts.

(7) A caselet with a diagram.

(8) Bar Graph and Line Graph.

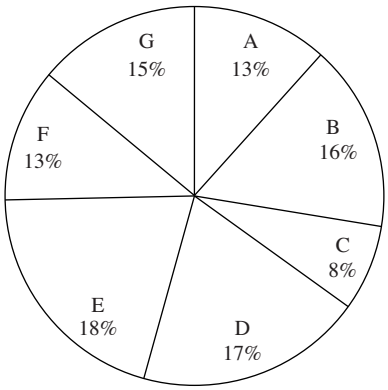
The following exercises are the examples of the combination of diagrams—

Exercise 1

Directions—Study the following diagrams carefully and answer the questions that follow—

Villages	% Population Below Poverty Line
A	45
B	52
C	38
D	58
E	46
F	49
G	51

Proportion of Population of Seven Villages in the Year of 1995



1. In 1996, the population of villages A and B is increased by 10% from the year 1995. If the population of village A in 1995 was 5000 and the percentage of population below poverty line in 1996 remains same as in 1995, Find approximately the population of village B below poverty line in 1996—
   (A) 4000 (B) 4500
   (C) 2500 (D) 3000
   (E) 3500
2. If in 1997 the population of village D is increased by 10% and the population of village G is reduced by 5% from 1995 and the population of village G in 1995 was 9000, what is the total population of villages D and G in 1997 ?
   (A) 19770
   (B) 19200
   (C) 18770
   (D) 19870
   (E) None of these
3. If in 1995 the total population of the seven villages together was 55,000 approximately,

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what will be population of village F in that year below poverty line ?

(A) 3000 (B) 2500
(C) 4000 (D) 3500
(E) 4500

4. If the population of village C below poverty line in 1995 was 1520, what was the population of village F in 1995 ?

(A) 4000 (B) 6000
(C) 6500 (D) 4800
(E) None of these

5. The population of village C was 2000 in 1995. What will be the ratio of population of village C below poverty line to that of the village E below poverty line in that year ?

(A) 207 : 76 (B) 76 : 207
(C) 152 : 207 (D) Data inadequate
(E) None of these

Answers with Explanation

1. (E) Population of village B in 1995

$$= 5000 \times \frac{16}{13}$$

$$= 6150 \text{ (App.)}$$

$\therefore$ Population of village B in 1996

$$= 6150 \times \frac{110}{100}$$

$$= 6750$$

$\therefore$ Population below poverty line

$$= 52\% \text{ of } 6750$$

$$= \frac{52 \times 6750}{100}$$

$$= 3500 \text{ (App.)}$$

2. (A) Population of village D in 1995

$$= 9000 \times \frac{17}{15}$$

$$= 10{,}200$$

Population of village D in 1997

$$= 10200 \times \frac{110}{100}$$

$$= 11220$$

Population of village G in 1997

$$= 9000 \times \frac{95}{100}$$

$$= 8550$$

$\therefore$ Total population

$$= 11220 + 8550$$

$$= 19770$$

3. (D) Population of village F below poverty line

$$= 55000 \times \frac{13}{100} \times \frac{49}{100}$$

$$= 3500 \text{ (App.)}$$

4. (C) Population of village F in 1995

$$= 1520 \times \frac{100}{38} \times \frac{13}{8}$$

$$= 6500$$

5. (B) Population of village C below poverty line

$$= 2000 \times \frac{38}{100} = 760$$

Population of village E below poverty line

$$= \frac{2000}{8} \times 18 \times \frac{46}{100}$$

$$= 2070$$

$\therefore$ The required ratio

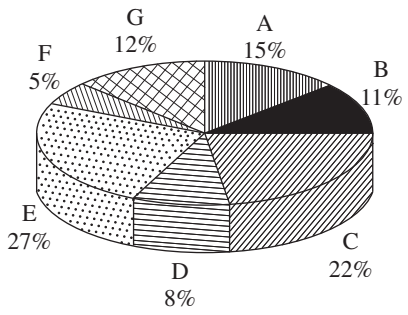
$$= \frac{760}{2070}$$

$$\Rightarrow 76 : 207$$

Exercise 2

Directions—Seven companies A, B, C, D, E, F and G are engaged in production of two items I and II. The comparative data about production of these items by the seven companies is given in the following graph and table. Study them carefully and answer the questions given below—

Percentage of the Total Production Produced by the Seven Companies



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Cost of the total production (both items together) by seven companies = Rs. 25 crores
Ratio of production between items I and II and the per cent profit earned for the two items

Company	Ratio of Production		Per cent Profit Earned	
	Item I	Item II	Item I	Item II
A	2	3	25	20
B	3	2	32	35
C	4	1	20	22
D	3	5	15	25
E	5	3	28	30
F	1	4	35	25
G	1	2	30	24

1. What is the total cost of the production of item 1 by companies A and C together in Rs. crore ?
   (A) 9·25 (B) 5·9
   (C) 4·1625 (D) 4·9
   (E) None of these

2. What is the amount of profit earned by company D on item II ?
   (A) Rs. 3·125 cr (B) Rs. 31·25 cr
   (C) Rs. 3·125 lakhs (D) Rs. 31·25 lakhs
   (E) None of these

3. Cost of production of item I by company F is what per cent of the cost of production of item II by company D ?
   (A) 16% (B) 33·33%
   (C) 66·67% (D) 12·5%
   (E) None of these

4. What is total profit earned by company G for items I and II together ?
   (A) Rs. 78 lakhs (B) Rs. 1·62 cr
   (C) Rs. 7·8 cr (D) 16·2 lakhs
   (E) None of these

5. What is the ratio of the cost of production of item I by company A to the cost of production of item I by company D ?
   (A) 3 : 5 (B) 1 : 2
   (C) 2 : 1 (D) 2 : 3
   (E) None of these

6. What is the total of profit earned by company B on production of item I and the profit earned by company A on production of item II ?
   (A) Rs. 9·78 cr (B) Rs. 97·8 lakhs
   (C) Rs. 52·8 lakhs (D) Rs. 5·28 cr
   (E) None of these

7. The cost of production of both items together by company E is equal to the total cost of production of both items together by which of the two companies ?
   (A) C and D (B) B and G
   (C) A and D (D) C and F
   (E) A and B

8. What is the total of the cost of production of item I by company A and the cost of production of item II by company B ?
   (A) Rs. 2·6 cr (B) Rs. 26 lakh
   (C) Rs. 3·35 cr (D) Rs. 33·65 lakh
   (E) None of these

Answers with Explanation

1. (B) Total cost $= \frac{2}{5} \times \frac{15}{100} \times 25 + \frac{4}{5} \times \frac{22}{100} \times 25$
   $= 1{\cdot}5 + 4{\cdot}4$
   $= 5{\cdot}9$ cr.

2. (D) Amount of profit earned by company D on item II
   $= \frac{5}{8} \times \frac{8}{100} \times 25 \times \frac{25}{100}$
   $= 31{\cdot}25$ lakh

3. (E) Cost of production of item I by company F
   $= \frac{1}{5} \times \frac{5}{100} \times 25$
   $= 0{\cdot}25$ cr
   Cost of production of item II by company D
   $= \frac{5}{8} \times \frac{8}{100} \times 25$
   $= 1{\cdot}25$ cr
   $\therefore$ Reqd. % $= \frac{0{\cdot}25}{1{\cdot}25} \times 100$
   $= 20\%$

4. (A) Total profit earned by company G
   $= \frac{1}{3} \times \frac{12}{100} \times 25 \times \frac{30}{100}$
   $+ \frac{2}{3} \times \frac{12}{100} \times 25 \times \frac{24}{100}$

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$= 0{\cdot}3 + 0{\cdot}48$

$=$ Rs. 78 lakh

5. (C) Required ratio $= \dfrac{\frac{2}{5} \times \frac{15}{100} \times 25}{\frac{3}{8} \times \frac{8}{100} \times 25}$

$= 2 : 1$

6. (B) Required profit

$= \dfrac{3}{5} \times \dfrac{11}{100} \times 25 \times \dfrac{32}{100} + \dfrac{3}{5} \times \dfrac{15}{100} \times 25 \times \dfrac{20}{100}$

$= 0{\cdot}528 + 0{\cdot}450$

$=$ Rs. 97·8 lakh

7. (D) 8. (A)

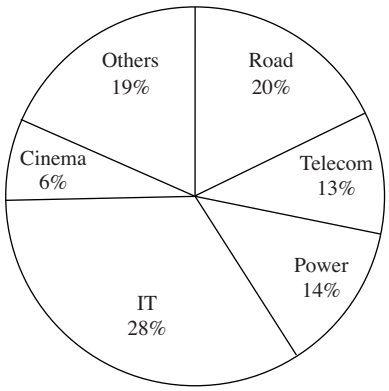
Exercise 3

Directions—Study the following table and Pie Chart carefully and answer the questions that follow—

FDI in Indian States During the Year 1999-2000

States	UP	Delhi	Karnataka	Maharashtra	Kerala	MP	AP
FDI (In Rs. Cr.)	500	400	600	550	580	520	650

The Investment in Different Sectors



1. The ratio of investment of UP to the state of AP in power sector is—
   (A) 10 : 13 (B) 13 : 10
   (C) 10 : 21 (D) 21 : 10
2. What is the ratio between the investment in IT of AP and in other of UP ?
   (A) 65 : 63 (B) 182 : 95
   (C) 63 : 65 (D) 95 : 182
3. The total investment in Road sector by these states is—
   (A) 800 cr (B) 720 cr
   (C) 760 cr (D) 700 cr
4. The FDI in cinema sector in Delhi is what per cent less than that in Kerala in others ?
   (A) 40% (B) 80%
   (C) 50% (D) 60%
5. FDI in Maharashtra in Telecom sector is what per cent of that in AP in IT sector ?
   (A) 42 (B) 32
   (C) 62 (D) 22

Answers with Explanation

1. (A) The required ratio

$= \dfrac{\frac{500 \times 14}{100}}{\frac{650 \times 14}{100}} = \dfrac{500}{650} = \dfrac{10}{13}$

$\Rightarrow 10 : 13$

2. (B) The required ratio

$= \dfrac{28\% \text{ of } 650}{19\% \text{ of } 500} = \dfrac{182}{95}$

$= 182 : 95$

3. (C) The investment in Road sector

$= \dfrac{20}{100} \times (500 + 400 + 600 + 550 + 580 + 520 + 650)$

$= \dfrac{20}{100} \times 3800$

$= 760$ cr

4. (B) FDI in cinema by Delhi

$= 6\%$ of $400 = 24$ cr

FDI in others by Kerala

$= 19\%$ of $580 = 110{\cdot}20$ cr

$\therefore$ Required % $= \left(\dfrac{110{\cdot}20 - 24}{110{\cdot}20}\right) \times 100$

$= 80\%$ (App.)

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5. (A) The required %

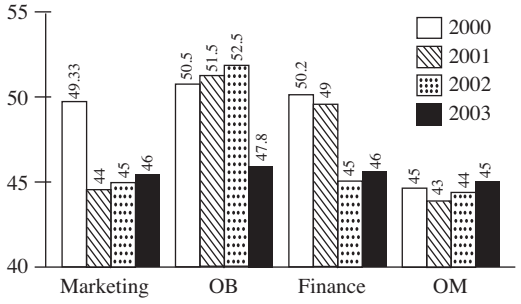
$$= \frac{13\% \text{ of } 550}{28\% \text{ of } 650} \times 100$$

$$= 42\% \text{ (App.)}$$

Exercise 4

Directions—On the basis of the following information, answer the questions given below—

A management institute was established on January 1, 2000 with 3, 4, 5 and 6 faculty members in the Marketing. Organizational Behaviour (OB), Finance and Operations Management (OM) areas respectively, to start with. No faculty member retired or joined the institute in the first three months of the year 2000. In the next four years the institute recruited one faculty member in each of the four areas. All these new faculty members, who joined the institute subsequently over the years, were 25 years old at the time of their joining the institute. All of them joined the institute on April 1. During these four years, one of the faculty members retired at the age of 60. The following diagram gives the areawise average age (in terms of number of completed year) of faculty members as on April 1 of 2000, 2001, 2002 and 2003.



1. In which year did the new faculty member join the Finance area ?
   (A) 2000 (B) 2001
   (C) 2002 (D) 2003

2. What was the age of the new faculty member who joined the OM area, as on April 1,2003 ?
   (A) 25 (B) 26
   (C) 27 (D) 28

3. From which area did the faculty member retire ?
   (A) Finance (B) Marketing
   (C) OB (D) OM

4. Professors Naresh and Devesh, two faculty members in the marketing area, who have been with the institute since its inception, share a birthday which fulls on 20th November one was born in 1947 and the other one in 1950, on April 1, 2005, what was the age of the third faculty member who has been in the same area since inception ?
   (A) 47 (B) 50
   (C) 51 (D) 52

Answers with Explanation

1. (C) In finance, the average age of faculty members in the year 2000 is 50·2 years. There are five faculty members

$\therefore$ Total age of 5 members $= 50{\cdot}2 \times 5$

$= 251$ years

In the year 2001, average age is 49 years.

Hence a retirement takes place whose age is 60 years.

Therefore, $$\frac{251 + \left(\begin{array}{c}\text{Enhancement}\\ \text{of age of all the}\\ \text{five members}\end{array}\right) - 60}{5 - 1 \text{ (Retirement)}}$$

$$= \frac{251 + 5 - 60}{4}$$

$$= 49 \text{ years}$$

In the year 2002, a new member of 25 years join the finance area as

$$\frac{49 \times 4 + 4(\text{Enhancement of age}) + 25}{4 + 1 \text{ (New joining)}}$$

$$= (196 + 4 + 25)/5$$

$$= 45 \text{ years}$$

2. (C) In 2000

Total age of 6 members 45×6

$= 270$ years in 2001

Total age of 6 members $= 270 + 6$

$= 276$ years

But as given in the diagram the average age is decreasing. It means a new member joins whose age is 25 years.

Thus, $\frac{270 + 6 + 25}{7} = \frac{301}{7}$

$= 43$ (which has been given)

In 2001 new member's age is 25 years after two years his age would be 27 years.

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3. (A) As shown in the solution of Q. no. of 1.

4. (D) From the time of inception of Marketing area there were three members in which Professors Naresh and Devesh whose date of birth of 20 Nov. 1947 and 20 Nov., 1950.

The exact age one member on 1 Jan., 2000 was—

1 Jan., 2000
–20 Nov., 1947
52years, 41 days

The exact age of other member on 1 Jan., 2002 was

1 Jan., 2000
–20 Nov., 1950
49 years, 41 days

Total age of both the members
= (52 years + 41 days)
+ (49 years + 41 days)
= 101 years 82 days

Total age of all the three members
49·33 □ 3 = 148 years

Age of third member on 1 Jan., 2002
= 46 years, 273 days

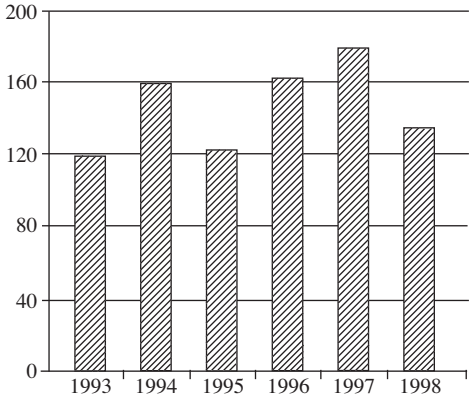
Age of third member on April 1
46 years + 273 days + 5 years and 92 days
= 52 years

Exercise 5

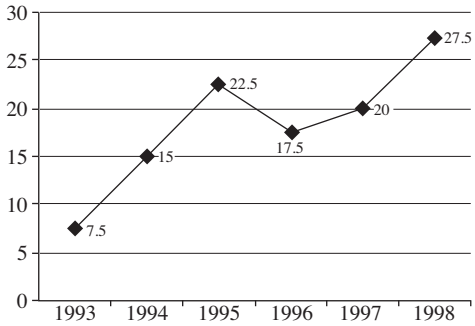
Directions—Study the following graphs carefully and answer the questions given below—

Income of a Company

(In Rs. lakhs)



Percentage Profit Over the Years



1. In which of the following years was the amount of profit the maximum ?
(A) 1997 (B) 1994
(C) 1993 (D) 1995
(E) None of these

2. Approximately what was the average expenditure of the given years ?
(A) Rs. 110 lakhs
(B) Rs. 130 lakhs
(C) Rs. 120 lakhs
(D) Rs. 140 lakhs
(E) Data inadequate

3. In which of the following years was the increase/decrease in per cent profit from the previous year the minimum ?
(A) 1994 (B) 1996
(C) 1997 (D) 1995
(E) None of these

4. Approximately what was the expenditure in 1994 ?
(A) Rs. 120 lakhs
(B) Rs. 160 lakhs
(C) Rs. 140 lakhs
(D) Rs. 180 lakhs
(E) None of these

5. If the profit percentage in 1997 was 25, what would have been the expenditure in that years ?
(A) Rs. 130 lakhs
(B) Rs. 148 lakhs
(C) Rs. 120 lakhs
(D) Rs. 152 lakhs
(E) None of these

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Answers with Explanation

1. (E) By the use of direct formula for Profit

$$= \text{Income}\left[1 - \frac{100}{100 + \%\ \text{profit}}\right]$$

We see that the profit is maximum in 1998.

2. (B) Total expenditure

$$= 120 \times \frac{10}{107{\cdot}5} + 160 \times \frac{100}{115} + 130 \times \frac{100}{122{\cdot}5} + 170 \times \frac{100}{117{\cdot}5} + 190 \times \frac{100}{120} + 150 \times \frac{100}{127{\cdot}5}$$

$$= \text{Rs. } 777{\cdot}51 \text{ lakh}$$

$$\therefore \quad \text{Average} = \frac{777{\cdot}51}{6}$$

$$\approx \text{Rs. } 130 \text{ lakh}$$

3. (A) Per cent profit increase/decrease from the previous year

1994	1995	1996	1997	1998
100	50	(–) 22·22	14·28	37·5

4. (C) Expenditure in 1994 $= 160 \times \frac{100}{115}$

$$\approx 140 \text{ lakh}$$

5. (D) Expenditure in 1997 $= 190 \times \frac{100}{125}$

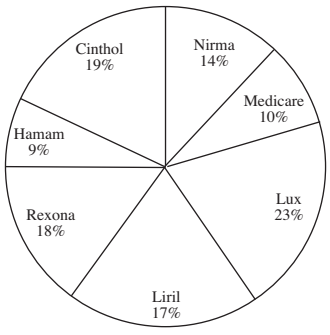
$$\approx 152 \text{ lakh}$$

Exercise 6

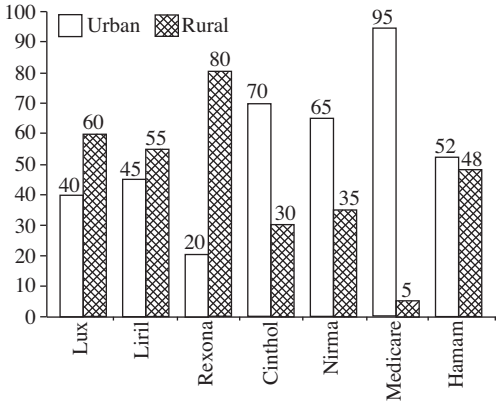
Directions—Study the following diagrams of Pie-chart and bar graph carefully and answer the questions given below—

Production of Soaps in India

(Total production 100000 units per month)



Percentage Selling in Rural and Urban Areas



1. What is the difference between the sale of Lux in urban areas and that of Cinthol in rural areas ?
   (A) 3500 units (B) 4000 units
   (C) 4500 units (D) 2000 units
2. Which company sells maximum number of soaps in urban areas ?
   (A) Rexona (B) Medicare
   (C) Nirma (D) Hamam
3. What per cent of the total number of soaps sales in rural areas ?
   (A) 62 (B) 57
   (C) 49 (D) 55
4. What is the difference between the sale of Nirma and Rexona in urban areas ?
   (A) 5500 (B) 5600
   (C) 6000 (D) 5800
5. How many Medicare soaps sell in rural areas ?
   (A) 400 (B) 700
   (C) 500 (D) 600

Answers with Explanation

1. (A) The required difference

$$= 100000\left(\frac{23}{100} \times \frac{40}{100} - \frac{19}{100} \times \frac{30}{100}\right)$$

$$= \frac{100000}{100 \times 100}(920 - 570)$$

$$= 3500 \text{ units}$$

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2. (B) Sale of Hamam $= 100000 \times \frac{9}{100} \times \frac{52}{100}$

$= 4680$

Sale of Medicare $= 100000 \times \frac{10}{100} \times \frac{95}{100}$

$= 9500$

Sale of Nirma $= 100000 \times \frac{14}{100} \times \frac{65}{100}$

$= 9100$

Sale of Rexona $= 100000 \times \frac{18}{100} \times \frac{20}{100}$

$= 3600$

3. (C) The required %

$$= \left(\frac{23}{100} \times \frac{60}{100} + \frac{17 \times 55}{100 \times 100} \times \frac{18 \times 80}{100 \times 100} + \frac{19 \times 30}{100 \times 100} + \frac{14 \times 35}{100 \times 100} + \frac{10 \times 5}{100 \times 100} + \frac{9 \times 48}{100 \times 100}\right) \times 100$$

$$= \left(\frac{1380 + 935 + 1440 + 570 + 490 + 50 + 432}{10000}\right) \times 100$$

$= 49\%$ (App.)

4. (A) The required difference

$$= \frac{100000}{100 \times 100}(14 \times 65 - 18 \times 20)$$

$= 10 \times (910 - 360)$

$= 5500$

5. (C) The required sell

$= 100000 \times \frac{10}{100} \times \frac{5}{100}$

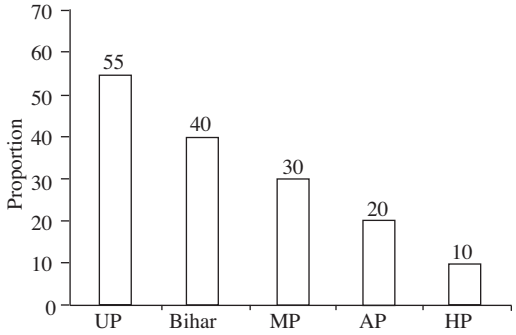
$= 500$

Exercise 7

Directions—Study the following graphs carefully and answer the questions that follow—

States	% Population Below Poverty Line
UP	40
Bihar	50
MP	60
AP	30
HP	30

Proportion of Population of the States in 1994



1. If the population of AP below poverty line is 2 crore, what will be the population of MP in 1994 ?
   (A) 10 crore
   (B) 20 crore
   (C) 12 crore
   (D) 18 crore
2. If the population of all the states together in 1994 was 50 crore, what will be the population of Bihar below poverty line in 1994 ?
   (A) 2 crore
   (B) 10 crore
   (C) 7 crore
   (D) 5 crore
3. What is the % population below poverty line in the state of HP ?
   (A) 30 (B) 40
   (C) 60 (D) 50

Answers with Explanation

1. (A) The required population

$= 2 \times \frac{100}{30} \times \frac{30}{20}$

$= 10$ crore

2. (B) The required population

$= 50 \times \frac{40}{100} \times \frac{50}{100}$

$= 10$ crore

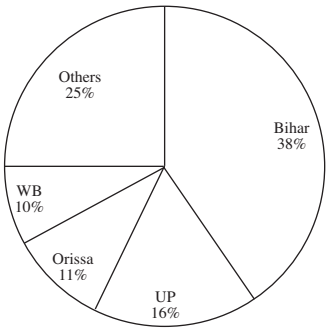
3. (A)

Exercise 8

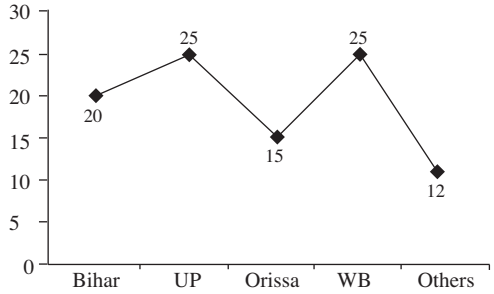
Directions—Study the following charts carefully and answer the questions that follow—

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The Percentage of Number of Students Passed in PO Examination from Different Parts of the Country in 1999



Percentage of Students who Passed their Graduation in 1999



1. If in 1999 the total passed candidates from different parts of the country was 650, then how many non-fresher candidates from Bihar passed the examination in 1999 ?
   (A) 200 (B) 195
   (C) 198 (D) 204
   (E) 188

2. If in 1999 total no. of freshers from WB was 160, then how many non-fresher candidates passed the exam from Others ?
   (A) 1398 (B) 1588
   (C) 640 (D) 1408
   (E) Can't be determined

3. If total passed candidates from UP in 1999 was 112, what is the ratio between the no. of freshers from Bihar and that of non-fresher from Orissa ?
   (A) 760 : 187 (B) 187 : 760
   (C) 40 : 11 (D) 11 : 40
   (E) None of these

4. If there is an increase of 10% and 20% candidates from Bihar and Others in the year 2000 respectively, and the number of total passed candidates from Orissa in 1999 was 77, what would be the **approximate** total passed candidates from Bihar and Others in 2000 ?
   (A) 210 (B) 480
   (C) 450 (D) 550
   (E) 500

Answers with Explanation

1. (C) Number of non-fresher candidates from Bihar
$$= 650 \times \frac{38}{100} \times \frac{80}{100} = 198$$

2. (D) Number of non-fresher candidates from Others
$$= \frac{160}{25} \times 100 \times \frac{25}{10} \times \frac{88}{100}$$
$$= 1408$$

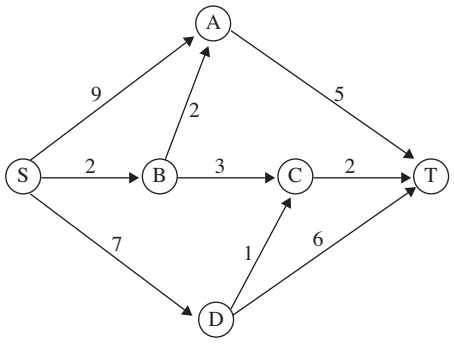
3. (E) Required ratio $= \dfrac{112 \times \frac{38}{16} \times \frac{20}{100}}{112 \times \frac{11}{16} \times \frac{85}{100}}$
$$= 152 : 187$$

4. (E) Total passed candidates in 2000
$$= \frac{38}{11} \times 77 \times \frac{110}{100} + \frac{25}{11} \times 77 \times \frac{120}{100}$$
$$\approx 500$$

Exercise 9

Directions—Answer the following questions on the basis of the information given below—

A significant amount of traffic flows from point S to point T in the one-way street-network shown below. Points A, B, C and D are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street.



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Motorists travelling from point S to point T would obviously take the route for which the total cost of travelling is the minimum. If two or more routes have the same least travel cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among all the least cost routes.

The government can control the flow of traffic only by levying appropriate toll at each junction. For example, if a motorist takes the route S-A-T (using junction) A alone), then the total cost of travel would be Rs. 14 (*i.e.*, Rs. 9- Rs. 5) plus the toll charged at junction A.

1. If the government wants to ensure that no traffic flows on the street from D to T, while equal amount of traffic flows through junctions A and C, then a feasible set of toll charged (in rupees) at junctions A, B, C and D respectively to achieve this goal is—
   (A) 1, 5, 3, 3 (B) 1, 4, 4, 3
   (C) 1, 5, 4, 2 (D) 0, 5, 2, 3
   (E) 0, 5, 2, 2

2. If the government wants to ensure that all motorists travelling from S to T pay the same amount (fuel costs and toll combined) regardless of the route they choose and the street from to C is under repairs (and hence unusable), then a feasible set of toll charged (in rupees) at junctions A, B, C and D respectively to achieve this goals is—
   (A) 2, 5, 3, 2 (B) 0, 5, 3, 1
   (C) 1, 5, 3, 2 (D) 2, 3, 5, 1
   (E) 1, 3, 5, 1

3. If the government wants to ensure that the traffic at S gets evenly distributed along streets from S to A, from S to B, and from S to D, then a feasible set of toll charged (in rupees) at junctions A, B, C and D respectively to achieve this goal is—
   (A) 0, 5, 4, 1 (B) 0, 5, 2, 2
   (C) 1, 5, 3, 3 (D) 1, 5, 3, 2
   (E) 0, 4, 3, 2

4. If the government wants to ensure that all routes from S to T get the same amount of traffic, then a feasible set of toll charged (in rupees) at junctions A, B, C and D respectively to achieve this goals is—
   (A) 0, 5, 2, 2 (B) 0, 5, 4, 1
   (C) 1, 5, 3, 3 (D) 1, 5, 3, 2
   (E) 1, 5, 4, 2

5. The government wants to device a toll policy such that the total cost to the commuters per trip is minimized. The policy should also ensure that not more than 70 per cent of the total traffic passes through junction B. The cost incurred by the commuter travelling from point S to point T under this policy will be—
   (A) Rs. 7 (B) Rs. 9
   (C) Rs. 10 (D) Rs. 13
   (E) Rs. 14

Answers with Explanation

S. No.	Possible Route	Final Cost (Rs.)
1.	S – A – T	9 + 5 = Rs. 14
2.	S – B – A – T	2 + 2 + 5 = Rs. 9
3.	S – B – C – T	2 + 3 + 2 = Rs. 7
4.	S – D – C – T	7 + 1 + 2 = Rs. 10
5.	S – D – T	7 + 6 = Rs. 13

1. (E) Travelling cost should be higher along S – D – T route and should be equal along remaining route. In any condition and from all the options the toll charges of D either 2 or 3 and 2 is minimum. From options if toll charges are A – 0, B – 5, C – 2 and D – 2, total travelling cost along all routes will be Rs. 14. while along S – D – T will be Rs. 15.

For Verification		Fuel Cost	Toll Charges	Total
Route 1	S-A-T	(9 + 5) +	(0)	= 14
Route 2	S-B-A-T	(2 + 2 + 5) +	(5 + 0)	= 14
Route 3	S-B-C-T	(2 + 3 + 2) +	(5 + 2)	= 14
Route 4	S-D-C-T	(7 + 1 + 2) +	(2 + 2)	= 14

2. (B) There are four possible routes
   S-A-T (14), S-B-A-T (9) S-D-C-T (10) and S-D-T (13).
   By taking Toll Tax, also the options B and C can be considered. In B cost Rs. 14 and in C cost Rs. 15.
   Since, minimum fare should be considered. The correct option is (B). Option B (0, 5, 3, 1) is possible route S-B-A-T = 9 + 5 = 14.
3. (D) If toll charges are A-1, B-5, C-3, D-2. Total travelling cost along all routes will be same Rs. 15.

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Possible Route	Fuel Cost	Toll Charges	Total
S-A-T	(9 + 5) +	1	Rs. 15
S-B-C-T	(2 + 3 + 2) +	(5 + 3)	Rs. 15
S-D-T	(7 + 6) +	2	Rs. 15

4. (D) Toll charges 1 5 3 2
A B C D

Total travelling cost along all routes will be Rs. 15.

	Fuel Cost	Toll Charges	Total
(1) S-A-T	(9 + 5) +	1	Rs. 15
(2) S-B-C-T	(2 + 3 + 2) +	(5 + 1)	Rs. 15
(3) S-B-C-T	(2 + 3 + 2) +	(5 + 3)	Rs. 15
(4) S-D-C-T	(7 + 1 + 2) +	(2 + 3)	Rs. 15
(5) S-D-T	(7 + 6) +	2	Rs. 15

5. (C) For cost minimizing, we have to consider that the toll tax should minimum at all junctions. All traffic will flow along S-B-A-T *i.e.*, 100% not possible. All traffic will flow along S-B-C-T. *i.e.*, 100% not possible. Let toll tax at D and C be zero at B be Rs. 3 and A and B together be Rs. 1. Then 50% of traffic will pass through B.

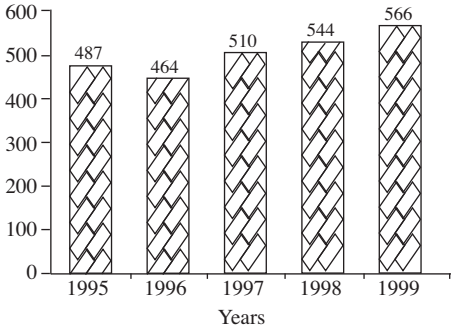
Route S-B-C-T = 2 + 3 + 2 + (Toll 3) = 10

Route S-B-A-T = 2 + 2 + 5 (Toll 1) = 10

Exercise 10

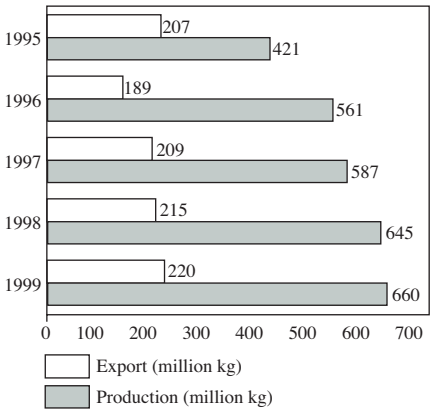
Directions—Study the following graphs carefully and answer the questions that follow—

Per Capita Availability of Tea (gms) in Chaidesh



(**Note**—Availability is defined as production less export.)

Production and Export of Tea (Chaidesh)



1. In which year during the period 1996-1999 was Chaidesh's export of tea, as a proportion tea produced, the highest ?
(A) 1996 (B) 1997
(C) 1998 (D) 1999
(E) 1995

2. In which of the following years was the population of Chaidesh the lowest ?
(A) 1995 (B) 1996
(C) 1997 (D) 1999
(E) None of these

3. The area under tea cultivation continuously decreased in all four years from 1996 to 1999, by 10%, 7%, 4% and 1% respectively. In which year was tea productivity (production per unit of area) the highest ?
(A) 1999 (B) 1998
(C) 1997 (D) 1996
(E) None of these

Answers with Explanation

1. (B) % of tea export in respect of production in 1996

$$= \frac{189 \times 100}{561} = 33{\cdot}69$$

% of tea export in respect of production in 1997

$$= \frac{209 \times 100}{587} = 35{\cdot}61$$

% of tea export in respect of production in 1998

$$= \frac{215 \times 100}{645} = 33{\cdot}33$$

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% of tea export in respect of production in 1999

$$= \frac{220 \times 100}{660} = 33{\cdot}33$$

$\therefore$ It is highest during 1997.

2. (A) Population in 1995

$$= \frac{(421 - 207) \times 1000}{487}$$

$$= 439{\cdot}425 \text{ million}$$

Population in 1996

$$= \frac{(561 - 189) \times 1000}{464}$$

$$= 801{\cdot}724 \text{ million}$$

Population in 1997

$$= \frac{(587 - 209) \times 1000}{510}$$

$$= 741{\cdot}176 \text{ million}$$

and Population in 1999

$$= \frac{(660 - 220) \times 1000}{566}$$

$$= 777{\cdot}385 \text{ million}$$

$\therefore$ It is the lowest in 1995.

3. (A) Let the area in 1995 under cultivation be 100.

$\therefore$ Area in 1996 $= 100 - 10 = 90$

$\therefore$ Production of tea per unit area in 1996

$$= \frac{561}{90} = 6{\cdot}23 \text{ million kg}$$

$$\text{Area in 1997} = \frac{90(100 - 7)}{100} = 83{\cdot}7$$

$\therefore$ Production of tea per unit area in 1997

$$= \frac{587}{83{\cdot}7} = 7{\cdot}01 \text{ million kg}$$

$$\text{Area in 1998} = 83{\cdot}7 \times \frac{(100 - 4)}{100}$$

$$= 80{\cdot}352$$

$\therefore$ Production of tea per unit area in 1998

$$= \frac{645}{80{\cdot}352}$$

$$= 8{\cdot}027 \text{ million kg}$$

$$\text{and area in 1999} = 8{\cdot}352 \times \frac{(100 - 1)}{100}$$

$$= 79{\cdot}548$$

$\therefore$ Production of tea per unit area in 1999

$$= \frac{660}{79{\cdot}548}$$

$$= 8{\cdot}297 \text{ millions kg}$$

$\therefore$ It is the higher in 1999.

Exercise 11

Directions—Study the following information carefully and answer the questions that follow—

The profitability of a company is defined as the ratio of its operating profit to its operating income, typically expressed in percentage. The following two charts show the operating income as well as the profitability of six companies in the Financial Years (F.Ys) 2001-02 and 2002-2003.

Chart I

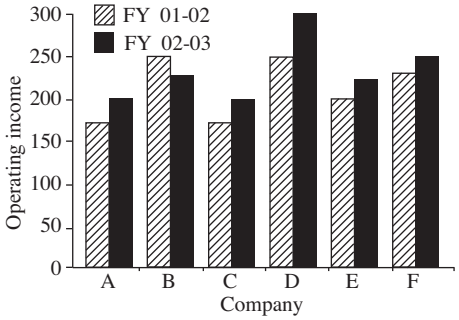
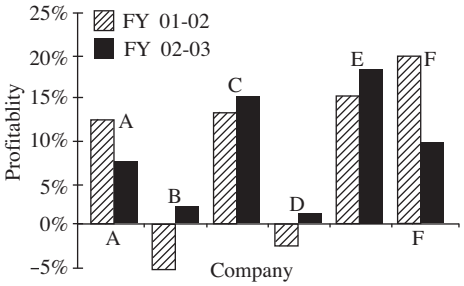
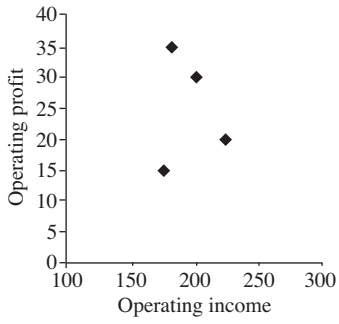


Chart II



The operating profits of four of these companies are plotted against their respective operating income figures for the F.Y. 2002-03, in the third chart given below—

Operating Profit Vs. Operating Income



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1. What is the approximate average operating profit, in F.Y. 2001-2002, of the two companies excluded from the third chart ?
   (A) –7·5 crore
   (B) 3·5 crore
   (C) 25 crore
   (D) Cannot be determined
2. Which company recorded the highest operating profit in F.Y. 2002-03 ?
   (A) A (B) C
   (C) E (D) F
3. Which of the following statements is NOT true ?
   (A) The company with the third lowest profitability in F.Y. 2001-02 has the lowest operating income in F.Y. 2002-03
   (B) The company with the highest operating income in the two financial years combined has the lowest operating profit in F.Y. 2002-03
   (C) Companies with a higher operating income in F.Y. 2001-02, than in F.Y. 2002-03 have higher profitability in F.Y. 2002-03 than in F.Y. 2001-02
   (D) Companies with profitability between 10% and 20% in F.Y. 2001-02 also have operating incomes between 150 crore and 200 crore in F.Y. 2002-03
4. The average operating profit in F.Y. 2002-03, of companies with profitability exceeding 10% in F.Y. 2002-03, is approximately—
   (A) 17·5 crore (B) 25 crore
   (C) 27·5 crore (D) 32·5 crore

Answers with Explanation

1. (A) Operating profit for A in 2002-2003
$$= \frac{8 \times 185}{100} = \text{Rs. } 14{\cdot}8 \text{ crore}$$
Operating profit for B in 2002-2003
$$= \frac{2 \times 220}{100} = \text{Rs. } 4{\cdot}4 \text{ crore}$$
Operating profit for C in 2002-2003
$$= \frac{15 \times 200}{100} = \text{Rs. } 30 \text{ crore}$$
Operating profit for D in 2002-2003
$$= \frac{1 \times 290}{100} = \text{Rs. } 2{\cdot}9 \text{ crore}$$
Operating profit for E in 2002-2003
$$= \frac{17{\cdot}5 \times 200}{100}$$
$$= \text{Rs. } 35 \text{ crore}$$
and operating profit for F in 2002-2003
$$= \frac{9 \times 220}{100}$$
$$= \text{Rs. } 19{\cdot}8 \text{ crore}$$
$\therefore$ In the third chart two companies B and D are excluded
Now, operating profit for B in 2001-2002
$$= -\frac{4 \times 240}{100}$$
$$= \text{Rs. } -9{\cdot}6 \text{ crore}$$
and operating profit for D in 2001-2002
$$= -\frac{2 \times 250}{100}$$
$$= \text{Rs. } (-5) \text{ crore}$$
$\therefore$ Required average
$$= \frac{-9{\cdot}6 - 5}{2} = -\frac{-14{\cdot}6}{2}$$
$$= -7{\cdot}3$$
$$= \text{Rs. } -7{\cdot}5 \text{ crore}$$
2. (C) From the third chart it is clear that the company E recorded the highest operating profit in F. Y. 2002-2003.
3. (D) The companies A, C and E are such whose profitability is between 10% and 20% in F.Y. 2001-02. But the operating income of the company C in F.Y. 2002-03 is not between Rs. 150 crore and Rs. 200 crore. Hence, this statement is not true.
4. (D) The companies C and E are such whose profitability is more than 10%.
   $\therefore$ Average operating profit of the companies C and E in F.Y. 2002-03
$$= \frac{30 + 35}{2}$$
$$= \text{Rs. } 32{\cdot}5 \text{ crore}$$

Exercise 12

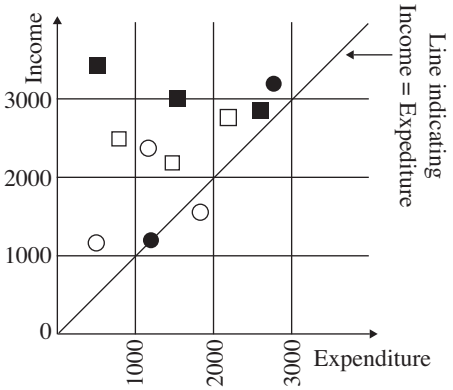
Directions—Answer the following questions on the basis of the information given below—

The data points in the figure below represent monthly income and expenditure data of individual members of the Ahuja family (■), the Bose

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family (□), the Coomar family (◯), and the Dubey family (●). For the following questions, savings is defined as—

Saving = Income – Expenditure



1. Which family has the highest average expenditure ?
   (A) Ahuja (B) Bose
   (C) Coomar (D) Dubey
2. Which family has the lowest average income ?
   (A) Ahuja (B) Bose
   (C) Coomar (D) Dubey
3. Which family has the lowest average savings ?
   (A) Ahuja (B) Bose
   (C) Coomar (D) Dubey
4. The highest amount of savings accrues to a member of which family ?
   (A) Ahuja (B) Bose
   (C) Coomar (D) Dubey

Answers with Explanation

1. (D) Average expenditure of Ahuja
$$= \frac{700 + 1700 + 2600}{3}$$
$$= 1700 \text{ (App.)}$$
Average expenditure of Bose
$$= \frac{800 + 1700 + 2400}{3}$$
$$= 1600 \text{ (App.)}$$
Average expenditure of Coomar
$$= \frac{400 + 1100 + 1900}{3}$$
$$= 1100 \text{ (App.)}$$
and average expenditure of Dubey
$$= \frac{1200 + 2800}{2}$$
$$= 2000 \text{ (App.)}$$
$\therefore$ Dubey's family has the highest average expenditure.

2. (C) Average income of Ahuja family
$$= \frac{3300 + 3000 + 2800}{3}$$
$$= 3000 \text{ (App.)}$$
Average income of Bose family
$$= \frac{2400 + 2200 + 2800}{3}$$
$$= 2500 \text{ (App.)}$$
Average income of Coomar family
$$= \frac{1100 + 2200 + 1600}{3}$$
$$= 1600 \text{ (App.)}$$
and average income of Dubey family
$$= \frac{1300 + 3200}{2}$$
$$= 2250 \text{ (App.)}$$
$\therefore$ Coomar family has lowest average income.

3. (D) Average saving of Ahuja family
$$= 3000 - 1700$$
$$= 1300$$
Average saving of Bose family
$$= 2500 - 1600$$
$$= 900$$
Average saving of Coomar family
$$= 1600 - 1100$$
$$= 500$$
and average saving of Dubey family
$$= 2250 - 2000$$
$$= 250$$
Hence, Dubey family has the lowest average savings.

4. (A) Approximate amount saving of Ahuja
$$= (3300 + 3000 + 2800) - (700 + 1700 + 2000)$$
$$= 4100$$
Approximate amount saving of Bose
$$= (2400 + 2200 + 2800) - (800 + 1700 + 2400)$$
$$= 2500$$

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Approximate amount saving of Coomar
= (1100 + 2200 + 1600)
– (400 + 1100 + 1900)
= 1500

Approximate amount saving of Dubey
= (1300 + 3200) – (1200 + 2800)
= 500

From the above, we can see that Ahuja's amount of saving is the highest.

Exercise 13

Directions—Study the following information carefully and answer the questions that follow—

The students of a school have an option to study only Hindi, only Sanskrit or a composite subject Hindi and Sanskrit. Out of the 175 students in the school, boys and girls are in the ratio of 3 : 4 respectively. 40% of boys have opted for only Hindi; 44% of the students have opted for only Sanskrit. Out of the total number of girls 32% have opted for the composite subject. The number of boys who opted for only Sanskrit and that for composite subject are in the ratio of 2 : 1 respectively.

1. What is the ratio between the number of boys who have opted for only Hindi and the number of girls who have opted for the composite subject respectively ?
(A) 15 : 16 (B) 10 : 7
(C) 10 : 9 (D) 11 : 12
(E) None of these

2. How many boys have opted for the composite subject ?
(A) 30 (B) 15
(C) 21 (D) 32
(E) None of these

3. How many girls have opted for only Sanskrit ?
(A) 72 (B) 47
(C) 51 (D) 77
(E) None of these

Answers with Explanation

No. of boys = $\frac{3}{7} \times 175 = 75$

No. of girls = 175 – 75 = 100

No. of boys who opt only Hindi
= 40% of 75 = 30

Remaining boys = 75 – 30 = 45

Numbers of boys who opt only Sanskrit
= $\frac{2}{3} \times 45 = 30$

Numbers of boys who opt composite subjects
= 45 – 30 = 15

Total no. of students who opt only Sanskrit
= 44% of 175 = 77

No. of girls who opt only Sanskrit
= 77 – 30 = 47

No. of girls who opt composite subjects = 32

No. of girls who opt Hindi only
= 100 – (32 + 47) = 21

1. (A) From above, the required ratio
= 30 : 32
⇒ 15 : 16

2. (B) 3. (B)

●●

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8 Data Sufficiency

Data sufficiency—Data sufficiency means—'the data, that are given to us to find any solution, are sufficient or not.' Questions that are based on data sufficiency are only to judge the sufficiency of their statements, not to show their ultimate solutions. Generally, In data sufficiency, questions are given followed by two or three statements. These two or three statements contain some pieces of information or the data by which the questions may be solved. We are required to judge whether the given information or the data are sufficient or not to find the solutions of the questions.

The questions, that are on the pattern of Data Sufficiency, do not cover the new topics of any kind. Generally, they cover only the topics that are already in running, *e.g.* simple and compound interest, percentage, profit and loss, Time and work, Number system, Ratio and proportion, and the topics of Algebra etc.

These questions are judged by their own methods of processing or the observations.

Example 1. The following example has a question of **Number System** and three statements labelled I, II and III.

For this question, we are required to judge that the given statements are sufficient or not to find the solution or the answer of the given question.

It may be—

(i) Only statement I is sufficient

(ii) Only statement II is sufficient

(iii) Only statement III is sufficient

(iv) Only statements I and II are sufficient

(v) Only statements II and III are sufficient

(vi) Only statements I and III are sufficient

(vii) All the three statements are required to find the solution

(viii) None of the above statements is sufficient.

Question—What is the two digit number ?

I. The number obtained by interchanging the digits is more than the original number by 9.

II. Sum of the digits is 7.

III. Difference between the digits is 1.

(A) I and III are only sufficient.

(B) I and II are only sufficient.

(C) II and III are only sufficient.

(D) All I, II and III are sufficient.

(E) Question cannot be answered even with the information in all the three statements.

Solution : The answer of the question is (B). By the statements I and II, we can find the required two digits number, while with the help of statements II and III, we can find only the two digits, not the two-digits number.

Example 2. The following example has a question of **Profit and Loss** and two statements-labelled I and II.

For this question, we are required to judge the sufficiency of the given statements to find the required solution.

Question—By selling a product at 20% profit, how much profit was earned ?

(I) The difference between cost and selling price is Rs. 40.

(II) The selling price is 120% of the cost price.

Give Answer as :

(A) If the data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.

(B) If the data in statement II alone are sufficient to answer the question, while the data in statement, I alone are not sufficient to answer the question.

(C) If the data either in statement I alone or in statement II alone are sufficient to answer the question.

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(D) If the data even in both the statements I and II together are not sufficient to answer the question.

(E) If the data in both the statements I and II together are necessary to answer the question.

Solution : The answer of the question is (A).

To answer the question, we need one of the following—

(i) Cost price of the product.

(ii) Selling price of the product.

(iii) Difference of the selling price and the cost price.

From the statement I. We can get the required profit because profit = selling price – cost price.

From the statement II. It is the restatement because when profit earned is 20%, then obviously selling price will be 120% of the cost price.

Hence, only the statement I alone is sufficient.

Exercise 1

Directions—Each of the questions below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient or not to answer the question.

Read both the statements carefully and give the answer as—

(A) If the data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question.

(B) If the data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question.

(C) If the data either in statement I alone or in statement II alone are sufficient to answer the question.

(D) If the data even in both the statements I and II together are not sufficient to answer the question.

(E) If the data in both the statements I and II together are necessary to answer the question.

1. What is the difference between the two digits in a two digit number ?
   I. The sum of the two digits is 8.
   II. $\frac{1}{5}$ of that number is 15 less than $\frac{1}{2}$ of 44.
2. Which is the smaller of the two numbers ?
   I. The difference between these two numbers is one third of the largest number.
   II. The sum of these two numbers is 30.
3. What is the value of $m - n$? 37 ?
   I. M is the largest possible six digit number and n is the smallest possible six digit numbers.
   II. The difference between m and n is known.
4. What is the original number ?
   I. Sum of the digits of a number is 10. The ratio between the two digits is 1 : 4.
   II. Product of two digits of a number is 16. Quotient of the two digits is 4.
5. The difference between the two digits of a number is 6. What is the number ?
   I. The digit at the units place is bigger than the other digit.
   II. The sum of the two digits is 12.
6. X, Y and Z are integers. Is X an odd number ?
   I. An odd number is obtained when X is divided by 5.
   II. (X + Y) is an odd number.
7. What is a two digit number ?
   I. The number obtained by interchanging the digits is smaller than the original number by 63.
   II. Sum of the digits is 11.
8. A, B and C are integers. Is B an even number ?
   I. (A + B) is an odd number.
   II. (C + B) is an odd number.
9. What is the two digit number where the digit at the unit's place is smaller ?
   I. The difference between the two digits is 5.
   II. The sum of the two digits is 7.
10. A, B and C are positive integers. Is their product an even number ?
    I. A is an even number.
    II. The product of A and B is an even number and that of A and C is also an even number.

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11. What will be the cost of the second necklace ?
    I. The cost of the first necklace is $\frac{1}{5}$ more than the second and the cost of the third necklace is $\frac{2}{5}$ more than the second. The total cost of all the three necklaces is Rs. 1,20,000.
    II. The cost of the first necklace is $\frac{2}{5}$ more than the second. The cost of the third necklace is the least and total cost of all the three necklaces is Rs. 1,20,000.
12. What will be the average weight of the remaining class ?
    I. Average weight of 30 children out of total 46 in the class is 22·5 kg and that of the remaining children is 29·125 kg. A child having weight more than 40 kg is excluded.
    II. Average weight of a class of 46 children is 23·5 kg. A child weighting 46 kg is dropped out.
13. How many marks did Prakash obtain in Mathematics ?
    I. Prakash secured on an average 55 per cent marks in Mathematics, Physics and Chemistry together.
    II. Prakash secured 10 per cent more than the average in Mathematics.
14. What is the average monthly income per family member.
    I. Each male earns Rs. 1,250 a month, each female earns Rs. 1,050 a month.
    II. Ratio of males to females in the family is 2 : 1.
15. How many children are there in the group ?
    I. Average age of this group of children is 16 years. The total of ages of all the children in the group is 240 years.
    II. The total of ages of all the children in the group and the teacher is 262 years. The teacher's age is six years more than the average age of the children.
16. What is the average age of the children in a class ?
    I. The age of the teacher is as many years as the number of children.
    II. The average age increases by 1 year if teacher's age is also included.
17. What is the present age of the mother ?
    I. Father's age is eight years more than the Mother's age. Father got married at the age of 28 years.
    II. Present age of the father is 30 years. Four years back the ratio of Mother's age to Father's age was 12 : 13.
18. What was the ratio between the ages of P and Q four years ago ?
    I. The ratio between the present ages of P and Q is 3 : 4.
    II. The ratio between the present ages of Q and R is 4 : 5.
19. What is Sudha's present age ?
    I. Sudha's present age is five times her son's present age.
    II. Five years ago her age was twenty-five times her son's age that time.
20. What was the population of State 'A' in 1999 ?
    I. Population of the State increases every year by 20% and its population in 1997 was 1,20,000.
    II. Population of State A in 1997 was twice that of State B in the same year.
21. What was the population of State 'A' in 1999 ?
    I. Population of State 'A' increases every year by 20%.
    II. Population of State 'A' in 1999 was 172·8% of its population in 1996.
22. How many children are there in the class ?
    I. Numbers of boys and girls are in the respective ratio of 3 : 4.
    II. Number of girls is 18 more than the number of boys.
23. By selling a product for Rs. 100, how much profit was earned ?
    I. 20% profit would have been earned, if it had been sold for Rs. 90.
    II. The profit was one-third of the purchase price.

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24. What was the cost price of the suitcase purchased by Samir ?
    I. Samir got 20 per cent concession on the labelled price.
    II. Samir sold the suitcase for Rs. 2000 with 25 per cent profit on the labelled price.

25. What is the rate of simple interest per annum ?
    I. The sum triples in 20 years at simple interest.
    II. The difference between the sum and the simple interest earned after 10 years is Rs. 1000.

26. What is the sum which earned interest ?
    I. The total simple interest was Rs. 7000 after 7 years.
    II. The total of sum and simple interest was double of sum after 5 years.

27. What percentage rate of simple interest per annum did Ashok pay to Sudhir ?
    I. Ashok borrowed Rs. 8000 from Sudhir for four years.
    II. Ashok returned Rs. 8800 to Sudhir at the end of two years and settled the loan.

28. What is the rate of interest p.c.p.a. ?
    I. Difference between compound interest and simple interest on an amount of Rs. 10,000 for two years is Rs. 225.
    II. The amount doubles itself on simple interest in $6\frac{2}{3}$ years.

29. What was the total compound interest on a sum after three years ?
    I. The interest after one years was Rs. 100 and the sum was Rs. 1,000.
    II. The difference between simple and compound interest on a sum of Rs. 1,000 at the end of two years was Rs. 10.

30. A train crosses another train running in the opposite direction in x seconds. What is the speed of the train ?
    I. Both the trains are running at the same speed.
    II. The first train is *y* cm long.

31. A train crosses a signal post in X seconds. What is the length of the train ?
    I. The train crosses a platform of 100 metres in Y seconds.
    II. The train is running at the speed of 80 km/hr.

32. Train 'A' running at a certain speed crosses another train 'B' running at a certain speed in the opposite direction in 12 seconds. What is the length of train 'B' ?
    I. The length of both the trains together is 450 metres.
    II. Train 'A' is slower than train 'B'.

33. What is the speed of a running train ?
    I. The train crosses a signal post in 6 seconds.
    II. The train crosses another train running in the opposite direction in 15 seconds.

34. A train crosses another train running in the opposite direction in x seconds. What is the speed of the train ?
    I. Both the trains have the same length and are running at the same speed.
    II. One train crosses a pole in 5 seconds.

35. What is the speed of the boat in still water ?
    I. It takes 2 hours to cover the distance between A and B downstream.
    II. It takes 4 hours to cover the distance between A and B upstream.

36. What is the speed of a boat ?
    I. The boat covers a distance of 48 km in 6 hours while running upstream.
    II. It covers the same distance in 4 hours while running downstream.

37. What is the area of a circle ?
    I. The circumference of the circle is 308 metres.
    II. The radius of the circle is 28 metres.

38. The area of a square is equal to that of a circle. What is the circumference of the circle ?
    I. The diagonal of the square is *x* inches.
    II. The side of the square is *y* inches.

39. What is the cost of the laying carpet in a rectangular hall ?
    I. Cost of the carpet is Rs. 450 per square metre.
    II. Perimeter of the hall is 50 metres.

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40. What is the capacity of a cylindrical tank ?
    I. Radius of the base is half of its height, which is 28 metres.
    II. Area of the base is 616 square metres and height is 28 metres.

Answers with Explanation

1. (B) Let the two-digit number is $10x + y$, then

$$\text{From II, } \frac{1}{5}(10x + y) = \frac{44}{2} - 15$$
$$= 7$$
$$\therefore \text{ The number } = 35$$
$$\therefore \text{ The required difference } = 5 - 3$$
$$= 2$$

Hence, statement II alone is sufficient.

2. (E) Let the two numbers be x and y, then

$$\text{I. } x - y = \frac{1}{3}x$$
$$\Rightarrow 2x - 3y = 0$$
$$\text{II. } x + y = 30$$

Hence, statements I and II together are necessary to answer the question.

3. (A) I. $M = 999999$
$N = 100000$
$\therefore 999999 = 100000 \div 37$
$= 999999 - 2702{\cdot}70$
$= 997293{\cdot}30$

$\Rightarrow$ Value can be found.

II. '$m - n$= Known' is not sufficient because neither the value of 'm' is known nor the value of 'n' is known, Therefore, we cannot find the value of '$m - n \div 37$' by this statements.

4. (D) Let the original number be $10x + y$.

From I. $\Rightarrow$ Case I.

$$x + y = 10$$
$$x : y = 1 : 4$$
$$\therefore x = 2$$
$$y = 8$$
$$\therefore \text{ The number } = 10 \times 2 + 8$$
$$= 28$$

Case II. $x + y = 10$
$y : x = 1 : 4$
$\Rightarrow x = 8$
$y = 2$
$\therefore$ The number $= 82$

From II. Case I.

$$xy = 16$$
$$\frac{x}{y} = 4$$
$$\Rightarrow x = 8$$
$$y = 2$$
$$\therefore \text{ The number } = 10 \times 8 + 2$$
$$= 82$$

Case II. $xy = 16$

$$\frac{y}{x} = 4$$
$$\Rightarrow x = 2$$
$$y = 8$$
$$\therefore \text{ The number } = 28$$

From both the statements, we can get two numbers 28 and 82. Therefore the original number cannot be determined.

5. (E) Let the digits are x and y assuming $x > y$.
We have $x - y = 6$
I. x occupies unit's place.
II. $x + y = 12$

With the help of information in the question and in statement II, we can find the value of x and y easily, but to determine the number we will need the help of statement I.

6. (A) The statement I alone is sufficient to answer the question because we know that whenever any odd number is divided by any odd number. It gives an odd number.

7. (E) Both the statements I and II together are necessary to answer the question.

8. (D) **From I.** A + B is odd
$\Rightarrow$ If A is an even number, then B will be an odd number or *vice-versa*.
From II. C + B is odd
$\Rightarrow$ If B is an even number, then C will be an odd number or *vice-versa*.
Therefore, even by combining the two statements together, we are not able to say that B is an even integer.

9. (E) Let the two digit number is $10x + y$, where $x > y$
I. $x - y = 5$
II. $x + y = 7$

By combining both the statements together, the value of x and y can be determined.

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Hence, both the statements together are necessary to answer the question.

10. (C) Either the statement I alone or the statement II alone is sufficient to answer the question.

11. (A) From the statement I, the ratio of the costs of first, second and third necklaces is 6 : 5 : 7. Therefore the price of second necklace can be found.

12. (B) The statement II alone is sufficient.

13. (D) 14. (E) 15. (A) 16. (D)

17. (B) From the statement I, we can determine the ages of father and mother at the time of marriage only

Statement II.

$$\Rightarrow \quad \frac{M-4}{F-4} = \frac{12}{13}$$

$$\Rightarrow \quad 13M - 52 = 12F - 48$$

$$\Rightarrow \quad M = 28 \text{ years}$$

Therefore, only the statement II alone is sufficient.

18. (D) 19. (E) 20. (A)

21. (D) The population of the state A for a given year is not given in any of the statements. When we start with the statement I, we will get the statement II. Therefore, both the statements I and II together are not sufficient.

22. (E) I. $\Rightarrow$ The ratio of boys and Girls

$$= 3 : 4$$

From the statements I and II together

$$4K - 3K = 18$$

$$\Rightarrow \quad K = 18$$

$$\therefore \quad 4K + 3K \Rightarrow 7 \times 18$$

$$= 126$$

Therefore, both the statements are necessary to answer.

23. (C) I. $\quad C.P. = 90 \times \frac{100}{120}$

$$= \text{Rs. } 75$$

$$\therefore \quad \text{Profit} = 100 - 75$$

$$= \text{Rs. } 25$$

II. $\quad SP = CP + \text{Profit}$

$$\Rightarrow \quad x + \frac{x}{3} = 100$$

$$\Rightarrow \quad x = \frac{100 \times 3}{4} = 75$$

$$\therefore \quad \text{Profit} = \frac{75}{3}$$

$$= \text{Rs. } 25$$

Therefore, either the statement I or the statement II alone is sufficient to answer the question.

24. (E) Combining both the statements together, we can get the required value.

25. (A) From the statement I.

$$R = (3-1) \times \frac{100}{20}$$

$$= 10\%$$

II. Here, the sum is not given. Therefore, this statement cannot be applied. Statement I alone is sufficient to answer the question.

26. (E) From the statements I, we can calculate the SI after 5 years, combining with the statement II, we can get the value of sum, *i.e.*,

$$(P + 5000) = 2P$$

$$\Rightarrow \quad P = \text{Rs. } 5000$$

27. (E) Combining both the statements together,

$$\text{Rate of interest} = \frac{800}{2 \times 8000} \times 100$$

$$= 5\%$$

Therefore, both the statements are necessary to answer the question.

28. (C) 29. (C) 30. (D)

31. (C) Either the statement I or the statement II is sufficient to answer the question.

32. (D) 33. (D) 34. (D)

35. (D) Let the distance between A and B is D km and the speed of the boat and current in still water are x km/hr and y km/hr respectively.

I. $\quad D = (x + y) \times 2$

II. $\quad D = (x - y) \times 4$

Both the statements are not sufficient to answer the question.

36. (E) Here, both the statements are important for the speed of the boat (V_B) and that of water flow (V_W).

I. $\quad V_B - V_W = \frac{48}{6} = 8 \qquad \ldots(i)$

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II. $V_B + V_W = \frac{48}{4} = 12$ ...(ii)

By solving equations (i) and (ii), we can find the required answer.

37. (C) **From I.**

$$\text{Radius of circle} = \frac{308 \times 7}{2 \times 22}$$

$$= 49 \text{ m}$$

$$\therefore \quad \text{Area of circle} = \frac{22}{7} \times 49 \times 49$$

$$= 7546 \text{ m}^2$$

From II. $\text{Area of circle} = \frac{22}{7} \times 28 \times 28$

$$= 2464 \text{ m}^2$$

Hence, either I alone or the II alone is sufficient to answer the question.

38. (C) We can get the answer by either of the statements.

39. (D) To find out the cost of laying carpet, we need the following.

(i) Cost of carpet per square metre.

(ii) Area of the floor to be carpeted.

Both the statements I and II are not sufficient to answer the questions.

40. (C) The capacity of a cylindrical tank can be found out by the following formulas.

(i) Area of the base × height.

(ii) $\pi r^2 h$ where r is the radius of the cylinder and h is the height of the cylinder. Statement I gives the value or r and h. Hence, this alone is sufficient to answer the question.

Again, statement II gives the information about the area of the base and the height. Hence, this statement is also sufficient to answer the question.

Exercise 2

Directions—The following questions are accompanied by three statements I, II and III. You have to determine which statement/statements is/are sufficient to answer the questions.

1. What is the two-digit number ?
   - I. Sum of the digits is 17.
   - II. Difference between the number and the number obtained by interchanging the digits is 9.
   - III. Digit in the unit's place is bigger than the digit in the ten's place by 1.
   - (A) Only I and II
   - (B) Only I and III
   - (C) Only II and III
   - (D) All I, II and III
   - (E) Any two of the above statements
2. What is the sum of two numbers ?
   - I. The bigger of these two numbers is 6 more than the smaller number.
   - II. 40% of the smaller number is equal to 30% of the bigger number.
   - III. The ratio between half of the bigger number and one-third of the smaller number is 2 : 1.
   - (A) Only II and III together are required
   - (B) Only I and II together are required
   - (C) Any two of I, II and III together are required
   - (D) All I, II and III together are required
   - (E) None of these
3. What is the difference between two numbers X and Y ?
   - I. X is 20 per cent more than another number Z.
   - II. Y is 20 per cent less than Z.
   - III. The sum of Y and Z is 72.
   - (A) Only I and II are required
   - (B) Only I and III are required
   - (C) All I, II and III together are required
   - (D) Any two of I, II and III are required
   - (E) Even with all I, II and III together the answer cannot be arrived at
4. What is this two-digit number ?
   - I. The number obtained by interchanging the digits is more than the original number by 9.
   - II. Sum of the digits is 7.
   - III. Difference between the digits is 1.
   - (A) I and III only
   - (B) I and II only
   - (C) II and III only
   - (D) All I, II and III
   - (E) Question cannot be answered even with the information in all the three statements.

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5. What is a two-digit number ?
   I. The difference between the two-digit number and the number formed by interchanging the digits is 27.
   II. The difference between the two digit is 3.
   III. The digit at unit's place is less than that at ten's place by 3.
   (A) Only I and II
   (B) Only I and either II or III
   (C) Only I and III
   (D) All I, II and III
   (E) Even with all the three statements the answer cannot be given

6. What is the present age of Rohit ?
   I. After two years the ratio of the ages of Rohit and Amit will be 37 : 27.
   II. One-fourth of the sum of ages of Rohit and Amit is equal to five more of their age difference.
   III. Rohit is 10 years older than Amit.
   (A) Any of them
   (B) Only I and II together
   (C) Only II and III together
   (D) Only III
   (E) Any two of them

7. What will be the ratio between Ramesh's and Anand's ages after 7 years—
   I. The ratio between their present ages is 7 : 8.
   II. The difference between their ages after eight years will 5 years.
   III. Four years ago the ratio between their ages was 5 : 7.
   (A) II only
   (B) III only
   (C) Any two of the three
   (D) I, II and III are all required
   (E) None of these

8. What is Sangita's present age ?
   I. Five years ago, Sangita's age was double that of her son's age that time.
   II. Present ages of Sangita and her son are in the ratio of 11 : 6 respectively.
   III. Five years hence, the respective ratio of Sangita age and her son's age will become 12 : 7.
   (A) Only I and III
   (B) Only II and III
   (C) Only I and II
   (D) Any two of the three
   (E) None of the above

9. What is the present age of Subir ?
   I. The present age of Subir is half that of his father.
   II. After 5 years the ratio of Subir's age to his father's will be 6 : 11.
   III. Subir is 5 years younger than his brother.
   (A) Only I and II
   (B) Only I and III
   (C) Only II and III
   (D) All I, II and III
   (E) Even with all the three statements answer cannot be given

10. What is Sudha's present salary ?
   I. The salary increases every year by 15%
   II. Her salary at the time of joining was Rs. 10,000
   III. She had joined exactly 5 years ago.
   (A) II and III only
   (B) I and II only
   (C) All I, II and III
   (D) I and III only
   (E) None of the above

11. How many students are there in all in the institute of Arts, Commerce and Science ?
   I. 20% of the students study Science.
   II. The number of students studying Arts and Commerce are in the ratio of 3 : 5.
   III. The number of students studying Commerce is more than that of studying Science by 375.
   (A) II and III only
   (B) III and either I or II only
   (C) Any two of the three
   (D) All I, II and III
   (E) Question cannot be answered even with the information in all the three statements

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12. What is the monthly salary of Pravin ?
    I. Pravin earns Rs. 1,200 more than Amal.
    II. The ratio between Amal and Vimal's monthly salary is 5 : 3.
    III. Vimal earns Rs. 1,000 less than Amal.
    (A) Any two of I, II and III are required
    (B) Only I and II are required
    (C) Only II and III are required
    (D) All I, II and III together are required
    (E) None of these

13. What is the staff strength of Company 'X' ?
    I. Male and female employees are in the ratio of 2 : 3 respectively.
    II. Of the officer employees 80% are males.
    III. Total number of officer is 132.
    (A) I and III only
    (B) II and either III or I only
    (C) All I, II and III
    (D) Any two of the three
    (E) Question cannot be answered even with the information in all the three statements.

14. What is R's share of profit in a joint venture ?
    I. A started a business investing Rs. 80,000.
    II. R joined him after 3 months.
    III. P joined after 4 months with a capital of Rs. 1,20,000 and got Rs. 6,000 as his share of profit.
    (A) Only I and III are required
    (B) Only II and III are required
    (C) All I, II and III together are required
    (D) Even with all I, II and III, the answer cannot be found out
    (E) None of the above

15. What was the amount of profit earned ?
    I. 10% discount was offered on the labelled price.
    II. Had there been no discount, profit would have been 30%.
    III. Selling price was more than the cost price by 20%.
    (A) I and either II or III
    (B) Any two of the three
    (C) All I, II and III
    (D) Either I or II and III
    (E) Question cannot be answered

16. What was the profit earned on the cost price by Mahesh by selling an article ?
    I. He get 15% concession on labelled price in buying that article.
    II. He sold it for Rs. 3,060.
    III. He earned a profit of 2% on the labelled price.
    (A) Only I and II together are required
    (B) Only II and III together are required
    (C) Only either I or III and II together are required
    (D) Even with all I, II and III, the answer cannot be arrived at.
    (E) All I, II and III together are required

17. How many articles were sold ?
    I. Total profit earned was Rs. 1,596.
    II. Cost price per article was Rs. 632.
    III. Selling price per article was Rs. 765.
    (A) II and III only
    (B) I and II only
    (C) All I, II and III
    (D) Any two of the three
    (E) Question cannot be answered

18. What was the rate of compound interest on an amount of money ?
    I. The amount fetches a total of Rs. 945·75 as compound interest at the end of three years.
    II. The difference between the total simple interest and the total compound interest at the end of two years with the same rate of interest was Rs. 15.
    III. The ratio between the principal amount and the total simple interest at the end of three years is 20 : 3.
    (A) Only I and II are required
    (B) Only II and III are required
    (C) All I, II and III together are required
    (D) Even with all I, II and III, together the answer cannot be determined
    (E) None of these

19. What is the rate of interest pc, pa ?
    I. The amount doubles itself in 5 years on simple interest.

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II. Difference between the compound interest and the simple interest earned on this amount in two years is Rs. 400.
III. Simple interest earned per annum is Rs. 2000.
(A) Only I
(B) Only II and III
(C) Any two of the three
(D) All I, II and III
(E) Only I or only II and III

20. What is the speed of the train ?
I. The train crosses 300 metres long platform in 21 seconds.
II. The train crosses another stationary train of equal length in $19\frac{1}{2}$ seconds.
III. The train crosses a signal pole in $9\frac{3}{4}$ seconds.
(A) Only I and II
(B) Only II and either I or III
(C) Only I and either II or III
(D) Only III and either I or II
(E) None of the above

21. What is the speed of the train 'A' ?
I. Train A crosses 200-metre-long train B running in opposite direction in 20 seconds.
II. Speed of train B is 60 kmph.
III. Length of train A is twice that of train B.
(A) I and II only
(B) II and III only
(C) I and III only
(D) All I, II and III
(E) Question cannot be answered even with information in all three statements.

22. What is the speed of a train ?
I. The train crosses a signal pole in 18 secs.
II. The train crosses a platform of equal length on 36 secs.
III. Length of the train is 330 metres.
(A) I and III only
(B) II and III only
(C) I and II only
(D) III and either I or II only
(E) Any two of the three

23. In how many days can 10 women finish a work ?
I. 10 men can complete the work in 6 days.
II. 10 men and 10 women together can complete the work in $3\frac{3}{7}$ days.
III. If 10 men work for 3 days and thereafter 10 women replace them, the remaining work is completed in 4 days.
(A) Only I and II
(B) Any two of the three
(C) Only I and III
(D) Only II and III
(E) None of these

24. In how many days can a work be completed by A and B together ?
I. A alone can complete the work in 8 days.
II. If A alone works for 5 days and B alone works for 6 days, the work gets completed.
III. B alone can complete the work in 16 days.
(A) Any two of the three
(B) II and either I or III
(C) I and II only
(D) II and III only
(E) None of these

25. What is the area of the right-angled triangular garden ?
I. Perimeter of the garden is y cm.
II. Length of the diagonal side is x cm.
III. Perpendicular sides of the garden are in the ratio of 5 : 12.
(A) Only I and III or only II and III
(B) All I, II and III
(C) Any two of the three
(D) Only I and III
(E) None of these

26. What is the area of the right-angled triangle ?
I. The perimeter of the triangle is 30 cm.
II. The ratio between the base and the height of the triangle is 5 : 12.
III. The area of the triangle is equal to the area of a rectangle of length 10 cm.
(A) Only II and III together are required
(B) Only I and II together are required

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(C) Only either I or II and III together are required
(D) Only I and III together are required
(E) None of these

27. What is the area of the isosceles triangle ?
I. Perimeter of the triangle is 14 metres.
II. Base of the triangle is 14 metres.
III. Height of the triangle is 5 metres.
(A) I and II only
(B) II and III only
(C) I and II only or II and III only
(D) I and III only
(E) All I, II and III

28. What is the perimeter of a rectangular garden ?
I. The area of the garden is 2400 sq. metres.
II. The diagonal of the garden is 50 metres.
III. The ratio between the length and the breadth of the garden is 3 : 2.
(A) All I, II and III together are required
(B) Any two of I, II and III are sufficient
(C) Only I and II are required
(D) Only II and III are required
(E) None of these

29. The cost of carpeting a rectangular Hall will be how much ?
I. Perimeter of a rectangle is 60 m.
II. Angle between width and hypotenuse is 30°.
III. The cost of carpeting the surface floor is Rs. 125 per square metre.
(A) Only I and II
(B) Only II and III
(C) Only I and III or only II and III
(D) Question cannot be answered even with information in all three
(E) All the three statements I, II and III together are necessary for answering the question

30. What is the cost of flooring a rectangular hall ?
I. The length and the breadth of the hall are in the ratio of 3 : 2.
II. The length of the hall is 48 metres and the cost of flooring is Rs. 850 per square metre.
III. The perimeter of the hall is 160 metres and the cost of flooring is Rs. 850 per square metre.
(A) Only I and II
(B) Only I and III
(C) Only III
(D) Only I and either II or III
(E) Any two of the three

31. What is the cost of flooring a rectangular hall ?
I. Perimeter of the hall is 76 m.
II. Area of the hall is 336 m^2.
III. Cost of flooring per square metre is Rs. 550.
(A) I and III only
(B) II and III only
(C) Any two of the three
(D) All I, II and III
(E) None of these

32. How many marks did Arun get in English ?
I. Arun secured an average of 60 marks in four subjects including English.
II. He secured a total of 170 in English and Maths together.
III. He secured a total of 180 in Maths and Science together.
(A) All I, II and III together are required
(B) Only I and II together are required
(C) Only II and III together are required
(D) Only I and III together are required
(E) None of the above

33. How much marks was obtained by Mukesh in Geography ?
I. The average marks obtained by Mukesh in English, History and Geography was 65.
II. The difference between the marks obtained by Mukesh in English and History was 15.
III. The total marks obtained by Mukesh in Geography and Mathematics was 140.
(A) All I, II and III together are required
(B) Only I and III are required

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(C) Only II and III are required
(D) Even with all I, II and III together, the answer cannot be determined
(E) Any two of I, II and III are sufficient

34. Who earns most among M, N, P, Q and R ?
I. M earns less than P but not les than R.
II. Q earns more than M but not equal to N.
III. N earns more than M and R.
(A) Question cannot be answered even with information in all three statements
(B) I and II only
(C) Only I and II or only I and III
(D) Only I and III
(E) All the three statement I, II and III together are necessary for answering the question

35. What is the price of 1 dozen oranges ?
I. Price of 2 dozen oranges and 1 dozen banana is Rs. 110.
II. Price of 3 dozen apples and 1 dozen banana is Rs. 170.
III. Price of 1 dozen oranges and 1 dozen apples is Rs. 95.
(A) Only I and II or only I and III
(B) Only I and III or only II and III
(C) Only I and II or only II and III
(D) Only II and III
(E) All the three statements I, II and III are necessary for answering the question

36. What is the capacity of a cylindrical tank ?
I. The radius of the base is half of its height.
II. The area of the base is 616 sq. metres.
III. The height of the cylinder is 28 metres.
(A) Only I and II
(B) Only II and III
(C) Only I and III
(D) All I, II and III
(E) Any two of the three

Answers with Explanation

1. (E) Let the two digit number is $10x + y$. Then from—

I. $x + y = 17$

II. $(10x + y) - (10y + x) = 9$

III. $y = x + 1$

From the statements I, II and III any of the two statements are sufficient to find the required number. Hence, (E) is the required answer.

2. (E) Let the bigger and smaller numbers are x and y respectively.

From I. $x - y = 6$ …(i)

From II. 40% of y = 30% of x

$\Rightarrow$ $4y = 3x$ …(ii)

From III. $\frac{x}{2} : \frac{y}{3} = 2 : 1$

$\Rightarrow$ $3x = 4y$ …(iii)

We see that the equations (ii) and (iii) are the same. Hence, statement I and either statement II or III is required.

3. (C) 4. (B)

5. (E) Let the two-digit number is $10x + y$, then

From I. $|10x + y - 10y - x| = 27$

$\Rightarrow$ $|x - y| = 3$

From II. $|x - y| = 3$

From III. $x - y = 3$

Here, by taking any two, the values of x and y cannot be determined. Therefore, the answer is (E).

6. (E) Let the present ages of Rohit and Amit be x and y respectively.

From I. $\frac{x+2}{y+2} = \frac{37}{27}$

From II. $\frac{1}{4}(x + y) = S + (x - y)$

From III. $x - y = 10$

Here, by solving any two of the above, the values of x and y can be calculated.

7. (C) 8. (D)

9. (A) Let the present ages of Subir, his brother and his father be S, B and F respectively, then

From I. $S = \frac{F}{2}$

From II. $\frac{S+5}{F+5} = \frac{6}{11}$

From III. $B - S = 5$

Here, with the help of I and II together, the values of S and F can be determined.

10. (C) By combining all the three statements together, we can get the required answer.

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11. (D) Statements I and II give the percentage number of the students studying in different disciplines. Combining these with III, the total number of students can be determined.

12. (D) 13. (E) 14. (D)

15. (E) None of the statements gives the amount of labelled price or the S.P. So, even by combining all the statements together, the question cannot be answered.

16. (E) From the statements I and II,

$$\text{Labelled price} = \frac{3060 \times 100}{85}$$

$$= \text{Rs. } 3{,}600 \quad \ldots(i)$$

Combining (i) and statement III,

$$\text{Profit} = 3600 \times \frac{2}{100}$$

$$= \text{Rs. } 72 \quad \ldots(ii)$$

Combining (ii) and statement II

$$\text{C.P.} = 3{,}600 - 72$$

$$= \text{Rs. } 2{,}988$$

$$\therefore \quad \text{Profit \%} = \frac{72 \times 100}{2988}$$

$$= 2{\cdot}40\%$$

Hence, all the statements are required to answer the question.

17. (C)

18. (E) From the statement III alone we can find out the rate of interest.

19. (E) **From I.** Rate of interest

$$= \frac{(2-1) \times 100}{5}$$

$$= 20\%$$

From II and III.

Rate of interest (For 2 years only)

$$= \frac{2 \times \text{dff. in C.I. and S.I.}}{\text{S.I.}}$$

$$= \frac{2 \times 400}{4000} \times 100$$

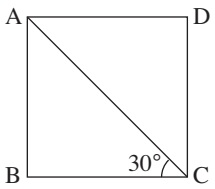
$$= 20\%$$

Hence, either I alone or the statements II and III together can provide the required answer.

20. (C) 21. (D) 22. (D) 23. (B) 24. (A)
25. (A) 26. (B) 27. (B) 28. (B)

29. (E) **From I.** $2(L + B) = 60$

$$\therefore \quad L + B = 30 \quad \ldots(i)$$



From II. In Δ ABC,

$$\tan 30° = \frac{L}{B}$$

$$\Rightarrow \quad L : B = \sqrt{3} : 1$$

Combining statements I and II, we can get the values of L and B, *i.e.*,

$$L = 19\text{m}$$

$$B = 11\text{m}$$

$$\therefore \quad \text{Area of rectangle} = 19 \times 11$$

$$= 209 \text{ m}^2$$

From III. Cost = Rs. 125 per m^2

$\therefore$ All the three statements I, II and III together are necessary for answering the question.

30. (E) With the help of any two statements, the length and the breadth can be determined and combining this with the cost per square metre, we can get the total cost of flooring the rectangular hall.

31. (B) 32. (E) 33. (D)

34. (A) **From I.**

$$P > M, M > R \text{ or } M = R$$

From II. $Q > M, Q > N$ or $Q < N$

From III. $N > \frac{M}{R}$

Here, by combining any one with the other or even by combining all, we cannot reach any conclusion about who earns the most.

35. (E) Let the price of 1 dozen oranges, 1 dozen bananas, and 1 dozen apples by x, y and z respectively, then

From I. we have—

$$2x + y = 110$$

From II. $32 + y = 170$

From III. $x + z = 95$

By combining all, we can get the required value.

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36. (E) To find the capacity of a cylindrical tank, we need either radius of the tank or the area of the base and height of the cylinder. Therefore, any two of the three statements fulfill our require.

Exercise 3

Directions—(Q. 1–10) Each of the following problems comprises of a question followed by two statements labelled (I) and (II). Use these statements and generic mathematical knowledge to decide whether the given statements are sufficient to answer the question. Then mark your answer according to the following.

(A) if you can get the answer from (I) alone but not from (II) alone.

(B) if you can get the answer from (II) alone but not from (I) alone.

(C) if you can get the answer from both (I) and (II) together but not from (I) alone or (II) alone.

(D) if you cannot get the answer from (I) and (II) together but need more data.

1. Is Y greater than X ?
   I. $5X = 3K$
   II. $K = Y^2$

2. What is the two-digit number whose first digit is a and second digit is b. The number is greater than 9.
   I. $2a + 3b = 11a + 2b$
   II. The two-digit number is multiple of 19.

3. Is the radius of a circle greater than 4 ?
   I. The points with coordinates (2, 11) and (6, 4) are on the circle.
   II. The points with coordinates (2, 1) and (4, 4) are on the circle.

4. In a class of 49 students, all were offered to participate in 3 college activities, A, B and C. 38 of the students opted for at least one of the activities. How many of the 49 students opted for exactly two of the activities ?
   I. Twelve of the 49 students opted for all the three activities.
   II. Twenty of the 49 students opted for activity A.

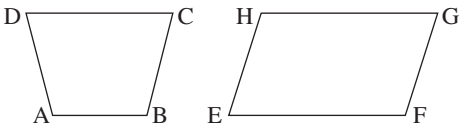
5. Shiva owns 100 shares of stock A and 150 shares of stock B. What is the total value of his stocks ?
   I. The value of each share of stock A is twice the value of each share of stock B.
   II. The total value of 4 shares of stock A and 6 shares of stock B is Rs. 750.

6. A list contains 16 consecutive integers. What is the smallest integer on the list ?
   I. If X is the largest integer on the list, then $(X + 128)^{1/3} = 4$.
   II. If X is the smallest integer on the list and Z is outside the list, then $16X^{-2} = Z^{-2}$.

7. For a particular size of paper, a copier machine makes copies of an original document at a constant rate. How many copies of one original A4 size document does the machine make per minute ?
   I. The machine takes twice as long to make one 11'' × 17'' copy as it takes to make one A4 size copy.
   II. The machine made 1000 copies of 11'' × 17'' documents last month.

8. Is $K^2 + K - 2 > 0$?
   I. $K < 1$
   II. $K > -1$

9. Which of the figures below has the larger area—



   I. The perimeter of ABCD is larger than the of EFGH.
   II. AC is longer than EG.

10. Did the share price of XYZ company's stock increase every week of the year 2001 ?
   I. The share price of XYZ company was Rs. 380 on January 1, 2001.
   II. The share price of XYZ company was Rs. 540 on January 1, 2002.

Directions—(Q. 11–15) Each of these questions is followed by two statements I and II. You have to decide whether the two statements are individually, severally or jointly sufficient to answer the given questions, and mark your answer as—

(A) If statement I alone is sufficient to answer the question.

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(B) If statement II alone is sufficient to answer the question.

(C) If both the statements are not sufficient to answer the question individually or collectively.

(D) If both the statements are individually or collectively sufficient to answer the given question.

11. The area of a rectangle is equal to the area of a circle. What is the length of the rectangle ?
   I. The diameter of the circle is 30 cm.
   II. The breadth of the rectangle is 24 cm.

12. Simran's marks in Geography are 16 more than the average marks obtained by her in Mathematics, Science, English and Hindi. What are her marks in Geography ?
   I. The maximum marks in each subject were 100.
   II. The total marks obtained by her in Mathematics, Science, English and Hindi were 250.

13. The speed of a 110 metres long running train 'X' is 45 per cent more than the speed of another 160 metres long train 'Z' running in opposite directions. What is the speed of the train 'Z' ?
   I. The two trains crossed each other in 6·5 seconds.
   II. The difference between the speeds of the two trains was 28 km/hour.

14. Aditi gave a part of money she had, to Geetanjali. Geetanjali in turn gave 30 per cent to what she got from Aditi to Deepti. How much money did Deepti get ?
   I. Aditi had Rs. 8000 with her.
   II. The difference between the amounts of Geetanjali and Deepti was Rs. 600.

15. The difference between the digits of a two-digit number is 4. What is the digit in the unit place in that number ?
   I. The difference between the number and the number obtained by interchanging the positions of the digits is 36.
   II. The sum of the digits of that number is 12.

Directions—(Q. 16–31) Each of these questions has a problem and two statements, labelled (I) and (II). Use the data given with other information to decide whether the statements are sufficient to answer the given problems. Choose the best alternative from (A), (B), (C) and (D) as—

(A) If you get the answer from (I) alone but not from (II) alone.

(B) If you get the answer from (II) alone but not from (I) alone.

(C) If you get the answer from both (I) and (II) together, but not from (I) alone or (II) alone.

(D) If either statement (I) alone or statement (II) alone suffices.

16. Is Amritha's age now is greater than Brindha's age ?
   I. Amritha is twice as old as she was 10 years ago.
   II. Brindha is half as old as she will be in 10 years.

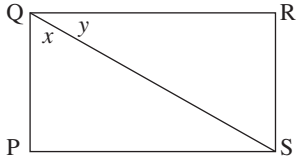
17. Is t an even integer ?
   I. If t is divided by 4, the result is an odd integer.
   II. The value of t is equal to 3 times an integer.

18. Guha has a total of 64 compact discs and casettes. How many compact discs does he have ?
   I. If he buys 10 more cassettes, he will have 58 cassettes.
   II. He has 3 times as many cassettes as compact discs.

19. What is the value of the ratio $p : q$?
   I. $3p = 2q$
   II. $2p + q = 6$

20. Is b always equal to 1 ?
   I. $\frac{5b^2}{7b^2} = \frac{5}{7}$
   II. b is any number except 0.

21. In the figure that follows, is $x > y$?



   I. PS > PQ
   II. PQRS is a parallelogram

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22. How tall is Nandini ?
    I. If she were 20 centimetres taller, then she would have been $1\frac{1}{2}$ times as tall as her younger brother.
    II. If she were half as tall, she would have been 70 centimetres shorter than she is now.

23. A man holding 7 cards in his hand. Four are "nines" and three are "fives". How many cards does he lay on the table ?
    I. He lays a card on the table if the number on the card is divisible by 3.
    II. He lays a card on the table if and only if the number on it is divisible by 3.

24. How much was the loss ?
    I. The cost is Rs. 300.
    II. The loss is 25 per cent of the selling price.

25. A man invests Rs. 50,000, part in bonds at per cent and the rest in stocks at 4 per cent, how much is invested in stock ?
    I. His total income from the two investments is Rs. 2,000.
    II. He invested Rs. 12,500 more in stocks than he did in bonds.

26. What is the value of the integer n ?
    I. $n^2 - 10n + 9 = 0$
    II. $\frac{1}{6} > \frac{1}{n-1} > \frac{1}{9}$

27. The towns A, B and C are on a straight line. Town C is between A and B. The distance from A to B 100 kilometres. How far is A from C—
    I. The distance from A to B is 25 per cent more than the distance from C to B.
    II. The distance from A to C is $\frac{1}{4}$ of the distance from C to B.

28. Is $\frac{x}{12}$ greater than $\frac{y}{40}$?
    I. $10x$ is greater than $3y$.
    II. $12x$ is smaller than $4y$.

29. John's house is 60 miles from the town. On Sunday, he went to town and returned home. How long did the entire trip take ?
    I. He travelled at a uniform rate for the round trip of 30 miles per hour.
    II. If John travelled 10 miles per hour faster, it would have taken him $\frac{3}{4}$ of the time for the round trip.

30. Is $(x + y)^2 < (x^2 + y^2)$?
    I. $xy < 0$ II. $x^2 < y^2$

31. What is the average of p, q, r, s and t in terms of m and n ?
    I. The average of p, q and r is m.
    II. The average of s and t is n.

Directions—(Q. 32–50) Each of the following problems has a question and two statements labelled (I) and (II). Use the data given in statements (I) and (II) together with other available information (such as the number of hours in a days, the definition of clockwise, mathematical facts, etc.) to decide whether the two given statements are sufficient to answer the respective question. Then mark your answer as—

(A) If statement (I) alone is sufficient to answer the question, but statement (II) alone is not sufficient.

(B) If statement (II) alone is sufficient, but statement (I) alone is not sufficient.

(C) If both the statements (I) and (II) together are sufficient, but neither statement alone is sufficient.

(D) If even both the statements (I) and (II) together are not sufficient to answer the question.

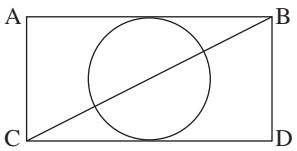
All numbers used in this section are total numbers. A figure given for a problem is intended to provide information consistent with that in the question, but not necessarily with the additional information contained in the statements.

32. How many chocolates can Sheena buy if she has to spend 20% of her budget on vegetables and 30% on groceries ?
    I. Sheena has Rs. 50 with her.
    II. Each chocolate costs 50 paise.

33. How long will it take for jeep to travel a distance of 250 km ?
    I. The relative speed of the jeep with respect to the car moving in the same direction at 40 kmph is 50 kmph.
    II. The car started at 3·00 a.m. in the morning.

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34. What is the perimeter of rectangle ABCD ?



I. Area of the circle is 78·5 sq. cm.
II. AB = 10 cm.

35. Find the value of algebraic expression $x^3y - \left(\frac{x^3}{y}\right)$ —
I. $x = 2$
II. $y = 1$

36. If n is a two-digit number (so $n = ba$ with digits b and a) then what is the last digit of a n ?
I. The number $3n$ is a three-digit number whose last digit is a.
II. The digit a is less than 7.

37. Is the number $\frac{M}{3}$ an odd integer ? (You may assume that $\frac{M}{3}$ is an integer)
I. M = 3K, where K is an integer.
II. M = 6J + 3, where J is an integer.

38. How many families in Jabalpur own exactly two phones ?
I. 75000 families in Jabalpur own at least one telephone.
II. 5000 families in Jabalpur own at least three telephones.

39. What is the value of $p^3 - q^3$?
I. $p^6 - q^6 = 0$ II. $q = 0$

40. How much does Sohan weigh ? Mohan weighs 70 kg—
I. Mohan's weight plus Shyam's weight is equal to Sohan's weight.
II. Sohan's weight plus Shyam's weight is equal to twice the Mohan's weight.

41. What was the value of the sales of the ABC Company in 1980 ?
I. The sales of the ABC Company increased by Rs. 1,00,000 each year form 1970 to 1980.
II. The value of the sales of the ABC Company doubled between 1970 and 1980.

42. Is p greater than 1 ? (You may assume that q is not equal to zero)
I. $\left(\frac{p}{q}\right)$ is greater than 1.
II. $\left(\frac{1}{q}\right)$ is less than 1.

43. How many litres of a chemical can be stored in a cylindrical tank if the radius of the tank is 5 metres ?

$$1 \text{ litre} = \frac{1}{1000} \text{ cubic metre}$$

I. The height of the tank is 5 m.
II. The temperature is 70 degrees Fahrenheit.

44. If $a^6 - b^6 = 0$, then what is the value of $a^3 - b^3$?
I. a is positive.
II. b is greater than 1.

45. If both the conveyer belts A and B are used, then they can fill a hopper with iron ore in one hour. How long will it take for the conveyer belt A to fill the hopper without conveyer belt B ?
I. Conveyer belt A moves twice as much iron ore as conveyer belt B.
II. Conveyer belt B would take more than 3 hours to fill the hopper without belt A.

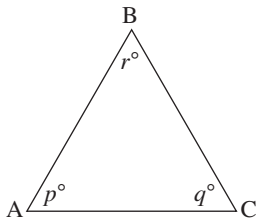
46. Is y larger than 1 ?
I. y is larger than 0
II. $y^2 - 4 > 0$.

47. A worker is hired for 6 days. He is paid Rs. 5 more for each day of work than he was paid for the preceding day. How much was he paid for the first day of the work ?
I. His total wages for 6 days were Rs. 900.
II. He was paid less than Rs. 100 on the first day.

48. A car originally, was sold for Rs. 2,00,000. After a month, the car was discounted $x\%$, and a month later, the car's price was discounted $y\%$. Is the car's price after the discounts less than Rs. 1,75,000 ?
I. $y = 10$
II. $x = 15$

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49. In triangle ABC, find r if AB = 5 and $q = 40$.



I. BC = 5

II. $r > p$

50. How much cardboard will it take to make an open cubical box with no top ?

I. The area of the bottom of the box is 4 square metres.

II. The volume of the box is 8 cubic metres.

Directions—(Q. 51–64) In these questions, a question is followed by two statements A and B. Use the data given in the statements A and B together to decide whether the statement or statements are sufficient to answer the given question. Choose your answer as—

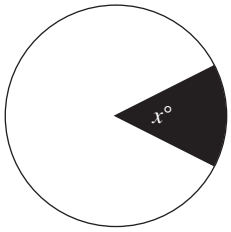
(A) If you can get the answer to the given question from statements A alone but not from B alone.

(B) If you can get the answer to the question from B alone but not from A alone.

(C) If both A and B together are required to answer the given question.

(D) If more data are needed.

51. What is the area of the shaded part of the circle ?



A. The radius of the circle is 4.

B. x is 60.

52. What was Ram Gopal's income in 1990 ?

A. His total income for 1988, 1989 and 1990 was Rs. 3,00,000.

B. He earned 20% more in 1989 than what he did in 1988.

53. Is a quadrilateral ABCD a square ?

A. A pair of adjacent sides are equal.

B. The angle enclosed by these equal adjacent sides is 90°.

54. A large corporation has 7,000 employees. What is the average yearly wage of an employee in the corporation ?

A. 4,000 of the employees are executive.

B. The total wage bill for the company each year is Rs. 77,000,000.

55. Is $x > y$?

A. $(x + y)^2 > 0$

B. x is positive.

56. How long will it take to travel from A and B ? It takes 4 hours to travel from A to B and back to A—

A. It takes 25% more time to travel from A to B than it does to travel from B to A.

B. C is midway between A and B and it takes 2 hours to travel from A to C and back to A.

57. What is $x + y + z$?

A. $x + y = 3$

B. $y + z = 2$

58. Is a number divisible by 9 ?

A. The number is divisible by 3.

B. The number is divisible by 27.

59. Is the integer K odd or even ?

A. K^2 is odd

B. 2K is even

60. Is x positive ?

A. $x^2 + 3x - 4 = 0$

B. $x > -2$

61. Is 2^n divisible by 8 ?

A. n is an odd integer.

B. n is an integer greater than 5.

62. Find $x + y$—

A. $x - y = 6$

B. $2x + 3y = 7$

63. How many books are on the bookshell f ?

A. The bookshelf is 12 feet long.

B. The average weight of each book is 800 gm.

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64. Is x greater than y ?
    A. $x = 2y$
    B. $x = y + 2$.

Directions—(Q. 65–82) Each of these questions is followed by two statements, labelled (P) and (Q), in which certain data are given. In these questions you do not actually have to compute an answer, but rather you have to decide whether the data given in the statements are sufficient for answering the given questions. Using the data given in the statements plus your knowledge of mathematics and everyday facts (such as the number of days in a month) you are to choose your answer as—

(A) If the statement (P) alone is sufficient but statement (Q) alone is not sufficient to answer the question asked.

(B) If the statement (Q) alone is sufficient but statement (P) alone is not sufficient to answer the question asked.

(C) If both the statements (P) and (Q) together are sufficient to answer the question asked but neither of the statements alone is sufficient.

(D) If the statements (P) and (Q) together are not sufficient to answer the question asked and additional data specific to the problem are needed.

65. On a certain auto race track, car's average speed is 160 MPH. What is the length of the track ?
    P. On straight sections, cars can go @ 100 MPH.
    Q. Average lap time (once around the track) is 1 minute 4 seconds.

66. How many tonnes to cement will be needed for the foundation of an apartment building ?
    P. The entire building will require 5000 tonnes of cement.
    Q. The volume of the cement needed for the foundation is 1000 cubic yards.

67. A horse ran 100 miles without stopping. What was its average speed in miles per hour ?
    P. The entire journey takes from 8 p.m. one day to 4 a.m. the following day.
    Q. The horse ran 20 miles per hour for the first 50 miles.

68. Is the side GF of the triangle GFD 5 inches long ?
    P. GD = FD
    Q. GD = 2 inches

69. A television set was originally priced at Rs. 25,000. What per cent discount was given on its original price ?
    P. The stores has 5 of these televisions sets left.
    Q. If the store were to sell all of the remaining television sets, it would receive Rs. 10,000 for them.

70. What is the cost of two kilos of apples ?
    P. Ten apples weigh 2·1 kilos on the average.
    Q. Ten kilos of apples cost Rs. 300.

71. Can truck A pass safely underneath an elevated highway 12 feet above the ground ?
    P. Truck B can pass safely underneath the highway.
    Q. Truck B is taller than Truck A.

72. How many words are listed in the 1280-pages dictionary ?
    P. Page 387 lists 50 words.
    Q. There are 2000 words listed under 'A'.

73. How many minutes does the clock lost a day ?
    P. The clock reads 6 : 00 when it is really 5 : 48.
    Q. The clock is 40 seconds fast each hour.

74. A gold ring weighs 1 gram. The ring is not of pure gold but is mixed with copper. What is the value of the metal in the ring ?
    P. Gold is worth Rs. 350 per gram.
    Q. 50% of the ring is due to copper.

75. Ramesh works 42 hours this week. How much did the earn ?
    P. Ramesh works 35 hours a week at the rate of Rs. 30 per hour.
    Q. Ramesh gets Rs. 40 per hour for overtime work.

76. City X has two libraries. Does the total number of books in both the libraries exceed 18,000 ?
    P. One library has twice as many books as the other library.
    Q. One library has 9,000 books.

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77. Can Usha buy the radio with Rs. 300 ?
    P. The radio now costs 5/6 of its former price.
    Q. After cutting the price of the radio, the store's profit has decreased by 1/2.

78. A circulation manager of a high school newspaper must deliver papers to students and teachers. Will an order of 3200 papers be sufficient ?
    P. There are 15 times as many students as teachers in the school.
    Q. 50 of the students belong to lower classes, not entitled to receive the newspaper.

79. What is width of the widest of the four rivers ?
    P. The most narrow river is 240 yards across.
    Q. The average narrow width is 570 yards across.

80. What is the length of the bed ?
    P. The sum of 2 different yardsticks measures the length exactly.
    Q. If stretched out fully, a man 6 feet 6 inches tall would not fit into the bed.

81. How many hits must a batter get to raise his batting average to 300 ?
    P. X has hit 140 in 10 hits.
    Q. X has hit 250 in 10 hits.

82. How many students in 12th class received over 80 marks in the Maths test ?
    P. The sum of all the marks of the class was 2400.
    Q. The class average in the test was 80 marks.

Directions—(Q. 83–115) Each of the questions below consists of a question and two statements numbered A and B given below it. You have to decide whether the data provided in the statements are sufficient/necessary to answer the question. Read both the statements and give answer as—

(A) If the data in statement A alone are sufficient to answer the question, while the data in statement B alone are not sufficient to answer the question.

(B) If the data in statement B alone are sufficient to answer the question while the data in statement A alone are not sufficient to answer the question.

(C) If the data either in statement A alone or in statement B alone are sufficient to answer the question.

(D) If the data even in both the statements A and B together are not sufficient to answer the question.

(E) If the data in both statements A and B together are necessary to answer the question.

83. What is the height of a circular cone ?
    A. The area of that cone is equal to the area of a rectangle whose length is 33 cm.
    B. The area of the base of that cone is 154 sq. cm.

84. What is the price of a table ?
    A. The total price of 3 chairs and 5 tables is Rs. 18,800.
    B. The total price of 6 chairs and 4 tables is Rs. 20,800.

85. What was the speed of a running train A ?
    A. The relative speed of train A and another train B running in opposite direction is 160 kmph.
    B. The train B crosses a signal post in 9 seconds.

86. What is the difference between the two digits in a two-digit number ?
    A. The sum of the two digits is 8.
    B. 1/5 of that number is 15 less than 1/2 of 44.

87. What is the monthly income of Q ?
    A. Q earns Rs. 6,000 more than R, who earns Rs. 3,000 less than P.
    B. The total monthly income of P and Q is Rs. 27,000.

88. What will be the compounded amount ?
    A. Rs. 200 were borrowed for 192 months at 6% compounded monthly.
    B. Rs. 200 were borrowed for 16 years at 6%.

89. What would have been the selling price per kg of rice ?
    A. 50 kg of rice was purchased for Rs. 3,350 and Rs. 150 was spent on transport.
    B. Profit earned as 5%.

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90. What will be ratio of men to women and children in the town ?
   A. Population in the town is 93,280 of which 56,100 are men.
   B. The ratio of men to children is 5 : 2 and women are double in number than the children.

91. What will be the average weight of the remaining class ?
   A. Average weight of 30 children out of total 46 in the class is 22·5 kg and that of remaining children is 29·125 kg. A child having weight more than 40 kg is excluded.
   B. Average weight of a class of 46 children is 23·5 kg. A child weighing 46 kg is dropped out.

92. What will be the number ?
   A. One-fifth of a number is equal to 20% of that number.
   B. Thirty-five per cent of number is $\frac{7}{20}$ of that number.

93. How many marks did Prakash obtain in Mathematics ?
   A. Prakash secured on an average 55 per cent marks in Mathematics, Physics and Chemistry together.
   B. Prakash secured 10 per cent more than the average in Mathematics.

94. What is the rate of compound interest on a sum of money ?
   A. The total compound interest at the end of two years is Rs. 820.
   B. The total simple interest at the same rate on Rs. 5,000 at the end of three years is Rs. 750.

95. Which is the smaller of the two numbers ?
   A. The difference between these two numbers is one-third of the largest number.
   B. The sum of these two numbers is 30.

96. What is the height of a right-angled triangle ?
   A. The area of the right-angled triangle is equal to the area of a rectangle whose breadth is 12 cm.
   B. The length of the rectangle is 18 cm.

97. What is the speed of a running train which takes 9 seconds to cross a signal post ?
   A. The length of the train is 90 metres.
   B. The train takes 27 seconds to cross a platform of 180 metres.

98. How many boys are there in the class ?
   A. The class has total 45 children and ratio of boys to girls is 4 : 5.
   B. The ratio of girls to boys is 4 : 5 and boys are nine more than the girls.

99. What is the average monthly income per family member—
   A. Each male earns Rs. 1,250 a month and each female earns Rs. 1,050 a month.
   B. Ratio of males to females in the family is 2 : 1.

100. What is the value of $m - n$ □ 37 ?
   A. m is the largest possible six-digit number and n is the smallest possible six-digit number.
   B. The difference between m and n is known.

101. What selling price should be marked on the article ?
   A. Discount of 5% is to be given and profit percentage should be double the discount. Purchase cost is in the range of Rs. 300—Rs. 400.
   B. 10% discount is to be allowed and 15% profit is to be obtained on the purchase cost of Rs. 200 of the article.

102. What is the cost of polishing the rectangular floor ?
   A. Room is 9 m long and 7m wide.
   B. Cost of polishing the floor of 10m by 5m is Rs. 112·50.

103. What will be the cost of painting of the inner wall of a room if the rate of painting is Rs. 20 per square metre ?
   A. Perimeter of the floor is 44 feet.
   B. Height of the wall of the room is 12 feet.

104. What is the ratio of the number of boys and girls in a school ?
   A. Number of boys is 40 more than the girls.

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B. Number of girls is 80 per cent of the number of boys.

105. What is the difference between two numbers ?
   A. First number is 60 per cent of the other number.
   B. 50 per cent of the sum of first and second numbers is 24.

106. What was the speed of the running train ?
   A. Length of the train was 120 metres.
   B. The train crossed the other train whose length was 180 m in 4 seconds.

107. What will be the compound interest after 3 years ?
   A. Rate of interest is 5 per cent.
   B. The difference between the total simple interest and the total compound interest after two years is Rs. 20.

108. What will, be the cost of the second necklace ?
   A. The cost of the first necklace is $\frac{1}{5}$ more than the second and the cost of the third necklace is $\frac{2}{5}$ more than the second. The total cost of all the three necklaces is Rs. 1,20,000.
   B. The cost of the first neclace is $\frac{2}{5}$ more than the second. The cost of the third necklace is the least and total cost of all the three necklaces is Rs. 1,20,000.

109. How many items did the distributor purchase ?
   A. The distributor purchased all the items for Rs. 4500.
   B. If the distributor had given Rs. 5 more for each item, he would have purchased 10 items less.

110. How long will it take to fill a tank ?
   A. One pipe can fill the tank completely in 3 hours.
   B. Second pipe can empty that tank in 2 hours.

111. What will be the area of a plot in sq. metres ?
   A. The length of that plot is $1\frac{2}{3}$ times the breadth of that plot.
   B. The diagonal of that plot is 30 metres.

112. How much minimum marks will be require to pass an examination ?
   A. Student A secured 32% marks in that examination and he failed by 1 mark. Student B secured 36% marks in the same examination and his marks was 1 more than the minimum pass marks.
   B. Student A secured 30% of full marks in the examination and he failed by 2 marks. If he had secured 5 more marks his percentage of marks would have been 40%.

113. What is the original number ?
   A. Sum of two digits of a number is 10. The ratio between the two digits is 1 : 4.
   B. Product of two digits of a number is 16. Quotient of the two digits is 4.

114. What is the rate of the compound interest ?
   A. A certain amount invested at the compound interest rate amounts to Rs. 1331.
   B. The amount was invested for a period of three years.

115. What is the present age of the mother ?
   A. Father's age is eight years more than the Mother's age Father got married at the age of 28 years.
   B. Present age of the Father is 30 years. Four years back the ratio of Mother's age to Father's age was 12 : 13.

Answers with Explanation

1. (D) From I. $X = \frac{3K}{5}$

From II. $y = \sqrt{K}$

If K = 1, $X = \frac{3}{5}$

$y = 1$

$\Rightarrow x < y$

If K = 2, $X = 1{\cdot}2$

$y = 1{\cdot}414$

$\Rightarrow X < y$

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If K = 3, $X = 1{\cdot}8$

$Y = 1{\cdot}732$

$\Rightarrow X > y$

If K = 4, $X = 2{\cdot}4$

$y = 2$

$\Rightarrow X > y$

$\therefore X > y$ for $K > 3$

$X < y$ for $K < 3$

K being a positive integer. The answer can not be determined from I and II together unless K is given.

2. (A) 3. (A) 4. (D)

5. (C) **From Statement I.**

Suppose the value of each share of stock A = Rs. x and the value of each share of stock B = Rs. y.

$\therefore x = 2y$

From the Statement II.

$4x + 6y = 750$

$\Rightarrow 14y = 750$

$\Rightarrow y = \frac{750}{14}$

$x = \frac{750}{7}$

$\therefore$ The total value of $100x + 150y$ can be found out.

$\Rightarrow$ Both the statements are necessary to answer the questions.

6. (A) **From the Statement I.**

$(x + 128)^{\frac{1}{3}} = 4$

$\Rightarrow x + 128 = 64$

$\Rightarrow x = -64$

$\Rightarrow -64$ is the largest integer, then -79 will be the smallest integer.

From Statement (II). The required value cannot be found out.

7. (D) 8. (D) 9. (D) 10. (D)

11. (D) **From the Statement I.**

Area of the circle $= \pi \times \left(\frac{30}{2}\right)^2$

$=$ Area of the rectangle

$\Rightarrow$ We cannot find the length of the rectangle from this.

From statement II, we can get the breadth of the rectangle.

Therefore, we can find the answer from the statements I and II collectively.

12. (D) 13. (D) 14. (B) 15. (D) 16. (C)

17. (A) 18. (D) 19. (A)

20. (D) **From the Statement I.**

$\frac{5b^2}{7b^2} = \frac{5}{7}$

b can have any real number except 0.

Hence, b is not always equal to 1.

From the Statement II.

Clearly, b is not always equal to 1.

Therefore, either statement I alone or statement II alone sufficient.

21. (A) **From the Statement I.**

PS > PQ

$\Rightarrow$ PS > SR

$\Rightarrow$ Angle subtended on Q by PS > Angle subtended on Q by SR.

$\Rightarrow x > y$

Statement II cannot provide the required answer. So, statement I is alone sufficient.

22. (B) 23. (D) 24. (C) 25. (D) 26. (A)

27. (D) 28. (D) 29. (D) 30. (A) 31. (C)

32. (C) 33. (A) 34. (C) 35. (B) 36. (D)

37. (B) 38. (D) 39. (D) 40. (C) 41. (C)

42. (D) 43. (A) 44. (C) 45. (A)

46. (C) I. $y > 0$

$\Rightarrow 0 < y < 2$

II. $y^2 > 4$

$\Rightarrow -2 < y < 2$

47. (A) Let he was paid Rs. x per day, then

$x + (x + 5) + (x + 10) + (x + 15) + (x + 25) + (x + 20) = 900$

$\Rightarrow 6x = 900 - 75 = 825$

$\therefore x = 137{\cdot}50$

48. (B) According to I,

The price of the car after the Ist month = Rs. 1,70,000 and the cost price after the discounts was less than Rs. 1,75,000.

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49. (A) According to I,

$$AB = BC = 5 \text{ cm}$$

$\therefore \quad \angle q = \angle p$

Since, $\angle q = 40°$

$\Rightarrow \quad \angle p = 40°$

$\Rightarrow \quad r = 100°$

50. (None) **From statement I.**

Area of the open cubical box with no top

$= 4 \times 5$

$= 20 \text{ m}^2$

From statement II.

Edge of the given cubical box

$= 2$ metres

$\therefore$ Area of the open cubical box with no top

$= 6 \times (2)^2 - 4 = 20 \text{ m}^2$

We can get the answer from statement I alone and also statement II alone.

Therefore, the answer should be either statement I or statement II is sufficient.

51. (C) Combining both the statements (A) and (B).

Area of the circle $\pi r^2 = 16\pi$

$\therefore \quad 360° = 16\pi$

$\therefore$ The required area $= 60°$

$$= \frac{8\pi}{3}$$

52. (D)	53. (D)	54. (B)	55. (D)	56. (A)
57. (D)	58. (B)	59. (A)	60. (C)	61. (B)
62. (C)	63. (D)	64. (C)	65. (D)	66. (D)
67. (A)	68. (C)	69. (C)	70. (B)	71. (C)
72. (D)	73. (B)	74. (D)	75. (C)	76. (D)
77. (D)	78. (D)	79. (D)	80. (D)	81. (D)
82. (D)	83. (D)	84. (E)	85. (D)	86. (B)
87. (E)	88. (C)	89. (E)	90. (B)	91. (B)
92. (D)	93. (D)	94. (B)	95. (B)	96. (D)
97. (C)	98. (C)	99. (E)	100. (A)	101. (B)
102. (E)	103. (D)	104. (B)	105. (E)	106. (E)
107. (C)	108. (A)	109. (E)	110. (D)	111. (E)
112. (C)	113. (D)	114. (D)	115. (B)	

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9 Permutation and Combination

In this chapter, we shall learn some basic counting techniques which will enable us to answer the questions without actually listing the objects or the things arrangement.

In fact, these techniques will be useful in determining the number of different ways of arranging and selecting the objects without actually listing them. First of all, we study some fundamental principles and notations.

Fundamental Principles of Counting

(a) **Addition principle**—If an event can occur in m different ways and a second event in n different ways, then either of the two events can occur in $(m + n)$ ways provided only one event can occur at a time.

(b) **Multiplication principle**—If an event can occur in m different ways and if corresponding to each way of occurring of this event, there are n different ways of the second event, then both the events can occur simultaneously in $(m \times n)$ different ways.

Remark—The above principle can be extended for any finite number of events. If there are p different ways, the third event can occur corresponding to each of $(m \times n)$ ways of occurring of the first two events, then the three events can occur simultaneously in $(m \times n \times p)$ diff. ways and so on.

For example, we want to find the number of four letter words with or without meaning, which can be formed out of the letters of the word ROSE, where the repetition of the letters is not allowed, then there are as many words as there are ways of filling in 4 vacant places by the 4 letters, keeping in mind that the repetition is not allowed. The first place can be filled in 4 different ways by anyone of the 4 letters R, O, S and E. Following which, the second place can be filled in by any one of the remaining 3 letters in 3 different ways, following which the third place can be filled in 2 different ways, following which, the fourth place can be filled in 1 way.

Therefore, the number of ways in which the 4 places can be filled, by the multiplication principle, is $4 \times 3 \times 2 \times 1 = 24$.

Hence, the required number of words is 24.

If the repetition of the letters is allowed, each of the 4 vacant places can be filled in succession in 4 different ways. Hence, the required number of words

$$= 4 \times 4 \times 4 \times 4$$
$$= 256$$

Principle of Factorial Notation

The continued product of first n natural numbers from 1 to n is called the factorial n and is denoted by the symbol $n\,!$ or $\lfloor n$ and is read as factorial n.

i.e., $n\,!$ or $\lfloor n = 1 \times 2 \times 3 \ldots \times (n-1) \times (n)$

or, $\lfloor n = n \times (n-1) \times \ldots \times 3 \times 2 \times 1$

e.g., $\lfloor 5 = 5 \times 4 \times 3 \times 2 \times 1$

1. $\lfloor (n+1) = (n+1) \lfloor n$
2. $\lfloor 1 = 1$
3. $\lfloor 0 = 1$

Permutation

A permutation is an arrangement in a definite order of a number of objects taken some or all at a time.

The number of permutations of n different objects taken r at a time, where $0 < r \le n$ and the objects do not repeat is $n(n-1)\,(n-2) \ldots (n-r+1)$, which is denoted by $p(n, r)$ or ${}^{n}p_{r}$,

where $${}^{n}p_{r} = \frac{\lfloor n}{\lfloor (n-r)}, 0 \le r \le n$$

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1. If $r = n$

$${}^{n}p_{n} = \lfloor n$$

2. $${}^{n}p_{0} = 1$$

Theorem I. The number of all the permutations of n different objects taken all at a time, is given by $p(n, n)$ or ${}^{n}p_{n}$.

$${}^{n}p_{n} = n(n-1)(n-2) \ldots 3 \times 2 \times 1$$
$$= \lfloor n$$

Theorem II—The number of all permutations of n different objects taken r at a time. $(1 \le r \le n)$ is

$$P(n, r) = {}^{n}P_{r} = n(n-1)(n-2)\ldots\ldots (n-r+1)$$
$$= \frac{n!}{(n-r)!}$$

Theorem III—The number of all permutations of n different objects taken r at a time, when a particular object is never taken in each arrangement is given by ${}^{n-1}P_{r}$.

Permutation of Repeated Objects—To find the number of permutations (X) of n objects taken all at a time, when p of them are alike and of one kind, q of them are alike and of second kind while all others being different.

$$X = \frac{n!}{p!\,q!}$$

Theorem—Let $p_1 + p_2 + p_3 + \ldots\ldots + p_r = n$

The number of permutations of n things, of which p_1 are alike of one kind, p_2 are alike of second kind, p_3 are alike of third kind, ……p_r are alike of rth kind is given by

$$X = \frac{n!}{p_1!\,p_2!\ldots\ldots p_r!}$$

Theorem—Let there be r objects to be arranged, allowing repetition.

Let first object occur exactly p_1 times, second p_2 times etc.

Then the total no. of permutations of these r objects to the above condition.

$$= \frac{(p_1 + p_2 + \ldots\ldots + p_r)!}{p_1!\,p_2!\ldots\ldots p_r!}$$

Circular permutations—If we consider arrangement of objects in the form of a circle, instead of a line; such permutation are called circular permutation.

Let any one of the n objects be fixed at the first place. Now, the remaining $(n-1)$ objects can be arranged among themselves in $(n-1)!$ ways.

Hence, the required number of ways $= (n-1)!$

Difference between clockwise and anti-clockwise arrangement—The number of circular permutation of n things in which clockwise and anti-clockwise arrangement give rise to different permutations is $(n-1)!$. For example, 4 person seated around a table is $(4-1)! = 3!$ because with respect to table clockwise and anticlockwise arrangement are distinct.

If anticlockwise and clockwise order of arrangements are not distinct *e.g.,* arrangement of beads in necklace, arrangement of flowers in a garland etc., then the number of circular permutations of n distinct items is $\frac{1}{2}\{(n-1)!\}$.

Combination

A selection which can be formed by taking some or all of number of objects at a time irrespective of the order of their arrangements, is called a combination.

Theorem—The number of all combinations of n distinct objects taken r at a time, is denoted by symbol ${}^{n}C_{r} = \frac{n!}{(n-r)!\,r!} = C(n-r)$.

Since, each combination consists of r different things, which can be arranged among themselves in $\lfloor r$ ways.

$$\therefore \quad {}^{n}C_{r} \times \lfloor r = {}^{n}P_{r}$$

Properties of ${}^{n}C_{r}$ or C (n, r)

(I) $${}^{n}C_{r} = {}^{n}C_{n-r}$$

(II) Let n and r be non-negative integers such that $r \le n$, then

$${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r} \quad \text{(Pascal's rule)}$$

(III) If $1 \le r \le n$, then

$$n \cdot {}^{n-1}C_{r-1} = (n-r+1)\,{}^{n}C_{r-1}$$

(IV) $${}^{n}C_{x} = {}^{n}C_{y}$$

$\Rightarrow$ either $x = y$ or $x + y = n$

(V) If n is even then the greatest value of ${}^{n}C_{r}$ is ${}^{n}C_{n/2}$ or ${}^{2n}C_{n}$.

(VI) If n is odd, then the greatest value of ${}^{n}C_{r}$ $(0 \le r \le n)$ is

$${}^{n}C_{\frac{n+1}{2}} \quad \text{or} \quad {}^{n}C_{\frac{n-1}{2}}$$

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Division into two groups—The number of ways in which $(m + n)$ things can be divided into two groups containing m and n things respectively

$$^{m+n}C_m \text{ or } ^{m+n}C_n = \frac{(m+n)!}{m!\,n!}$$

Remark—If we have to distribute $(m + n)$ items among two persons in the group containing m and n elements. Then the total no. ways is given by

$$= \frac{(m+n)!}{m!\,n!} \times \lfloor 2$$

Division into groups of equal size—The number of ways in which mn different items can be divided equally into m groups, each containing n objects and the order of the group is not important is

$$\frac{(mn)!}{(n!)^m} \times \frac{1}{m!}$$

The number of ways in which mn different items can be divided equally into m groups, each containing n objects and the order of group is important is

$$\left(\frac{mn!}{(n!)^m} \times \frac{1}{m!}\right) m! = \frac{mn!}{(n!)^m}$$

Generalisation—The number of ways in which $(p + q + r)$ things can be divided into three groups each containing p, q and r things respectively :

$$^{p+q+r}C_p \times {}^{q+r}C_q \times {}^rC_r$$
$$= \frac{(p+q+r)!}{p!\,(q+r)!} \times \frac{(q+r)!}{q!\,r!} \times 1$$
$$= \frac{(p+q+r)!}{p!\,q!\,r!}$$

Similarly, this result can be extended to the case of dividing a given number of things into more than three groups.

Selection of different things—The total number of ways in which a selection can be made out of $(p + q + r)$ things of which p are alike of one kind, q are alike of another kind and r are alike of a third kind hence the required no. of ways

$$= (p + 1)(q + 1)(r + 1) - 1.$$

The total number of ways in which a selection can be made out of $(p + q + r)$ things of which p are alike of one kind, q are alike of another kind and the remaining r are all different.

Hence, the required number of ways is given by $= (p + 1)(q + 1)\,2^r - 1$.

Points to Remember

1. The number of permutation of n different things taken r at a time, where repetition is not allowed $= {}^np_r$.
2. The number of permutations of n different things, taken r at a time, where repetition is allowed $= n^r$.
3. The number of permutations of n objects, where p objects are of the same kind and rest are all different $= \frac{\lfloor n}{\lfloor p}$.
4. The number of permutations of n objects taken all at a time, where p_1 objects are of first kind, p_2 objects are of the second kind, ……p_r objects are of the rth kind and rest, if any, are all different $= \frac{n!}{p_1! \times p_2! \times \ldots p_r!}$.
5. The number of combinations of n different things taken r at a time, is denoted by ${}^nC_r$ and is given by.
$$^nC_r = \frac{n!}{r!(n-r)!}$$
$$0 \le r \le n.$$
6. The number of selection methods of r persons out of n persons $= {}^nC_r$.

Exercise

Directions—Study the following problems carefully and choose the correct alternative given below.

1. If you have 4 flags of different colours, how many different signals can be made, if a signal requires the use of 2 flags one below the other ?
   (A) 12 (B) 15
   (C) 18 (D) 20
   (E) None of these
2. If $\frac{1}{\lfloor 8} + \frac{1}{\lfloor 9} = \frac{x}{\lfloor 10}$, the value of x will be—
   (A) $\lfloor 10$ (B) $\lfloor 100$
   (C) 100 (D) 1000
   (E) None of these
3. If $r = n$, what will be the value ${}^np_r$?
   (A) $n(n-1)$ (B) $\lfloor n$

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(C) n^2 (D) n^r
(E) None of these

4. What is the number of permutations of the letters of the word INSTITUTE ?
(A) 30240 (B) 3024
(C) 2430 (D) 7560
(E) None of these

5. How many 4 digit numbers can be formed by using the digits 1 to 9, if repetition of digits is not allowed ?
(A) 3024 (B) 3032
(C) 2430 (D) 2824
(E) None of these

6. The number of permutations of the letters of the word 'ALLAHABAD' will be—
(A) 2530 (B) 7560
(C) 6075 (D) 3025
(E) None of these

7. How many numbers lying between 100 and 1000 can be made with the digits 0 to 5. If the repetition of digits is not allowed ?
(A) 60 (B) 80
(C) 100 (D) 120
(E) None of these

8. What will be the value of n such that ${}^{n}p_5 = 42\ {}^{n}p_3$ where $n > 4$?
(A) 10 (B) 12
(C) 15 (D) 100
(E) None of these

9. Find the number of different 8 letter arrangements that can be made from the letters of the word DAUGHTER so that all the vowels do not occur together—
(A) 3600 (B) 36000
(C) 40000 (D) 46000
(E) None of these

10. In how many ways can 4 red, 3 green and 2 blue discs be arranged in a row if the discs of the same colour are indistinguishable ?
(A) 1260 (B) 1200
(C) 1500 (D) 1560
(E) None of these

11. What will be the number of arrangements of the letters of the word 'INDEPENDENCE', if the words begin with I and end in P ?
(A) 12000 (B) 12500
(C) 13000 (D) 12600
(E) None of these

12. In the word 'INDEPENDENCE' how many number of arrangements can be made, if all the vowels always occur together ?
(A) 16800 (B) 16000
(C) 15000 (D) 17800
(E) None of these

13. Find the value of ${}^{n}C_{17}$, if
${}^{n}C_9 = {}^{n}C_8$
(A) 10 (B) 1
(C) 100 (D) 1000
(E) None of these

14. A committee of 5 persons is to be constituted from a group of 4 men and 5 women. In how many ways can this be done ?
(A) 120 (B) 122
(C) 126 (D) 130
(E) None of these

15. A committee of 3 persons is to be made from a group of 2 men and 3 women. How many of these committees would consist of 1 man and 2 women ?
(A) 6 (B) 8
(C) 10 (D) 12
(E) None of these

16. Find the number of ways of choosing 4 cards from a pack of 52 playing cards—
(A) 270225 (B) 270725
(C) 370725 (D) 320225
(E) None of these

17. What is the number of ways of choosing 4 cards from a pack of 52 cards, if they are of the same suit ? Given that there are four suits as diamond, club, spade and heart and there are 13 cards of each suit—
(A) 2860 (B) 3060
(C) 2560 (D) 3080
(E) None of these

18. Find the number of ways of choosing 4 cards from a pack of 52 playing cards. If two cards should be red and another two should be black—
(A) 90225 (B) 105625
(C) 105725 (D) 925225
(E) None of these

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19. A bag contains 5 black and 6 red balls. Determine the number of ways in which 2 black and 3 red balls can be selected—
(A) 100 (B) 200
(C) 300 (D) 250
(E) None of these

20. How many words, with or without meaning, each of 3 vowels and 2 consonants can be formed from the letters of the word INVOLUTE ?
(A) 2880 (B) 3000
(C) 2820 (D) 2580
(E) None of these

21. Find the number of permutations from the letters of the word ELORA—
(A) 120 (B) 150
(C) 180 (D) 100
(E) None of these

22. Find the number of permuations from the letters a, e, i, o, u if the repetition is allowed—
(A) 3000 (B) 2580
(C) 3125 (D) 3185
(E) None of these

23. How many words can be made from the word 'EXAMINATION' by taking together all ?
(A) 4989600 (B) 498000
(C) 4009600 (D) 459600
(E) None of these

24. 5 questions are printed in a question paper. A student wants to select 4 questions out of them, how many ways can he select the questions ?
(A) 3 (B) 5
(C) 10 (D) 15
(E) None of these

25. There are 7 points at a plane, out of them, any three are non-collinear. How many lines can be drawn to meet the points ?
(A) 19 (B) 21
(C) 23 (D) 25
(E) None of these

26. There are 10 points at a plane, out of them 4 points are collinear and the rest are non-collinear. How many triangles can be made by joining these points ?
(A) 96 (B) 100
(C) 116 (D) 122
(E) None of these

27. How many telephone numbers of 6 digits can be made from the digits 0 to 9, the numbers should be begin with 63 and the repetition of digits is not allowed ?
(A) 1680 (B) 1700
(C) 1720 (D) 1800
(E) None of these

28. What will be the 50th word from the letters of the word 'AGAIN', if the words are written as in a dictionary ?
(A) NAAGI (B) NAAIG
(C) NGIAA (D) NGAAI
(E) None of these

29. In how many ways can 5 girls and 3 boys be seated in a row so that no two boys are together ?
(A) 14400 (B) 144000
(C) 12200 (D) 12502
(E) None of these

Answers with Explanation

1. (A) We have $${}^{n}p_{r} = \frac{\lfloor n}{\lfloor n-r}$$

$$\therefore \quad {}^{4}p_{2} = \frac{\lfloor 4}{\lfloor 4-2} = \frac{\lfloor 4}{\lfloor 2}$$

$$= \frac{4 \times 3 \times 2 \times 1}{2 \times 1}$$

$$= 4 \times 3 = 12$$

2. (C) $$\frac{1}{\lfloor 8} + \frac{1}{\lfloor 9} = \frac{x}{\lfloor 10}$$

$$\Rightarrow \quad \frac{1}{\lfloor 8} + \frac{1}{9 \times \lfloor 8} = \frac{x}{10 \times 9 \times \lfloor 8}$$

$$\Rightarrow \quad 1 + \frac{1}{9} = \frac{x}{10 \times 9}$$

$$\Rightarrow \quad \frac{10}{9} = \frac{x}{10 \times 9}$$

$$\Rightarrow \quad x = 100$$

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3. (B) We have, $^{n}p_{r} = \frac{\lfloor n}{\lfloor n-2}$

If $r = n$, $\frac{\lfloor n}{\lfloor n-n} = \frac{\lfloor n}{\lfloor 0} \Rightarrow \lfloor n$,

because $\lfloor 0 = 1$.

4. (A) Here, there are 9 objects (letters) in which I appears 2 times and T appears 3 times and rest are all different.

Therefore, the required number of arrangements

$$= \frac{9!}{2!\,3!}$$
$$= \frac{9\cdot8\cdot7\cdot6\cdot5\cdot4\cdot3!}{2\times1\times3!}$$
$$= 72\times7\times6\times10$$
$$= 30240$$

5. (A) There will be as many 4 digit numbers as there are permutations of 9 different digits taken 4 at a time.

$\therefore$ The required 4 digit numbers

$$= {}^{9}p_{4} = \frac{9!}{(9-4)!}$$
$$= \frac{9!}{5!}$$
$$= \frac{9\times8\times7\times6\times5!}{5!}$$
$$= 9\times\ 8\times7\times6$$
$$= 3024$$

6. In this question, the total letters are 9 in which there are 4A's, 2L's and rest are all different.

Therefore, the required number of permutations

$$= \frac{\lfloor 9}{\lfloor 4\ \lfloor 2}$$
$$= \frac{9\times8\times7\times6\times5\times\lfloor 4}{\lfloor 4\ \times2\times1}$$
$$= 9\times8\times7\times3\times5$$
$$= 7560$$

7. (C) Every number between 100 and 1000 is a 3 digit number and we have all the 6 digits as 0 to 5. First, we have to count the permutations of 6 digits taken 3 a time and this number would be ${}^{6}p_{3}$. But these permutations will include those also where 0 is at the 100's place. But the numbers like 092, 052 …… etc are such numbers which are actually 2 digit numbers and hence the number of such numbers has to be subtracted from ${}^{6}p_{3}$ to get the required number. To get the number of such numbers, we have to fix 0 at the 100's place and rearrange the remaining 5 digits taking 2 at a time and this number is ${}^{5}p_{2}$.

Therefore, the required number

$$= {}^{6}p_{3} - {}^{5}p_{2}$$
$$= \frac{\lfloor 6}{\lfloor 6-3} - \frac{\lfloor 5}{\lfloor 5-2}$$
$$= \frac{\lfloor 6}{\lfloor 3} - \frac{\lfloor 5}{\lfloor 3}$$
$$= 4\cdot5\cdot6 - 4\cdot5$$
$$= 100$$

8. (A) We have, $^{n}p_{5} = 42\ {}^{n}p_{3}$

$\Rightarrow n(n-1)(n-2)(n-3)(n-4) = 42\ n(n-1)(n-2)$

Since, $n > 4$, so $n(n-1)(n-2) \neq 0$.

Therefore, by dividing both sides by $n(n-1)(n-2)$, we get

$$(n-3)(n-4) = 42$$
$$\Rightarrow\quad n^2 - 7n - 30 = 0$$
$$n^2 - 10n + 3n - 30 = 0$$
$$(n-10)(n+3) = 0$$
$$n - 10 = 0$$
$$\Rightarrow\quad n = 10$$
$$n + 3 = 0$$
$$\Rightarrow\quad n = -3$$

As n cannot be negative, so $n = 10$.

9. (B) If we want to count those arrangements in which all the vowels do not occur together, we, first have to find all possible arrangements of 8 letters taken all at a time and that can be done in $\lfloor 8$ ways. Then, we have to substract from this number, the number of permutations in which the vowels are always together.

$\because$ The number of permutations in which the vowels are always together will be $\lfloor 6\ \times \lfloor 3$.

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$\therefore$ The required number

$$= \lfloor 8 - \left(\lfloor 6 \times \lfloor 3 \right)$$
$$= \lfloor 6 \ (7 \times 8 - 6)$$
$$= 2 \times \lfloor 6 \ (28 - 3)$$
$$= 50 \times \lfloor 6$$
$$= 50 \times 720$$
$$= 36000$$

10. (A) The total number of discs are
$$4 + 3 + 2 = 4$$
Out of 9 discs, 4 are of the first kind (red), 3 are of the second kind (green) and 2 are of the third kind (blue).
Therefore, the number of arrangements
$$= \frac{\lfloor 9}{\lfloor 4 \ \lfloor 3 \ \lfloor 2}$$
$$= \frac{9 \times 8 \times 7 \times 6 \times 5 \times \lfloor 4}{\lfloor 4 \ \times 3 \times 2 \times 1 \times 2 \times 1}$$
$$= 1260$$

11. (D) There are 12 letters in the word INDEPENDENCE in which N appears 3 times, E appears 4 times and D appears 2 times and rest are all different.
Let us fix I and P at the extreme ends, we are left with 10 letters.
$\therefore$ The required number of arrangements
$$= \frac{\lfloor 10}{\lfloor 3 \ \lfloor 4 \ \lfloor 2}$$
$$= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times \lfloor 4}{3 \times 2 \times 1 \times \lfloor 4 \ \times 2 \times 1}$$
$$= 12600$$

12. (A) There are 12 letters in the word INDPENDENCE in which N appears 3 times, E appears 4 times and D appears 2 times and the rest all different. There are 5 vowels in the given word. Since, they have to always occur together, we treat them as a single object as 'EEEEI' for the time being. Thus, the total objects in this case will be 8 in which there are 3 Ns and 2 Ds that can be rearranged in $\frac{8!}{3!\,2!}$ ways.
Corresponding to each of these arrangements, the 5 volwels E, E, E, E and I can be rearranged in $\frac{5!}{4!}$ ways.
Therefore, by the principle of multiplication, the required number of arrangements
$$= \frac{8!}{3!\,2!} \times \frac{5!}{4!}$$
$$= \frac{8 \times 7 \times 6 \times 5 \times 4 \times \lfloor 3}{\lfloor 3 \ \times 2 \times 1} \times \frac{5 \times \lfloor 4}{\lfloor 4}$$
$$= 8 \times 7 \times 6 \times 5 \times 2 \times 5$$
$$= 16800$$

13. (B) We have, ${}^{n}C_9 = {}^{n}C_8$
$$\Rightarrow \frac{\lfloor n}{\lfloor 9 \ \lfloor (n-9)} = \frac{\lfloor n}{\lfloor n-8 \ \lfloor 8}$$
$$\Rightarrow \frac{1}{9} = \frac{1}{n-8}$$
$$\Rightarrow n - 8 = 9$$
$$\Rightarrow n = 17$$
$$\therefore {}^{n}C_{17} \Rightarrow {}^{17}C_{17}$$
$$= 1$$

14. (C) Here, order does not matter. Therefore, we need to count the combinations. There will be as many committees as there are combinations of 9 different persons taken 5 at a time.
Hence, the required number of ways
$${}^{9}C_5 = \frac{\lfloor 9}{\lfloor 5 \ \lfloor 4}$$
$$\Rightarrow = \frac{9 \times 8 \times 7 \times 6 \times \lfloor 5}{\lfloor 5 \ \times 4 \times 3 \times 2 \times 1}$$
$$\Rightarrow 9 \times 2 \times 7 = 126$$

15. (A) 1 man can be selected from 2 men in ${}^{2}C_1$ ways and 2 women can be selected from 3 women in ${}^{3}C_2$ ways. Therefore, the required number of committees will be
$$= {}^{2}C_1 \times {}^{3}C_2$$
$$= \frac{\lfloor 2}{\lfloor 1 \ \lfloor 1} \times \frac{\lfloor 3}{\lfloor 2 \ \lfloor 1}$$
$$= 6$$

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16. (B) There will be be as many ways of choosing 4 cards from 52 cards as there are combinations of 52 different things, taken 4 at a time. Therefore, the required number of ways

$$^{25}C_4 = \frac{\lfloor 52}{\lfloor 4 \lfloor 48}$$

$$= \frac{49 \times 50 \times 51 \times 52}{1 \times 2 \times 3 \times 4}$$

$$= 270725$$

17. (A) There are four suits-diamond, club, spade and heart and there are 13 cards of each suit. Therefore, there are $^{13}C_4$ ways of choos-ing 4 diamonds. Similarly, there are $^{13}C_4$ ways of choosing 4 clubs, $^{13}C_4$ ways of choosing 4 spades and $^{14}C_4$ ways of choosing 4 hearts.

Therefore, the required number of ways

$$= {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4$$

$$= 4 \times \frac{\lfloor 13}{\lfloor 4 \lfloor 9}$$

$$= \frac{4 \times 13 \times 12 \times 11 \times 10 \times \lfloor 9}{4 \times 3 \times 2 \times 1 \times \lfloor 9}$$

$$= 2860$$

18. (B) There are 26 red cards and 26 black cards in a pack of 52 playing cards.

Therefore, the required number of ways

$$= {}^{26}C_2 \times {}^{26}C_2$$

$$= \frac{\lfloor 26}{\lfloor 2 \lfloor 24} \times \frac{\lfloor 26}{\lfloor 2 \lfloor 24}$$

$$= \frac{26 \times 25 \times \lfloor 24}{2 \times 1 \times \lfloor 24} \times \frac{26 \times 25 \times \lfloor 24}{2 \times 1 \times \lfloor 24}$$

$$= 325 \times 325$$

$$= 105625$$

19. (B) Number of ways of selection

$$= {}^5C_2 \times {}^6C_3$$

$$\Rightarrow \quad = \frac{\lfloor 5}{\lfloor 2 \lfloor 3} \times \frac{\lfloor 6}{\lfloor 3 \lfloor 3}$$

$$= \frac{5 \times 4 \lfloor 3}{2 \times 1 \times \lfloor 3} \times \frac{6 \times 5 \times 4 \lfloor 3}{3 \times 2 \times 1 \lfloor 3}$$

$$= 10 \times 20$$

$$= 200$$

20. (A) In the word INVOLUTE, there are 4 vowels, namely I, O, E, U and 4 consonants, namely N, V, L and T.

The number of ways of selecting 3 vowels out of $4 = {}^4C_3 = 4$

The number of ways of selecting 2 consonants out of $4 = {}^4C_2 = 6$

Therefore, the number of combinations of 3 vowels and 2 consonants is $4 \times 6 = 24$

Now, each of these 24 combinations has 5 letters which can be arranged among themselves in $\lfloor 5$ ways.

Therefore, the required number of different words is

$$24 \times \lfloor 5 = 24 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$= 2880$$

21. (A) The required number

$$= {}^5p_5$$

$$= \frac{\lfloor 5}{\lfloor 5-5} = \frac{\lfloor 5}{\lfloor 0}$$

$$= 120$$

22. (C) We have 5 letters and 5 places

$\therefore$ The required number of permutations

$$= 5^5$$

$$= 5 \times 5 \times 5 \times 5 \times 5$$

$$= 3125$$

23. (A) Number of words

$$= \frac{\lfloor 11}{\lfloor 2 \lfloor 2 \lfloor 2}$$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 2 \times 1 \times 2 \times 1}$$

$$= 4989600$$

24. (B) The required ways $= {}^5C_4$

$$= \frac{\lfloor 5}{\lfloor 4 \lfloor 5-4} = \frac{5 \times \lfloor 4}{\lfloor 4 \lfloor 1}$$

$$= 5$$

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25. (B) The required number of lines

$$= {}^{n}C_2$$

$${}^{n}C_2 = \frac{\lfloor n}{\lfloor n-2 \ \lfloor 2}$$

$$= \frac{n(n-1)\lfloor n-2}{\lfloor n-2 \ \lfloor 2}$$

$$= \frac{n(n-1)}{2}$$

$$= \frac{7(7-1)}{2} = \frac{7 \times 6}{2}$$

$$= 21$$

26. (C) The number of triangles

$$= {}^{n}C_3 - {}^{p}C_3$$

$$= \frac{1}{6}[n(n-1)(n-2) - p(p-1)(p-2)]$$

$$= \frac{1}{6}(10(10-1)(10-2) - 4(4-1)(4-2)]$$

$$= \frac{1}{6}(720 - 24)$$

$$= \frac{1}{6} \times 696$$

$$= 116$$

27. (A) The required numbers

$$= {}^{8}p_4$$

$$= \frac{\lfloor 8}{\lfloor 8-4}$$

$$= \frac{8 \times 7 \times 6 \times 5 \times \lfloor 4}{\lfloor 4}$$

$$= 8 \times 7 \times 6 \times 5$$

$$= 1680$$

28. (B) To get the number of words starting with A, we fix the letter A at the extreme left position, then we rearrange the remaining 4 letters taken all at a time. There will be as many arrangements of these 4 letters taken 4 at a time as there are permutations of 4 different things taken 4 at a time. Hence, the number of words starting with A

$$= \lfloor 4 = 24$$

Then, starting with G, the number of words

$$= \frac{\lfloor 4}{\lfloor 2} = 12$$

As after placing G at the extreme left position we are left with the letters A, A, I and N. Similarly, there are 12 words starting with the next letter I. Hence, the total number of words so far obtained

$$= 24 + 12 + 12$$

$$= 48$$

$\therefore$ The 49th word will NAAGI and the required 50th words will be NAAIG.

29. (A) Let us first seat the 5 girls. This can be done in $\lfloor 5$ ways. For each such arrangement, the three boys can be seated only at the cross marked palces. There will be 6 cross marked places and the three boys can be seated in ${}^{6}p_3$ ways. Hence, by multiplication principle, the required number of ways

$$= \lfloor 5 \times {}^{6}p_3$$

$$= \lfloor 5 \times \frac{\lfloor 6}{\lfloor 3}$$

$$= 5 \times 4 \times 3 \times 2 \times 1 \times 6 \times 5 \times 4$$

$$= 14400$$

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10 Probability Theory

Experiment, Outcomes, Events

Experiment—A process of measurement or observations.

Randomness—A chance effect, where one cannot predict the result exactly.

Trial—Single performance of an experiment.

Outcome (Sample points)—Results of an experiment.

Sample spaces—Set of all possible outcomes (sample points) of an experiment.

Events—Subsets of sample space.

Simple event—Subsets of sample space that contain one outcome only, *e.g.*

An experiment is rolling a die, getting any number from 1 to 6 (uncertainty) is randomness. 1, 2, 3, 4, 5, 6 are outcomes of experiment.

$S = \{1, 2, 3, 4, 5, 6\}$ is known as sample space

(i) $\{1\}, \{2\}, \ldots \{6\}$ are simple events

(ii) $\{1, 3, 5\} \equiv$ Odd number

$\{2, 4, 6\} \equiv$ Even number

Getting odd number or even number is an event.

Union, Intersection, Complements of Events

Let S be a sample space and A, B, C, … are subsets (events) of S.

(1) **Union** $A \cup B = \{x : x \in A \text{ or } x \in B \text{ or } x \in A \text{ and B both}\}$

(2) **Intersection**

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

If $A \cap B = \phi$, then A and B are called mutually exclusive events.

(3) **Complement**

$$A^C = \{x \in S \text{ and } x \notin A\}$$

(a) $A \cap A^C = \phi$

(b) $A \cup A^C = S$

Probability—The probability of an event A of an experiment is a measure, how frequently A is about to occur if we make many trials.

Definition 1. If the sample space of an experiment consists of finitely many outcomes (points), that are equally likely, then the probability P(A) of an event A is

$$P(A) = \frac{\text{Number of points in A}}{\text{Number of points in S}} = \frac{N(A)}{N(S)}$$

and $P(S) = 1$

Definition 2. Given a sample space S, with each event A of S ($A \subset S$), there is associated a number P(A), called probability of A, such that following axioms of probability are satisfied

(i) For every $A \subset S$

$$0 \le P(A) \le 1$$

(ii) For the entire sample space

$$P(S) = 1$$

(iii) For mutually exclusive events A and B ($A \cap B = \phi$) [Addition rule for mutually exclusive events]

$$P(A \cup B) = P(A) + P(B)$$

(iv) For mutually exclusive events, $A_1, A_2, \ldots$

$$P(A_1 \cup A_2 \cup A_3 \ldots) = P(A_1) + P(A_2) + \ldots$$

Some Basic Theorems for Probability

1. **Complementation rule**—For an event A and its complement A^C in sample space S,
$$P(A) = 1 - P(A^C)$$
2. **Addition rule for mutually exclusive events**—For mutually exclusive events $A_1, \ldots, A_m$, in a sample space S,
$$P(A_1 \cup A_2 \cup \ldots \cup A_m) = P(A_1) + P(A_2) + \ldots + P(A_m)$$
3. **Addition rule for arbitrary events**—For events A and B in a sample space,
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

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4. $P(\phi) = 0$, $A \subset B$ then $P(A) \le P(B)$.
5. $0 \le P(A) \le 1$, for all events A.

Conditional probability—The probability of an event B under the condition that an event A occurs. (probability of B given A)

$$P(B/A) = \frac{P(A \cap B)}{P(A)}, \quad P(A) \neq 0$$

Multiplication rule—If A and B are events in a sample space S

and $P(A) \neq 0$ and $P(B) \neq 0$

then $P(A \cap B) = P(A)\,P(B/A) = P(B)\,P(A/B)$

Independent events—If events A and B are such that

$$P(A \cap B) = P(A)\,P(B)$$

If event A and B are independent events and $P(A) \neq 0, P(B) \neq 0$, then

$$P(A/B) = P(A) \text{ and } P(B/A) = P(B)$$

Events $A_1, A_2, \ldots A_n$ are independent if

$$P(A_1 \cap A_2 \cap \ldots \cap A_n) = P(A_1)\,P(A_2) \ldots P(A_n)$$

Rules of Total Probability

Partition—Let S be a sample space, $P_1, \ldots, P_n$ are n-subsets of S such that

(i) $P_i \cap P_j = \phi$ for all $i \neq j$

(ii) $P_1 \cup P_2 \cup \ldots \cup P_n = S$

then $P_1, \ldots, P_n$ forms partition of S.

Rule of elimination (or rule of total probability)—If the events $B_1, \ldots, B_n$ constitutes a partition of sample space S and $P(B_i) \neq 0$ for $i = 1, \ldots, n$, then for any event A in S,

$$P(A) = \sum_{i=1}^{n} P(B_i) \cdot P(A/B_i)$$

Bayes' theorem—If $B_1, \ldots, B_n$ constitutes a partition of the sample space S and $P(B_i) \neq 0$, for $i = 1, 2, \ldots, n$ then for any event A in S such that $P(A) \neq 0$,

$$P(B_r|A) = \frac{P(B_r) \cdot P(A/B_r)}{\sum_{i=1}^{n} P(B_i) \cdot P(A/B_i)}, \quad r = 1, \ldots, n$$

Permutation and Combination

Permutations—A permutation of given things (elements or objects) is an arrangement of these things in some order.

1. **Different things**—The number of permutations of n-things taken all at a time is

$$n! = 1.2.3.\ldots.n$$

2. **Classes of equal things**—If n given things can be divided into classes of a like things differing from class to class, then the number of permutations of these things taken all at time is

$$\frac{n!}{n_1!\, n_2! \ldots n_k!}, \quad (n_1 + n_2 + \ldots + n_k = n)$$

where n_j is the number of things in the j-th class. for $j = 1, 2, \ldots k$

3. The number of different permutations of n-different things taken k at a time without repetitions is

$$n\,(n-1) \ldots (n-k+1) = \frac{n!}{(n-k)!} \text{ and}$$

with repetition it is n^k.

Combination—A selection of one or more things without regard of order

Theorem—The number of different combinations of n things, k at a time, without repetitions is

$${}^{n}C_k = \binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

$$= \frac{n\,(n-1) \ldots (n-k+1)}{1.2 \ldots k}$$

and the number of those combinations where repetitions is allowed

$$= \binom{n+k-1}{k} \Rightarrow n+k-1\ C_k$$

combination of n different things, k at a time without repetition is the number of sets that can be made up from n-given things, each set containing k-different things and no sets are equal.

The factorial—

$$0! = 1$$

$$(n+1)! = (n+1)\,n!$$

and for large n, $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$

(Stirling formula $e = 2{\cdot}718\ldots$)

$$n \to \infty$$

Binomial coefficients—

$$\binom{a}{k} = \frac{a\,(a-1) \ldots (a-k+1)}{k!}$$

$(k \ge 0$, integer)

$$\binom{a}{0} = \binom{0}{0} = 1$$

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$$\binom{n}{k} = \binom{n}{n-k}$$

$(n \geq 0, 0 \leq k \leq n)$

$$\binom{a}{k} + \binom{a}{k+1} = \binom{a+1}{k+1}$$

$(k \geq 0, \text{integer})$

$$\binom{-m}{k} = (-1)^k \binom{m+k-1}{k}$$

$(k \geq 0, \text{integer } m > 0)$

$$\sum_{s=0}^{n-1} \binom{k+s}{k} = \binom{n+k}{k+1} \quad (k \geq 0, n \geq 1$$

both integer)

$$\sum_{k=0}^{r} \binom{p}{k}\binom{q}{r-k} = \binom{p+q}{r}$$

Bionomial theorem—

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^n\, b^{n-k}$$

Random Variables

Random variables—If S is a sample space with probability measure and X is a real valued function defined over the elements of S, then X is called a random variable.

Discrete random variable—If we can count a random variable.

Continuous random variable—If we can measure random variable.

Discrete Random Variables

Probability distribution—If X is a discrete random variable, the function $f(x) = P(X = x)$ for each x, within the range of X is called the probability distribution of X.

Probability distribution function—For a discrete random variable, the function given by $F(x) = P(X \leq x) = \sum_{t \leq x} f(x)$ for $-\infty < x < \infty$, where $f(t)$ is the value of probability distribution of X at t, is called the distribution function (cumulative distribution) of X.

Important Theorems

1. A function can serve as the probability distribution of a discrete random variable X iff its values, $f(x)$ satisfies the conditions.
   (i) $f(x) \geq 0$ with each value within its domain.
   (ii) $\sum_x f(x) = 1$, where the summation extends over all the values within its domain.
2. The value of $F(x)$ of the distribution function of a discrete random variable X, satisfies the conditions
   (i) $F(-\infty) = 0$ and $F(+\infty) = 1$
   (ii) If $a \leq b$, then $F(a) \leq F(b)$ for any real number a and b.
3. If the range of a random variable X consists of the values $x_1 < x_2 < \ldots < x_n$
   then $f(x_1) = F(x_1)$
   and $f(x_i) = F(x_i) - F(x_{i-1})$, $i = 2, 3, \ldots, n$

Continuous Random Variables

Probability density function—A function with values $f(x)$, defined over the set of all real numbers, is called probability density function of the continuous random variable X iff

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx,$$

for any real constants a and b, $a \leq b$.

Probability distribution function—For a continuous random variable and the value of its probability density at t is $f(t)$, the function is given by

$$F(x) = P(X \leq x) = \int_{-\infty}^{b} f(t)\, dt,$$

for $-\infty < x < \infty$

is called the distribution function (cumulative distribution) of X.

Important Theorems

1. If X is a continuous random variable and a, b are real constants, $a \leq b$, then
   $$P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = P(a < X < b)$$
2. A function can serve as probability density of a continuous random variable X if its values, $f(x)$ satisfying the conditions.
   (i) $f(x) \geq 0$ for $-\infty < x < \infty$
   (ii) $\int_{-\infty}^{\infty} f(x)\, dx = 1$
3. If $f(x)$ and $F(x)$ are the values of the probability density and the distribution function of X at x, then
   $$P(a \leq X \leq b) = F(b) - F(a)$$

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for any real constant a and b, $a \le b$ and

$$f(x) = \frac{d\,F(x)}{dx}$$

whenever derivative exists.

Expectation and Moments

Expected value—If X is a discrete random variable and $f(x)$ is the value of its probability distribution at x, the *expected value* of X is

$$E(X) = \sum x\,.f(x)$$

For a continuous random variable X and $f(x)$, the value of its probability density at x, the *expected value* of X is,

$$E(X) = \int_{-\infty}^{\infty} x\,.f(x)\,dx$$

Some Important Theorems

1. If X is a discrete random variable and $f(x)$ is the value of its probability distribution at x, the expected value of g (X) is given by
$$E[g(X)] = \sum_{x} g(x)\,.f(x)$$
2. For the continuous random variable X, and $f(x)$, the value of its probability density at x, the expected value of g (X) is given by
$$E[g(X)] = \int_{-\infty}^{\infty} g(x)\,.f(x)\,dx$$
3. If a and b are constants, then
$$E(a\,X + b) = a\,E(X) + b$$
4. If a is a constant, then
$$E(aX) = a\,E(X)$$
5. If b is a constant, then
$$E(b) = b$$
6. If $c_1, c_2, \ldots, c_n$ are constants, then
$$E\left[\sum_{i=1}^{n} c_i\,g_i(X)\right] = \sum_{i=1}^{n} c_i E[g_i(X)]$$

Moment—The r-th moment about the origin of a random variable X, μ'_r, is the expected value of X^r.

For discrete X,

$$\mu'_r = E(X^r) = \sum_{x} x^r f(x)$$

for $r = 0, 1, 2, \ldots$

For continuous X,

$$\mu'_r = E(X^r) = \int_{-\infty}^{\infty} x^r\,.f(x)\,dx$$

Mean—The first moment about the origin is called mean of X, (it is the expected value of X).

Moment about the mean—The r-th moment about the mean of a random variable X, μ_r is the expected value of $(X - \mu)^r$.

For discrete X,

$$\mu_r = E[(X-\mu)^r] = \sum (x-\mu)^r\,.f(x)$$

$r = 0, 1, 2, \ldots$

For continuous X,

$$\mu_r = E[(X-\mu)^r] = \int_{-\infty}^{\infty} (x-\mu)^r\,.f(x)\,dx$$

Second moment—The second moment about the mean μ is called the variance of the distribution of X (variance of X), σ^2 or (var (X))

$$\sigma^2 = E[(X-\mu)^2]$$

Standard deviation—The positive square root of the variance is called the standard deviation.

Some Important Theorems

1. $$\sigma^2 = \mu_2' - \mu^2$$
2. If X has the variance σ^2, then
$$\text{var}(a\,X + b) = a^2\sigma^2 = a^2\,\text{var}(X)$$
3. **Chebyshev's theorem**—If μ and σ are the mean and the standard deviation of a random variable X, then for any positive constant k, the probability is at least $1 - \frac{1}{k^2}$ that will take on a value within k standard deviations of the mean.
$$P(|X-\mu| < k\sigma) \ge 1 - \frac{1}{k^2}$$

Moment Generating Functions

The *moment-generating function* of a random variable X, where it exist, is,

For discrete X,

$$M_X(t) = E(e^{tX}) = \sum_{x} e^{tx}\,.f(x)$$

For continuous X,

$$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx}\,.f(x)\,dx$$

Some Important Theorems

1. $$\mu_r' = \left.\frac{d^r\,M_X(t)}{dt^r}\right|_{t=0}$$
2. If a and b are constants, then
   (i) $M_{X+a}(t) = E[e^{(X+a)t}] = e^{at}\,.\,M_X(t)$
   (ii) $M_{bX(t)} = E[e^{bXt}] = M_X(bt)$

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(iii) $M_{\frac{X+a}{b}}(t) = E\left[\exp\left(\frac{X+a}{b}\right)\right]$

$= e^{(a/b)t} \cdot M_X\left(\frac{t}{b}\right)$

Probability Generating Function

The probability generating function (*pgf*) of a random variable X is defined by

$$P_X(t) = p_0 + p_1 t + p_2 t^2 + \ldots + p_n t^n + \ldots$$

$$= \sum_{n=0}^{\infty} (p_n t^n) = E(t^X)$$

Probability Distribution

Uniform distribution—A random variable that takes on *k*-different values with equal probability then it has a uniform (discrete) distribution.

The discrete uniform probability distribution is given by

$$f(x) = \frac{1}{k}$$

$$x = x_1, x_2, \ldots x_k, x_i \neq x_j \quad i \neq j$$

and have mean, $\mu = \sum_{i=1}^{k} \frac{x_i}{k}$

and variance, $\sigma^2 = \sum_{i=1}^{k} \frac{(x_i - \mu)^2}{k}$

Bernoulli distribution—If an experiment has two possible outcomes; success and failure with probabilities p and $q = 1 - p$ then the number of successes 0 or 1 has a Bernoulli distribution.

The Bernoulli distribution is given by

$$f(x, p) = p^x (1-p)^{1-x}$$

$$= p^x q^{1-x}, \qquad x = 0, 1$$

Mean $\mu = p$

and variance, $\sigma^2 = pq$.

Binomial distribution—A random variable X has a Binomial distribution and is a Binomial random variable iff its distribution is given by

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, \ldots n$$

where $p + q = 1$

The number of successes in n-trials is a random variable, having Binomial distribution with parameters p and n. The term $b(x; n, p)$ is a successive term in Binomial expansion $[p + q]^n$.

Mean $\mu = np$

Variance $\sigma^2 = npq$

Moment generating function

$$M_X(t) = (q + pe^t)^n$$

Poisson distribution—The Poisson distribution is given by

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

The mean $\mu = \lambda$

Variance $\sigma^2 = \lambda$

and $M_X(t) = e^{\lambda(e^t - 1)}$

Normal distribution—The probability density function for normal distribution is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \qquad (\sigma > 0)$$

where

(1) μ is the mean and σ the standard deviation.

(2) $\frac{1}{\sigma\sqrt{2\pi}}$ is a constant factor that makes the area under the curve equal to 1.

i.e. $\int_{-\infty}^{\infty} f(x)\, dx = 1$

(3) The curve of $f(x)$ is symmetric with respect to $x = \mu$

for $x = 0 = \mu$, it is symmetric with respect to y-axis $x = 0$ (bell shaped).

(4) The exponential function tends to zero very fast, the faster the function is the smaller the standard deviation σ is.

The probability distribution function

$$F(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{u-\mu}{\sigma}\right)^2\right] du$$

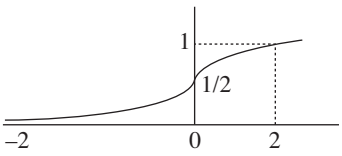
For standard normal distribution ($\mu = 0$ and $\sigma = 1$).

The distribution function,

$$\phi(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp\left(-\frac{u^2}{2}\right) du$$

$$f(x) = \frac{1}{2\pi} \exp\left(-\frac{x}{2}\right)$$

The curve of $\phi(z)$ is S-shaped, increases monotone from 0 to 1 and intersect the vertical axis at 1/2.



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Some Important Theorems

1. **Use of the normal table**—The distribution function F(x) of the normal distribution with any μ and σ is related to the standard normal distribution function $\phi(z)$ by

$$F(x) = \phi\left(\frac{x-\mu}{\sigma}\right)$$

2. **Normal probabilities for intervals**—The probability that a normal random variable X with mean μ and standard deviation σ assume any value in an interval $a < x \le b$ is

$$P(a < X \le b) = F(b) - F(a) = \phi\left(\frac{b-\mu}{\sigma}\right) - \phi\left(\frac{a-\mu}{\sigma}\right)$$

3. **Limit theorem of De Moivre and Laplace**—

(i) For large n,

$$f(x) \sim f^*(x) \qquad (x = 0, 1, 2, \ldots)$$

Here, f is probability function of Binomial distribution

and $f^*(x) = \dfrac{1}{\sigma\sqrt{2\pi}}$

$$= \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right] \quad (\sigma > 0)$$

The normal distribution with mean μ and variance σ^2 equal to the mean and variance of Binomial distribution

Here $\mu = np$

and $\sigma^2 = npq$

(the mean and variance of Binomial distribution)

(ii) $P(a \le X \le b) = \sum_{x=a}^{b} \binom{n}{x} p^x q^{n-x}$

$$\sim \phi(\beta) - \phi(\alpha)$$

where $\alpha = \dfrac{a - \mu - 0{\cdot}5}{\sigma}$

and $\beta = \dfrac{a - \mu + 0{\cdot}5}{\sigma}$

Distributions of Several Random Variables

Discrete Variable

Joint probability distribution—If X and Y are discrete random variables, the function given by $f(x, y) = P(X = x, Y = y)$ for each pair of values (x, y) with the range X and Y is called a joint probability distribution of X and Y.

Theorem—A bivariate function can serve as the joint probability distribution of a pair of discrete random variable X and Y iff its values $f(x, y)$ satisfies.

(a) $f(x, y) \ge 0$

(b) $\sum_x \sum_y f(x, y) \le 1$

Joint probability distribution function—If X and Y are discrete random variables, the function is given by

$$F(x, y) = P(x \le X, y \le Y) = \sum_{s \le x} \sum_{t \le y} f(s, t), \quad x, y \in (-\infty, \infty)$$

where $f(s, t)$ is the value of joint probability distribution of X and Y at (s, t), is called joint probability distribution function (joint cumulative distribution of X and Y).

Continuous Variables

Joint probability density function—A bivariate function with values $f(x, y)$ defined over XY plane is called a joint probability density function of continuous random variables X and Y iff

$$P[(X, Y) \in A] = \iint_A f(x, y)\, dx\, dy$$

for any region A in XY plane.

Theorem—A bivariate function can serve as joint probability density function of a pair of continuous random variables X and Y if its values $f(x, y)$ satisfies,

(a) $f(x, y) \ge 0$

(b) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$

Joint probability distribution function—If X and Y are continuous random variables, the function is given by

$$F(x, y) = P(X \le x, Y \le y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(s, t)\, ds\, dt$$

where $f(s, t)$ is the value of joint probability density of X and Y at (s, t), is called joint probability distribution function of X and Y.

Marginal Distribution

(a) If X and Y are discrete random variables and $f(x, y)$ is the value of joint probability distribution at (x, y), the function given by

$$g(x) = \sum_y f(x, y)$$

for each x, within the range of X, is called marginal (discrete) distribution of X.

$$h(y) = \sum_x f(x, y)$$

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for each y within the range of Y is called marginal distribution of Y.

(b) If X and Y are continuous random variables and $f(x, y)$ is the value of joint probability distribution at (x, y) the function given by

$$g(x) = \int_{-\infty}^{\infty} f(x, y)\, dy$$

for each x within the range of X, is called marginal (continuous) distribution of X.

$$h(y) = \int_{-\infty}^{\infty} f(x, y)\, dx,$$

for each y within the range of Y is called marginal distribution of Y.

Expected Value

(a) If X and Y are discrete random variables, $f(x, y)$ is the value of joint probability distribution at (x, y). The *expected value* of g (X, Y) is

$$E[g(X, Y)] = \sum_x \sum_y g(x, y) f(x, y)$$

(b) If X and Y are continuous random variables, $f(x, y)$ is the value of joint probability distribution at (x, y). The expected value of g (X, Y) is

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y)\, dx\, dy$$

Product moment about the origin—

$$\mu'_{r,s} = E(X^r, Y^s)$$

$$= \sum_x \sum_y x^r y^s f(x, y), \text{ (X, Y discrete)}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^r y^s f(x, y)\, dx\, dy$$

(X, Y continuous)

Product moment about the mean—

$$\mu_{r,s} = E[(X - \mu_X)^r (Y - \mu_Y)^s]$$

$$= \sum_x \sum_y (x - \mu_X)^r (y - \mu_Y)^s f(x, y)$$

(X, Y discrete)

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)^r (y - \mu_Y)^s f(x, y)\, dx\, dy$$

(X, Y continuous)

Covariance—

$\mu_{1,1}$ is called covariance of X and Y

$$\sigma_{XY} = \text{cov}(X, Y) = \mu_{1,1}$$

$$= E[(X - \mu_X)(Y - \mu_Y)]$$

$$= \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f(x, y),$$

(X, Y discrete)

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y)\, dx\, dy$$

(X, Y continuous)

Theorems

1. $$\sigma_{XY} = \mu'_{1,1} - \mu_X \mu_Y$$
$$= E(XY) - E(X) E(Y)$$

2. If X and Y are independent
$$E(XY) = E(X) E(Y)$$
and $$\sigma_{XY} = 0$$

Conditional expectation—

$$E(u(X)|y) = \sum_x u(x) f(x|y) \text{ (X, Y discrete)}$$

$$E(u(X)|y) = \int_{-\infty}^{\infty} u(x) f(x|y)\, dy$$

(X, Y continuous)

X is a random variable, $f(x|y)$ is the value of conditional probability distribution X given $Y = y$ at x, then expectation of $u(x)$ given by $Y = y$.

Here $$\mu_{X/y} = E(X|y)$$

and $$\sigma^2_{X|y} = E[(X - \mu_{X|y})^2|y]$$
$$= E(X^2|y) - \mu^2_{X|y}$$

Several Random Variables

Theorems

1. If $X_1, \ldots, X_n$ are n-independent random variables, then
$$E(X_1 \cdot X_2 \cdot X_3 \cdot \ldots X_n) = E(X_1) E(X_2) \ldots E(X_n)$$

2. If $X_1, X_2, \ldots, X_n$ are random variables and
$$Y = \sum_{i=1}^{n} a_i X_i,$$
where $a_1, \ldots, a_n$ are constants, then
$$E(Y) = \sum_{i=1}^{n} a_i E(X_i)$$
$$\text{var}(Y) = \sum_{i=1}^{n} a_i^2 \text{var}(X_i) + 2 \sum_{i<i} \sum a_{ij} \cdot \text{cov}(X_i X_j)$$

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3. If $X_1, \ldots, X_n$ are independent and

$$Y = \sum_{i=1}^{n} a_i X_i$$

then $$\text{var}(Y) = \sum_{i=1}^{n} a_i^2 \text{var}(X_i)$$

4. If $X_1, \ldots, X_n$ are random variables and

$$Y_1 = \sum_{i=1}^{n} a_i X_i$$

and $$Y_2 = \sum_{i=1}^{n} b_i X_i$$

where $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ are constants then

$$\text{cov}(Y_1, Y_2) = \sum_{i=1}^{n} a_i b_i \text{ var}(X_i) + \sum_{i<j} \sum (a_i b_j + a_j b_i) \cdot \text{cov}(X_i, X_j)$$

5. If $X_1, X_2, \ldots, X_n$ are independent,

$$Y_1 = \sum_{i=1}^{n} a_i X_i$$

and $$Y_2 = \sum_{i=1}^{n} b_i X_i$$

then $$\text{cov}(Y_1, Y_2) = \sum_{i=1}^{n} a_i b_i \text{ var}(X_i)$$

Points to Remember

- **Sample space**—The set of all possible outcomes.
- **Sample points**—Elements of sample space.
- **Event**—A subset of the sample space.
- **Impossible event**—The empty set.
- **Sure event**—The whole sample space.
- **Complementary event or 'not event'**—The set A′ or S – A.
- **Event A or B**—The set $A \cup B$
- **Event A and B**—The set $A \cap B$
- **Event A and not B**—The set A – B.
- **Mutually exclusive event**—A and B are mutally exclusive if $A \cap B = \phi$.
- **Exhaustive and mutually exclusive events**—Events $E_1, E_2, \ldots, E_n$ are mutually exclusive and exhaustive if $E_1 \cup E_2 \cup \ldots \cup E_n = S$ and $E_i \cap E_j = \phi \; \forall \; i \neq j$.
- **Probability**—Number $P(\omega_i)$ associated with sample point ω_i such that
  (i) $0 \leq P(\omega_i) \leq 1$
  (ii) $\Sigma P(\omega)_i)$ for all $\omega_i \in S = 1$
  (iii) $P(A) = \Sigma P(\omega_i)$ for all $\omega_i \in A$. The number $P(\omega_i)$ is called probability of the outcome ω_i.
- **Equally likely outcomes**—All outcomes with equal probability.
- **Probability of an event**—For a finite sample space with equally likely outcomes.
  Probability of an event $P(A) = \frac{n(A)}{n(S)}$, where $n(A)$ = number of elements in the set A, $n(S)$ = number of elements in the set S.
- If A and B are any two events, then
  P(A or B) = P(A) + P(B) – P(A and B)
  Equivalently, $P(A \cup B)$
  $= P(A) + P(B) - P(A \cap B)$
- If A and B are mutually exclusive, then
  P(A or B) = P(A) + P(B)
- If A is any event, then
  P(not A) = 1 – P(A)

Exercise

Directions—Study the following questions carefully and answer the questions given below—

1. Two coins, a one rupee coin and a two rupee coin, are tossed once. What will be the sample space for them ?
   (A) (HH, HT, TH, TT)
   (B) (TH, TT, HT)
   (C) (HH, TT)
   (D) (HT, TH)
   (E) None of these

2. A person is noting down the number of accidents along a busy highway during a year. What will be the sample space for this case ?
   (A) (0, 1, 2) (B) (1, 2, 3, 4)
   (C) (0, 1, 2, 3, …) (D) (1, 2, …)
   (E) None of these

3. Consider the experiment in which a coin is tossed repeatedly until the head comes up. Find the sample space—
   (A) (H, TH, TTH. HH)

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(B) (H, TH, TTH, TTTH, TTTTH)
(C) (H, TH, TTH, TTTH, …)
(D) (TH, TTH)
(E) None of these

4. Consider the experiment of rolling a die. If A be the event of getting a prime number and B be the event of getting an odd number. Find out the set of events 'A but not B'—
(A) (2) (B) (3)
(C) (5) (D) (2, 3)
(E) None of these

5. Consider the experiment of rolling a die, if A be the event of getting an even number and B be the event of getting an odd number, find out the set of events 'A and B'—
(A) ( ) (B) (2)
(C) (2, 3) (D) (3, 5)
(E) None of these

6. One card is drawn from a well shuffled deck of 52 cards. If each outcome is equally likely, calculate the probability that the card will not be an ace-card—
(A) $\frac{13}{12}$ (B) $\frac{1}{4}$
(C) $\frac{12}{13}$ (D) $\frac{5}{4}$
(E) None of these

7. One card is drawn from a deck of 52 cards. If each outcome is equally likely. Find the probability that the card will not be a black card—
(A) $\frac{1}{2}$ (B) $\frac{3}{4}$
(C) $\frac{1}{4}$ (D) $\frac{5}{4}$
(E) None of these

8. A bag contains 9 discs of which 4 are red, 3 are blue and 2 are yellow. The discs are similar in shape and size. A disc is drawn at random from the bag. Find the probability that it will be red—
(A) $\frac{9}{4}$ (B) $\frac{4}{9}$
(C) $\frac{1}{9}$ (D) $\frac{5}{9}$
(E) None of these

9. A bag contains 15 discs of which 7 are red, 5 are blue and 3 are yellow. The discs are similar in shape and size. A disc is drawn at random from the bag. Find the probability that it will be either red or blue—
(A) $\frac{4}{5}$ (B) $\frac{5}{4}$
(C) $\frac{7}{15}$ (D) $\frac{1}{3}$
(E) None of these

10. Two students Ram and Shyam appeared in an examination. The probability that Ram will qualify the examination is 0·05 and that Shyam will qualify the examination is 0·10. The probability that both will qualify the examination is 0·02. Find the probability that both Ram and Shyam will not qualify the examination—
(A) 0·82 (B) 0·87
(C) 0·92 (D) 0·97
(E) None of these

11. If two dice are thrown up together and the addition of the both is 6, what will be the probability of it ?
(A) $\frac{1}{12}$ (B) $\frac{5}{36}$
(C) $\frac{36}{5}$ (D) $\frac{1}{5}$
(E) None of these

12. 3 cards are drawn from a deck of 52 cards. What will be the probability of being the king of all the 3 cards ?
(A) $\frac{3}{1675}$ (B) $\frac{5}{17150}$
(C) $\frac{1}{25}$ (D) $\frac{3}{16575}$
(E) None of these

13. 3 cards are drawn from a deck of 52 cards. What will be the probability of being the red betel card of the 3 cards ?
(A) $\frac{11}{850}$ (B) $\frac{11}{950}$
(C) $\frac{5}{12}$ (D) $\frac{5}{17}$
(E) None of these

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14. The probability of an event is $\frac{2}{3}$. What will be the probability of that event not to be outcome ?

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{2}{5}$

(E) None of these

15. A question of Maths is given to 3 students to solve. The probability of being solved by each is $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$ respectively. If every student tries to solve, what will be the probability of being solved ?

(A) $\frac{3}{4}$ (B) $\frac{1}{4}$

(C) $\frac{2}{3}$ (D) $\frac{1}{5}$

(E) None of these

16. One bag contains 5 red and 7 white balls and another one contains 3 red and 12 white balls. If one ball is drawn out either of the two bags, what will be the probability of being red ball ?

(A) $\frac{37}{120}$ (B) $\frac{49}{120}$

(C) $\frac{1}{4}$ (D) $\frac{1}{5}$

(E) None of these

17. Two students Anil and Ashima appeared in an examination. The probability that Anil will qualify the examination is 0·05 and that Ashima will qualify the examination is 0·10. The probability that both will qualify the examination is 0·02. Find the probability that only one of them will qualify the examination ?

(A) 0·97 (B) 0·32

(C) 0·11 (D) 0·13

(E) None of these

18. A committee of two persons is selected from two men and two women. What is the probability that the committee will have no man ?

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{5}$

(E) None of these

19. Find the probability that when a hand of 7 cards is drawn from a well shuffled deck of 52 cards, it contains all kings—

(A) $\frac{1}{3577}$ (B) $\frac{3}{7735}$

(C) $\frac{5}{7}$ (D) $\frac{1}{7735}$

(E) None of these

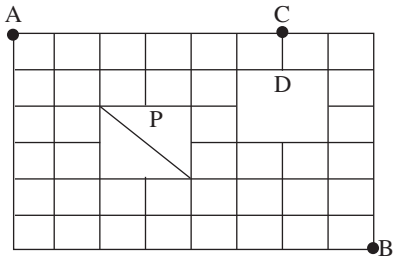
20. In a race competition, there are five teams A, B, C, D and E. What is the probability that A, B and C finish first, second and third respectively ?

(A) $\frac{1}{60}$ (B) $\frac{1}{70}$

(C) $\frac{1}{35}$ (D) $\frac{2}{5}$

(E) None of these

21. The figure below shows the plan of a town. The streets are at right angles to each other. A rectangular park (P) is situated inside the town with a diagonal road running through it. There is also a prohibited region (D) in the town—



Neelam rides her bicycle from her house at A to her office at B, taking the shorest path. Then the number of possible shortest paths that she can choose is—

(A) 60 (B) 75

(C) 45 (D) 90

(E) None of these

22. A bag contains 50 balls and 150 marbles. Half of the both have been marked red. If one item is drawn from the bag, what will be the probability of it as being marked red or the ball ?

(A) $\frac{5}{8}$ (B) $\frac{1}{4}$

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(C) $\frac{2}{3}$ (D) $\frac{8}{5}$

(E) None of these

23. The probability of speaking the truth of Ram is $\frac{3}{4}$% and the probability of speaking the truth of Shyam is $\frac{4}{5}$%. In how many ways % will they be opposite to each other to speak a term ?

(A) 25% (B) 30%

(C) 35% (D) 40%

(E) None of these

24. Both Ram and Shyam speak the truth in 10% conditions. In how many conditions (in %) will they be opposite to each other to speak a term ?

(A) 18% (B) 20%

(C) 22% (D) 25%

(E) None of these

25. Reeta, Geeta and Seeta throw up a die respectively until the victory can be gained by getting the odd digit. Find the probability of their victory if the game continues to the infinity—

(A) $\frac{4}{7}, \frac{2}{7}, \frac{1}{7}$ (B) $\frac{3}{7}, \frac{2}{7}, \frac{1}{7}$

(C) $\frac{10}{7}, \frac{2}{7}, \frac{4}{7}$ (D) $\frac{5}{3}, \frac{3}{5}, \frac{5}{8}$

(E) None of these

26. Four teams participate in a competition. The probabilities of winning to each are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}$ respectively. What will be the probability of winning of any team ?

(A) $\frac{7}{60}$ (B) $\frac{77}{60}$

(C) $\frac{70}{60}$ (D) $\frac{5}{60}$

(E) None of these

27. In a city, 40% persons read the newspaper of English, 75% persons read the news paper of Hindi and 20% persons read both the news papers. What is the percentage of those who do not read the newspaper ?

(A) 15% (B) 10%

(C) 5% (D) 2%

(E) None of these

28. In a certain lottery 10,000 tickets are sold and ten equal prizes are awarded. What is the probability of not getting a prize if you buy one ticket ?

(A) $\frac{999}{1000}$ (B) $\frac{888}{1000}$

(C) $\frac{997}{1000}$ (D) $\frac{999}{10,000}$

(E) None of these

Answers with Explanation

1. (A) The coins are distinguishable in the sense that we can speak of the first coin and the second coin. Since either coin can turn up Head (H) or Tail (T), the possible outcomes may be

Heads on both coins = (H, H) = HH

Head on first coin and Tail on the other = (H, T) = HT

Tail on first coin and Head on the other = (T, H) = TH

Tail on both coins = (T, T) = TT

∴ The required sample space = (HH, HT, TH, TT)

2. (C) The number of accidents along a busy highway during the year of observation can be either 0 or 1 or 2 or 3 or some other positive integer.

Therefore, the sample space

(S) = (0, 1, 2, 3, …)

3. (C) In this experiment head may come up on the first toss, or the 2nd toss, or the 3rd toss, and so on till head is obtianed.

Hence, the required sample space = (H, TH, TTH, TTTH, …)

4. (A) Here, S = (1, 2, 3, 4, 5, 6)

A = (2, 3, 5)

B = (1, 3, 5)

∴ A but not B = A – B = (2)

5. (A) Here, S = (1, 2, 3, 4, 5, 6)

A = (2, 4, 6)

B = (1, 3, 5)

∴ A and B = A ∩ B = ( ) ⇒ Empty set

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6. (C) Assuming that the event 'card drawn is an ace' is A.
Therefore, 'card drawn is not an ace' should be A'.

$$\therefore \quad P(A') = 1 - P(A)$$
$$= 1 - \frac{4}{52}$$
$$= 1 - \frac{1}{13} = \frac{12}{13}$$

7. (A) Let C denote the event 'card drawn is black card'.
Therefore, number of elements in the set

$$C = 26$$
$$\therefore \quad P(C) = \frac{26}{52} = \frac{1}{2}$$

$\Rightarrow$ Probability of a black card

$$= \frac{1}{2}$$

$\therefore$ The event 'card drawn is not a black card' will be C'.

$$\therefore \quad P(C') = 1 - P(C)$$
$$= 1 - \frac{1}{2} = \frac{1}{2}$$

$\Rightarrow$ Probability of not a black card

$$= \frac{1}{2}$$

8. (B) There are 9 discs in all so that the total number of possible outcomes is 9. If R is the event of the disc drawn is red,
The number of red discs

$$n(R) = 4$$
$$\therefore \quad P(R) = \frac{4}{9}$$

9. (A) There are 15 discs in all so that the total number of possible outcomes is 15. If A denotes the event of the disc drawn is red and B denotes an event of the disc drawn is blue.
The number of red and blue discs respectively

$$n(A) = 7$$
$$n(B) = 5$$
$$\therefore \quad P(A) = \frac{7}{15}$$
$$P(B) = \frac{5}{15}$$

The event 'either red or blue' will be described by the set 'A or B'

$$\therefore \quad P(A \text{ or } B) = P(A \cup B)$$
$$= P(A) + P(B)$$
$$= \frac{7}{15} + \frac{5}{15} = \frac{12}{15}$$
$$= \frac{4}{5}$$

10. (B) Let A and B denote the events that Ram and Shyam will qualify the examination respectively, then—

$$P(A) = 0{\cdot}05$$
$$P(B) = 0{\cdot}10$$
and $$P(A \cap B) = 0{\cdot}02$$

Now, the event 'both Ram and Shyam will not qualify the examination' can be expressed as A' $\cap$ B'.
Here A' $\Rightarrow$ 'not A,' *i.e.*, Ram will not qualify the examination and B' $\Rightarrow$ 'not B', *i.e.*, Shyam will not qualify the examination.

$$\Rightarrow \quad A' \cap B' = (A \cup B)'$$
$$\therefore \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$\Rightarrow \quad P(A \cup B) = 0{\cdot}05 + 0{\cdot}10 - 0{\cdot}02$$
$$= 0{\cdot}13$$

Therefore, $P(A' \cap B')$
$$= P(A \cup B)'$$
$$= 1 - P(A \cup B)$$
$$= 1 - 0{\cdot}13$$
$$= 0{\cdot}87$$

11. (B) The required probability

$$= \frac{(x-1)}{6^2} = \frac{(6-1)}{6^2}$$
$$= \frac{5}{36}$$

12. (D) The required probability

$$= \frac{4}{52} \times \frac{4}{51} \times \frac{2}{50}$$
$$= \frac{3}{16575}$$

13. (A) The required probability

$$= \frac{13}{52} \times \frac{12}{51} \times \frac{11}{20}$$
$$= \frac{11}{850}$$

14. (B) The required probability

$$= 1 - \frac{2}{3} = \frac{1}{3}$$

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15. (A) The probability of question being solved

$$= \left[1 - \frac{(a-1)(b-1)(c-1)}{abc}\right]$$

$$= \left[1 - \frac{(2-1)(3-1)(4-1)}{2 \times 3 \times 4}\right]$$

$$= \frac{3}{4}$$

16. (A) The required probability

$$= \frac{1}{2}\left[\frac{a}{(a+b)} + \frac{c}{(c+d)}\right]$$

$$= \frac{1}{2}\left[\frac{5}{(5+7)} + \frac{3}{(3+12)}\right]$$

$$= \frac{1}{2}\left[\frac{5}{12} + \frac{3}{15}\right]$$

$$= \frac{1}{2} \times \frac{111}{12 \times 15}$$

$$= \frac{36}{120}$$

17. (C) Let E and F denote the events that Anil and Ashima will qualify the examination respectively. We have—

$$P(E) = 0{\cdot}05$$
$$P(F) = 0{\cdot}10$$
$$P(E \cap F) = 0{\cdot}02$$

Now, the event only one of them will qualify the examination is same as the event either Anil will qualify, and Ashima will not qualify or Anil will not qualify and Ashima will qualify, *i.e.*,

$E \cap F'$ or $E' \cap F$, where $E \cap F'$ and $E' \cap F$ are mutually exclusive.

Therefore, P (only one of them will qualify)

$$= P(E \cap F' \text{ or } E' \cap F)$$
$$= P(E \cap F') + P(E' \cap F)$$
$$= P(E) - P(E \cap F) + P(F) - P(E \cap F)$$
$$= 0{\cdot}05 - 0{\cdot}02 + 0{\cdot}10 - 0{\cdot}02$$
$$= 0{\cdot}11$$

18. (A) The total number of persons

$$= 2 + 2 = 4$$

Out of these 4 persons, two can be selected in ${}^4C_2$ ways.

Now, No men in the committee of two, means there will be two women in the committee. Out of two women, two can be selected in ${}^2C_2 = 1$ way.

Therefore, $P(\text{No man}) = \frac{{}^2C_2}{{}^4C_2} = \frac{1 \times 2 \times 1}{4 \times 3}$

$$= \frac{1}{6}$$

19. (D) The total number of possible hands

$$= {}^{52}C_7$$

Number of hands with all kings

$$= {}^4C_4 \times {}^{48}C_3$$

($\because$ Other 3 cards must be chosen from the rest 48 cards.)

$\therefore$ The probability of all kings

$$= \frac{{}^4C_4 \times {}^{48}C_3}{{}^{52}C_7}$$

$$= \frac{1 \times \frac{\lfloor 48}{\lfloor 3 \ \lfloor 48}}{\frac{\lfloor 52}{\lfloor 7 \ \lfloor 48}}$$

$$= \frac{\frac{48 \times 47 \times 46}{3 \times 2 \times 1}}{\frac{52 \times 51 \times 50 \times 49 \times 48 \times 47 \times 46}{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}}$$

$$= \frac{1}{7735}$$

20. (A) If we consider the sample space consisting of all finishing orders in the first three places, we will have ${}^5p_3$, *i.e.*,

$$\frac{\lfloor 5}{\lfloor 5-3} = 5 \times 4 \times 3$$

= 60 sample spaces, each with a probability of $\frac{1}{60}$.

Now, A, B and C finish first, second and thrid respectively. Then, there is only one finishing order for this, *i.e.*, ABC.

Therefore, P(A, B and C finish first, second and third respectively) = $\frac{1}{60}$.

21. (D) Neelam has to take path XY

$$\text{A to Y} = {}^4C_2$$
$$= 6 \text{ possibilities}$$
$$\text{Y to B} = {}^6C_2$$
$$= 15 \text{ possibilities}$$

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Therefore, in all possibilities

$= 6 \times 15$

$= 90$

22. (A) The required probability

$$= \left[\frac{2A + B}{2(A + B)}\right]$$

$$= \left[\frac{2 \times 50 + 150}{2(50 + 150)}\right]$$

$$= \frac{250}{400}$$

$$= \frac{5}{8}$$

23. (C) The required percentage of probability

$$= \left[\frac{a(d - c) + c(b - a)}{bd}\right] \times 100\%$$

$$= \left[\frac{3(5 - 4) + 4(4 - 3)}{4 \times 5}\right] \times 100\%$$

$$= \frac{7}{20} \times 100\%$$

$$= 35\%$$

24. (A) The required probability %

$$= \left[2n - \frac{n^2}{50}\right]\%$$

$$= \left[2 \times 10 - \frac{(10)^2}{50}\right]\%$$

$$= \left(20 - \frac{100}{50}\right)\%$$

$$= 18\%$$

25. (A) The probability of victory for Reeta

$$\frac{2^2}{2^3 - 1} = \frac{4}{7}$$

The probability of victory for Geeta

$$= \frac{2}{2^3 - 1} = \frac{2}{7}$$

The probability of victory for Seeta

$$= \frac{1}{2^3 - 1} = \frac{1}{7}$$

26. (B) The required probability

$$= \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}$$

$$= \frac{154}{120} = \frac{77}{60}$$

27. (C) The required %

$= [100 + z - (x + y)]\%$

$= [100 + 20 - (40 + 75)]\%$

$= [120 - 115]\%$

$= 5\%$

28. (A) The required probability

$$= \frac{{}^{9990}C_1}{{}^{10000}C_1}$$

$$= \frac{\frac{9990}{1}}{\frac{10000}{9999}}$$

$$= \frac{\frac{9990}{9989}}{\frac{10000}{9999}}$$

$$= \frac{9990}{10000}$$

$$= \frac{999}{1000}$$

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Miscellaneous Exercise

Directions—(Q. 1–5) Study the following table carefully and answer the questions that follow—

Centrewise and Postwise Number of Candidates

Centre ↓ Post →	Officer	Clerk	Field Officer	Supervisor	Specialist
Bangalore	2000	5000	50	2050	750
Delhi	15000	17000	160	11000	750
Mumbai	17000	19500	70	7000	900
Hyderabad	3500	20000	300	9000	1150
Kolkata	14900	17650	70	1300	1200
Lucknow	11360	15300	30	1500	650
Chennai	9000	11000	95	1650	500

1. In Kolkata, number of Specialist Officers is **approximately** what per cent of Officers ?
   (A) 8·7 (B) 9
   (C) 6·5 (D) 8
   (E) 6·9

2. What is the difference between total number of Officers and Clerks ?
   (A) 29680 (B) 34180
   (C) 32690 (D) 28680
   (E) None of these

3. In Chennai, the number of Clerks is **approximately** how much per cent more than that of Officers ?
   (A) 18 (B) 22
   (C) 20 (D) 2
   (E) None of these

4. Which centre has 300% more number of Clerks as compared to those in Bangalore ?
   (A) Lucknow (B) Mumbai
   (C) Hyderabad (D) Chennai
   (E) None of these

5. Which centre has the highest number of candidates ?
   (A) Delhi (B) Kolkata
   (C) Hyderabad (D) Mumbai
   (E) None of these

Directions—(Q. 6–10) Study the information carefully to answer the following questions.

In an organization consisting of 750 employees, the ratio of males of females is 8 : 7 respectively. All the employees work in five different departments *viz*. HR, Management, PR, IT and Recruitment. 16% of the females work in management department. 32% of males are in HR department. One fifth of the females are in the department of recruitment. The ratio of males to females in the management department is 3 : 2 respectively. 20% of the total numbers of employees are in PR department. Females working in recruitment are 50% of the males working in the same department. 8% of the males are in IT department. The remaining males are in PR department. 22% of the females work in HR department and the remaining females are working in IT department.

6. What is the total number of females working in the IT and recruitment department together?
   (A) 147 (B) 83
   (C) 126 (D) 45
   (E) None of these

7. What is the number of females working in the HR department ?
   (A) 77 (B) 70
   (C) 56 (D) 134
   (E) None of these

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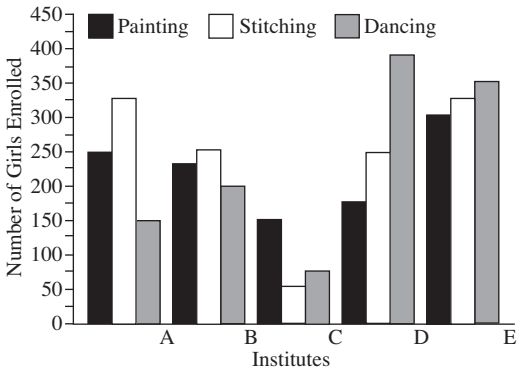
8. Number of males working in HR department forms approximately what per cent of total number of the employees in the organization ?
(A) 20 (B) 28
(C) 32 (D) 9
(E) 17

9. Number of males working in PR department forms what per cent of the number of females working in the same department ? (rounded off to two digits after decimal)
(A) 22·98 (B) 16·68
(C) 11·94 (D) 6·79
(E) 27·86

10. What is the total number of employees working in the management department ?
(A) 128 (B) 77
(C) 210 (D) 140
(E) None of these

Directions—(Q. 11–15) Study the graph carefully to answer the questions that follow—

Number of Girls Enrolled in Different Hobby Classes in Various Institutes in a Year



11. What is the respective ratio of total number of girls enrolled in painting in the institutes A and C together to those enrolled in stitching in the institutes D and E together ?
(A) 5 : 4 (B) 5 : 7
(C) 16 : 23 (D) 9 : 8
(E) None of these

12. Number of girls enrolled in stitching in institute B forms approximately what per cent of the total number of girls enrolled in stitching in all the institutes together ?
(A) 29 (B) 21
(C) 33 (D) 37
(E) 45

13. What is the respective ratio of total number of girls enrolled in painting, stitching and dancing from all the institutes together ?
(A) 44 : 48 : 47 (B) 43 : 47 : 48
(C) 44 : 47 : 48 (D) 47 : 48 : 44
(E) None of these

14. Number of girls enrolled in dancing in institute A forms what per cent of total number of girls enrolled in all the hobby classes together in that institute ? (rounded off to two digits after decimal)—
(A) 23·87 (B) 17·76
(C) 31·23 (D) 33·39
(E) 20·69

15. What is the total number of girls enrolled in painting from all the institutes together ?
(A) 1150 (B) 1200
(C) 1275 (D) 1100
(E) None of these

Directions—(Q. 16–20) Study the following table carefully to answer the questions that follow—

Number of Students Studying in Different Faculties in Seven Institutions

Institution	Faculty				
	Art	Commerce	Science	Engineering	Management
A	125	187	216	98	74
B	96	152	198	157	147
C	144	235	110	164	127
D	165	138	245	66	36
E	215	196	287	86	66
F	184	212	195	112	97
G	225	206	182	138	89

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16. What is the percentage of students studying science with respect to the total number of students studying in the institute G ?
(A) 17·20 (B) 12·70
(C) $21\frac{1}{3}$ (D) $21\frac{2}{3}$
(E) None of these

17. Out of the total students of the institute 'D', **approximately** what percentage of students study Management ?
(A) 9 (B) 8
(C) 12 (D) 10
(E) 5

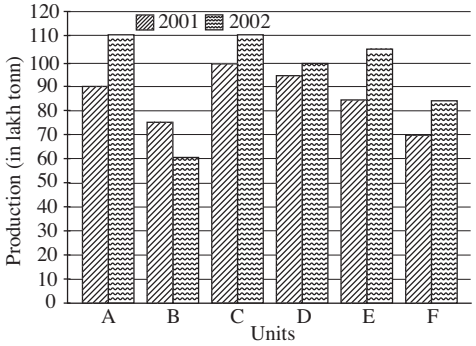
18. The total number of students studying Arts in institutes A, B and C together is **approximately** what per cent of the total number of students studying commerce in institutes D, E, F and G together ?
(A) 50 (B) 45
(C) 42 (D) 55
(E) 53

19. What is the percentage of students studying Engineering in institute C with respect to the total students of all institutions studying Engineering ? (rounded to the nearest integer)
(A) 19 (B) 20
(C) 18 (D) 21
(E) None of these

20. In which institution, the percentage of students studying Commerce with respect to the total students of the institution is maximum ?
(A) F (B) E
(C) C (D) A
(E) None of these

Directions—(Q. 21–25) Study the following diagram carefully and answer the questions that follow—

Production (in lakh tonns) of Six Units of a Company in 2001 and 2002



21. What is the average production of all the units (in lakh tons) for the year 2002 ?
(A) 89 (B) 92
(C) 87 (D) 95
(E) None of these

22. Average production of three units A, B and C in 2001 is what per cent of the average production of units D, E and F in 2002 ? (rounded off to two digits after decimal)
(A) 109·43 (B) 90·37
(C) 91·38 (D) 106·43
(E) None of these

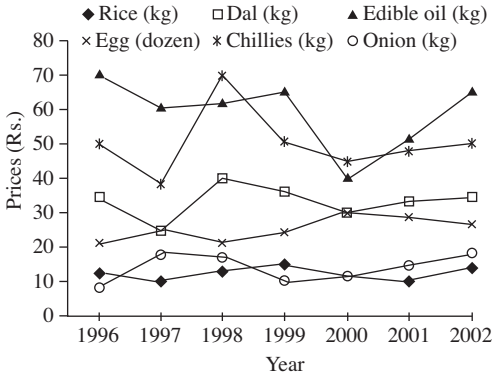
23. What is the ratio of total production for two years together for unit B to that for C ?
(A) 17 : 13 (B) 13 : 17
(C) 11 : 13 (D) 19 : 13
(E) None of these

24. Total production for two years together by unit F is what per cent of the total production of the two years together by unit D ? (rounded off to two digits after decimal)
(A) 79·49 (B) 78·49
(C) 78·47 (D) 79·29
(E) None of these

25. What is the total production of units C, D and E together for both the years ? (in lakh tonns)
(A) 495 (B) 595
(C) 545 (D) 515
(E) None of these

Directions—(Q. 26–28) Answer the questions on the basis of the data presented in the figure below—

Mid-Year Price of Essential Commodities



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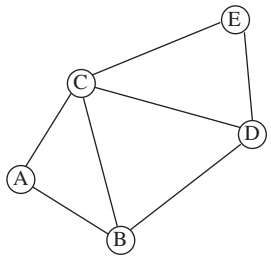
26. During 1996-2002, the number of commodities that exhibited a net overall increase and a net overall decrease, respectively, were—
    (A) 3 and 3
    (B) 2 and 4
    (C) 4 and 2
    (D) 5 and 1

27. The number of commodities that experienced a price decline for two or more consecutive years is—
    (A) 2 (B) 3
    (C) 4 (D) 5

28. For which commodities did a price increase immediately follow a price decline only once in this period ?
    (A) Rice, Edible oil and Dal
    (B) Egg and Dal
    (C) Onion only
    (D) Egg and Onion

Directions—(Q. 29–30) Answer the questions on the basis of the following information.

Shown below is the layout of major streets in a city.



Two days (Thursday and Friday) are left for campaigning before a major election, and the city administration has received requests from five political parties for taking out their processions along the following routes.

Congress : A-C-D-E
BJP : A-B-D-E
SP : A-B-C-E
BSP : B-C-E
CPM : A-C-D

Street B-D cannot be used for a political procession on Thursday due to a religious processions. The district administration has a policy of not allowing more than one procession to pass along the same street on the same day. However, the administration must allow all parties to take out their processions during these two days.

29. Congress procession can be allowed—
    (A) Only on Thursday
    (B) Only on Friday
    (C) On either day
    (D) Only if the religious procession is cancelled

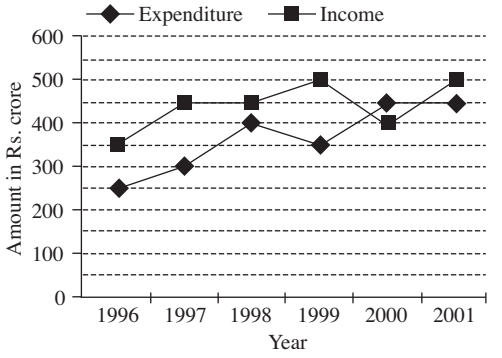
30. Which of the following is NOT true ?
    (A) Congress and SP can take out their processions on the same day
    (B) The CPM procession cannot be allowed on Thursday
    (C) The BJP procession can only take place on Friday
    (D) Congress and BSP can take out their processions on the same day

Directions—(Q. 31–35) Study the following graphs carefully and answer the questions that follow—

Income and Expenditure of Company 'X' During the Period 1996 to 2001

Profit/Loss = Income – Expenditure

$$\%\text{Profit/Loss} = \frac{\text{Income} - \text{Expenditure}}{\text{Expenditure}} \times 100$$



31. What is the average profit earned (in crore Rs) in the given years ?
    (A) $83\frac{1}{2}$ (B) 600
    (C) $113\frac{2}{3}$ (D) 200
    (E) None of these

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32. What **approximately** is the per cent profit earned during the year 1999 ?
(A) 48 (B) 43
(C) 52 (D) 49
(E) None of these

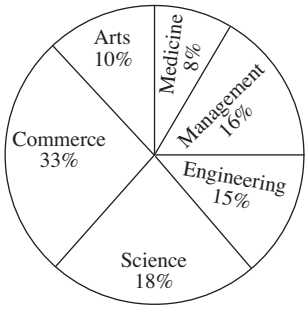
33. Which of the following years has the maximum per cent increase/decrease in income from the previous year ?
(A) 2000 (B) 1999
(C) 1997 (D) 2001
(E) 1997 and 1999

34. What is the percentage increase in expenditure from 1997 to 1998 ?
(A) 25
(B) $33\frac{1}{3}$
(C) $33\frac{2}{3}$
(D) 30
(E) None of these

35. What is the average income (in crore Rs) for the given years ?
(A) $336\frac{2}{3}$
(B) 280
(C) 450
(D) $366\frac{2}{3}$
(E) None of these

Directions—(Q. 36–40) Study the following information carefully and answer the questions that follow—

Distribution of Students Studying Different Discipline In a University



Total Number of Students = 8000

	Ratio of Male : Female		
	Male		**Female**
Arts	2	:	3
Medicine	1	:	1
Management	9	:	7
Engineering	7	:	5
Science	4	:	5
Commerce	3	:	5

36. Number of female students studying Management is what per cent of the total number of students in the University ?
(A) 27 (B) 12
(C) 9 (D) 8
(E) None of these

37. What is the total number of female students studying Engineering and Medicine ?
(A) 1280 (B) 5000
(C) 820 (D) 480
(E) None of these

38. What is the total number of students studying Commerce ?
(A) 1280
(B) 1440
(C) 1650
(D) 2640
(E) None of these

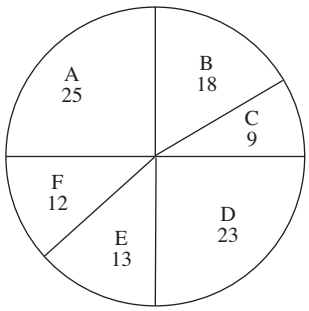
39. What is the difference between the number of students studying Arts and Science ?
(A) 480
(B) 640
(C) 800
(D) 320
(E) None of these

40. How many male students are studying Arts ?
(A) 320
(B) 480
(C) 800
(D) 720
(E) None of these

Directions—(Q. 41–45) The following Pie-chart represents the domestic expenditure of a family in per cent. Study the chart and answer the questions that follow—

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The Total Monthly Income of the Family is Rs. 33,650



A : Expenditure on food.
B : Expenditure on house rent.
C : Expenditure on entertainment.
D : Expenditure on education and maintenance of children.
E : Medical and miscellaneous expenditure.
F : Deductions towards provident fund.

41. The house rent per month is—
(A) Rs. 6000 (B) Rs. 6152
(C) Rs. 6057 (D) Rs. 6048

42. The annual savings in the form of Provident Fund would be—
(A) Rs. 48,456 (B) Rs. 48,540
(C) Rs. 44,856 (D) Rs. 45,480

43. After provident fund deductions and payment of house rent, the total monthly income of the family remains—
(A) Rs. 23,545 (B) Rs. 24,435
(C) Rs. 23,555 (D) Rs. 25,355

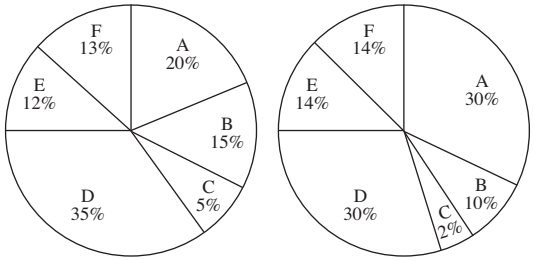
44. The total amount per month the family spends on food and entertainment combined together, is—
(A) Rs. 11,432
(B) Rs. 11,441
(C) Rs. 12,315
(D) Rs. 12,443

45. Had there been no children in the family what would have been the total savings of the family, including that by provident fund ?
(A) Rs. 12,667·50
(B) Rs. 12,625·50
(C) Rs. 11,727·50
(D) Rs. 11,777·50

Directions—(Q. 46–50) Study the following information to answer the given questions—

Percentage of students in various courses (A, B, C, D, E, F) in Pie chart I and Percentage of girls in Pie chart II. Total students : 1200 (800 girls + 400 boys)

Chart-I **Chart-II**



46. For course D, what is the respective ratio of boys and girls ?
(A) 3 : 4 (B) 4 : 5
(C) 3 : 5 (D) 5 : 6
(E) None of these

47. For which pair of courses is the number of boys the same ?
(A) E and F
(B) A and D
(C) C and F
(D) B and D
(E) None of these

48. For course E, the number of girls is how much per cent more than the number of boys for course E ?
(A) 250 (B) 350
(C) 150 (D) 80
(E) None of these

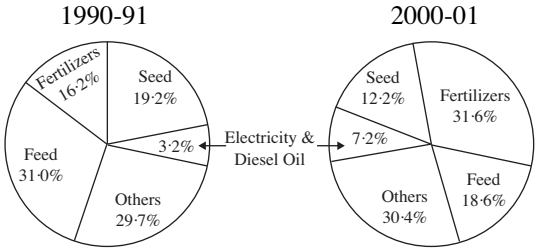
49. For which course is the number of boys the minimum ?
(A) E (B) F
(C) C (D) A
(E) None of these

50. How many girls are there is course C ?
(A) 44 (B) 16
(C) 40 (D) 160
(E) None of these

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Directions—(Q. 51–56) Study the following Pie charts carefully to answer the questions that follow—

Major Inputs Used in Agriculture



Total = Rs. 7659 crore Total = Rs. 14610 crore

51. The total expenditure on electricity and diesel oil in the year 2000-01 exceeded the similar expenditure in 1990-91 by approximately Rs.—
(A) 815 crore (B) 950 crore
(C) 1000 crore (D) 2000 crore

52. The actual input in fertilizers in the year 2000-01 exceeded the input in the year 1990-91 by approximately—
(A) 1·5 times (B) 2 times
(C) 2·75 times (D) 4 times

53. The total input in fertilizers and feed in the year 1990-91 amounted to approximately Rs.—
(A) 3800 crore (B) 3900 crore
(C) 4000 crore (D) 3650 crore

54. The input in the Feed in the year 2000-01 from that in the year 1990-91 has approximately—
(A) Decreased by 55%
(B) Increased by 15%
(C) Increased by 40%
(D) Decreased by 30%

55. It was proposed to increase the input in the Feed to 25% of the total input for the year 2000-01. Approximately, how much reduction in fertilizers input will be required to keep the total input and the percentage wise other inputs the same ?
(A) Rs. 3000 crore
(B) Rs. 2000 crore
(C) Rs. 1000 crore
(D) Rs. 900 crore

56. In terms of actual financial input in electricity and diesel, the increase in the year 2000-01 as compared to 1990-91 was roughly—
(A) 2 times
(B) 3 times
(C) 4 times
(D) The same

Directions—(Q. 57–60) Study the following information carefully to answer the questions that follow—

As per the latest budget proposals, income tax : From the total salary income, 1/3 of the income subject to a maximum of Rs. 20,000 may be deducted as incidentals. From the dividend income, the total dividend income or Rs. 15,000, whichever is lower, may be deducted. The balance after these two deductions is taxable income. The first Rs. 40,000 are tax free. Upto Rs. 60,000, the tax is 10% of the income above 40,000. Above Rs. 60,000 and upto Rs. 1,50,000, it is Rs. 2000 plus 20% of the income over Rs. 60,000. Above 1,50,000, it is Rs. 20,000 plus 30% of the income above 1,50,000. From of tax so calculated, $\frac{1}{5}$ of the investments, subject to a maximum of Rs. 5,000, may be deducted as rebate. For an individual above the age of 65 years, Rs. 10,000 may be deducted from the tax so calculated.

57. If a person has a salary income of Rs. 1,00,000, dividend income of Rs. 50,000 and invests Rs. 20,000, then his income tax must be—
(A) 7400 (B) 9000
(C) 13,000 (D) 15,500

58. What is the maximum income that a person can have, on which no tax is due, if he plans his dividends and investments judiciously ?
(A) Rs. 78,000 (B) Rs. 98,000
(C) Rs. 1,15,000 (D) Rs. 1,25,000

59. In Q 58. what is the maximum income if he has no dividend income and maximum investment—
(A) 75,000 (B) 85,000
(C) 90,000 (D) 1,00,000

60. In Q. 57. if the person is above 65 years of age, then what is the limit ?
(A) 1,20,000 (B) 1,40,000
(C) 1,60,000 (D) 1,80,000

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Directions—(Q. 61–66) Each question is followed by three statements. You will have to study the question and all the three statements given and decide whether any information provided in the statements is sufficient or not for answering the question—

61. What is the amount saved by Sahil per month from his salary ?
    I. Sahil spends 25% of his salary on food, 35% on medicine and education.
    II. Sahil spends Rs. 4000 per month on food and 15% on entertainment and saves the remaining amount.
    III. Sahil spends Rs. 2500 per month on medicine and education and saves the remaining amount.
    (A) II only (B) III only
    (C) II and III both (D) II or III only
    (E) Question cannot be answered even with the information given in all three statements

62. What is the average salary of 15 employees ?
    I. Average salary of 7 clerical cardre (out of the 15 employees) employees is Rs. 8500.
    II. Average salary of 5 clerical cadre (out of the 15 employees) employees is Rs. 10,000.
    III. Average salary of the 3 sub-staff employees.
    (A) None (B) Only I
    (C) Only II (D) Only III
    (E) Question cannot be answered even with the information given in all three statements

63. What is the ratio of the present ages of Rohan and his father ?
    I. Five years ago Rohan's age was one-fifth of his father's age that time.
    II. Two years ago the sum of the ages of Rohan and his father was 36.
    III. The sum of the ages of Rohan, his mother and his father is 62.
    (A) I only (B) I and II only
    (C) III only (D) II or III only
    (E) I or III only

64. What will be the sum of the ages of father and the son after five years ?
    I. Father's present age is twice son's present age.
    II. After ten years the ratio of father's age to the son's age will become 12 : 7.
    III. Five years ago the difference between the father's age and son's age was equal to the son's present age.
    (A) I or II only
    (B) II or III only
    (C) I or III only
    (D) III only
    (E) I or II or III only

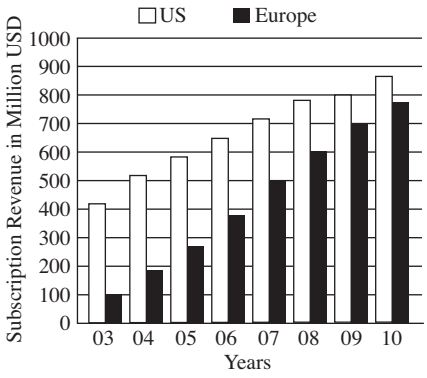
65. What will be the share of P in the profit earned by P, Q and R together ?
    I. P, Q and R invested total amount of Rs. 25,000 for a period of two years.
    II. The profit earned at the end of two years is 30%.
    III. The amount invested by Q is equal to the amount invested by P and R together.
    (A) I only
    (B) II only
    (C) III only
    (D) All I, II and III are required to answer the question
    (E) Question cannot be answered even with the information given in all three statements

66. P, Q and R together invested an amount of Rs. 20,000 in the ratio of 5 : 3 : 2. What was the per cent profit earned by them at the end of one year ?
    I. Q's share in the profit is Rs. 2400.
    II. The amount of profit received by P is equal to the amount of profit received by Q and R together.
    III. The amount of profit received by Q and R together is Rs. 4000.
    (A) II and I or III only
    (B) I or III only
    (C) I and II both
    (D) II and III both
    (E) Information in all the three statements is required to answer the question

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Directions—(Q. 67–70) Answer the following questions based on the information give below—

The bar chart below shows the revenue received, in million US Dollars (USD), from subscribes to a particular Internet service. The data covers the period 2003 to 2007 for the United State (US) and Europe. The bar chart also shows the estimated revenues from subscription to this service for the period 2008 to 2010.



67. The difference between the estimated subscription in Europe in 2008 and what it would have been if it were computed using the percentage growth rate of 2007 (over 2006), is closet to—
(A) 50 (B) 80
(C) 20 (D) 10
(E) 0

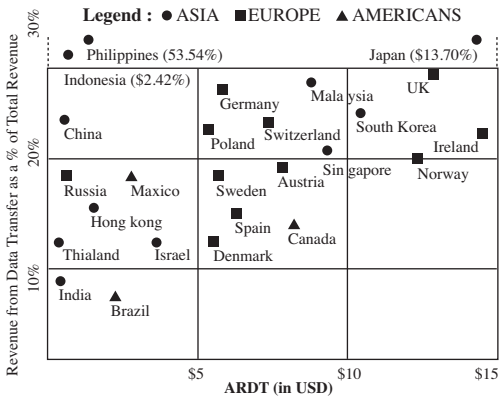
68. In 2003, sixty per cent of subscribers in Europe were men. Given that women subscribers increase at the rate of 10 per cent per annum and men at the rate of 5 per cent per annum, what is the approximate percentage growth of subscribers between 2003 and 2010 in Europe ? The subscription prices are volatile and may change each year—
(A) 62 (B) 15
(C) 78 (D) 84
(E) 50

69. Consider the annual per cent change in the gap between subscription revenues in the US and Europe. What is the year in which the absolute value of this change is the highest ?
(A) 03-04 (B) 05-06
(C) 06-07 (D) 08-09
(E) 09-10

70. While the subscription in Europe has been growing steadily towards that of the US, the growth rate in Europe seems to be declining. Which of the following is closet to the per cent change in growth rate of 2007 (over 2006) relative to the growth rate of 2005 (over 2004) ?
(A) 17 (B) 20
(C) 35 (D) 60
(E) 100

Directions—(Q. 71–73) Answer the following questions based on the information give below—

Telecom operators get revenue from transfer of data and voice. Average revenue received from transfer of each unit of data is known as ARDT. In the diagram below, the revenue received from data transfer as percentage of total revenue received and the ARDT in US Dollars (USD) are given for various countries.



71. It was found that the volume of data transfer in India is the same as that of Singapore. Then which of the following statements is true ?
(A) Total revenue is the same in both countries
(B) Total revenue in India is about 2 times that in Singapore
(C) Total revenue in India is about 4 times that of Singapore
(D) Total revenue in Singapore is about 2 times that of India
(E) Total revenue in Singapore is about 4 times that of India

72. It is expected that by 2010, revenue from Data transfer as a percentage of total revenue

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will triple for India and double for Sweden. Assume that in 2010, the total revenue in India is twice that of Sweden and that the volume of data transfer is the same in both the countries. What is the percentage increase of ARDT in India if there is no change in ARDT in Sweden ?

(A) 400% (B) 550%
(C) 800% (D) 950%
(E) Cannot be determined

73. If the total revenue received is the same for the pairs of countries listed in the choices below, choose the pair that has approximately the same volume of data transfer—

(A) Philippines and Austria
(B) Canada and Poland
(C) Germany and USA
(D) UK and Spain
(E) Denmark and Mexico

Directions—(Q. 74–75) Cities A and B are in different time zones. A is located 3000 km East of B. The table below describes the schedule of an airline operating non-stop flights between A and B. All the times indicated are local and on the same day—

Departure		Arrival	
City	Time	City	Time
B	8·00 am	A	3·00 pm
A	4·00 pm	B	8·00 pm

Assume that planes cruise at the same speed in both directions. However, the effective speed is influenced by a steady wind blowing from East to West at 50 km per hour.

74. What is the times difference between A and B ?

(A) 1 hour and 30 minutes
(B) 2 hours
(C) 2 hours and 30 minutes
(D) 1 hour
(E) Cannot be determined

75. What is the plane's cruising speed in km per hour ?

(A) 700 (B) 550
(C) 600 (D) 500
(E) Cannot be determined

Directions—(Q. 76–80) Answer the following questions based on the information given below—

A low-cost airline company connects ten Indian cities, A to J. The table below gives the distance between a pair of airports and the corresponding price charged by the company. Travel is permitted only from a departure airport to an arrival airport. The customers do not travel by the route where they have to stop at more than two intermediate airports.

Sector No.	Airport of Departure	Airport of Arrival	Distance between the Airports (km)	Price (Rs.)
1	A	B	560	670
2	A	C	790	1350
3	A	D	850	1250
4	A	E	1245	1600
5	A	F	1345	1700
6	A	G	1350	2450
7	A	H	1950	1850
8	B	C	1650	2000
9	B	H	1750	1900
10	B	I	2100	2450
11	B	J	2300	2275
12	C	D	460	450
13	C	F	410	430
14	C	G	910	1100
15	D	E	540	590
16	D	F	625	700
17	D	G	640	750
18	D	H	950	1250
19	D	J	1650	2450
20	E	F	1250	1700
21	E	G	970	1150
22	E	H	850	875
23	F	G	900	1050
24	F	I	875	950
25	F	J	970	1150
26	G	I	510	550
27	G	J	830	890
28	H	I	790	970
29	H	J	400	425
30	I	J	460	540

76. What is the lowest price, in rupees, a passenger has to pay for travelling by the shortest route from A to J ?

(A) 2275 (B) 2850
(C) 2890 (D) 2930
(E) 3340

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77. The company plans to introduce a direct flight between A and J. The market research results indicate that all its existing passengers travelling between A and J will use this direct flight if it is priced 5% below the minimum price that they pay at present. What should the company charge approximately, in rupees, for this direct flight ?
(A) 1991 (B) 2161
(C) 2707 (D) 2745
(E) 2783

78. If the airports C, D and H are closed down owing to security reasons, what would be the minimum price, in rupees, to be paid by a passenger travelling from A to J ?
(A) 2275 (B) 2615
(C) 2850 (D) 2945
(E) 3190

79. If the prices include a margin of 10% over the total cost that the company incurs, what is the minimum cost per kilometer that the company incurs in flying from A to J ?
(A) 0·77 (B) 0·88
(C) 0·99 (D) 1·06
(E) 1·08

80. If the prices include a margin of 15% over the total cost that the company incurs, which among the following is the distance to be covered in flying from A to J that minimizes the total cost per kilometer for the company ?
(A) 2170 (B) 2180
(C) 2315 (D) 2350
(E) 2390

Directions—(Q. 81–82) Answer the following questions on the basis of the information given below—

An airline has a certain free luggage allowance and charges for excess luggage at a fixed rate per kg. Two passengers, Raja and Praja have 60 kg of luggage between them, and are charged Rs. 1200 and Rs. 2400 respectively for excess luggage. Had the entire luggage belonged to one of them, the excess luggage charge would have been Rs. 5400.

81. What is the weight of Praja's luggage ?
(A) 20 kg (B) 25 kg
(C) 30 kg (D) 35 kg
(E) 40 kg

82. What is the free luggage allowance ?
(A) 10 kg (B) 5 kg
(C) 20 kg (D) 25 kg
(E) 30 kg

Directions—(Q. 83–85) Each question is followed by three statements. You will have to study the question and all the three statements given and decide whether any information provided in the statements is sufficient or not to answer the question.

83. What is the rate of interest pcpa ?
I. The amount becomes Rs. 11,025 at compound interest after 2 years.
II. The same amount with simple interest becomes Rs. 11,000 after two years.
III. The amount invested is Rs. 10,000.
(A) I or II or III only (B) I or II only
(C) II and III only (D) I or III only
(E) All I, II and III are required to answer the question

84. The difference between the compound interest and the simple interest at the same rate on a certain amount at the end to two years is Rs. 12·50—
What is the rate of interest ?
I. Simple interest for two years is Rs. 500.
II. Compound interest for two years is Rs. 512·50.
III. Amount on simple interest after two years becomes Rs. 5,500.
(A) I or II only
(B) I or III only
(C) III only
(D) III and either I or II
(E) Any two of I, II and III

85. What is the total compound interest earned at the end of three years ?
I. Simple interest earned on that amount at the same rate and for the same period is Rs. 4500.
II. The rate of interest is 10 p.c.p.a.
III. Compound interest for three years is more than the simple interest for that period by Rs. 465.
(A) Only I and II
(B) Only II and III

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(C) Only I and III
(D) Any two of the three
(E) Either II or III only

Directions—(Q. 86–90) Answer the following questions on the basis of the information given below—

In the word 'INDEPENDENCE', there are 12 letters in all. The letter N appears 3 times, letter E appears 4 times, letter D appears 2 times and the rest letters are all different.

86. What is the total number of arrangements of all the letters of the word—'INDEPENDENCE' ?
(A) 1663200 (B) 1663000
(C) 1553200 (D) 1773200
(E) None of these

87. How many words can be made from the letters of the word 'INDEPENDENCE' if the words start with P ?
(A) 140600 (B) 138600
(C) 138700 (D) 142600
(E) None of these

88. In how many ways can the total arrangements be made from the word INDEPENDENCE, if all the vowels always occur together ?
(A) 16820 (B) 15820
(C) 16800 (D) 17800
(E) None of these

89. In how many ways can the total arrangements be made from the letters of the word 'INDEPENDENCE' if the vowels never occur together ?
(A) 1646400 (B) 1746400
(C) 1656800 (D) 1946400
(E) None of these

90. How many words can be made from the letters of the word—'INDEPENDENCE' if the words begin with I and end in P ?
(A) 12500 (B) 12300
(C) 12700 (D) 12600
(E) None of these

Directions—(Q. 91–96) Answer the following questions on the basis of the information given below—

You have a complete pack of 52 playing cards in which there are four suits—diamond, club, spade, and heart and there are 13 cards of each suit. You are required to choose 4 cards from this complete pack of 52 cards.

91. What is the number of ways of choosing 4 cards from this pack of 52 cards ?
(A) 270725 (B) 271725
(C) 268725 (D) 525725
(E) None of these

92. How many ways can you choose the 4 cards from the complete pack of 52 cards, if all the 4 cards will be of the same suit ?
(A) 3060 (B) 2860
(C) 3160 (D) 2880
(E) None of these

93. In how many ways can the 4 cards be choosen, if all the 4 cards belong to four different suits ?
(A) 13^3 (B) 3^{13}
(C) 13^4 (D) 4^{13}
(E) None of these

94. How many ways can you choose the 4 cards from the complete pack of 52 cards, if all the 4 cards will be face cards ?
(A) 495 (B) 500
(C) 525 (D) 485
(E) None of these

95. In how many ways can the 4 cards be choosen, if two are red cards and two are black cards ?
(A) 105725 (B) 105625
(C) 107625 (D) 109625
(E) None of these

96. What is the number of ways of choosing 4 cards from the pack 52 cards, if cards are of the same colour ?
(A) 29900 (B) 28900
(C) 29999 (D) 26900
(E) None of these

Directions—(Q. 97–99) Answer the following questions on the basis of the information given below—

A group consists of 4 girls and 7 boys. A team of 5 members is to be selected out of the given group.

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97. How many ways can be team be made, if the team has no girl ?
(A) 21 (B) 27
(C) 25 (D) 30
(E) None of these

98. How many ways can the team be selected, if the team has at least one boy and one girl ?
(A) 541 (B) 341
(C) 441 (D) 221
(E) None of these

99. How many ways can the team be selected, if the team has the least 3 girls ?
(A) 71 (B) 81
(C) 89 (D) 91
(E) None of these

Directions—(Q. 100–106) Study the following questions carefully and answer them accordingly—

100. A die is thrown, what is the probability of it that it will be a prime number ?
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$
(C) $\frac{1}{4}$ (D) $\frac{2}{5}$
(E) None of these

101. In an entrance test that is graded on the basis of two examinations, the probability of a randomly chosen student pussing the first examination is 0·8 and the probability of passing the second examination is 0·7. The probability of passing atleast one of them is 0·95. What is the probability of passing bath ?
(A) 0·33 (B) 0·44
(C) 0·55 (D) 0·66
(E) None of these

102. A die is thrown, Find the probability that the number greater than or equal to 3 will appear—
(A) $\frac{1}{3}$ (B) $\frac{2}{3}$
(C) $\frac{1}{6}$ (D) $\frac{1}{4}$
(E) None of these

103. Three coins are tossed once. What will be the probability of getting atleast 2 heads ?
(A) $\frac{1}{3}$ (B) $\frac{1}{4}$
(C) $\frac{1}{5}$ (D) $\frac{1}{2}$
(E) None of these

104. The number lock of a suitcase has 4 wheels, each labelled with ten digits, *i.e.*, from 0 to 9. The lock opens with a sequence of four digits with no repeats. What is the probability of a persons getting the right sequence to open the suitcase ?
(A) $\frac{1}{5040}$ (B) $\frac{1}{5050}$
(C) $\frac{1}{5055}$ (D) $\frac{1}{3040}$
(E) None of these

105. Out of 100 persons, two sections of 40 and 60 are formed. It you and your friend are among the 100 persons. What is the probability that you both enter the same section ?
(A) $\frac{15}{33}$ (B) $\frac{16}{33}$
(C) $\frac{17}{33}$ (D) $\frac{19}{33}$
(E) None of these

106. How many numbers greater than 1000000 can be formed by using the digits 1, 2, 0, 2, 4, 2, 4 ?
(A) 360 (B) 380
(C) 372 (D) 480
(E) None of these

Directions—(Q. 107–110) Answer the following questions on the basis of the information given below—

'A bag contains a balls, in which 4 are red, 3 are blue and 2 are yellow. The balls are similar in shape and size.'

107. A ball is drawn at random from the bag. Find the probability that it will be yellow—
(A) $\frac{1}{9}$ (B) $\frac{2}{3}$
(C) $\frac{2}{9}$ (D) $\frac{1}{3}$
(E) None of these

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108. A ball is drawn at random from the bag. Find the probability that it will not be blue ?

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$

(C) $\frac{3}{2}$ (D) $\frac{1}{9}$

(E) None of these

109. A ball is drawn at random from the bag. Calculate the probability that it will be blue ?

(A) $\frac{1}{9}$ (B) $\frac{1}{3}$

(C) $\frac{2}{3}$ (D) $\frac{2}{9}$

(E) None of these

110. A ball is drawn at random from the bag. Calculate the probability that it will be either red on blue—

(A) $\frac{7}{9}$ (B) $\frac{5}{9}$

(C) $\frac{2}{9}$ (D) $\frac{1}{9}$

(E) None of these

Directions—(Q. 111–115) Study the following table carefully and answer the questions that follow—

Number of Students of the Standard B.A.(I) Participating in Different Games

Games	Class-B.A.(I) (Sections)					
	A	B	C	D	E	Total
Hockey	8	4	8	4	8	32
Football	8	8	12	12	12	52
Chess	8	8	8	4	4	32
Table Tannis	12	16	12	8	12	60
Badminton	8	12	8	12	12	52
Total No. of Boys	44	48	48	40	48	228

Note—(1) Every student, whether boy or the girl, of each section of the standard B.A.(I) participats in a game.

(2) In each section, the number of girls participating in each game is 25% of the number of boys participating in each game.

(3) Each student (whether boy or the girl) participates in one and only one game.

111. What should be the total number of students in the college if all the boys of section A together with all the girls section B and section C were to be equal to 25% of the total number of students ?

(A) 272 (B) 656

(C) 560 (D) 340

(E) None of these

112. If boys of section E participating in chess together with girls of section B and section C participating in Table Tannis and Hockey respectively are selected for a course at the college of sports, what per cent of the students will get this advantage approximately ?

(A) 3·51 (B) 10·52

(C) 13·5 (D) 9·80

(E) None of these

113. All the boys of section D passed the annual examination but a few girls failed. If all the boys and girls who passed and entered into the next class are in the ratio of boys to girls as 5 : 1, what would be the number of girls who failed in section D ?

(A) 1 (B) 4

(C) 2 (D) 3

(E) None of these

114. Girls playing which of the following games need to be combined to yield a ratio of boys to girls of 4 : 1, If all boys playing chess and Badminton are combined ?

(A) Hockey and Badminton

(B) Hockey and Football

(C) Table Tennis and Hockey

(D) Badminton and Table Tennis

(D) None of these

115. If for a social work, every boy of section D and section C is paired with a girl of the same sections, what percentage of the boys of these two section cannot participate in social work ?

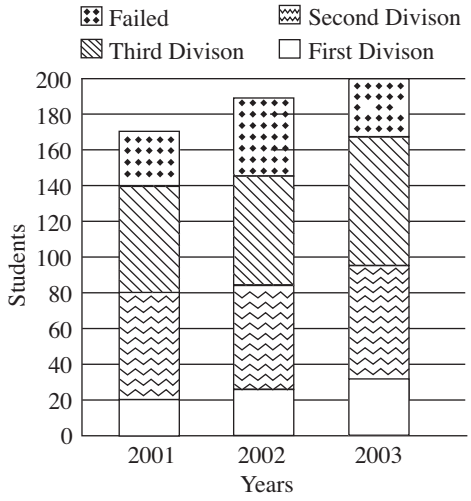
(A) 60 (B) 65

(C) 72 (D) 75

(E) None of these

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Directions—(Q. 116–120) The following sub-divided bar diagram depicts the result of M.Sc. students of a college for the years, 2001 to 2003. Study the bar diagram carefully and answer the questions that follow—



116. How many per cent students passed in Ist division in 2001 ?
(A) $11\frac{3}{7}\%$ (B) $12\frac{17}{13}\%$
(C) 33% (D) 22%
(E) None of these

117. In which year the college had the best result for M.Sc. ?
(A) 2003
(B) 2002
(C) 2001
(D) Cannot be determined
(E) None of these

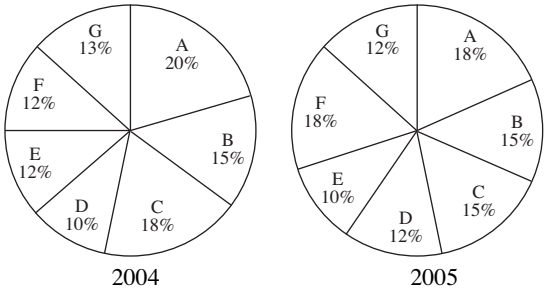
118. What was the pass percentage in the year of 2001 ?
(A) 78·2% (B) 82·3%
(C) 80% (D) 33%
(E) None of these

119. What is the aggregate pass percentage during the three years ?
(A) 62%
(B) 34%
(C) 45%
(D) 80·5%
(E) None of these

120. What is the percentage of students in 2003 over 2001 ?
(A) $117\frac{11}{17}\%$
(B) 33%
(C) $35\frac{11}{17}\%$
(D) $17\frac{11}{17}\%$
(E) None of these

Directions—(Q. 121–125) Study the given pie charts carefully and answer the questions that follow—

Number of Students in Different Disciplines in an Institution for the Years 2004 and 2005



121. In how many disciplines the number of students has decreased from 2004 to 2005 ?
(A) 2 (B) 3
(C) 4 (D) 1
(E) None of these

122. By how much per cent approximately the number of students of discipline B has increased from 2004 to 2005 ?
(A) 11·27% (B) 12·5%
(C) 14·3% (D) $7\frac{8}{7}\%$
(E) None of these

123. What is the maximum difference of the number of students for the same discipline for the two years ?
(A) 58
(B) 120
(C) 135
(D) 115
(E) None of these

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124. In which of the following pairs of disciplines, the difference between the number of students for the same discipline for the two years is equal to that for the other discipline for the two years ?

(A) B and F (B) A and E
(C) E and G (D) G and C
(E) None of these

125. The number of students which has increased for F from 2004 to 2005 is how many times the number of students for F in 2004 ?

(A) 0·7 (B) 0·5
(C) 0·9 (D) 1·1
(E) None of these

Answers

1. (D)	2. (C)	3. (B)	4. (C)	5. (D)
6. (B)	7. (A)	8. (E)	9. (C)	10. (D)
11. (C)	12. (B)	13. (A)	14. (E)	15. (D)
16. (D)	17. (E)	18. (A)	19. (B)	20. (C)
21. (D)	22. (C)	23. (E)	24. (A)	25. (B)
26. (C)	27. (D)	28. (C)	29. (A)	30. (D)
31. (A)	32. (B)	33. (D)	34. (B)	35. (C)
36. (E)	37. (C)	38. (D)	39. (B)	40. (A)
41. (C)	42. (A)	43. (C)	44. (B)	45. (D)
46. (A)	47. (C)	48. (A)	49. (D)	50. (B)
51. (A)	52. (C)	53. (D)	54. (B)	55. (D)
56. (B)	57. (B)	58. (A)	59. (C)	60. (C)
61. (D)	62. (A)	63. (C)	64. (E)	65. (E)
66. (A)	67. (A)	68. (A)	69. (D)	70. (C)
71. (E)	72. (C)	73. (D)	74. (D)	75. (B)
76. (D)	77. (B)	78. (C)	79. (B)	80. (D)
81. (D)	82. (E)	83. (A)	84. (E)	85. (D)
86. (A)	87. (B)	88. (C)	89. (A)	90. (D)
91. (A)	92. (B)	93. (C)	94. (A)	95. (B)
96. (A)	97. (A)	98. (C)	99. (D)	100. (A)
101. (C)	102. (B)	103. (D)	104. (A)	105. (C)
106. (A)	107. (C)	108. (A)	109. (B)	110. (A)
111. (A)	112. (A)	113. (C)	114. (B)	115. (D)
116. (A)	117. (A)	118. (B)	119. (D)	120. (A)
121. (A)	122. (C)	123. (B)	124. (B)	125. (A)

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